

# SGA 5 — Exposé I

## Dualizing Complexes

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### Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of “continuous” coefficients the existence of dualizing complexes presents no difficulties, in the case of “discrete” coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne’s biduality theorem (SGA 4<sup>1/2</sup>, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as “exchangers of functors.”

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a “good” regular prescheme  $X$ , for example on an excellent noetherian regular prescheme of characteristic zero<sup>1</sup>, the complex  $(\mathbb{Z}/n\mathbb{Z})_X$  is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a “good” prescheme  $X$  endowed with a closed point  $x$  whose residue field is separably closed, and for a constructible sheaf  $F$ , one has a perfect pairing between  $\underline{H}_x^i(F')$  and  $\underline{H}_x^{-i}(F)$ , where  $F'$  is the dual of  $F$ , i.e.  $F' = R\underline{\mathrm{Hom}}(F, K_X)$ , where  $K_X$  is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex  $(\mathbb{Z}/n\mathbb{Z})_X$  is dualizing ( $n$  prime to the residual characteristics).

## 1 Definition and formal properties of dualizing complexes

We fix a coefficient ring  $A = \mathbb{Z}/n\mathbb{Z}$ . If  $X$  is a prescheme, we denote by  $A_X$  the constant sheaf on  $X_{\text{ét}}$  with value  $A$ , and by  $D(X)$  the derived category of the category of  $A_X$ -modules. We denote, on the other hand, by  $D_c(X)$  the full triangulated subcategory of  $D(X)$  formed by the complexes  $F$  such that  $H^i(F)$  is a constructible  $A_X$ -module for every  $i$ . We put  $D_c^*(X) = D^*(X) \cap D_c(X)$ , where  $*$  =  $\emptyset$ ,  $+$ ,  $-$ , or  $b$ .

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<sup>1</sup>This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the “purity theorem.”

### Definition 1.1

An  $A_X$ -module  $I$  is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible*  $A_X$ -module  $F$  and every  $i > 0$ .

### Remarks 1.2

- a) Contrary to what happens in the case of “continuous” coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let  $X = \mathrm{Spec}(\mathbb{R})$  and  $A = \mathbb{Z}/2\mathbb{Z}$ : the sheaf  $A_X$  is quasi-injective, but is not injective, nor even of finite injective dimension, since  $H^*(X; A_X) = H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$  is a polynomial algebra on one generator of dimension 1.
- b) Let  $I$  be a quasi-injective  $A_X$ -module. It is false in general that the relation  $\underline{\mathrm{Ext}}^i(F, I) = 0$  for  $i > 0$ , valid for constructible  $F$ , is valid for every  $F$ . Indeed, take  $A = \mathbb{F}_\ell$ . Let  $k$  be a field,  $\bar{k}$  a separable closure of  $k$ , and  $f : \bar{x} = \mathrm{Spec}(\bar{k}) \rightarrow X = \mathrm{Spec}(k)$  the canonical morphism, and suppose that, for every connected étale covering  $X' \rightarrow X$ , there exists a connected étale covering  $X'' \rightarrow X'$  such that  $[X'' : X']$  is divisible by  $\ell$  (take, for example,  $k = \mathbb{Q}$ ). Consider the exact sequence

$$(*) \quad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow  $\varepsilon$ . In view of the hypothesis, the section of  $\mathrm{Ext}^1(F, A_X)$  defined by  $(*)$  does not vanish above any  $U$  étale over  $X$ ; hence  $\underline{\mathrm{Ext}}^1(F, A_X)_{\bar{x}} \neq 0$ . But one checks trivially that  $A_X$  is quasi-injective.

### Proposition 1.3 (cf. [H] 1.7.6)

Let  $X$  be a prescheme and  $N \in \mathbb{Z}$ . The following conditions on  $K \in \mathrm{ob} D^+(X)$  are equivalent:

- (i)  $K$  is isomorphic, in  $D(X)$ , to a bounded complex  $I$  such that  $I^i$  is quasi-injective for every  $i$  and zero for  $i > N$ ;
- (ii) for every  $F \in \mathrm{ob} D_c^b(X)$  such that  $H^i(F) = 0$  for  $i < 0$ , one has  $\underline{\mathrm{Ext}}^i(F, K) = 0$  for  $i > N$ ;
- (iii) for every constructible  $A_X$ -module  $F$ , one has  $\underline{\mathrm{Ext}}^i(F, K) = 0$  for  $i > N$ .

*Proof.* (i)  $\Rightarrow$  (ii). Let  $F \in \mathrm{ob} D_c^b(X)$  be such that  $H^i(F) = 0$  for  $i < 0$ , and let  $I \in \mathrm{ob} D^+(X)$  be bounded and such that  $I^i$  is quasi-injective for every  $i$  and zero for  $i > N$ ; we show that  $\underline{\mathrm{Ext}}^i(F, I) = 0$  for  $i > N$ . Arguing by induction on the number of indices  $i$  such that  $H^i \neq 0$ , one is reduced, by dévissage, to the case where  $N = 0$  and  $I$  is concentrated in degree 0. One then has a biregular spectral sequence

$$E_2^{pq} = \underline{\mathrm{Ext}}^p(H^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since  $I$  is quasi-injective,  $E_2^{pq} = 0$  for  $p > 0$  and every  $q$ , and hence one has  $E_2^{0i} = \underline{\mathrm{Hom}}(H^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$ , which gives the conclusion.

(ii)  $\Rightarrow$  (iii) is trivial.

(iii)  $\Rightarrow$  (i) We may suppose that  $K$  is bounded below and has injective components. The hypothesis, applied to  $F = A_X$ , implies that  $H^i(K) = 0$  for  $i > N$ . We may therefore replace  $K$  by an isomorphic complex  $K'$ , bounded, and such that  $K'^i$  is injective for  $i < N$  and zero for  $i > N$ . It is then observed that  $K'^{!N}$  is quasi-injective, which completes the proof.

### Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis “ $F \in \text{ob } D_c^b(X)$ ” is replaced by “ $F \in \text{ob } D_c^+(X)$ ”.

### Definition 1.4

The *quasi-injective dimension* of  $K$ , denoted  $\dim . q. inj(K)$ , is the lower bound (in  $\overline{\mathbb{R}}$ ) of the integers  $N$  satisfying the equivalent conditions of (1.3).

If  $K$  is of finite quasi-injective dimension, the functor  $R \underline{\text{Hom}}(\_, K) : D(X) \rightarrow D(X)$  induces a functor  $D_c^b(X) \rightarrow D^b(X)$ .

We shall see below that if  $X$  is smooth of dimension  $d$  over a field  $k$  and  $n$  is prime to the characteristic of  $k$ , then  $\dim . q. inj(A_X) = 2d$ .

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

### Proposition 1.5

Let  $X$  be a locally noetherian prescheme and  $K \in \text{ob } D^+(X)$ . Suppose that there exists an integer  $N$  such that  $\text{cd}_{\mathbf{n}}(X) \leq N$ , where  $\mathbf{n}$  is the set of prime divisors of  $n$  (notation of SGAA X 1). Then, for  $N' \in \mathbb{Z}$ , one has

$$\dim . q. inj(K) \leq N' \implies \dim . inj(K) \leq N + N'.$$

We shall use, in the proof, the following

### Lemma 1.5.1

Let  $C$  be an abelian category satisfying (AB5) and possessing a family of generators  $(U_i)_{i \in I}$ . Let  $K \in \text{ob } D^+(C)$  and  $N \in \mathbb{Z}$ . The following conditions are equivalent:

- (i)  $\dim . inj(K) \leq N$ ;
- (ii) for every  $i \in I$  and every quotient  $F$  of  $U_i$ , one has

$$\text{Ext}^n(F, K) = 0 \quad \text{for } n > N.$$

*Proof.* It suffices to prove (ii)  $\Rightarrow$  (i). By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if  $K' \in \text{ob}(C)$  is such that  $\text{Ext}^1(F, K') = 0$  every time  $F$  is a quotient of a  $U_i$ , then  $K'$  is injective. In other words, one has to prove that, if every morphism from a subobject

$V$  of a  $U_i$  into  $K'$  extends to  $U_i$ , then  $K'$  is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

*Proof of (1.5).* Suppose  $\dim .q.inj(K) \leq N'$  and prove  $\dim .inj(K) \leq N + N'$ . Since the category of  $A_X$ -modules admits as generators the  $A_{U,X}$  where  $U$  is étale of finite type over  $X$ , it suffices, by (1.5.1), to prove that  $\text{Ext}^i(X; F, K) = 0$  for  $i > N + N'$  when  $F$  is a quotient of such an  $A_{U,X}$ , hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\text{Ext}}^q(F, K)) \Rightarrow \text{Ext}^*(X; F, K).$$

The hypothesis  $\text{cd}_n(X) \leq N$  (resp.  $\dim .q.inj(K) \leq N'$ ) implies  $E_2^{pq} = 0$  for  $p > N$  (resp.  $q > N'$ ). Hence  $E_2^{pq} = 0$  for  $p + q > N + N'$ , whence the conclusion.

### Remark 1.5.2

The hypothesis of (1.5) is verified (with  $N = 2d$ ) if  $X$  is a prescheme of finite type and dimension  $d$  over a separably closed field  $k$ , with  $n$  prime to  $\text{char}(k)$  (SGAA X 4).

### Proposition 1.6

Let  $f : X \rightarrow Y$  be a separated quasi-finite morphism of noetherian preschemes, and let  $K \in \text{ob } D^+(Y)$ . Then, for  $N \in \mathbb{Z}$ , one has

$$\dim .q.inj(K) \leq N \quad \implies \quad \dim .q.inj(R^!f(K)) \leq N.$$

*Proof.* According to the “Main theorem” (EGA IV 8.12.6), there exists a factorization of  $f$  as  $f = uf'$ , where  $f'$  is an open immersion and  $u$  is a finite morphism. We have  $R^!f = R^!f' R^!u$ , which reduces us to examining separately the case of an open immersion and that of a finite morphism. If  $f$  is an open immersion, the assertion is trivial (since every constructible sheaf on  $X$  is then induced by a constructible sheaf on  $Y$ ). Suppose therefore that  $f$  is a finite morphism, and let  $F$  be a constructible  $A_X$ -module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_* \underline{\text{Ext}}^i(F, R^!f(K)) \xrightarrow{\sim} \underline{\text{Ext}}^i(f_* F, K).$$

Since  $f_*(F)$  is constructible (SGAA IX 2.14), it follows that, if  $\dim .q.inj(K) \leq N$ , then  $f_* \underline{\text{Ext}}^i(F, R^!f(K)) = 0$  for  $i > N$ , hence  $\underline{\text{Ext}}^i(F, R^!f(K)) = 0$  for  $i > N$  (SGAA VIII 5.5), as required.

Let now  $X$  be a prescheme, and let  $K \in \text{ob } D^+(X)$ . Denote by  $\underline{D}_K$  the functor  $R \underline{\text{Hom}}(-, K)$ . We shall define, for  $F \in \text{ob } D(X)$ , a functorial homomorphism  $F \rightarrow \underline{D}_K \underline{D}_K F$  (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that  $K$  is formed of injectives, which allows us to “remove the  $R$ ”. The identity morphism  $\underline{\text{Hom}}(F, K) \rightarrow \underline{\text{Hom}}(F, K)$  defines, by the formula dear to Cartan, a homomorphism  $F \otimes \underline{\text{Hom}}(F, K) \rightarrow K$ , whence, by a second application of the said formula, a homomorphism  $F \rightarrow \underline{\text{Hom}}(\underline{\text{Hom}}(F, K), K)$ , which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair  $(F, K)$  is *bidualizing*, or *satisfies biduality*, or again that  $F$  is *reflexive* relative to  $K$ , if the canonical homomorphism  $F \rightarrow \underline{D}_K \underline{D}_K F$  is an isomorphism.

**From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed lo**

### Definition 1.7

Let  $X$  be a prescheme. A complex  $K \in \text{ob } D^+(X)$  is said to be *dualizing* if the following conditions are satisfied:

- (i)  $K$  is of finite quasi-injective dimension;
- (ii) for every  $F \in \text{ob } D_c^-(X)$ , one has  $\underline{D}_K(F) \in \text{ob } D_c^+(X)$ ;
- (iii) every  $F \in \text{ob } D_c^b(X)$  is reflexive relative to  $K$ .

### Remark 1.8

- a) By a standard dévissage (truncating  $F$  and using the fact that  $D_c(X)$  is triangulated), one sees that condition (ii) is equivalent to  
(ii bis): for every constructible  $A_X$ -module  $F$  and every  $i \in \mathbb{Z}$ ,  $\underline{\text{Ext}}^i(F, K)$  is constructible.
- b) Condition (ii bis) implies  $K \in \text{ob } D_c^+(X)$ . Contrary to the example of coherent sheaves (EGA 0<sub>III</sub> 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the “good” cases, for example when one has on  $X$  resolution of singularities and purity (in particular if  $X$  is an excellent prescheme of characteristic zero).
- c) By dévissage on  $F$ , one sees that condition (iii) is equivalent to (iii bis): every constructible  $A_X$ -module  $F$  is reflexive relative to  $K$ .
- d) If  $K \in \text{ob } D^+(X)$  satisfies (i) and (ii) (resp. is dualizing), the functor  $\underline{D}_K$  induces a functor (resp. an equivalence of categories)  $(D_c^b(X))^\circ \rightarrow D_c^b(X)$ .
- e) If  $K \in \text{ob } D^+(X)$  is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every  $F \in \text{ob } D_c(X)$ , one has  $\underline{D}_K(F) \in D_c(X)$ , and  $\underline{D}_K$  induces a functor  $(D_c^+(X))^\circ \rightarrow D_c^-(X)$ .  
If moreover  $K$  is dualizing, every  $F \in \text{ob } D_c(X)$  is reflexive relative to  $K$ , and  $\underline{D}_K$  induces equivalences of categories

$$(D_c(X))^\circ \xrightarrow{\sim} D_c(X), \quad (D_c^-(X))^\circ \xrightarrow{\sim} D_c^+(X), \quad (D_c^+(X))^\circ \xrightarrow{\sim} D_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that  $K$  be of finite injective dimension.

We now turn to the “exchange formulas.” We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9, and also SGA 6, I, 5. Retain only the following point: if  $X$  is a prescheme and  $G \in \text{ob } D^-(X)$ , the condition  $\underline{\text{Tor}}_i(F, G) = 0$  for  $i > N$  and all  $A_X$ -modules  $F$  is equivalent to the same condition with  $F$  constructible, by SGA A, Exposé IX, 2.9 and the commutation of tensor product with filtered inductive limits. Thus there is no need to introduce a notion of “quasi-tor-dimension.”

### Lemma 1.9, cf. [H] II.4.3.

- a) Let  $X$  be a prescheme and let  $F, G \in \text{ob } D_c^-(X)$ . One has  $F \otimes^L G \in \text{ob } D_c(X)$  if  $F \in \text{ob } D_c^-(X)$ , or if  $G$  has finite tor-dimension.

b) Let  $f : X \rightarrow Y$  be a morphism of preschemes and let  $G \in \text{ob } D_c^-(Y)$ . Then  $f^*(G) \in \text{ob } D_c(X)$ .

*Proof.* Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on  $(F, G)$  implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\text{Tor}}_{-p}(\mathbf{H}^{q_1}(F), \mathbf{H}^{q_2}(G)) \Rightarrow \mathbf{H}^*(F \otimes^{\mathbf{L}} G)$$

is biregular. Hence one is reduced to the case where  $F$  and  $G$  are concentrated in degree 0, that is, are constructible  $A_X$ -modules. The question is local, so one may suppose  $X$  noetherian. Then  $F$  admits, by SGA A, Exposé IX, 2.7, a left resolution  $F'$  in which the  $F'^i$  are of the form  $A_{U,X}$  with  $U$  étale and of finite type over  $X$ . We have

$$F \otimes^{\mathbf{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the  $A_{U,X} \otimes G$  are constructible.  $\square$

**Notation 1.10.** If  $X$  is a prescheme, we denote by  $D^b(X)_{\text{torf}}$ , respectively  $D^b(X)_{\text{qinj}}$ , the full subcategory of  $D(X)$  consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

**Proposition 1.11, cf. [H] V.2.6.** Let  $X$  be a prescheme and  $K \in \text{ob } D^+(X)$ .

a) For  $F, G \in \text{ob } D^-(X)$ , respectively for  $F \in \text{ob } D(X)$  and  $G \in \text{ob } D^b(X)_{\text{torf}}$ , there is a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathbf{L}} G) \xrightarrow{\sim} \mathbf{R} \underline{\text{Hom}}(F, \underline{D}_K(G)).$$

b) If  $K$  satisfies conditions (i) and (ii) of Definition 1.7, then

$$G \in \text{ob } D_c^b(X)_{\text{torf}} \implies \underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{qinj}}$$

and  $\underline{D}_K(G)$  satisfies condition 1.7(ii).

c) If  $K$  is dualizing, then for  $F \in \text{ob } D_c^-(X)$  and  $G \in \text{ob } D_c^b(X)$ , respectively for  $F \in \text{ob } D(X)$  and  $G \in \text{ob } D_c^b(X)$  with  $\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{torf}}$ , there is a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathbf{L}} \underline{D}_K(G)) \xrightarrow{\sim} \mathbf{R} \underline{\text{Hom}}(F, G).$$

Moreover,

$$\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{torf}} \implies G \in \text{ob } D_c^b(X)_{\text{qinj}},$$

and  $G$  satisfies condition 1.7(ii).

d) If  $K$  is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

*Proof.* Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let  $G \in \text{ob } D_c^b(X)_{\text{torf}}$ . If  $F$  is a variable constructible  $A_X$ -module, then the cohomology objects of  $F \otimes^{\mathbf{L}} G$  are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent

of  $F$ . The same therefore holds for the cohomology objects of  $\underline{D}_K(F \otimes^{\mathbf{L}} G)$ , and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to  $\underline{D}_K(G)$  in place of  $G$ , together with the reflexivity of  $G$ . The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that  $\underline{D}_K(G) \in \text{ob } D_c^b(X)_{\text{qinj}}$  and satisfies condition 1.7(ii). Then, as  $F$  varies through constructible modules, the cohomology objects of  $\underline{D}_K(F \otimes^{\mathbf{L}} G)$  are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of  $\underline{D}_K \underline{D}_K(F \otimes^{\mathbf{L}} G)$ ; but

$$F \otimes^{\mathbf{L}} G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^{\mathbf{L}} G)$$

because  $K$  is dualizing and of finite injective dimension, cf. 1.8(d). Hence  $G$  has finite tor-dimension. The converse implication in (c) is proved analogously.  $\square$

**Remarks 1.11.1.**

- a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that  $K$  be dualizing.
- b) One expresses the preceding proposition pictorially by saying that the dualizing functor  $\underline{D}_K$  exchanges the functors  $\otimes^{\mathbf{L}}$  and  $\mathbf{R}\underline{\text{Hom}}$ , as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when  $K$  has finite injective dimension.

**Proposition 1.12.** Let  $Y$  be a noetherian prescheme, let  $f : X \rightarrow Y$  be a morphism of finite type, and let  $K \in \text{ob } D^+(Y)$ . Suppose that  $n$  is prime to the residual characteristics of  $Y$ , and put

$$K_X = \mathbf{R}^! f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

- a) For  $F \in \text{ob } D^-(X)$  there is a functorial isomorphism

$$\mathbf{R}f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathbf{R}f_*(F). \quad (\text{i})$$

If, moreover,  $K_X$  and  $K_Y$  are dualizing,<sup>2</sup> then for  $F \in \text{ob } D_c^b(X)$  there is a functorial isomorphism

$$\mathbf{R}f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathbf{R}f_*(F). \quad (\text{ii})$$

- b) For  $F \in D_c^-(Y)$  there is a functorial isomorphism<sup>3</sup>

$$\mathbf{R}^! f \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X f^*(F). \quad (\text{i})$$

If, moreover,  $K_X$  and  $K_Y$  are dualizing, then for  $F \in D_c^b(Y)$  there is a functorial isomorphism

$$f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \mathbf{R}^! f F. \quad (\text{ii})$$

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<sup>2</sup>We shall see below that under fairly general conditions  $K_Y$  dualizing implies  $K_X$  dualizing.

<sup>3</sup>If the functor  $\mathbf{R}^! f$  has finite cohomological dimension, as for example when  $Y$  is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for  $F \in \text{ob } D_c(Y)$ .

*Proof.* The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying  $\underline{D}_Y$  to both sides of (a)(i), written with argument  $\underline{D}_X F$ , and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that  $Rf_*(\underline{D}_X(F)) \in \text{ob } D_c(Y)$ .

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying  $\underline{D}_X$  to both sides of (b)(i), written with argument  $\underline{D}_Y F$ , and using 1.9(b).  $\square$

**Scholium 1.12.1.** In figurative language, the dualizing functors  $\underline{D}_X$  and  $\underline{D}_Y$  exchange the usual and unusual direct images,  $Rf_*$  and  $Rf_!$ , and also the usual and unusual inverse images,  $f^*$  and  $R^!f$ .

**Corollary 1.13.** Under the hypotheses of Proposition 1.12, suppose in addition that  $f$  is proper, and let  $F \in \text{ob } D^-(X)$ . If  $(F, K_X)$  is bidualizing, then  $(Rf_*F, K_Y)$  is bidualizing. Conversely, if  $f$  is finite and  $(Rf_*F, K_Y)$  is bidualizing, then  $(F, K_X)$  is bidualizing.

*Proof.* Suppose  $(F, K_X)$  is bidualizing. Applying  $Rf_*$  to the canonical isomorphism  $F \xrightarrow{\sim} \underline{D}_X^2 F$ , one obtains

$$\begin{aligned} Rf_*F &\xrightarrow{\sim} Rf_*\underline{D}_X\underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y Rf_*\underline{D}_X F && \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y \underline{D}_Y Rf_*F && \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose  $f$  finite and  $(f_*F, K_Y)$  bidualizing, since  $Rf_* = f_*$  for  $f$  finite. We prove that the canonical morphism  $F \rightarrow \underline{D}_X^2 F$  is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_*F \longrightarrow f_*\underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_*F \longrightarrow \underline{D}_Y^2 f_*F,$$

which completes the proof.  $\square$

**Lemma 1.14.** Let  $f : X \rightarrow Y$  be a finite morphism of preschemes.

- a) For a morphism  $u : F \rightarrow G$  in  $D(X)$ ,  $u$  is an isomorphism if and only if  $f_*u$  is an isomorphism.
- b) For  $F \in \text{ob } D(X)$ , one has  $F \in \text{ob } D_c(X)$  if and only if  $f_*F \in \text{ob } D_c(Y)$ .

*Proof.* For (a), since  $f_*$  is exact, one may restrict to the case where  $F$  and  $G$  are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of  $f_*F$  and  $f_*G$ , SGA A, Exposé VIII, 5.5.

For (b), again one may restrict to the case where  $F$  is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency:  $f_*F$  constructible implies  $F$  constructible.

The question is local upstairs, hence a fortiori downstairs, so we may suppose  $Y$  and hence  $X$  affine. By EGA IV, 17.16.6, there exists a finite family  $(Y_i)$  of affine subschemes of  $Y$ , pairwise disjoint and with union  $Y$ , such that, if  $X_i = X \times_Y Y_i$ , the induced morphism  $f_i : X_i \rightarrow Y_i$  factors as

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{v_i} Y_i,$$

where  $u_i$  is finite radicial and surjective and  $v_i$  is finite étale. Put  $F|_{X_i} = F_i$ . By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F_i) \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for  $(f_i, F_i)$ . Put  $F'_i = u_{i*}(F_i)$ , so that  $f_{i*}(F_i) = v_{i*}(F'_i)$ . It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that  $F_i$ , respectively  $F'_i$ , is constructible if and only if  $F'_i$ , respectively  $v_{i*}(F'_i)$ , is constructible. This proves the claim.  $\square$

**Corollary 1.15.** Let  $f : X \rightarrow Y$  be a separated quasi-finite morphism of noetherian preschemes. Let  $K_Y \in \text{ob } D^+(Y)$ , and put, as in Proposition 1.12,

$$K_X = R^!f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y}.$$

If  $K_Y$  is dualizing, then  $K_X$  is dualizing.

*Proof.* We already know, by Proposition 1.6, that if  $K_Y$  has finite quasi-injective dimension, then the same is true of  $K_X$ . We show that if  $K_Y$  satisfies condition (ii), respectively condition (iii), of Definition 1.7, then  $K_X$  satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6,  $f$  factors as  $f = f'i$ , where  $i$  is an open immersion and  $f'$  is finite. Since

$$R^!f \simeq R^!i R^!f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If  $f$  is an open immersion, the assertion is trivial, since a constructible sheaf on  $X$  is induced by a constructible sheaf on  $Y$ , and formation of  $R\text{Hom}$  commutes with restriction to an open.

Suppose then that  $f$  is finite. It follows from Corollary 1.13 that if  $K_Y$  satisfies (iii), then  $K_X$  satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if  $K_Y$  satisfies (ii), then  $K_X$  satisfies (ii). Let  $F \in \text{ob } D_c^-(X)$ . We must show that  $\underline{D}_X(F) \in \text{ob } D_c^+(X)$ . By the finiteness theorem,  $f_*F \in \text{ob } D_c^-(Y)$ . Hence, by the hypothesis on  $K_Y$ ,

$$\underline{D}_Y f_*F \in \text{ob } D_c^+(Y).$$

But

$$\underline{D}_Y f_*F \xrightarrow{\sim} f_*\underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof.  $\square$

## 2. Uniqueness of the dualizing complex

In this section we suppose that  $A$  is local, that is,

$$A = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where  $\ell$  is prime.

**Theorem 2.1, cf. [H] V.3.1.** Let  $X$  be a connected locally noetherian prescheme, and let  $K$  be a dualizing complex on  $X$  in the sense of Definition 1.7. Let also  $K' \in \text{ob } D_c^+(X)$ . Then  $K'$  is dualizing if and only if there exist an invertible  $A_X$ -module  $L$  and an integer  $r$ , necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^L L[r].$$

*Proof.* Let  $L$  be an invertible  $A_X$ -module and  $r$  an integer, and put  $L^\vee = \underline{\text{Hom}}(L, A_X)$ . For  $F \in \text{ob } D(X)$  and  $G \in \text{ob } D^+(X)$  there are canonical isomorphisms

$$\text{R } \underline{\text{Hom}}(F \otimes^L L^\vee[-r], G) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G \otimes^L L[r]) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G) \otimes^L L[r]. \quad (*)$$

It follows immediately that, if  $K' \simeq K \otimes^L L[r]$ , then  $K'$  is dualizing.

Moreover, if we write  $\underline{D}_X = \underline{D}$  and  $\underline{D}_{K'} = \underline{D}'$ , then for  $F \in \text{ob } D(X)$  the isomorphism  $(*)$  gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^L L^\vee[-r].$$

Since  $\underline{D}K \simeq A_X$ , one obtains

$$\underline{D}'K \xrightarrow{\sim} L^\vee[-r],$$

which shows that  $L^\vee[r]$ , hence  $L$  and  $r$ , are determined essentially uniquely by  $K'$ .

Conversely, suppose that  $K'$  is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \text{R } \underline{\text{Hom}}(K, K').$$

We shall prove:

- (a)  $K' \xrightarrow{\sim} K \otimes^L P$ ,
- (b)  $P \xrightarrow{\sim} L[r]$  for an invertible  $A_X$ -module  $L$ .

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for  $F \in \text{ob } D_c(X)$  there is an isomorphism

$$F \otimes^L P \xrightarrow{\sim} \underline{D}'\underline{D}F. \quad (**)$$

Applying  $(**)$  to  $F = K$  gives

$$K \otimes^L P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying  $(**)$  to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^L P = \underline{D}'\underline{D}\underline{D}\underline{D}'A_X \xrightarrow{\sim} \underline{D}'\underline{D}'A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. □

**Lemma 2.2, cf. [H] V.3.3.** Let  $X$  be a connected locally noetherian prescheme, and let  $P, P' \in \text{ob } D_c^-(X)$  be such that

$$P \otimes^L P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible  $A_X$ -module  $L$  and an integer  $r$  such that

$$P \xrightarrow{\sim} L[r], \quad P' \xrightarrow{\sim} L^\vee[-r].$$

*Proof.* The proof has two steps.

First suppose that  $X = \text{Spec}(k)$ , with  $k$  separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0<sub>I</sub>, 5.4.3, if  $M$  and  $M'$  are finite-type  $A$ -modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then  $M$  and  $M'$  are invertible and  $M' \simeq M^\vee$ . This is where the hypothesis that  $A$  is local is used.

We now pass to the general case. By the special case just treated, for every  $x \in X$  there exists an integer  $r(x)$  such that

$$P_{\bar{x}} \simeq A_{\bar{x}}[r(x)], \quad P'_{\bar{x}} \simeq A_{\bar{x}}[-r(x)].$$

Equivalently,  $H^i(P_{\bar{x}}) = 0$ , respectively  $H^i(P'_{\bar{x}}) = 0$ , except for  $i = -r(x)$ , respectively  $i = r(x)$ , and the corresponding nonzero cohomology modules are isomorphic to  $A_{\bar{x}}$ . It remains to show that the function  $x \mapsto r(x)$  is locally constant.

Assume this for the moment. Since  $X$  is connected,  $r$  is constant, say equal to  $r_0$ . Then

$$H^i(P) = 0 \ (i \neq -r_0), \quad H^i(P') = 0 \ (i \neq r_0),$$

and

$$H^{-r_0}(P) \otimes H^{r_0}(P') \xrightarrow{\sim} A.$$

Put  $L = H^{-r_0}(P)$  and  $L' = H^{r_0}(P')$ . Let  $u : \bar{x} \rightarrow \bar{y}$  be a specialization morphism, and let

$$u^* : L_{\bar{y}} \rightarrow L_{\bar{x}}, \quad u'^* : L'_{\bar{y}} \rightarrow L'_{\bar{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\bar{y}} \otimes L'_{\bar{y}} & \xrightarrow{\sim} & A_{\bar{y}} \\ \downarrow u^* \otimes u'^* & & \downarrow \\ L_{\bar{x}} \otimes L'_{\bar{x}} & \xrightarrow{\sim} & A_{\bar{x}}. \end{array} \quad (*)$$

Choose a basis  $e$ , respectively  $f$ , of  $L_{\bar{x}}$ , respectively  $L_{\bar{y}}$ , and let  $e'$ , respectively  $f'$ , be the dual basis of  $L'_{\bar{x}}$ , respectively  $L'_{\bar{y}}$ , meaning that  $e \otimes e'$  and  $f \otimes f'$  map to 1. If  $u^*(f) = \lambda e$  and  $u'^*(f') = \lambda' e'$ , the commutativity of  $(*)$  gives  $\lambda\lambda' = 1$ . Hence  $u^*$  is an isomorphism, and  $u'^*$  is the contragredient isomorphism. By SGA A, Exposé IX, 2.13, it follows that  $L$  and  $L'$  are locally constant, hence of finite presentation as  $A_X$ -modules. Consequently, for every  $x \in X$  the canonical map

$$\underline{\text{Hom}}(L, F)_{\bar{x}} \longrightarrow \text{Hom}(L_{\bar{x}}, F_{\bar{x}})$$

is an isomorphism for every  $A_X$ -module  $F$ . Since the geometric fibres of  $L$  are invertible,  $L$  itself is invertible. Similarly  $L'$  is invertible, and  $L' \simeq L^\vee$ . Hence

$$P \simeq L[-r_0], \quad P' \simeq L^\vee[r_0].$$

It remains only to prove that  $r$  is locally constant. For  $m \in \mathbb{Z}$ , the condition  $r(x) = m$  is equivalent to

$$H^m(P')_{\bar{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function  $r$  is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that  $X$  is local. Let  $x$  be its closed point and  $U$  the open complement.

We prove that  $r$  is constant. By induction on the dimension of  $X$ , we may already suppose that  $r(y)$  is constant and equal to  $r_0$  for  $y \in U$ ; the point is to prove  $r(x) = r_0$ .

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with  $L$  an invertible  $A_U$ -module. After translating degrees we may suppose  $r_0 = 0$ . On the other hand  $r(x) \neq 0$ ; write  $r(x) = a > 0$ . Let

$$i : U \rightarrow X, \quad j : x \rightarrow X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \quad (**)$$

By hypothesis the only nonzero cohomology objects of  $P$  are  $H^0(P)$  and  $H^{-a}(P)$ , and

$$H^0(P) \xrightarrow{\sim} i_!L.$$

It follows that  $(**)$  splits naturally, via the map  $P \rightarrow H^0(P)$ , hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^L P' &\xrightarrow{\sim} i_!(P|_U) \otimes^L P' \oplus j_*j^*P \otimes^L P' \\ &\xrightarrow{\sim} i_!(P|_U \otimes^L P'|_U) \oplus j_*(j^*P \otimes^L j^*P') \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since  $i_!A_U \oplus j_*A_x$  is plainly not isomorphic to  $A_X$ ; already the degree-zero cohomology sheaves are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1.  $\square$

### 3. Existence of dualizing complexes

#### 3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a “good” regular prescheme  $X$ , the sheaf  $A_X$  is dualizing, where  $n$  is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

**3.1.1. Singular locus.** Let  $X$  be a prescheme, locally noetherian. We shall say that  $X$  satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

- (i) for every sub-prescheme  $Y$  of  $X$ , the set  $\text{Reg}(Y)$  of regular points of  $Y$  is open in  $Y$ ;
- (ii) for every closed integral sub-prescheme  $Y$  of  $X$ , the set  $\text{Reg}(Y)$  contains a non-empty open subset of  $Y$ .

The condition *(reg)* is satisfied if  $X$  is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over  $\mathbb{Z}$  (EGA IV, 6.12.6); more generally, it is satisfied if  $X$  is excellent (EGA IV, 7.8.6).

**3.1.2. Dévissage of constructible sheaves.** Let  $X$  be a noetherian prescheme, and let  $C$  be a strictly full subcategory of the category  $\text{Cons}(X)$  of constructible  $A_X$ -modules. Suppose that  $C$  is stable under direct factors and extensions. The following conditions are equivalent:

- (i)  $C = \text{Cons}(X)$ ;
- (ii)  $C$  contains sheaves of the form  $i_!(F)$ , where  $i : Y \rightarrow X$  is the immersion of an integral sub-prescheme and  $F$  is locally constant on  $Y$ ;
- (iii)  $C$  contains sheaves of the form  $f_*i_!(A'_U)$ , where  $i : U \rightarrow Y$  is an open subset of an integral prescheme,  $f : Y \rightarrow X$  is a finite morphism such that the generic point of  $Y$  is separable over its image, and  $A' = \mathbb{Z}/\ell\mathbb{Z}$ , for  $\ell$  a prime divisor of  $n$ .

If  $X$  satisfies *(reg)* (3.1.1), these conditions are again equivalent to the following:

- (i bis) the analogue of (ii), with  $Y$  moreover regular;
- (ii bis) the analogue of (iii), with  $U$  regular.

If, in addition,  $X$  is affine, then in (ii), respectively in (ii bis), one may suppose  $Y$  closed in an open subset of the form  $D(f)$ , with  $f \in \Gamma(X, \mathcal{O}_X)$ .

*Proof.* The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction.  $\square$

**3.1.3. Cohomological dimension.** Let  $X$  be a locally noetherian prescheme. We denote by  $(\text{cdloc})_n$  the following condition. For every sub-prescheme  $Y$  of  $X$ , every geometric point  $\bar{y}$  of  $Y$ , and every

$$f \in \Gamma(\tilde{Y}(\bar{y}), \mathcal{O}_{\tilde{Y}, \bar{y}}),$$

where  $\tilde{Y}(\bar{y})$  denotes the strict localization of  $Y$  at  $\bar{y}$  (SGA A, Exposé VIII, 7.1), one has

$$\text{cd}_{\mathbf{n}}(\tilde{Y}(\bar{y}) - V(f)) \leq \dim \mathcal{O}_{Y, \bar{y}} \quad (= \dim \mathcal{O}_{Y, \bar{y}} \text{ by EGA IV, 6.1.3}),$$

where  $\mathbf{n}$  is the set of prime divisors of  $n$ .

The condition  $(\text{cdloc})_n$  is satisfied if  $X$  is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).<sup>4</sup>

**3.1.4. Absolute cohomological purity.** Let  $X$  be a regular prescheme, and let  $Y$  be a regular closed sub-prescheme of  $X$ , of codimension  $d$  at every point. One says that the pair  $(X, Y)$  satisfies the absolute cohomological purity theorem, relative to  $n$ , if

$$\underline{H}_Y^i(A_X) = 0 \quad (i \neq 2d),$$

and if the “local fundamental class” homomorphism (SGA 4<sup>1/2</sup>, Cycle 2.2)

$$A_Y \longrightarrow \underline{H}_Y^{2d}((\mu_n)_X^{\otimes d})$$

is an isomorphism.

One says that the absolute cohomological purity theorem is true on  $X$  if every pair  $(U, Y)$  as above satisfies it, where  $U$  is an open subset of  $X$ .

Absolute purity is true on  $X$  if  $X$  is smooth over a perfect field of characteristic prime to  $n$  (SGA A, Exposé XVI, 3.9), or if  $X$  is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if  $X$  is regular of dimension 1 (cf. 5.1).<sup>5</sup>

### 3.1.5. Hironaka-style resolution of singularities.

- a) Let  $X$  be a regular prescheme, and let  $D \geq 0$  be a divisor on  $X$ . One says that  $D$  has *strict normal crossings* if there exists a finite family  $(f_i)_{i \in I}$  of elements of  $\Gamma^*(X, \mathcal{O}_X)$  such that

$$D = \sum_{i \in I} \text{Div}(f_i)$$

(EGA IV, 21), and such that, for every  $x \in \text{Supp}(D)$ , the germs  $(f_i)_x$  which lie in  $\mathfrak{m}_x$  form part of a regular system of parameters. One says that  $D$  has *normal crossings* if, locally for the étale topology,  $D$  has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

- b) Let  $X$  be a locally noetherian prescheme. We shall say that  $X$  is *strongly desingularizable* if the following condition is satisfied: for every finite morphism  $f : Y \rightarrow X$ , where  $Y$  is integral and its generic point is separable over its image, and every regular open subset  $U$  of  $Y$ , there exists a proper birational morphism  $h : Y' \rightarrow Y$ , with  $Y'$  regular, such that  $h$  induces an isomorphism from  $U' = h^{-1}(U)$  onto  $U$ , and  $Y' - U'$  is a normal crossings divisor.

Here are cases where  $X$  is strongly desingularizable:

- a)  $X$  is excellent of characteristic zero;
- b)  $X$  is locally noetherian, regular, and of dimension  $\leq 1$ ;

<sup>4</sup>Added in 1977: or, by a recent result of G. Ofer, if  $X$  is excellent, regular, and of dimension 2.

<sup>5</sup>Added in 1977: or, by a recent result of G. Ofer, if  $X$  is excellent, regular, and of dimension 2.

c)  $X$  is of finite type over  $\mathbb{Z}$  or over a perfect field, and of dimension  $\leq 2$ .<sup>6</sup>

**3.1.6. Cohomological finiteness.** Let  $f : X \rightarrow Y$  be a morphism of preschemes, and let  $F \in \text{ob } D_c^+(X)$ . One has  $Rf_*(F) \in \text{ob } D_c^+(Y)$  in the following cases:

- a)  $f$  is proper and of finite presentation (SGA A, Exposé XIV, 1.1);
- b) “when resolution of singularities and the purity theorem are available”, in particular:
  - (i) if  $X$  and  $Y$  are locally of finite type over a field  $k$  of characteristic zero and  $f$  is a  $k$ -morphism of finite type (SGA A, Exposé XVI, 5.1);
  - (ii) if  $Y$  is excellent of characteristic zero and  $f$  is of finite type (SGA A, Exposé XIX, 5.1);
  - (iii) if  $X$  is locally of finite type over a field and  $\dim(X) \leq 2$  (SGA A, Exposé XIX, 5.1).

### 3.2. The quasi-injective dimension of $A_X$

**Proposition 3.2.1.** Let  $X$  be a regular prescheme whose residual characteristics are prime to  $n$ . Suppose that  $X$  satisfies (reg) (3.1.1) and (cdloc) $_n$  (3.1.3), and that absolute purity is true on  $X$  (3.1.4). Then, for every constructible  $A_X$ -module  $F$  and every  $x \in X$ , one has

$$\underline{\text{Ext}}^i(F, A_X)_x = 0 \quad \text{for } i > 2 \dim \mathcal{O}_{X,x}. \quad (*)$$

*Proof.* The question being local, we may suppose  $X$  affine. By dévissage (3.1.2), it is enough to verify (\*) for  $F = i_!(G)$ , where  $i : Y \rightarrow X$  is the immersion of a regular integral sub-prescheme, closed in an open subset  $D(f)$ ,  $f \in A(X)$ , and  $G$  is a locally constant constructible sheaf on  $Y$ . We must compute

$$\underline{\text{Ext}}^q(F, A_X) = H^q(R \underline{\text{Hom}}(F, A_X)).$$

The projection formula for  $i$  (SGA A, Exposé XVIII) gives

$$R \underline{\text{Hom}}(i_! G, A_X) \xrightarrow{\sim} Ri_* R \underline{\text{Hom}}(G, R^! i(A_X)).$$

By purity,

$$R^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes d}[-2d],$$

where  $d = \text{codim}(Y, X)$ . Put  $T = (\mu_n)_Y^{\otimes d}$ . Since  $G$  is locally constant constructible, for every geometric point  $\bar{y}$  of  $Y$  and every  $p$  one has

$$\underline{\text{Ext}}^p(G, T)_{\bar{y}} \xrightarrow{\sim} \text{Ext}^p(G_{\bar{y}}, T_{\bar{y}}).$$

But  $T_{\bar{y}}$  is injective, being isomorphic to  $A = \mathbb{Z}/n\mathbb{Z}$ . Hence  $\underline{\text{Ext}}^p(G, T) = 0$  for  $p \neq 0$ , and

$$R \underline{\text{Hom}}(G, T) \xrightarrow{\sim} \underline{\text{Hom}}(G, T).$$

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<sup>6</sup>Added in 1977: more generally, if  $X$  is excellent noetherian of dimension  $\leq 2$ , by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

Thus

$$\mathrm{R} \underline{\mathrm{Hom}}(F, A_X) \xrightarrow{\sim} \mathrm{R} i_*(\mathcal{H})[-2d], \quad \mathcal{H} = \underline{\mathrm{Hom}}(G, T).$$

Let  $Z$  be the schematic closure of  $Y$  in  $X$  (EGA I, 9.5.10). Then  $Y = Z \cap D(f)$  (EGA 0<sub>I</sub>, 2.1.6). Denote by  $j : Y \rightarrow Z$  and  $k : Z \rightarrow X$  the canonical immersions. We have  $\mathrm{R} i_* = k_* \mathrm{R} j_*$ , so we are reduced to computing  $\mathrm{R} j_*(\mathcal{H})$ . If  $\bar{x}$  is a geometric point of  $Z$ , let  $\tilde{Z} = \mathrm{Spec}(\mathcal{O}_{Z, \bar{x}})$  be the strict localization, let  $\tilde{f}$  be the image of  $f$  in  $\mathcal{O}_{Z, \bar{x}}$ , let  $\tilde{Y} = \tilde{Z} \times_Z Y = \tilde{Z} - V(\tilde{f})$ , and let  $\tilde{\mathcal{H}}$  be the inverse image of  $\mathcal{H}$  on  $\tilde{Y}$ . By SGA A, Exposé VIII, 5.2,

$$\mathrm{R}^q j_*(\mathcal{H})_{\bar{x}} \xrightarrow{\sim} \mathrm{H}^q(\tilde{Y}, \tilde{\mathcal{H}}),$$

hence

$$\underline{\mathrm{Ext}}^q(F, A_X)_{\bar{x}} \xrightarrow{\sim} \mathrm{H}^{q-2d}(\tilde{Y}, \tilde{\mathcal{H}}).$$

The condition  $(\mathrm{cdloc})_n$  implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z, \bar{x}}.$$

Using

$$d = \mathrm{codim}_{\bar{x}}(Z, X) \leq \dim \mathcal{O}_{X, \bar{x}}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z, \bar{x}} = \dim \mathcal{O}_{X, \bar{x}} - \mathrm{codim}_{\bar{x}}(Z, X)$$

(EGA 0<sub>IV</sub>, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\underline{\mathrm{Ext}}^q(F, A_X)_x = 0 \quad (q > 2 \dim \mathcal{O}_{X, x}),$$

as required.  $\square$

**Remark 3.2.2.** The bound obtained in 3.2.1 is the best possible. Indeed, let  $x$  be a point of  $X$ , and let  $Y$  be the closed integral sub-prescheme of  $X$  whose generic point is  $x$ . Since  $X$  satisfies (reg), there exists a regular open subset  $U$  of  $Y$  containing  $x$ . By purity,

$$\underline{\mathrm{Ext}}^{2d}(A_U, A_X)|_U = (\mu_n)_U^{\otimes d},$$

where  $d = \dim \mathcal{O}_{X, x} = \mathrm{codim}(Y, X)$ .

**Corollary 3.2.3.** Let  $X$  be a regular prescheme of dimension  $d$ . If  $X$  is smooth over a field  $k$  of characteristic prime to  $n$ , or if  $X$  is excellent of characteristic zero, then  $X$  satisfies the conclusion of 3.2.1; consequently

$$\dim . q . \mathrm{inj}(A_X) \leq 2d.$$

*Proof.* If  $X$  is excellent of characteristic zero, or if  $k$  is perfect of characteristic prime to  $n$  and  $X$  is smooth over  $k$ , then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when  $X$  is smooth over an arbitrary field  $k$ , as follows immediately after the base change  $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$ , where  $k'$  is a perfect closure of  $k$  (SGA A, Exposé VIII, 1.1).  $\square$

**Lemma 3.2.4.** Let  $X$  be a prescheme satisfying the conclusion of 3.2.1 and of dimension  $\leq d$ . If there exists an integer  $N$  such that  $X$  has cohomological dimension  $\leq N$ , then

$$\dim . \mathrm{inj}(A_X) \leq 2d + N.$$

If, moreover, for every closed subset  $Y$  of  $X$ , the implication

$$\text{codim}(Y, X) \geq p \quad \Rightarrow \quad \text{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim \cdot \text{inj}(A_X) \leq N.$$

*Proof.* The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for  $F$  constructible on  $X$ , the relation

$$\text{Ext}^i(X; F, A_X) = 0 \quad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\text{Ext}}^q(F, A_X)) \Rightarrow \text{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves  $\underline{\text{Ext}}^{2i}(F, A_X)$  and  $\underline{\text{Ext}}^{2i-1}(F, A_X)$  are concentrated in codimension  $\geq i$ . Therefore  $E_2^{p, 2i} = E_2^{p, 2i-1} = 0$  for  $p > N - 2i$ , by the hypothesis. Hence  $E_2^{pq} = 0$  for  $p + q > N$ , and the assertion follows.  $\square$

**Corollary 3.2.5.** If  $X$  is a smooth prescheme of dimension  $d$  over a separably closed field  $k$  of characteristic prime to  $n$ , then

$$\dim \cdot \text{inj}(A_X) = 2d.$$

*Proof.* By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence  $\dim \cdot \text{inj}(A_X) \leq 2d$ . The equality follows from purity: if  $x$  is a closed point of codimension  $d$  in  $X$ , then

$$\text{Ext}^{2d}(X, A_x, A_X) = H_x^{2d}(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

$\square$

### 3.3. Constructibility of the sheaves $\underline{\text{Ext}}^i(F, A_X)$

**Proposition 3.3.1.** Let  $X$  be a prescheme.

a) The following conditions are equivalent:

(i) for  $F \in \text{ob } D_c^-(X)$  and  $G \in \text{ob } D_c^+(X)$ , one has

$$\mathbf{R} \underline{\text{Hom}}(F, G) \in \text{ob } D_c^+(X);$$

(i bis) for every pair  $(F, G)$  of constructible  $A_X$ -modules, one has  $\mathbf{R} \underline{\text{Hom}}(F, G) \in \text{ob } D_c^+(X)$ , i.e. the sheaves  $\underline{\text{Ext}}^i(F, G)$  are constructible for all  $i$ ;

(ii) for every open immersion  $i : U \rightarrow X$  and every constructible  $A_U$ -module  $F$ , the sheaves  $\mathbf{R}^p i_*(F)$  are constructible for all  $p$ ;

(ii bis) for every immersion  $i : Y \rightarrow X$  and every constructible  $A_Y$ -module  $F$ , the sheaves  $\mathbf{R}^p i_*(F)$  are constructible for all  $p$ ;

- (ii ter) for every immersion  $i : Y \rightarrow X$  and every locally constant constructible  $A_Y$ -module  $F$ , the sheaves  $R^p i_*(F)$  are constructible for all  $p$ ;
- (iii) for every finite morphism, respectively every closed immersion,  $f : Y \rightarrow X$ , and every  $F \in \text{ob } D_c^+(X)$ , one has  $R^! f(F) \in \text{ob } D_c^+(Y)$ .

b) These conditions are satisfied in each of the following two cases:

- (i)  $\dim(X) \leq 1$ ;
- (ii) resolution of singularities and purity are available on  $X$  in the sense of 3.1.6(b).

*Proof.* The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis)  $\Rightarrow$  (ii). Let  $\bar{F}$  be a constructible  $A_X$ -module extending  $F$ , for example  $\bar{F} = i_! F$ . By duality (SGA A, Exposé XVIII, 3.1.10),

$$Ri_* F = Ri_* R \underline{\text{Hom}}(A_U, i^! \bar{F}) \xrightarrow{\sim} R \underline{\text{Hom}}(i_! A_U, \bar{F}),$$

and this proves the claim.

(ii)  $\Rightarrow$  (ii bis). Factor  $i$  as a closed immersion followed by an open immersion.

(ii bis)  $\Rightarrow$  (i bis). The question is local, so we may suppose  $X$  affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for  $F = i_! M$ , where  $i : Y \rightarrow X$  is the immersion of a sub-prescheme and  $M$  is a locally constant constructible  $A_Y$ -module. By the projection formula (SGA A, Exposé XVIII),

$$R \underline{\text{Hom}}(i_! M, G) \xrightarrow{\sim} Ri_* R \underline{\text{Hom}}(M, Ri^! G),$$

and by a spectral sequence we are reduced to proving that

$$R \underline{\text{Hom}}(M, Ri^! G) \in \text{ob } D_c^+(Y).$$

First prove  $Ri^! G \in \text{ob } D_c^+(Y)$ . After factoring  $i$ , we may suppose that  $Y$  is closed. Let  $j : U = X - Y$  be the complementary open immersion. Applying  $R \underline{\text{Hom}}(\_, G)$  to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle. Since  $R \underline{\text{Hom}}(A_X, G) = G$ , and

$$R \underline{\text{Hom}}(A_{U,X}, G) \xrightarrow{\sim} Rj_* j^! G$$

belongs to  $D_c^+(X)$  by hypothesis, it follows that

$$R \underline{\text{Hom}}(A_Y, G) \xrightarrow{\sim} i_* Ri^! G$$

belongs to  $D_c^+(X)$ . Resolving  $M$  locally on  $Y$ , for the étale topology, by free  $A_Y$ -modules, another dévissage gives

$$R \underline{\text{Hom}}(M, Ri^! G) \in \text{ob } D_c^+(Y).$$

This proves the implication.

(ii bis)  $\Rightarrow$  (ii ter) is trivial.

(ii ter)  $\Rightarrow$  (ii). Let  $i : U \rightarrow X$  be an open immersion and let  $F$  be a constructible  $A_U$ -module. We must prove  $Ri_*F \in \text{ob } D_c^+(X)$ . The question being local, suppose  $X$  noetherian and argue by noetherian induction on  $X$ . We may suppose  $U \neq \emptyset$ . Then there exists a non-empty open subset  $j : V \rightarrow U$  such that  $j^*F$  is locally constant (SGA A, Exposé IX, 2.4). Let  $M$  be the mapping cylinder of

$$F \longrightarrow Rj_*j^*F.$$

Applying  $Ri_*$  gives a distinguished triangle

$$Ri_*F \longrightarrow Ri_*Rj_*j^*F \longrightarrow Ri_*M \xrightarrow{[1]}.$$

Condition (ii ter) implies

$$Ri_*Rj_*j^*F \xrightarrow{\sim} R(ij)_*j^*F \in \text{ob } D_c^+(X).$$

Consequently,

$$Rj_*j^*F \xrightarrow{\sim} i^*R(ij)_*j^*F \in \text{ob } D_c^+(U).$$

Thus  $M \in \text{ob } D_c^+(U)$ , since  $F \in \text{ob } D_c^+(U)$ . Moreover, the cohomology of  $M$  is supported on a closed subset  $Z$  of  $U$  distinct from  $U$ . Applying the induction hypothesis to  $U \cap \overline{Z} \rightarrow \overline{Z}$ , we deduce  $Ri_*M \in \text{ob } D_c^+(X)$ , and hence  $Ri_*F \in \text{ob } D_c^+(X)$ .

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where  $X$  is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b).  $\square$

### 3.4. The biduality theorem on regular preschemes

**Theorem 3.4.1, cf. [H], V, 2.2.** Let  $X$  be a regular locally noetherian prescheme of finite dimension. Suppose that  $X$  is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and (cdloc) $_n$  (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over  $X$ . Then  $A_X$  is dualizing.

*Proof.* We must prove that  $A_X$  satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible  $A_X$ -module  $F$  we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X \underline{D}_X F, \quad \underline{D}_X = R\underline{\text{Hom}}(\ , A_X),$$

is an isomorphism. The question being local, suppose  $X$  noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_*i_!(A'_U),$$

where  $i : U \rightarrow Y$  is the inclusion of a regular open subset of an integral prescheme,  $f : Y \rightarrow X$  is finite and the generic point of  $Y$  is separable over its image, and  $A' = \mathbb{Z}/\ell\mathbb{Z}$  for a prime divisor  $\ell$  of  $n$ .

Because  $X$  is strongly desingularizable, there is a proper birational morphism  $h : Y' \rightarrow Y$ , with  $Y'$  regular, inducing an isomorphism from  $U' = h^{-1}U$  to  $U$ , and such that  $Y' - U'$  is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} Rh_*i'_!(A'_{U'}),$$

where  $i' : U' \rightarrow Y'$  is the inclusion. Hence, with  $g = fh$ ,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* R h_* i'_!(A'_{U'}) \xrightarrow{\sim} R g_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), R^! g(A_X))$$

is bidualizing. Locally immersing  $Y'$  in a prescheme  $X'$  smooth over  $X$ , and applying purity on  $X'$ , one sees that  $R^! g(A_X)$  is locally isomorphic to  $A_{Y'}$ , up to a translation of degrees. We are therefore reduced to the following special case.

**Lemma 3.4.2.** Let  $X$  be a locally noetherian regular prescheme on which absolute purity is true. Let  $Y = Y_1 \cup \cdots \cup Y_p$  be a strict normal crossings divisor on  $X$ , and put  $U = X - Y$ . Then the pair  $(A'_{U,X}, A_X)$  is bidualizing.

*Proof.* By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that  $(A'_Y, A_X)$  is bidualizing. We argue by induction on  $p$ .

If  $p = 1$ , let  $i : Y \rightarrow X$  be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, R^! i(A_X))$$

is bidualizing. By purity,

$$R^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes 1}[-2].$$

Thus the canonical homomorphism  $A'_Y \rightarrow \underline{D}_Y \underline{D}_Y(A'_Y)$  is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for  $A$ -modules.

Suppose  $p \geq 2$ , and put  $Y' = Y_1 \cup \cdots \cup Y_{p-1}$ . We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis,  $(A'_{Y'}, A_X)$  is bidualizing. By the five lemma, we are reduced to biduality for  $(A'_{(Y-Y'),X}, A_X)$ . But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since  $(A'_{Y_p}, A_X)$  is bidualizing, it remains to prove the same for  $(A'_{Y_p \cap Y'}, A_X)$ . Applying 1.11 again and using purity, we are finally reduced to biduality for this pair on  $Y_p$ , where the induction hypothesis applies.  $\square$

This completes the proof of Theorem 3.4.1.

**Corollary 3.4.3.** Let  $S$  be an excellent noetherian regular prescheme of characteristic zero. Then  $A_S$  is dualizing. Moreover, if  $f : X \rightarrow S$  is a morphism of finite type, then  $R^! f(A_S)$  is dualizing.

*Proof.* The first assertion follows because  $S$  satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that  $R^! f(A_S)$  has finite quasi-injective dimension. Cover  $S$  by finitely many affine open subsets  $S_i$  such that  $X_i = f^{-1}(S_i)$  is a finite union of affine open subsets

$X_{ij}$  immersed as closed subschemes in preschemes  $X'_{ij}$  smooth of finite dimension over  $S_i$ . From 3.2.3 and 1.6 it follows that

$$\dim . q. \operatorname{inj}(\mathbf{R}^! f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on  $X$  for the étale topology, we may suppose that  $f$  factors as  $f = f' i$ , where  $f' : X' \rightarrow S$  is smooth of relative dimension  $d$ , and  $i : X \rightarrow X'$  is a closed immersion. Then

$$\mathbf{R}^! f(A_S) \simeq \mathbf{R}^! i \mathbf{R}^! f'(A_S).$$

But  $\mathbf{R}^! f'(A_S)$  is locally isomorphic to  $A_{X'}$ , up to the translation by  $2d$ , hence is dualizing by the first assertion. By 1.15,  $\mathbf{R}^! f(A_S)$  is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7.  $\square$

**Corollary 3.4.4.** Let  $k$  be a field, and let  $f : X \rightarrow S = \operatorname{Spec}(k)$  be a morphism locally of finite type, with  $\dim(X) \leq 2$ . Then  $\mathbf{R}^! f(A_S)$  is dualizing.

*Proof.* Replacing  $k$  by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose  $k$  perfect. Cover  $X$  by affine open subsets  $X_i$  of finite type over  $k$ . Since  $\dim(X_i) \leq 2$ , the normalization lemma makes  $X_i$  finite over a scheme  $X'_i$  smooth over  $S$  and of dimension  $\leq 2$ . By 1.15 this reduces us to the case where  $f$  is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that  $A_X$  is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1).  $\square$

**Remark 3.4.5.** We shall see below that if  $X$  is regular of dimension  $\leq 1$ , then  $A_X$  is dualizing.

## 4. Local duality

The object of this section is to present the notion of “biduality” developed in the preceding sections in a form closer to geometric intuition.

### 4.1

Let  $X$  be a prescheme, let  $i : x \rightarrow X$  be a closed point whose residue field is separably closed, and let  $K_X$  be a dualizing complex on  $X$  (1.7). We know (1.15) that  $K_{\{x\}} = \mathbf{R}^! i(K_X)$  is a dualizing complex on  $x$ , hence, by (2.1) and (3.4.4), isomorphic to  $A[r]$  for some  $r \in \mathbb{Z}$  (if  $A$  is local). We shall say that  $K_X$  is *normalized at  $x$*  if  $r = 0$  and if an isomorphism  $K_{\{x\}} \xrightarrow{\sim} A$  has been given.

For example, if  $X$  is smooth of relative dimension  $d$  over the spectrum of a separably closed field, the complex  $(\mu_n)_X^{\otimes d}[2d]$  is, by the purity theorem (SGA XVI 3, SGA 4<sup>1/2</sup> Cycle 2.3), canonically normalized at every closed point of  $X$ .

## 4.2

Let  $X$  be a prescheme,  $i : Y \rightarrow X$  a closed sub-prescheme, and  $K_X$  a dualizing complex on  $X$ . Then (1.15)  $K_Y = R^!i(K_X)$  is a dualizing complex on  $Y$ , and one has the *complementary induction formula* (1.12 b) (ii)), for  $F \in \text{ob } D_c^b(X)$ ,

$$(4.2.1) \quad i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y R^!i(F),$$

where  $\underline{D}_X = R \underline{\text{Hom}}(\_, K_X)$  and  $\underline{D}_Y = R \underline{\text{Hom}}(\_, K_Y)$ .

Apply (4.2.1) in particular in the case where  $Y = \{x\}$  is a closed point of  $X$  with separably closed residue field,  $K_X$  being normalized at  $x$  (4.1). Taking into account that  $A_x$  is injective, (4.2.1) is then written

$$(4.2.2) \quad \underline{D}_X(F)_x \xrightarrow{\sim} \text{Hom}^*(R^!i(F), A).$$

Noting that  $R^!i(F)$  is also written, by definition,  $R\Gamma_x(F)$ , one therefore has a perfect pairing in  $D(A)$ :

$$(4.2.2)' \quad \underline{D}_X(F)_x \times R\Gamma_x(F) \longrightarrow A,$$

giving rise to perfect pairings between finite groups:

$$(4.2.2)'' \quad \underline{H}^i(\underline{D}_X(F))_x \times \underline{H}_x^{-i}(F) \longrightarrow A.$$

For example, let  $X$  be a prescheme smooth of dimension  $d$  over a separably closed field, and let  $x \in X$  be a closed point. Then, for  $F \in \text{ob } D_c^b(X)$ , one has a canonical perfect duality between finite groups:

$$(4.2.2)''' \quad \text{Ext}^{2d-i}(F, \mu_n^{\otimes d})_x \times \underline{H}_x^i(F) \longrightarrow A.$$

### Exercise 4.2.3

Let  $X$  be a strictly local scheme (SGA VIII 4.2), with closed point  $x$ , and let  $i : x \rightarrow X$  be the canonical morphism. Let  $K$  be a dualizing complex on  $X$ , normalized at  $x$ . Denote by

$$\mathbf{L} \otimes_{i?} : D^b(A) \longrightarrow D^-(X)$$

the functor defined by

$$\mathbf{L} \otimes_{i?}(M) = M \otimes_A K.$$

(If  $K$  has finite tor-dimension over  $A$ ,  $\mathbf{L} \otimes_{i?}$  extends to a functor from  $D^+(A)$  to  $D(X)$ .)

Show that there exists, for  $L \in D^-(X)$  and  $M \in D^b(A)$ , a canonical functorial isomorphism

$$(*) \quad R \underline{\text{Hom}}(L, \mathbf{L} \otimes_{i?}(M)) \xrightarrow{\sim} R \underline{\text{Hom}}(R^!i(L), M),$$

(assuming of course that  $R^!i : D^-(X) \rightarrow D^-(A)$  is defined ...).

*Hint:* First define a “trace” morphism  $R^!i \otimes_{i?}(M) \rightarrow M$ ; then, to check that  $(*)$  is an isomorphism, reduce by dévissage to the case where  $L \in \text{ob } D_c^b(X)$  and  $M = A$ , and apply (4.2.2). Note that, for  $M = A$ ,  $(*)$  reduces to the perfect pairing (“local duality”, cf. [H] V 6.2)

$$R\Gamma_x(L) \times R \underline{\text{Hom}}(L, K) \longrightarrow A,$$

or

$$\underline{H}_x^i(L) \times \text{Ext}^{-i}(L, K) \longrightarrow A.$$

### 4.3

We shall now generalize (4.2.2) to the case of a geometric point  $\bar{x}$  of  $X$  localized at a point  $x$  which is not necessarily closed. For this we need a few preliminaries.

**Lemma 4.3.** Let  $X$  be a prescheme,  $X'$  the strict localization of  $X$  at a geometric point  $\bar{x}$ , and  $f : X' \rightarrow X$  the canonical map. Then, for  $F \in \text{ob } D_c^-(X)$  and  $G \in \text{ob } D^+(X)$ , the canonical functorial homomorphism (SGA VI I 3.7)

$$f^* \text{R} \underline{\text{Hom}}(F, G) \longrightarrow \text{R} \underline{\text{Hom}}(f^* F, f^* G)$$

is an isomorphism.

*Proof.* As usual, one reduces to  $F$  and  $G$  concentrated in degree 0. Since the question is local in a neighborhood of  $x$ , one may suppose  $X$  affine. Then  $F$  admits (SGA IX 2.7) a left resolution by sheaves of the form  $i_!(A_U)$ , where  $i : U \rightarrow X$  is étale of finite presentation; hence, by dévissage, one is reduced to  $F = i_!(A_U)$ . Form the cartesian square

$$\begin{array}{ccc} U & \xleftarrow{g} & U' \\ i \uparrow & & \uparrow i' \\ X & \xleftarrow{f} & X' \end{array}$$

One has canonical isomorphisms, trivial special cases of duality (SGA XVIII 3.1.10):

$$\text{R} \underline{\text{Hom}}(F, G) \xleftarrow{\sim} \text{R} i_* i^* G, \quad \text{R} \underline{\text{Hom}}(f^* F, f^* G) \xleftarrow{\sim} \text{R} i'_* g^* i^* G.$$

Under these isomorphisms, the canonical arrow

$$f^* \text{R} \underline{\text{Hom}}(F, G) \longrightarrow \text{R} \underline{\text{Hom}}(f^* F, f^* G)$$

is identified with the base-change arrow (SGA XVII)

$$f^* \text{R} i_* i^*(G) \longrightarrow \text{R} i'_* g^* i^*(G).$$

But this is an isomorphism, as one sees at once by passing to the limit (SGA VII 5.11).

**Lemma 4.4.** Let  $X$  be a prescheme, and let  $K \in \text{ob } D_c^+(X)$  be a complex satisfying conditions (i) and (ii) of (1.7). In order that  $K$  be dualizing, it is necessary and sufficient that, for every strict localization  $f : \bar{X}(\bar{x}) \rightarrow X$ ,  $f^* K$  be dualizing.

*Proof.* Use the fact that every constructible sheaf on a strict localization  $\bar{X}(\bar{x})$  is induced by a constructible sheaf on an étale  $\bar{x}$ -pointed scheme over  $X$  (SGA IX 2.7.4), and apply (4.3).

### 4.5

Let  $X$  be a prescheme,  $\bar{x}$  a geometric point of  $X$ , and  $f : X' = \bar{X}(\bar{x}) \rightarrow X$  the corresponding strict localization.

Let  $Y$  be the integral closed sub-prescheme which is the closure of  $x$  in  $X$ ; denote by  $g : \bar{x} \rightarrow Y$  and  $i : Y \rightarrow X$  the canonical maps, and form the cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{i} & X \\ g \uparrow & & \uparrow f \\ \bar{x} & \xrightarrow{i'} & X' \end{array}$$

( $\bar{x}$  is both the closed point of  $X'$  and the strict localization of  $Y$  at  $\bar{x}$ ).

We shall denote by

$$\mathrm{R}\Gamma_{\bar{x}} : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(\bar{x})$$

the functor defined by

$$(4.5.1) \quad \mathrm{R}\Gamma_{\bar{x}} = g^* \mathrm{R}^! i.$$

Using the well-known relation between the functors  $\mathrm{R}^! i$  and  $\mathrm{R}j_* j^*$ , where  $j : U \rightarrow X$  is the open complement of  $Y$ , and applying passage to the limit (SGA VII 5.11), one obtains a canonical isomorphism

$$(4.5.2) \quad \mathrm{R}\Gamma_{\bar{x}} \xrightarrow{\sim} \mathrm{R}^! i' f^*.$$

When  $x$  is a closed point with separably closed residue field, the functor  $\mathrm{R}\Gamma_{\bar{x}}$  coincides with the usual functor  $\mathrm{R}\Gamma_x$  (SGA V 4.3), which justifies the notation.

Let  $K_X \in \mathrm{ob} \mathrm{D}^+(X)$ , and put  $K_Y = \mathrm{R}^! j(K_X)$ ,  $K_{X'} = f^* K_X$ , and  $K_{\bar{x}} = \mathrm{R}\Gamma_{\bar{x}}(K_X)$  (warning:  $K_{\bar{x}}$  is not the fiber of  $K_X$  at  $\bar{x}$ !). Suppose that  $K_X$  is dualizing. Then, by (1.15) and (4.4), the same is true of  $K_{X'}$ ,  $K_Y$ , and  $K_{\bar{x}}$ . Hence (2.1) one has  $K_{\bar{x}} \simeq A[d]$  for some  $d \in \mathbb{Z}$ .

We shall say that  $K_X$  is *normalized at  $\bar{x}$*  if  $d = 0$  and if an isomorphism  $K_{\bar{x}} \simeq A$  has been chosen. In the case where  $x$  is closed with separably closed residue field, this recovers the notion introduced in (4.1).

Let  $K_X$  be a dualizing complex, and put  $\underline{D}_X = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_X)$ ,  $\underline{D}_Y = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_Y)$ , etc. Let  $F \in \mathrm{ob} \mathrm{D}_c^b(X)$ ; let us compute  $(\underline{D}_X F)_{\bar{x}}$ . One has:

$$\begin{aligned} (\underline{D}_X F)_{\bar{x}} &\xrightarrow{\sim} g^* i^* \underline{D}_X F \\ &\xrightarrow{\sim} g^* \underline{D}_Y \mathrm{R}^! i F && \text{(by the complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} g^* \mathrm{R}^! i(F) && \text{(by (4.3)), that is,} \end{aligned}$$

$$(4.5.3) \quad (\underline{D}_X F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}\Gamma_{\bar{x}}(F).$$

Of course, one could have made an analogous computation by passing through  $X'$  instead of  $Y$ ; one would then have obtained a canonical isomorphism

$$(\underline{D}_X F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}^! i' f^*(F),$$

compatible with (4.5.3) by means of the identification (4.5.2).

If moreover  $K_X$  is normalized at  $\bar{x}$ , one obtains perfect pairings generalizing (4.2.2)' and (4.2.2)'':

$$(4.5.3)' \quad \mathrm{H}_Y^i(F)_{\bar{x}} \times \mathrm{R}^{-i} \Gamma_{\bar{x}}(F) \longrightarrow A,$$

$$(4.5.3)'' \quad \mathrm{H}^i(\underline{D}_X(F))_{\bar{x}} \times \mathrm{H}_{\bar{x}}^{-i}(F) \longrightarrow A,$$

where  $H_{\bar{x}}^i = H^i R\Gamma_{\bar{x}}$ .

Conversely, let  $K_X \in \text{ob } D^+(X)$  satisfy conditions (i) and (ii) of (1.7), and suppose that  $K_{\bar{x}} = R\Gamma_{\bar{x}}(K_X)$  is dualizing for every geometric point  $\bar{x}$  of  $X$ . Then, for every  $\bar{x}$ , one can define (cf. 1.12 b) (ii)) a canonical arrow

$$(4.5.3)^{\text{bis}} \quad \underline{D}_X(F)_{\bar{x}} \longrightarrow \underline{D}_{\bar{x}} R\Gamma_{\bar{x}}(F) \quad (F \in \text{ob } D_c^b(X)).$$

If this is an isomorphism for every  $\bar{x}$  and every  $F$ , then  $K_X$  is dualizing. Indeed, it remains to verify that the canonical arrow

$$F \longrightarrow \underline{D}_X \underline{D}_X(F)$$

is an isomorphism for every  $F \in \text{ob } D_c^b(X)$ . At an arbitrary geometric point  $\bar{x}$ , one has

$$\begin{aligned} F_{\bar{x}} &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_{\bar{x}}(F_{\bar{x}}) && (\text{since } K_{\bar{x}} \text{ is dualizing}) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} R\Gamma_{\bar{x}} \underline{D}_X(F) && (\text{by (4.3) and the induction formula 1.12 b) (i)}) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_X(F)_{\bar{x}} && (\text{by hypothesis}), \end{aligned}$$

and therefore, modulo a compatibility check,  $(F, K_X)$  is bidualizing, which proves the assertion. In summary:

**Proposition 4.5.4.** Let  $X$  be a prescheme, and let  $K_X \in \text{ob } D^+(X)$  be a complex satisfying conditions (i) and (ii) of (1.7). In order that  $K_X$  be dualizing, it is necessary and sufficient that, for every geometric point  $\bar{x}$  of  $X$ , the complex  $K_{\bar{x}}$  be dualizing and the canonical arrow  $(4.5.3)^{\text{bis}}$  be an isomorphism for every  $F \in \text{ob } D_c^b(X)$ .

**Remark 4.5.5.** The existence of dualizing complexes should be regarded, in a sense, as an important complement to the global duality theorem. Indeed, the latter expresses that, for a separated morphism of finite type  $f : X \rightarrow Y$  ( $Y$  quasi-compact), one has a canonical isomorphism

$$\underline{D}_Y R_! f(F) \xrightarrow{\sim} Rf_* \underline{D}_X(F)$$

(with the notation of 1.12). Its use for computing  $R_! f(F)$  (when  $F \in \text{ob } D_c^b(X)$ ) requires information about  $\underline{D}_X(F)$ . When  $K_X$  is dualizing, this information is supplied by the isomorphism (4.5.3), which makes possible, at least in theory, a computation of the fibers of  $\underline{D}_X(F)$  as invariants of “purely spatial” cohomology.

**Exercise 4.5.6.** Let  $X$  be a prescheme endowed with a dualizing complex (1.7). The family of functors (cf. (4.5.3))

$$H_{\bar{x}}^i : D_c^b(X) \longrightarrow (A\text{-modules})$$

( $i$  running through  $\mathbb{Z}$  and  $\bar{x}$  through the set of geometric points of  $X$ ) is “conservative”, i.e. a morphism  $F \rightarrow G$  in  $D_c^b(X)$  such that the maps  $H_{\bar{x}}^i(F) \rightarrow H_{\bar{x}}^i(G)$  are isomorphisms, is an isomorphism.

## 4.6

Let  $X$  be a prescheme,  $j : Y \rightarrow X$  a closed sub-prescheme, and  $i : \bar{x} \rightarrow Y$  a geometric point of  $Y$ . Let, on the other hand,  $K_X$  be a dualizing complex on  $X$ , normalized at  $\bar{x}$  (cf. (4.5)). We saw in (4.5) that, if  $Y$  is the closure of  $x$ , one has, for  $F \in \text{ob } D_c^b(X)$ , a perfect pairing

$$R^! j(F)_{\bar{x}} \times \underline{D}_X(F)_{\bar{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Introduce the factorization of  $i$  as

$$\bar{x} \xrightarrow{i'} \{\bar{x}\} \xrightarrow{i''} Y.$$

For  $F \in \text{ob } D_c^b(X)$ , one has canonical isomorphisms:

$$\begin{aligned} R\Gamma_{\bar{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} R\Gamma_{\bar{x}} \underline{D}_Y R^! j(F) && \text{(complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} i'^* R^! i'' \underline{D}_Y R^! j(F) && \text{(definition of } R\Gamma_{\bar{x}} \text{ (4.5.1))} \\ &\xrightarrow{\sim} i'^* \underline{D}_{\bar{x}} i''^* R^! j(F) && \text{(ordinary induction (1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} i'^* R^! j(F) && \text{(by 4.3)), i.e. finally a} \end{aligned}$$

canonical isomorphism

$$(4.6.1) \quad R^! j(F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} R\Gamma_{\bar{x}} j^* \underline{D}_X(F),$$

and hence perfect pairings

$$(4.6.1)' \quad R^! j(F)_{\bar{x}} \times R\Gamma_{\bar{x}} j^* \underline{D}_X(F) \longrightarrow A,$$

$$(4.6.1)'' \quad H_Y^i(F)_{\bar{x}} \times H_{\bar{x}}^{-i}(\underline{D}_X(F)|Y) \longrightarrow A.$$

**Example 4.6.2.** Let  $X$  be a regular prescheme, let  $\bar{x}$  be a geometric point of  $X$ , and put  $d = \dim(\mathcal{O}_{X, \bar{x}})$ . Set  $K_X = (\mu_n)_X^{\otimes d}[2d]$ . If  $X$  satisfies condition (reg) (3.1.1) and the purity theorem (3.1.4), then  $K_X$  is canonically normalized at  $\bar{x}$ . Thus, if  $K_X$  is dualizing, one has perfect pairings

$$H_Y^i(F)_{\bar{x}} \times H_{\bar{x}}^{2d-i}(F'|Y) \longrightarrow A,$$

for every closed subset  $Y$  containing  $x$  and every  $F \in \text{ob } D_c^b(X)$ , with  $F' = R\text{Hom}(F, (\mu_n)_X^{\otimes d})$ .

To finish, let us give a strictly local variant of (4.6.1).

## 4.7

Let  $X$  be a noetherian strictly local scheme (SGA VIII 4.2), with closed point  $x$ . Let  $Y$  be a non-empty closed subset of  $X$ , and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Finally, let  $K_X$  be a dualizing complex normalized at  $x$  (4.1), giving dualizing complexes “corresponding” on  $U, V, Y, x$  (and by hypothesis  $\underline{D}_x = R\text{Hom}(\_, A)$ ):

$$K_V = j^* K_X, \quad K_U = i^* K_V, \quad K_Y = R\Gamma_Y(K_X).$$

Let  $G \in \text{ob } D_c^b(U)$ , and put  $G' = \underline{D}_U(G)$ . We propose to define a natural pairing

$$(4.7.1) \quad R\Gamma(U, G) \times R\Gamma(V, i_! G')[-1] \longrightarrow A,$$

i.e. a functorial homomorphism

$$(4.7.1)' \quad R\Gamma(U, G) \otimes_A^{\mathbf{L}} R\Gamma(V, i_! G')[-1] \longrightarrow A.$$

(Since the ring  $A = \mathbb{Z}/\ell\mathbb{Z}$  has infinite cohomological dimension for  $\ell \geq 2$ , one knows how to define only the tensor products

$$\mathbf{L} \otimes : D^-(A) \times D^-(A) \longrightarrow D^-(A)$$

and

$$\mathbf{L} \otimes : D(A) \times D^b(A)_{\text{torf}} \longrightarrow D(A) \quad (*).$$

We therefore leave it to the reader to make suitable hypotheses ensuring that the two factors of  $\mathbf{L} \otimes$  appearing in (4.7.1)' are in  $D^b(A)$ , or that one of them has finite tor-dimension.)

First note that, by the complementary duality formula (1.12 a) (ii), one has a canonical isomorphism

$$(4.7.2) \quad i_!(G') \xrightarrow{\sim} \underline{D}_V(Ri_*G).$$

On the other hand, one has the ordinary duality isomorphism

$$(4.7.3) \quad R\text{Hom}(A_U, G) \xrightarrow{\sim} R\text{Hom}(A_V, Ri_*G).$$

By the definition of  $\underline{D}_V$  (and independently of the fact that  $K_V$  is dualizing), one has a canonical arrow

$$(4.7.4) \quad Ri_*G \otimes^{\mathbf{L}} \underline{D}_V(Ri_*G) \longrightarrow K_V.$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a (canonical) arrow

$$(4.7.5) \quad R\Gamma(U, G) \otimes_A^{\mathbf{L}} R\Gamma(V, i_!G') \longrightarrow R\text{Hom}(A_V, K_V).$$

But  $K_V = j^*K_X$ , and one has the duality isomorphism (SGA 3.1.10):

$$(4.7.6) \quad R\text{Hom}(A_V, K_V) \xrightarrow{\sim} R\text{Hom}(A_{V,X}, K_X).$$

Finally, the exact sequence

$$(4.7.7) \quad 0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0$$

gives a homomorphism of degree +1:

$$(4.7.8) \quad R\text{Hom}(A_{V,X}, K_X) \longrightarrow R\text{Hom}(A_x, K_X) \xrightarrow{\sim} A.$$

Composing (4.7.5), (4.7.6), and (4.7.8), one obtains the announced arrow (4.7.1)'.

(\*) (added in 1977) *Inexact: in fact it is easy to extend  $\mathbf{L} \otimes : D^-(A) \times D^-(A) \rightarrow D^-(A)$  to  $\mathbf{L} \otimes : D^-(A) \times D(A) \rightarrow D(A)$ .*

We shall now see that the pairing (4.7.1) is “perfect”, i.e. that the arrow (4.7.1)' identifies  $R\Gamma(V, i_!G')[-1]$  with  $\underline{D}_x R\Gamma(U, G)$ . Indeed suppose, as is allowed, that

$$G = (ji)^*F,$$

with  $F \in \text{ob } D_c^b(X)$ . Put  $F' = \underline{D}_X(F)$  and  $F'_{U,X} = (ji)_!G'$ . One has the exact sequence

$$(4.7.9) \quad 0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc}
 & \mathrm{R}\Gamma_x(F'_{U,X}) & \\
 \swarrow & & \nwarrow +1 \\
 \mathrm{R}\Gamma_x(F'|Y) & \xrightarrow{\quad\quad\quad} & \mathrm{R}\Gamma_x(F')
 \end{array} \tag{b}$$

By definition of  $\mathrm{R}\Gamma_x$ , one has

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R}\mathrm{Hom}(A_x, F'_{U,X});$$

hence, using (4.7.7) and the fact that  $(F'_{U,X})_x = \mathrm{R}\Gamma(X, F'_{U,X}) = 0$ , one has a canonical isomorphism

$$\mathrm{R}\mathrm{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But one has

$$\begin{aligned}
 \mathrm{R}\mathrm{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R}\mathrm{Hom}(j_! A_V, j_! i_! G') \\
 &\xrightarrow{\sim} \mathrm{R}\mathrm{Hom}(A_V, i_! G') \quad \text{by duality (SGA 3.1.10),}
 \end{aligned}$$

and hence finally a canonical isomorphism

$$(4.7.10) \quad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

On the other hand one has the exact sequence

$$(4.7.11) \quad 0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc}
 & \mathrm{R}\Gamma(U, F|U) & \\
 \swarrow & & \nwarrow +1 \\
 \mathrm{R}\Gamma_Y(F) & \xrightarrow{\quad\quad\quad} & \mathrm{R}\Gamma_X(F)
 \end{array} \tag{a}$$

By virtue of (4.7.1) and (4.7.10), one has a natural pairing

$$(4.7.12) \quad \mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A.$$

By virtue of (4.6.1), one has a perfect pairing

$$(4.7.13) \quad \mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A.$$

Finally, by the ordinary induction formula (1.12 b) (i)), one has a perfect pairing

$$(4.7.14) \quad \mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A.$$

The reader who has the patience to orient himself in this maze of “canonical arrows” will check that the pairings (4.7.12), (4.7.13), and (4.7.14) are “compatible” with the arrows of the triangles (a) and (b), and consequently that the pairings (4.7.12) and (4.7.1) are perfect.

The pairing of the triangles (a) and (b) gives rise to two exact sequences “transpose to one another”:

$$(4.7.15) \quad \cdots \longrightarrow \mathrm{H}_Y^i(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F) \longrightarrow \cdots$$

$$\cdots \longleftarrow H_x^{-i}(Y, F'|Y) \longleftarrow H_x^{-i}(X, F') \longleftarrow H_x^{-i-1}(V, F'_{U,V}) \longleftarrow \cdots.$$

In summary, we may state:

**Proposition 4.7.16.** Let  $X$  be a noetherian strictly local scheme, with closed point  $x$ . Let  $Y$  be a non-empty closed subset of  $X$ , let  $U = X - Y$ ,  $V = X - x$ , and let  $i : U \rightarrow V$  and  $j : V \rightarrow X$  be the canonical inclusions. Let  $K_X$  be a dualizing complex on  $X$  normalized at  $x$  (hence giving associated dualizing complexes on  $V, U, Y$ ). For  $G \in \text{ob } D_c^b(U)$ , one has a perfect pairing:

$$(*) \quad R\Gamma(U, G) \times R\Gamma(V, i_! G')[-1] \longrightarrow A$$

(where  $G' = \underline{D}_U(G)$ ). If  $G = F|U$ , where  $F \in \text{ob } D_c^b(X)$ , one has a canonical isomorphism

$$(**) \quad R\Gamma(V, i_! G')[-1] \xrightarrow{\sim} R\Gamma_x(F'_{U,X})$$

(where  $F' = \underline{D}_X(F)$  and  $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$ ), and a perfect pairing between the distinguished triangles

$$\begin{aligned} R\Gamma_Y(F) &\longrightarrow R\Gamma(X, F) \longrightarrow R\Gamma(U, F|U) \xrightarrow{+1}, \\ R\Gamma_x(F'_{U,X}) &\longrightarrow R\Gamma_x(F') \longrightarrow R\Gamma_x(F'|Y) \xrightarrow{+1}. \end{aligned}$$

compatible with the pairing  $(*)$  and the isomorphism  $(**)$ , and giving rise to two exact sequences transpose to one another (4.7.15).

**Remark 4.7.17.** In the particular case where  $Y = \{x\}$ , the pairing (4.7.1) is defined under the sole hypothesis that  $K_X$  is normalized at  $x$  (and not necessarily dualizing), i.e. such that  $R\Gamma_x(K_X) \simeq A$  (the isomorphism being given): the isomorphism (4.7.2) (which used biduality) then reduces to the identity. Since the pairing (4.7.14) is in any case perfect, one sees therefore that local duality on  $X$  (in the form (4.2.2)') is equivalent to the “global” duality on  $U = X - x$ , i.e. to the assertion that the pairing (4.7.1)

$$R\Gamma(U, G) \times R\Gamma(U, G')[-1] \longrightarrow A$$

is perfect, or equivalently that the pairings

$$H^i(U, G) \times H^{-i-1}(U, G') \longrightarrow A$$

are perfect.

Suppose that one has on  $X$  a dualizing complex  $K_X$  normalized at  $x$ . If  $X$  is excellent and  $U$  is regular of dimension  $d$  ( $x$  may therefore be an isolated singularity), one should be able to check that  $K_U = K_X|U$  is canonically isomorphic to  $(\mu_n)_U^{\otimes d}[2d]$  (this is easily checked, at any rate, if  $X$  is the strict localization of a prescheme locally of finite type over a field of characteristic 0). Then local duality on  $X$  implies that, for every locally constant constructible  $A_U$ -module  $G$ , one has perfect pairings:

$$H^i(U, G) \times H^{2d-1-i}(U, G^\circ) \longrightarrow A,$$

where  $G^\circ = \text{Hom}(G, (\mu_n)_U^{\otimes d})$ , corresponding formally to Poincaré duality on a scheme of dimension  $2d - 1$ .

## 5. Local duality on curves

**Theorem 5.1.** Let  $X$  be a noetherian regular prescheme of dimension 1. Suppose that  $n$  is prime to the residual characteristics of  $X$ . Then the purity theorem is true on  $X$  (3.1.4), and the complex  $A_X$  is dualizing (1.7).

*Proof.* Suppose purity has been proved. Since  $X$  plainly satisfies condition (reg) (3.1.1), and since condition (cdloc) $_n$  is verified by (SGA X 2.2), it follows from (3.2.1) that  $\dim .q.inj(A_X) = 2$ . On the other hand, condition (ii) of (1.7) is verified by (3.3.1 b) (i)) (which was proved assuming purity). Hence, by (4.4), we are reduced to proving that  $A_X$  is dualizing when  $X$  is strictly local. But, to prove purity, which is a local question in a neighborhood of a closed point, one may equally well suppose  $X$  strictly local. Finally, we are reduced to proving the theorem when  $X$  is strictly local.

Thus suppose that  $X$  is strictly local. Let  $x$  be its closed point, let  $(\pi = \text{car}(k(x)))$ , let  $\eta$  be its generic point, and let  $U = X - x = \text{Spec}(k(\eta))$ . Let  $G = \text{Gal}(k(\bar{\eta})/k(\eta))$  be the fundamental group of  $U$ . It is known [CL, Chap. IV § 2 Exer. 2] that one has a canonical exact sequence

$$(5.1.1) \quad 1 \longrightarrow P \longrightarrow G \longrightarrow H \longrightarrow 1,$$

where  $H = \prod_{\ell \neq \pi} \mathbb{Z}_\ell(1)$ ,  $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^\nu}$ , and  $P$  is a pro- $\pi$ -group. (Of course,  $H$  is isomorphic, but not canonically, to  $\prod_{\ell \neq \pi} \mathbb{Z}_\ell$ .)

It is also known (SGA VIII 2) that the étale cohomology of  $U$  is interpreted as the Galois cohomology of  $G$ , sheaves (respectively constructible sheaves) of  $A_U$ -modules corresponding to  $A$ -modules (respectively finite type  $A$ -modules) on which  $G$  acts continuously.

This being so, the  $H_x^i(A_X)$  are determined by the exact sequence

$$(5.1.2) \quad 0 \longrightarrow H_x^0(A_X) \longrightarrow H^0(X, A_X) \longrightarrow H^0(U, A_U) \longrightarrow H_x^1(A_X) \longrightarrow 0,$$

and the isomorphisms

$$(5.1.3) \quad H_x^i(A_X) \xrightarrow{\sim} H^{i-1}(U, A_U) \quad \text{for } i \geq 2.$$

But trivially  $H^0(X, A_X) = H^0(U, A_U) = A$ , so  $H_x^0(A_X) = H_x^1(A_X) = 0$ .

On the other hand, for a Galois  $G$ - $A$ -module  $M$ , one has the Hochschild–Serre spectral sequence

$$E_2^{pq} = H^p(H, H^q(P, M)) \implies H^{p+q}(G, M).$$

Since  $n$  is prime to  $\pi$  and  $P$  is a pro- $\pi$ -group, one has

$$H^q(P, M) = 0 \quad \text{for } q \neq 0,$$

hence  $E_2^{pq} = 0$  for  $q \neq 0$ , giving a canonical isomorphism

$$(5.1.4) \quad H^p(G, M) \xrightarrow{\sim} H^p(H, M^P).$$

By an analogous argument, one sees moreover that  $H^p(H, N)$  is identified with  $H^p(\widehat{\mathbb{Z}}(1), N)$  when the factor  $\widehat{\mathbb{Z}}(1)$  is made to act trivially on  $N$ . Thus, finally, with the preceding convention, one has a canonical isomorphism

$$(5.1.5) \quad H^p(G, M) \xrightarrow{\sim} H^p(\widehat{\mathbb{Z}}(1), M^P).$$

Now the Galois cohomology of  $\widehat{\mathbb{Z}}$  is well known (CG chap. I p. 31): for a  $G$ - $A$ -module  $N$ , one has

$$(5.1.6) \quad H^i(\widehat{\mathbb{Z}}, N) = 0 \quad \text{for } i \geq 2, \quad H^0(\widehat{\mathbb{Z}}, N) = N^{\widehat{\mathbb{Z}}}, \quad H^1(\widehat{\mathbb{Z}}, N) = N_{\widehat{\mathbb{Z}}}.$$

From (5.1.3), (5.1.5), and (5.1.6) one immediately deduces that

$$H^i(U, A_U) = 0 \quad \text{for } i \geq 2,$$

and

$$H^1(U, A_U) \xrightarrow{\sim} A.$$

This last isomorphism is not canonical, since it depends on the choice of an identification of  $\widehat{\mathbb{Z}}(1)$  with  $\widehat{\mathbb{Z}}$ . But, using the Kummer exact sequence, one finds (SGA XIX 1.3) a canonical isomorphism (given by the “local fundamental class”)

$$(5.1.7) \quad H^1(U, \mu_n) \xrightarrow{\sim} A.$$

Purity is therefore proved.

Let us now prove that  $A_X$  is dualizing. By (4.7.17), it is a matter of showing that, for every locally constant constructible  $A_U$ -module  $M$ , the pairing

$$(5.1.8) \quad H^i(U, M) \times H^{1-i}(U, \text{Hom}(M, A_U)) \longrightarrow H^1(U, A_U) \xrightarrow{\sim} (\mu_n)_x^{\otimes -1}$$

is a perfect duality. We shall interpret this pairing in terms of Galois cohomology. We shall use the following elementary lemma.

**Lemma 5.1.9.** Let  $D = \text{Hom}(\_, A)$  denote the Pontrjagin dualizing functor on the category of finite type  $A$ -modules. Let  $M$  be a finite type  $A$ -module on which a group  $G$  acts  $A$ -linearly.

(i) One has a canonical isomorphism

$$D(M^G) \xrightarrow{\sim} (DM)_G.$$

(ii) If  $G$  is a profinite group acting continuously on  $M$ , and if  $G$  has pro-order prime to  $n$ , then the canonical arrow

$$M^G \longrightarrow M_G$$

is an isomorphism.

*Proof.* By definition one has

$$M^G = \varprojlim_{s \in G} \text{Ker}(M \xrightarrow{s-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varprojlim_{s \in G} \text{Coker}(DM \xrightarrow{s-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

Let us prove (ii). Let  $M'$  be the  $A$ -module defined by the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

One has  $H^1(G, M') = 0$  because  $G$  has pro-order prime to  $n$ . The exact cohomology sequence therefore implies that the arrow  $M^G \rightarrow M_G$  is an epimorphism. But, applying the functor  $D$  and taking (i) into account, one deduces that it is a monomorphism, which completes the proof.

Applying (5.1.9) to  $M$  and  $P$ , one finds that  $\text{Hom}(M, A)^P$  is canonically identified with  $\text{Hom}(M^P, A)$ . If one chooses an isomorphism from  $\widehat{\mathbb{Z}}(1)$  to  $\widehat{\mathbb{Z}}$ , the pairing (5.1.8) is written, taking (5.1.5) into account, as

$$(5.1.8)' \quad H^i(\widehat{\mathbb{Z}}, M^P) \times H^{1-i}(\widehat{\mathbb{Z}}, D(M^P)) \longrightarrow H^1(\widehat{\mathbb{Z}}, A) \xrightarrow{\sim} A.$$

It follows immediately from (5.1.6) and (5.1.9(i)) that (5.1.8)' is a perfect pairing. This completes the proof of (5.1).

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## Appendix

by L. Illusie

The preceding theory extends easily to base Rings more general than the constant Ring  $\mathbb{Z}/n\mathbb{Z}$ . We shall briefly indicate how the theory is written for locally constant base Rings, killed by  $n > 0$ , and whose geometric fibres are noetherian rings. This is, moreover, only a provisional framework<sup>7</sup>: sooner or later one will have to globalize, by taking as “coefficients” on a prescheme  $X$  relative preschemes over the site  $(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z})$ .

### 1. Constructible sheaves

**1.1.** Let  $X$  be a prescheme, and let  $A_X$  be a sheaf of rings on  $X$  (for the étale topology), locally constant, whose geometric fibres are left noetherian rings. We denote by  $A^X$  the category of left

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<sup>7</sup>According to Grothendieck!

$A_X$ -Modules, and by  $D_A(X)$  (or  $D(X)$  if there is no risk of confusion) the corresponding derived category. The notion of a constructible left  $A_X$ -Module is defined as in (SGAA IX 2.3), by replacing the expression “locally constant with value of finite presentation” by “of finite presentation as an  $A_U$ -Module”.

We denote by  $D_c(X)$  the full subcategory of  $D(X)$  consisting of complexes  $E$  such that  $H^i(E)$  is a constructible  $A_X$ -Module for every  $i$ . It is a triangulated subcategory of  $D(X)$ .

**1.2.** Let  $E \in \text{ob } D(X)$ . We say (cf. Exp. II 3.11) that  $E$  is pseudo-coherent if, for every  $i$ ,  $H^i(E)$  is an  $A_X$ -Module of finite presentation (or locally constant constructible, if one prefers). We denote by  $D(X)_{\text{coh}}$  (or  $D_{\text{coh}}(X)$ ) the full subcategory of  $D(X)$  consisting of pseudo-coherent objects. It follows from (SGAA IX 2.1) that this is a triangulated subcategory of  $D_c(X)$ .

**Proposition 1.3** (dévissage lemma). Let  $X$  be a quasi-compact and quasi-separated prescheme (respectively a noetherian prescheme), and let  $C$  be a strictly full subcategory of the category  $\text{Cons}(X)$  of constructible left  $A_X$ -Modules. Suppose that  $C$  is stable under extensions. The following conditions are equivalent:

- (i)  $C = \text{Cons}(X)$ ;
- (ii)  $C$  contains the sheaves of the form  $i_!(F)$ , where  $i : Y \rightarrow X$  is a sub-prescheme of finite presentation (respectively an integral sub-prescheme), and  $F$  is locally constant constructible on  $Y$ .

If  $A_X$  is constant with value  $A$ , a left noetherian ring killed by an integer  $n > 0$ , the preceding conditions are equivalent to:

- (iii)  $C$  is stable under direct factors and contains the sheaves of the form  $f_*i_!(M_U)$ , where  $i : U \rightarrow Y$  is an open immersion (respectively an open subset of an integral prescheme),  $f : Y \rightarrow X$  is a finite morphism (respectively finite and such that the generic point of  $Y$  is separable over its image), and  $M_U$  is a left  $A_U$ -Module constant with value a monogenic  $A$ -module killed by a prime divisor of  $n$ .

Finally, if  $X$  is noetherian and satisfies condition (reg) (I 3.1.1), condition (ii) (respectively (iii)) is equivalent to (i) if one further assumes  $Y$  (respectively  $U$ ) regular.

*Proof.* The equivalence of conditions (i) and (ii) is proved as in the case where  $A_X$  is constant; see (SGAA IX 2.4 and 2.5).

Suppose that  $A_X$  is constant with value  $A$  and killed by  $n$ , and prove that (iii) implies (ii). Let  $E \in \text{ob } \text{Cons}(X)$ ; we show that  $E \in \text{ob } C$ . By the equivalence (i) $\Leftrightarrow$ (ii), we may suppose that  $E = j_!F$ , where  $j : Z \rightarrow X$  is the immersion of a sub-prescheme of finite presentation (respectively an integral one), and  $F$  is a constructible locally constant  $A_Z$ -Module, which we may even suppose, after a little more dévissage, to be trivialized by an étale covering  $g : Z' \rightarrow Z$  with group  $G$ . The sheaf  $F$  is therefore defined by a finite type  $A$ -module  $M$  on which  $G$  acts. If  $\ell$  is a prime divisor of  $n$ , the subgroup of  $\ell$ -torsion in  $M$  is a sub- $G$ - $A$ -module  $M_\ell$  of  $M$ , and

$$M = \bigoplus_{\ell} M_\ell,$$

where  $\ell$  runs over the set of prime divisors of  $n$ . Since  $C$  is stable under direct factors, we are reduced to the case where  $M$  is  $\ell$ -torsion. Let  $H$  be an  $\ell$ -Sylow subgroup of  $G$ , and let  $h : U = Z'/H \rightarrow Z$

be the corresponding intermediate covering. Since the degree of  $h$  is prime to  $\ell$ , the “trace method” (SGAA IX 5.1) shows that the adjunction map

$$F \longrightarrow h_* h^*(F)$$

is an isomorphism onto a direct factor; it is therefore enough to prove that  $j_! h_* h^*(F) \in \text{ob } C$ . By the “Main theorem” (EGA IV 8.12.6), there exists a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{i} & Y \\ h \downarrow & & \downarrow f \\ Z & \xrightarrow{j} & X, \end{array}$$

where  $i$  is an open immersion and  $f$  is finite. We have

$$j_! h_* h^*(F) \simeq f_* i_! h^*(F).$$

But  $h^*(F)$  is defined by  $M$  considered as an  $H$ - $A$ -module. Thus (iii) $\Rightarrow$ (ii) follows from the next lemma.

**Lemma 1.3.1.** Let  $A$  be a left noetherian ring,  $\ell$  a prime number,  $H$  a finite  $\ell$ -group, and  $M$  a left  $H$ - $A$ -module of finite type killed by  $\ell^\nu$  ( $\nu \geq 1$ ). There exists a finite filtration of  $M$  by sub- $H$ - $A$ -modules such that the successive quotients are monogenic  $A$ -modules killed by  $\ell$ , and on which  $H$  acts trivially.

*Proof.* The exact sequence of  $H$ - $A$ -modules

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0$$

reduces, by induction on  $\nu$ , to the case where  $M$  is killed by  $\ell$ . We show that then  $M$  has a finite filtration by sub- $H$ - $A$ -modules on whose successive quotients  $H$  acts trivially. Arguing by noetherian induction on the set of sub- $H$ - $A$ -modules of  $M$  having the stated property, we are reduced to proving that  $M \neq 0$  implies  $M^H \neq 0$ . But the kernel of the augmentation  $\mathbb{F}_\ell[H] \rightarrow \mathbb{F}_\ell$  is plainly nilpotent, and Nakayama’s lemma implies that every nonzero  $H$ - $\mathbb{F}_\ell$ -module has a nonzero invariant vector. This proves the assertion. Thus  $M$  has a finite filtration whose successive quotients are finite type  $A$ -modules killed by  $\ell$  and on which  $H$  acts trivially. By noetherian induction one may even arrange for the successive quotients to be monogenic  $A$ -modules, completing the proof of (1.3.1).

The last assertion of (1.3) is immediate and left to the reader.

## 2. Quasi-injective dimension

With the notation of (1.1), one defines, as in (I 1.1 and 1.4), the notions of a quasi-injective  $A_X$ -Module and of quasi-injective dimension of an object of  $D(X)$ . Propositions (I 1.3, 1.5 and 1.6) extend without change.

It is sometimes convenient to introduce a notion of pointwise quasi-injective dimension. Let  $x \in X$  and let  $K \in \text{ob } D^+(X)$ . We say that  $K$  has quasi-injective dimension  $\leq N$  at  $\bar{x}$  if, for every constructible  $A_X$ -Module  $F$ , one has

$$\underline{\text{Ext}}^i(F, K)_{\bar{x}} = 0 \quad \text{for } i > N.$$

The quasi-injective dimension of  $K$  at  $\bar{x}$ , denoted  $\dim .q.inj_{\bar{x}}(K)$ , is the greatest lower bound of the  $N$  satisfying the preceding condition. Of course,

$$\dim .q.inj(K) = \sup_{\bar{x}} \dim .q.inj_{\bar{x}}(K),$$

where  $\bar{x}$  runs through the set of geometric points of  $X$ .

It is interesting to compare this with the notion of strict topological dimension introduced by Verdier in [1]. We say that  $K \in \text{ob } D^+(X)$  has strict topological dimension  $\leq N$  at a geometric point  $\bar{x}$  of  $X$  if, for every sub-prescheme  $Y$  of  $X$ ,

$$\underline{\text{Ext}}^i(A_Y, K)_{\bar{x}} = \underline{H}_Y^i(K)_{\bar{x}} = 0 \quad \text{for } i > N.$$

The strict topological dimension of  $K$  at  $\bar{x}$ , denoted  $\dim .top_{\bar{x}}(K)$ , is the greatest lower bound of the  $N$  satisfying the preceding condition. Whereas quasi-injective dimension depends a priori on the base Ring, strict topological dimension depends only on the underlying complex of abelian sheaves.

One plainly has

$$\dim .top_{\bar{x}}(K) \leq \dim .q.inj_{\bar{x}}(K).$$

We shall see below examples where equality holds.

### 3. Dualizing complexes: definition and formal properties

Keep the preceding notation, but suppose  $A_X$  is commutative. The notions of bidualizing pair and dualizing complex are defined as in (I 1.7). The statements (I 1.9, 1.11, 1.14) remain valid without modification. The statements (I 1.12, 1.13, 1.15) are also valid, but on condition that  $A$  be killed by an integer  $n > 0$  prime to the residual characteristics of  $X$ . Finally, the uniqueness theorem (I 2.1) is valid provided one assumes  $\text{Spec}(A_{\bar{x}})$  connected for every geometric point  $\bar{x}$  of  $X$ .

### 4. Absolute purity

Return to the situation of absolute cohomological purity (I 3.1.4). Let  $X$  be a locally noetherian regular prescheme, let  $n > 0$  be an integer prime to the residual characteristics of  $X$ , and let  $i : Y \rightarrow X$  be a regular sub-prescheme of codimension  $d > 0$  at every point. To say that the pair  $(Y, X)$  satisfies absolute cohomological purity means, recall, that the “local fundamental class” homomorphism (SGA 4<sup>1/2</sup> Cycle 2.2)

$$(\mu_n^{\otimes d})_Y[-2d] \longrightarrow \text{Ri}^!(\mathbb{Z}/n\mathbb{Z})_X \quad (4.1)$$

is an isomorphism.

Now put on  $X$  a sheaf of rings  $A_X$  killed by  $n$ . For the moment set  $G = (\mathbb{Z}/n\mathbb{Z})_X$ . Then, for every  $F \in \text{ob } D_A^+(X)$ , one can define a canonical arrow in  $D_A^+(X)$ :

$$i^*(F) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L \text{Ri}^!(G) \longrightarrow \text{Ri}^!(F \otimes_{\mathbb{Z}/n\mathbb{Z}}^L G). \quad (4.2)$$

Indeed, by duality (SGAA XVIII 3.1.6), this is equivalent to defining an arrow

$$i_!(i^*(F) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L \text{Ri}^!(G)) \longrightarrow F \otimes_{\mathbb{Z}/n\mathbb{Z}}^L G;$$

we take this to be the composite of the Künneth arrow (SGAA XVII 5.4.2.2)

$$i_!(i^*(F) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L Ri^!(G)) \longrightarrow F \otimes_{\mathbb{Z}/n\mathbb{Z}}^L i_! Ri^!(G),$$

and the arrow deduced from the trace homomorphism (SGAA XVIII 3.1.6)

$$F \otimes_{\mathbb{Z}/n\mathbb{Z}}^L i_! Ri^!(G) \longrightarrow F \otimes_{\mathbb{Z}/n\mathbb{Z}}^L G.$$

Notice that in the definition of (4.2) only the fact that  $G$  and  $Ri^!(G)$  have finite Tor-dimension over  $\mathbb{Z}/n\mathbb{Z}$  was used. Suppose on the other hand that the functor  $i^!$  has finite cohomological dimension on the category of  $(\mathbb{Z}/n\mathbb{Z})_X$ -Modules: this is the case, for example, if  $X$  is of finite type over a field (SGAA XIV 3.1), or more generally if  $X$  is excellent of equal characteristic (SGAA XIX 6.1) when resolution of singularities is available, or again if  $X$  is of finite type over  $\text{Spec}(\mathbb{Z})$  (SGAA X 6). Then the arrow (4.2) is defined for all  $F \in \text{ob } D_A^+(X)$ .

Composing (4.1) and (4.2), one obtains, for  $F \in \text{ob } D_A^+(X)$ , a canonical arrow

$$i^*(F) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L (\mu_n^{\otimes d})_Y[-2d] \longrightarrow Ri^!(F). \quad (4.3)$$

Assume that  $A_X$  is locally constant, and let  $F \in \text{ob } D_A^+(X)$  be such that every  $H^i(F)$  is a locally constant  $A_X$ -Module. Then, if  $i^!$  has finite cohomological dimension, the arrow (4.2), and hence the arrow (4.3) if  $(Y, X)$  satisfies purity, is an isomorphism.

Indeed, by dévissage, one first reduces to the case where  $F$  is concentrated in degree zero. Since the question is local, one may suppose  $X$  affine and  $A_X$  and  $F$  constant. Resolving  $F$  on the left by free  $A_X$ -Modules and applying the hypothesis that  $i^!$  has finite cohomological dimension, one reduces to the case where  $F$  is free. By an immediate passage to the limit, one reduces to  $F$  free of finite type, and finally to  $F = A_X$ . It is therefore enough to prove the assertion when  $A_X = (\mathbb{Z}/n\mathbb{Z})_X$ . But the preceding dévissage reduces to  $F = (\mathbb{Z}/n\mathbb{Z})_X$ , and in that case (4.2) is the identity.

**Definition 4.4.** Given the Ring  $A_X$  (locally constant and killed by  $n$ ), we shall say that the pair  $(Y, X)$  satisfies cohomological purity in the strong sense if, for every  $F \in \text{ob } D_A^+(X)$  such that the  $H^i(F)$  are locally constant, the canonical arrow (4.3) is an isomorphism.

We shall say that the purity theorem is true in the strong sense on  $X$  if, locally for the étale topology, every pair  $(Y, X)$  as above satisfies purity in the strong sense. From what has been shown, for  $(Y, X)$  to satisfy purity in the strong sense it is enough that  $(Y, X)$  satisfy ordinary purity (I 3.1.) and that the functor  $i^!$  have finite cohomological dimension.

## 5. Bound for quasi-injective dimension

**Proposition 5.1.** Let  $X$  be a locally noetherian regular prescheme, equipped with a locally constant sheaf of rings  $A_X$ , killed by an integer  $n > 0$  prime to the residual characteristics of  $X$ , whose geometric fibres are left noetherian rings. Suppose that  $X$  satisfies conditions (reg) (I 3.1.1) and  $(\text{cdloc})_n$  (I 3.1.3), and that the purity theorem in the strong sense is true on  $X$  (4.4) (conditions all satisfied if  $X$  is smooth over a perfect field, or if  $X$  is excellent of characteristic zero).

Let  $F \in \text{ob } D_A^+(X)$  be such that the  $H^i(F)$  are locally constant, and let  $x$  be a point of  $X$  such that  $F_{\bar{x}} \neq 0$  and  $\dim .\text{inj}_{A_{\bar{x}}}(F_{\bar{x}}) < +\infty$ . Then

$$\dim .\text{q.inj}_{\bar{x}}(F) = 2 \dim(\mathcal{O}_{X,x}) + \dim .\text{inj}(F_{\bar{x}}),$$

with the notation of no. 2.

*Proof.* One merely repeats step by step the proof of (I 3.2.1). First show the inequality

$$\dim .q.\text{inj}_{\bar{x}}(F) \leq 2 \dim(\mathcal{O}_{X,x}) + \dim .\text{inj}(F_{\bar{x}}) = N,$$

that is, the relation

$$\underline{\text{Ext}}^i(E, F)_{\bar{x}} = 0 \quad \text{for } i > N \text{ and every constructible } A_X\text{-Module } E. \quad (*)$$

Since the question is local, by dévissage (1.3) one may restrict to checking (\*) for  $E = i_!(M)$ , where  $i : Y \rightarrow X$  is an integral regular sub-prescheme closed in an open  $D(f)$  (with  $f \in \Gamma(X, \mathcal{O}_X)$ ), of codimension  $d$  at every point of  $D(f)$ , and  $M$  is a locally constant constructible  $A_Y$ -Module ( $A_Y = i^* A_X$ ).

Now there is the duality isomorphism (SGAA XVIII)

$$\text{R}\underline{\text{Hom}}(i_!(M), F) \simeq \text{R}i_* \text{R}\underline{\text{Hom}}(M, \text{R}i^! F).$$

By the purity theorem, there is also the isomorphism (4.3)

$$i^*(F) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L (\mu_n^{\otimes d})_Y[-2d] \xrightarrow{\sim} \text{R}i^! F.$$

Let  $\bar{y}$  be a geometric point of  $Y$ . Since  $M$  is locally constant constructible, one has canonically

$$\text{R}\underline{\text{Hom}}(M, \text{R}i^! F)_{\bar{y}} \simeq \text{R Hom}(M_{\bar{y}}, (\text{R}i^! F)_{\bar{y}}),$$

hence, by (4.3),

$$\text{R}\underline{\text{Hom}}(M, \text{R}i^! F)_{\bar{y}} \simeq \text{R Hom}(M_{\bar{y}}, F_{\bar{y}})[-2d].$$

After replacing  $X$  by a connected neighbourhood of  $x$ , one has  $F_{\bar{x}} \simeq F_{\bar{y}}$  and  $A_{\bar{x}} \simeq A_{\bar{y}}$ , since  $A_X$  and the  $\text{H}^i(F)$  are locally constant. Thus

$$\text{Ext}^i(M_{\bar{y}}, F_{\bar{y}}) = 0 \quad \text{for } i > \dim .\text{inj}(F_{\bar{x}}),$$

and hence

$$\underline{\text{Ext}}^q(M, \text{R}i^! F) = 0 \quad \text{for } q > 2d + \dim .\text{inj}(F_{\bar{x}}).$$

Set  $G = \text{R}\underline{\text{Hom}}(M, \text{R}i^! F)$  and compute  $\text{R}i_*(G)_{\bar{x}}$ . Introduce, as in (I 3.2.1), the schematic closure  $Z$  of  $Y$  in  $X$ , giving a cartesian diagram of immersions

$$\begin{array}{ccc} Y & \longrightarrow & D(f) \\ j \downarrow & & \downarrow \\ Z & \xrightarrow{k} & X. \end{array}$$

Then  $\text{R}i_*(G) \simeq k_* \text{R}j_*(G)$ . Assume of course that  $x \in Z$ , since otherwise  $\text{R}i_*(G)_{\bar{x}} = 0$ . Let  $\bar{Z} = \text{Spec}(\mathcal{O}_{Z, \bar{x}})$  be the strict localization of  $Z$  at  $\bar{x}$ , let  $\bar{f}$  be the image of  $f$  in  $\mathcal{O}_{Z, \bar{x}}$ , let  $\bar{Y} = \bar{Z} \times_Z Y = \bar{Z} \cap D(f)$ , and let  $\bar{G}$  be the inverse image of  $G$  on  $\bar{Y}$ . By (SGAA VIII 5.2),

$$\text{R}i_*(G)_{\bar{x}} = \text{R}j_*(G)_{\bar{x}} \simeq \text{R}\Gamma(\bar{Y}, \bar{G}).$$

Condition (cdloc) $_n$  therefore implies, by a spectral sequence argument, that

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > 2d + \dim .\text{inj}(F_{\bar{x}}) + \dim(\mathcal{O}_{Z, x}).$$

But (cf. the proof of I 3.2.1)

$$\dim(\mathcal{O}_{Z,x}) = \dim(\mathcal{O}_{X,x}) - d \geq 0,$$

so

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > d + \dim(\mathcal{O}_{X,x}) + \dim \text{.inj}(F_{\bar{x}}),$$

and a fortiori for  $p > 2 \dim(\mathcal{O}_{X,x}) + \dim \text{.inj}(F_{\bar{x}})$ . This proves (\*).

It remains to prove that the bound (\*) is the best possible. Put  $f = \dim \text{.inj}(F_{\bar{x}})$ , and choose a finite type  $A_{\bar{x}}$ -Module  $M'$  such that

$$\text{Ext}^f(M', F_{\bar{x}}) \neq 0.$$

Let  $Z$  be the integral closed sub-prescheme of  $X$  which is the closure of  $x$ , and let  $Y = U \cap Z$  be a regular open subset of  $Z$  containing  $x$  and of codimension  $d = \dim(\mathcal{O}_{X,x})$  at every point. After possibly shrinking  $Y$ , there exists a locally constant constructible  $A_Y$ -Module  $M$  such that  $M_{\bar{x}} = M'$ . Applying the purity theorem, one finds

$$\underline{\text{Ext}}^{2d+f}(M, F|U)_{\bar{x}} \simeq \text{Ext}^f(M'_{\bar{x}}, F_{\bar{x}}) \neq 0,$$

which completes the proof.

If  $F$  is an  $A_X$ -Module such that  $F_{\bar{x}}$  is injective, then in the last part of the proof one may take  $M = A_Y$ ; hence:

**Corollary 5.1.1.** Under the hypotheses of (5.1), suppose that  $F$  is a locally constant  $A_X$ -Module such that  $F_{\bar{x}}$  is injective. Then

$$\dim \text{.q.inj}_{\bar{x}}(F) = \dim \text{.top}_{\bar{x}}(F) = 2 \dim(\mathcal{O}_{X,x}),$$

with the notation of no. 2.

**Remarks 5.1.2.** a) With the notation of (5.1), if  $F_{\bar{x}} = 0$ , then  $F = 0$  in a neighbourhood of  $x$ , hence  $\text{R}\underline{\text{Hom}}(E, F)_{\bar{x}} = 0$  for all  $E \in \text{ob D}^-(X)$ .

b) If  $\dim \text{.inj}(F_{\bar{x}}) = +\infty$ , then  $\dim \text{.q.inj}_{\bar{x}}(F) = +\infty$ ; indeed, for every  $p$  there exists a finite type  $A_{\bar{x}}$ -module  $M'$  such that  $\text{Ext}^p(M', F_{\bar{x}}) \neq 0$ , hence a constructible  $A_X$ -Module  $M$ , locally constant in a neighbourhood of  $x$ , such that  $\underline{\text{Ext}}^p(M, F)_{\bar{x}} \neq 0$ .

## 6. Constructibility of $\text{R}\underline{\text{Hom}}$

The equivalence of conditions a) of (I 3.5.1) remains valid when the Ring  $(\mathbb{Z}/n\mathbb{Z})_X$  is replaced by a commutative Ring, locally constant, whose geometric fibres are noetherian rings: nothing has to be changed in the proof.

The preceding conditions are verified when one has resolution of singularities and purity in the strong sense (4.4), in particular if  $X$  has dimension  $\leq 1$ , or if  $X$  is excellent of characteristic zero, or locally of finite type over a field and of dimension  $\leq 2$ .

## 7. Existence of dualizing complexes

In this number,  $X$  denotes a prescheme equipped with a commutative sheaf of rings  $A_X$ , locally constant, whose geometric fibres are noetherian rings.

**Lemma 7.1.** If  $F \in \text{ob } D_{\text{coh}}^-(X)$  and  $G \in \text{ob } D_{\text{coh}}^+(X)$ , then  $\text{R}\underline{\text{Hom}}(F, G) \in \text{ob } D_{\text{coh}}^+(X)$  (notation of (1.2)).

*Proof.* This is standard. One reduces, by a spectral sequence, to  $F$  and  $G$  concentrated in degree zero, and then argues as in (EGA 0<sub>III</sub> 12.3.3).

**Lemma 7.2.** Let  $g : X' \rightarrow X$  be a morphism of preschemes, and endow  $X'$  with the sheaf of rings  $A_{X'} = g^*(A_X)$ . Then, for  $F \in \text{ob } D_{\text{coh}}^-(X)$  and  $G \in \text{ob } D^+(X)$ , the canonical map

$$g^* \text{R}\underline{\text{Hom}}(F, G) \longrightarrow \text{R}\underline{\text{Hom}}(g^* F, g^* G)$$

is an isomorphism.

*Proof.* This too is standard: one reduces by dévissage to  $F = A_X$  (cf. EGA 0<sub>III</sub> 12.3.5).

**Remark 7.2.1.** The preceding lemmas are special cases of more general statements appearing in (SGA 6 I).

**Lemma 7.3.** Suppose  $X$  is quasi-compact, and let  $K \in \text{ob } D_{\text{coh}}^+(X)$  be such that, for every geometric point  $\bar{x}$  of  $X$ ,  $K_{\bar{x}}$  is a dualizing complex in the sense of [H] V 2.1. Then, for every  $F \in \text{ob } D_{\text{coh}}^b(X)$ , one has  $\text{R}\underline{\text{Hom}}(F, K) \in \text{ob } D_{\text{coh}}^b(X)$ , and  $F$  is reflexive relative to  $K$ .

*Proof.* It suffices, of course, to check this for  $F$  concentrated in degree zero. By (7.1),  $\text{R}\underline{\text{Hom}}(F, K) \in \text{ob } D_{\text{coh}}^+(X)$ . To prove boundedness, let  $\bar{x}$  be a geometric point of  $X$ . By (7.2),

$$\text{R}\underline{\text{Hom}}(F, K)_{\bar{x}} \simeq \text{R Hom}(F_{\bar{x}}, K_{\bar{x}}).$$

Putting  $\dim .\text{inj}(K_{\bar{x}}) = N(\bar{x})$ , one obtains

$$\text{Ext}^i(F, K)_{\bar{x}} = 0 \quad \text{for } i > N(\bar{x}).$$

Since the sheaves  $\underline{\text{Ext}}^i(F, K)$  are locally constant, there is an open neighbourhood  $U$  of  $x$  such that  $\underline{\text{Ext}}^i(F, K)|_U = 0$  for  $i > N(\bar{x})$ , and quasi-compactness of  $X$  gives the assertion. It remains to check that the canonical arrow

$$F \longrightarrow \text{R}\underline{\text{Hom}}(\text{R}\underline{\text{Hom}}(F, K), K)$$

is an isomorphism. It is enough to evaluate at a geometric point  $\bar{x}$  of  $X$ . In the commutative diagram

$$\begin{array}{ccc} F_{\bar{x}} & \longrightarrow & \text{R}\underline{\text{Hom}}(\text{R}\underline{\text{Hom}}(F, K), K)_{\bar{x}} \\ \text{Id} \downarrow & & \downarrow \\ F_{\bar{x}} & \longrightarrow & \text{R Hom}(\text{R Hom}(F_{\bar{x}}, K_{\bar{x}}), K_{\bar{x}}), \end{array}$$

the right vertical arrow is an isomorphism by (7.2), and the bottom horizontal arrow is an isomorphism because  $K_{\bar{x}}$  is dualizing; hence the top horizontal arrow is an isomorphism, q.e.d.

**Lemma 7.4.** Suppose  $X$  is a noetherian regular prescheme,  $A_X$  is killed by an integer  $n > 0$  prime to the residual characteristics of  $X$ , and the purity theorem in the strong sense (4.4) is true on  $X$ .

Let  $D$  be a divisor with normal crossings on  $X$  (I 3.1.5 a)), and denote by  $j : Y = V(D) \rightarrow X$  and  $i : X - Y = U \rightarrow X$  the canonical inclusions. Let  $K \in \text{ob } D_{\text{coh}}^+(X)$  be such that, for every geometric point  $\bar{x}$  of  $X$ ,  $K_{\bar{x}}$  is a dualizing complex for the ring  $A_{\bar{x}}$  (in the sense of [H] V 2.1). Then, for every  $E \in \text{ob } D_{\text{coh}}^b(X)$ , one has  $\text{R}\underline{\text{Hom}}(i_! i^*(E), K) \in \text{ob } D_c^b(X)$ , and the pair  $(i_! i^*(E), K)$  is bidualizing.

*Proof.* We may suppose  $E$  is concentrated in degree zero, i.e. a locally constant constructible  $A_X$ -Module. Since the question is local on  $X$  for the étale topology, we may suppose that (a)  $A_X$  is

constant with value  $A$ , and  $E$  is constant with value  $E$ ; and (b)  $D$  has strict normal crossings, say  $D = \sum_{i \in I} \text{div}(s_i)$  with  $s_i \in \Gamma(X, \mathcal{O}_X)$ .

Argue by induction on  $p = \text{card}(I)$ . For  $p = 0$ , the assertions follow from (7.3). Suppose  $I = \{1, \dots, p\}$ , put  $Y_i = V(s_i)$ ,  $Y' = V(\sum_{1 \leq i \leq p-1} \text{div}(s_i))$ , and  $U' = X - Y'$ . We get a cartesian diagram of immersions

$$\begin{array}{ccccc} \emptyset & \longrightarrow & U' \cap Y_p & \xrightarrow{i'_p} & Y_p \\ \downarrow & & j'_p \downarrow & & \downarrow j_p \\ U & \longrightarrow & U' & \xrightarrow{i'} & X. \end{array}$$

Consider the exact sequence

$$0 \longrightarrow E_{U,X} \longrightarrow E_{U',X} \longrightarrow E_{U' \cap Y_p, X} \longrightarrow 0.$$

By the induction hypothesis the theorem is true for  $(E_{U',X}, K)$ . For fixed  $K$ , the  $F$  such that  $\text{R}\underline{\text{Hom}}(F, K) \in \text{ob } D_c^b(X)$  and  $(F, K)$  is bidualizing form a triangulated subcategory of  $D_c(X)$ ; hence we are reduced to the case  $F = E_{U' \cap Y_p, X}$ . But

$$E_{U' \cap Y_p, X} = j_{p*} i'_{p!}(E_{U' \cap Y_p}),$$

and, by (I 1.13), it is equivalent to prove the theorem for  $(i'_{p!}(E_{U' \cap Y_p}), \text{R}j_p^!(K))$  on  $Y_p$ . By purity, however,

$$\text{R}j_p^!(K) \simeq j_p^*(K) \otimes_{\mathbb{Z}/n\mathbb{Z}}^L (\mu_n)_{Y_p}^{\otimes -1}[-2].$$

Thus  $\text{R}j_p^!(K) \in \text{ob } D_{\text{coh}}^+(Y_p)$  and, for every geometric point  $\bar{y}$  of  $Y_p$ ,  $(\text{R}j_p^!(K))_{\bar{y}}$  is dualizing. The induction hypothesis then applies.

**Theorem 7.5.** Assume the following:  $X$  is a noetherian regular prescheme;  $A_X$  is killed by an integer  $n > 0$  prime to the residual characteristics of  $X$ ;  $X$  is strongly desingularizable (I 3.1.5), satisfies conditions (reg) (I 3.1.1) and (cdloc) $_n$  (I 3.1.3); and the purity theorem in the strong sense (4.4) is true on every prescheme smooth over  $X$  (conditions satisfied if  $X$  is excellent of characteristic zero, or of dimension  $\leq 2$  and smooth over a perfect field).

Then, if  $K \in \text{ob } D_{\text{coh}}^+(X)$  is such that  $K_{\bar{x}}$  is dualizing (in the sense of [H] V 2.1) for every geometric point  $\bar{x}$  of  $X$ , the complex  $K$  is dualizing.

*Proof.* We must show that  $K$  satisfies conditions (i) to (iii) of (I 1.7). For every geometric point  $\bar{x}$  of  $X$ ,  $K_{\bar{x}}$  has finite injective dimension, and since  $K \in \text{ob } D_{\text{coh}}^+(X)$ , the function  $x \mapsto \dim \text{inj}(K_{\bar{x}})$  is locally constant; it is therefore bounded because  $X$  is noetherian. It follows from (5.1) that  $K$  has finite quasi-injective dimension.

By what was seen in no. 6, condition (ii) of loc. cit. is also satisfied.

It remains to prove condition (iii), i.e. that every  $F \in \text{ob } D_c^b(X)$  is reflexive relative to  $K$ . By dévissage we may suppose  $F$  is concentrated in degree zero. Since the question is local on  $X$  for the étale topology, we may suppose  $A_X$  constant with value  $A$ . Dévissage (1.3) then reduces us to  $F = f_* i_!(E_U)$ , where  $f : Y \rightarrow X$  is finite,  $Y$  is integral with generic point separable over its image,  $i : U \rightarrow Y$  is a nonempty regular open, and  $E_U$  is an  $A_U$ -Module constant with value a finite type  $A$ -module  $E$ . Since  $X$  is strongly desingularizable, there exists (I 3.1.5) a projective birational morphism  $h : Y' \rightarrow Y$  such that  $Y'$  is regular,  $h$  induces an isomorphism  $h^{-1}(U) \rightarrow U$ , and  $h^{-1}(U)$

is the complement of a normal crossings divisor in  $Y'$ . Identifying  $h^{-1}(U)$  with  $U$  via  $h$ , we have a commutative diagram

$$\begin{array}{ccccc} & & Y' & & \\ & i' \swarrow & \downarrow h & \searrow g & \\ U & \xrightarrow{i} & Y & \xrightarrow{f} & X. \end{array}$$

Consequently,

$$f_*i_!(E_U) \simeq f_*Rh_*i'_!(E_U) \simeq Rg_*i'_!(E_U) \simeq Rg_*i'_!i'^*(E_{Y'}).$$

By (I 1.13), it is enough to prove biduality for the pair  $(i'_!i'^*(E_{Y'}), Rg^!(K))$ . But, locally embedding  $Y'$  into a prescheme  $X'$  smooth over  $X$  and applying strong purity on  $X'$  (4.4), one finds that  $Rg^!(K)$  is locally isomorphic, up to degree shift, to  $g^*(K)$ . Hence  $Rg^!(K) \in \text{ob } D_{\text{coh}}^+(Y')$  and  $Rg^!(K)_{\bar{x}}$  is dualizing for every geometric point  $\bar{x}$  of  $Y'$ . It remains only to apply (7.4).

The preceding proof also shows that  $R\text{Hom}(F, K) \in \text{ob } D_c^b(X)$  (using the finiteness theorem for a proper morphism (SGAA XIV)), which will reassure the reader who did not trust the spiel of no. 6.

This completes the proof.

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## Bibliography

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SGA A = SGA 4.

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# SGA 5 — Exposé III

## LEFSCHETZ FORMULA

by A. Grothendieck, written up by L. Illusie

In this exposé we prove the Lefschetz formula in étale cohomology, for correspondences between complexes of sheaves, in Verdier’s formulation (cf. [18]). In an earlier version, distributed by the IHES about ten years ago, the formula was stated, under resolution hypotheses, but the required compatibilities were not proved. Deligne’s finiteness theorems (SGA 4<sup>1/2</sup>, Finiteness) made it possible to dispense with these hypotheses, provided one works with schemes separated and of finite type over a field. As for the compatibilities, the editor has verified them (painfully).

Let us say at once that the Lefschetz formula in question here, just like the trace formula for constant coefficients (SGA 4<sup>1/2</sup>, Cycle), which it generalizes, is essentially trivial: it only translates certain compatibilities in the formalism of duality. Roughly speaking, it says that, under certain conditions, the trace of an endomorphism of a hypercohomology complex  $R\Gamma(X, L)$ , where  $L$  is a complex of sheaves on  $X$ , can be calculated as an “integral” of certain cohomology classes on  $X \times X$ . In practice, the endomorphisms one considers are associated with correspondences  $C \subset X \times X$ , and the classes recovered on the product have support in  $C \cap \Delta$ , where  $\Delta$  is the diagonal; these classes are “local in nature” in a neighbourhood of the fixed points, which in principle facilitates their calculation. See 4.2 and 4.7 for precise statements. In fact, the calculation of these “local” classes in terms of more comprehensible invariants, without which the present formula would have no interest, has so far been undertaken only in very special situations (see 4.10, 4.11, and the following exposé, where in particular the case of the Frobenius correspondence on a curve is treated).

For lack of a good derived category of  $\ell$ -adic sheaves (Jouanolou’s thesis unfortunately not having been published), we work systematically with torsion coefficients (prime to the residual characteristics). Despite minor technical difficulties in the definition of traces, this point of view in fact leads to finer formulae, also more manageable insofar as, for the calculation of local terms, one wants to trivialize the sheaves involved by coverings (see notably (SGA 4<sup>1/2</sup>, Report), III B, and XII).

In no. 1, we fix the notation and recall various Künneth formulae from (SGA 4<sup>1/2</sup>, Finiteness), which play an essential role in what follows. We also introduce a certain number of fibred and cofibred categories, whose consideration later facilitates certain verifications of compatibilities. No. 2, which is very technical, may be skipped on first reading. Its main result (2.5) expresses a compatibility between external tensor product of  $R\mathrm{Hom}$ , direct image, and base change. In no. 3, we define cohomological correspondences between complexes, and show how they act on hypercohomology; the result quoted above makes it possible to describe this action in several equivalent ways. The core of the exposé is no. 4, where Verdier’s pairing between cohomological correspondences is defined and the Lefschetz formula is proved, in a relative situation appreciably more general than the one usually considered, with a view to later applications to computations of local terms. We give the most evident corollaries for correspondences with constant coefficients. In no. 5, we indicate rapidly how to generalize to correspondences between complexes certain usual constructions on correspondences with constant coefficients (cf. [12]), such as passage to the dual, composition, and external tensor product. Most of the verifications, analogous to those in the preceding numbers, have been omitted. Finally, in the appendix, we paraphrase the preceding theory in the context of coherent sheaves, using the formalism of duality of [8]. Here one may work over any noetherian base  $S$ , provided that, when necessary, one imposes hypotheses of properness, finite tor-dimension, and tor-independence on the  $S$ -schemes considered. We show that the very general Lefschetz formula thereby obtained contains as a special case the Woods Hole formula [1], and we point out some applications.

## Contents

1. Notation and recalls of Künneth formulae.
2. Functoriality of  $\mathrm{RHom}$  and external tensor products.
3. Cohomological correspondences.
4. Pairings of correspondences. Lefschetz formula.
5. Complements.
6. Appendix. Lefschetz formula for coherent sheaves.

## 2 Notation and recalls of Künneth formulae

### 1.1

We denote by  $S$  the spectrum of a field. Unless otherwise stated, the  $S$ -schemes considered in what follows will be  $S$ -schemes separated and of finite type.

We denote by  $A$  a commutative noetherian ring, annihilated by an integer invertible on  $S$ . If  $X$  is an  $S$ -scheme, we denote by  $D(X) = D(X, A)$  the derived category of complexes of  $A$ -modules on  $X$  (for the étale topology), and by  $D_{\mathrm{ctf}}(X)$  the full subcategory consisting of complexes of finite tor-dimension, with constructible cohomology. The category  $D_{\mathrm{ctf}}(X)$  is a triangulated subcategory of  $D(X)$ . If  $E, F \in \mathrm{ob} D_{\mathrm{ctf}}(X)$ , it follows readily from the definitions that  $E \otimes^L F \in \mathrm{ob} D_{\mathrm{ctf}}(X)$ . According to (SGA 4<sup>1/2</sup>, Finiteness Theorem 1.6, 1.7), one also has  $\mathrm{RHom}(E, F) \in \mathrm{ob} D_{\mathrm{ctf}}(X)$ .

### 1.2

Let  $f : X \rightarrow Y$  be an  $S$ -morphism. One has trivially

$$f^* D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

and, by (loc. cit.),

$$Rf_* D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \quad Rf_! D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \quad Rf^! D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

(the inclusion relative to  $Rf_!$  uses only the finiteness theorem for proper morphisms (SGA 4 XIV) and the sorites (SGA 4 XVII 5.2.11), and hence holds more generally for every compactifiable morphism whose fibres have bounded dimension). In what follows, we shall sometimes write  $f_*$ ,  $f_!$ ,  $f^!$  instead of  $Rf_*$ ,  $Rf_!$ ,  $Rf^!$ ; the hypotheses of 1.1 imply that these functors are defined on the whole derived category.

### 1.3

For  $f : X \rightarrow S$ , we shall put

$$K_X = f^! A,$$

and shall denote by  $D_X$  (or  $D$ ) the functor  $\mathrm{RHom}(-, K_X)$ . According to (SGA 4<sup>1/2</sup>, Finiteness Theorem 4.3), the complex  $K_X$  is dualizing: for  $E \in \mathrm{ob} D_{\mathrm{ctf}}(X)$ , one has canonically

$$E \xrightarrow{\sim} DDE.$$

## 1.4

Let  $f : X \rightarrow Y$  be an  $S$ -morphism,  $L \in \text{ob } D(X)$ ,  $M \in \text{ob } D(Y)$ . We put

$$\begin{aligned}\text{Hom}_f^{(1)}(L, M) &= \text{Hom}(f^*M, L) \quad (\simeq \text{Hom}(M, f_*L)), \\ \text{Hom}_f^{(2)}(L, M) &= \text{Hom}(L, f^*M), \\ \text{Hom}_f^{(3)}(L, M) &= \text{Hom}(f_!L, M) \quad (\simeq \text{Hom}(L, f^!M)), \\ \text{Hom}_f^{(4)}(L, M) &= \text{Hom}(f^!M, L).\end{aligned}$$

The elements of  $\text{Hom}_f^{(i)}(L, M)$  will be called *arrows of type  $i$  from  $L$  to  $M$  over  $f$*  (or  $f$ -arrows of type  $i$  from  $L$  to  $M$ ). The transitivity isomorphisms for the functors  $f^*$ ,  $f_*$ ,  $f_!$ ,  $f^!$  allow one to compose arrows of type  $i$  for fixed  $i$ , and one thus obtains certain fibred or cofibred categories over the category of  $S$ -schemes: for example, for  $i = 1$  (resp.  $i = 3$ ) one obtains the fibred and cofibred category considered in (SGA 4 XVII 4.1.3) (resp. (SGA 4 XVIII 3.1.13)), whose fibre over  $X$  is the category opposite to  $D(X)$  (resp.  $D(X)$ ).

## 1.5

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme,  $X = X_1 \times_S X_2$ , and  $L_i \in \text{ob } D^-(X_i)$ . We put

$$L_1 \otimes_S^L L_2 = \text{pr}_1^* L_1 \otimes^L \text{pr}_2^* L_2,$$

where  $\text{pr}_i : X \rightarrow X_i$  is the canonical projection.

## 1.6

Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \rightarrow Y = Y_1 \times_S Y_2,$$

$L_i \in \text{ob } D^-(X_i)$ ,  $M_i \in \text{ob } D^-(Y_i)$ . The evident canonical isomorphism

$$f_1^* M_1 \otimes_S^L f_2^* M_2 \xrightarrow{\sim} f^*(M_1 \otimes_S^L M_2) \quad (1.6.1)$$

allows one to define, for  $u_i \in \text{Hom}_{f_i}^{(1)}(L_i, M_i)$  (resp.  $\text{Hom}_{f_i}^{(2)}(L_i, M_i)$ ),

$$u_1 \otimes_S^L u_2 \in \text{Hom}_f^{(1)}(L, M) \quad (\text{resp. } \text{Hom}_f^{(2)}(L, M)), \quad (1.6.2)$$

where  $L = L_1 \otimes_S^L L_2$ ,  $M = M_1 \otimes_S^L M_2$ . If  $u_i, v_i$  are composable, one has

$$(v_1 u_1) \otimes_S^L (v_2 u_2) = (v_1 \otimes_S^L v_2)(u_1 \otimes_S^L u_2). \quad (1.6.3)$$

Taking  $M_i = f_{i*} L_i$ , and  $u_i$  of type 1, given by the identity of  $f_{i*} L_i$ , 1.6.2 gives a *Künneth arrow* (cf. SGA 4 XVII 5.4.1.4)

$$f_{1*} L_1 \otimes_S^L f_{2*} L_2 \longrightarrow f_*(L_1 \otimes_S^L L_2). \quad (1.6.4)$$

According to (SGA 4<sup>1/2</sup>, Finiteness Theorem 1.9), 1.6.4 is an isomorphism: by 1.6.3 one is indeed reduced to the case where  $X_2 = Y_2$ ,  $f_2 = 1$ , and one applies (loc. cit. App.). When  $f_1$  and  $f_2$  are proper, it is not necessary to invoke (loc. cit.): 1.6.4 is a special case of the Künneth isomorphism (SGA 4 XVII 5.4.3).

## 1.7

Choose a compactification  $f_i = p_i j_i$ , where  $j_i$  is an open immersion and  $p_i$  is a proper morphism. Composing the evident canonical isomorphism (where  $j = j_1 \times_S j_2$ )

$$j_{1!} L_1 \otimes_S^L j_{2!} L_2 \xrightarrow{\sim} j_!(L_1 \otimes_S^L L_2)$$

with the isomorphism 1.6.4 relative to  $(p_i, j_i, L_i)$ , one obtains a Künneth isomorphism (SGA 4 XVII 5.4.3)

$$f_{1!} L_1 \otimes_S^L f_{2!} L_2 \xrightarrow{\sim} f_!(L_1 \otimes_S^L L_2). \quad (1.7.1)$$

One checks easily (loc. cit.) that 1.7.1 does not depend on the compactifications chosen, and is compatible with the transitivity isomorphisms for composites  $g_i f_i$  (cf. (loc. cit. 5.2.4)). Thanks to 1.7.1, one can define, for  $u_i \in \mathrm{Hom}_{f_i}^{(3)}(L_i, M_i)$ ,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(3)}(L, M) \quad (1.7.2)$$

with  $L$  and  $M$  as in 1.6.2, and one has a transitivity formula analogous to 1.6.3.

For  $L_i = f_i^! M_i$ , and  $u_i$  given by the identity of  $L_i$ , 1.7.2 gives a Künneth arrow

$$f_1^! M_1 \otimes_S^L f_2^! M_2 \longrightarrow f^!(M_1 \otimes_S^L M_2). \quad (1.7.3)$$

### Proposition 1.7.4

The arrow 1.7.3 is an isomorphism.

By the transitivity formula, it suffices to treat the case where  $X_2 = Y_2$ ,  $f_2 = 1$ . By dévissage, one reduces to  $M_2 = g_* j_! A_V$ , where  $j : V \rightarrow U$  is an open immersion and  $g : U \rightarrow Y_2$  a finite morphism, and then, by (SGA 4 XVIII 3.1.12.3), to the case where  $M_2 = A_{Y_2}$ . By localization on  $X_2$ , one may suppose moreover that  $f_1 = g_1 j_1$ , where  $j_1$  is a closed immersion and  $g_1$  a smooth morphism purely of relative dimension  $n_1$ . By transitivity, it suffices to treat the cases

$$\text{a) } f_1 = g_1, \quad \text{b) } f_1 = j_1.$$

In case a), the trace morphism supplies by adjunction (SGA 4 XVIII 3.2.5) an isomorphism

$$f_1^* M_1(n_1)[2n_1] \xrightarrow{\sim} f_1^! M_1;$$

this is compatible with the base change  $Y_2 \rightarrow S$ , because the trace morphism is so (SGA 4 XVIII 2.9), whence 1.7.4 in this case. In case b), if  $i_1$  denotes the inclusion of the complementary open, one has an exact sequence

$$0 \longrightarrow j_{1*} Rj_1^! M_1 \longrightarrow M_1 \longrightarrow Ri_{1*} i_1^* M_1 \longrightarrow 0,$$

and the compatibility of  $Ri_{1*}$  with base change, a special case of 1.6.4, gives the conclusion.

Thanks to 1.7.3, one can define, for  $u_i \in \mathrm{Hom}_{f_i}^{(4)}(L_i, M_i)$ ,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(4)}(L, M) \quad (1.7.5)$$

( $L$  and  $M$  as in 1.6.2), with a transitivity formula analogous to 1.6.3.

On the other hand, with the notation of 1.3, 1.7.3 gives an isomorphism

$$K_{X_1} \otimes_S^L K_{X_2} \xrightarrow{\sim} K_X. \quad (1.7.6)$$

## 1.8

Let  $f : X \rightarrow Y$  be an  $S$ -morphism,  $L \in \text{ob } D(X)$ ,  $M \in \text{ob } D(Y)$ ,  $S'$  an  $S$ -scheme,  $P \in \text{ob } D(S')$ , and  $f' : X' \rightarrow Y'$  the morphism deduced from  $f$  by the base change  $S' \rightarrow S$ . The identity arrow of  $P$  may be regarded indifferently as an arrow of type  $i$  ( $1 \leq i \leq 4$ ) over the identity arrow of  $S'$ ; for  $u \in \text{Hom}_f^{(i)}(L, M)$  ( $1 \leq i \leq 4$ ), one may therefore, by 1.6.2, 1.7.2, 1.7.5, consider

$$u \otimes_S^L \text{Id}_P \in \text{Hom}_{f'}^{(i)}(L \otimes_S^L P, M \otimes_S^L P). \quad (1.8.1)$$

We shall sometimes write  $u \otimes_S^L P$  instead of  $u \otimes_S^L \text{Id}_P$ .

## 3 Functoriality of $\text{RHom}$ and external tensor products

### 2.1

Let  $f : X \rightarrow Y$  be an  $S$ -morphism,  $P \in \text{ob } D(Y)$ ,  $Q \in \text{ob } D^+(Y)$ . The evident canonical arrow at the level of complexes

$$f^* \mathcal{H}om^\bullet(P, Q) \longrightarrow \mathcal{H}om^\bullet(f^* P, f^* Q)$$

gives by derivation a canonical arrow

$$f^* \text{RHom}(P, Q) \longrightarrow \text{RHom}(f^* P, f^* Q). \quad (2.1.1)$$

Let  $L \in \text{ob } D(X)$ ,  $M \in \text{ob } D^+(X)$ . Put

$$\text{Hom}_f^{(i,j)}((L, M), (P, Q)) = \text{Hom}_f^{(i)}(L, P) \times \text{Hom}_f^{(j)}(M, Q), \quad (2.1.2)$$

with the notation of 1.4. For  $(u, v) \in \text{Hom}_f^{(2,1)}((L, M), (P, Q))$  one defines

$$\text{RHom}(u, v) \in \text{Hom}_f^{(1)}(\text{RHom}(L, M), \text{RHom}(P, Q)) \quad (2.1.3)$$

by composing 2.1.1 with the arrow  $\text{RHom}(f^* P, f^* Q) \rightarrow \text{RHom}(L, M)$  defined by  $u$  and  $v$ . For  $L = f^* P$ ,  $Q = f_* M$ ,  $u$  (resp.  $v$ ) given by the identity of  $L$  (resp.  $Q$ ),  $\text{RHom}(u, v)$  is the “trivial duality” isomorphism

$$\text{RHom}(P, f_* M) \xrightarrow{\sim} f_* \text{RHom}(f^* P, M). \quad (2.1.4)$$

For  $(u, v) \in \text{Hom}_f^{(3,4)}((L, M), (P, Q))$ , one defines

$$\text{RHom}(u, v) \in \text{Hom}_f^{(1)}(\text{RHom}(L, M), \text{RHom}(P, Q)) \quad (2.1.5)$$

as the composite of the arrows

$$\text{RHom}(P, Q) \xrightarrow{a} \text{RHom}(f_! L, Q) \xrightarrow{b} f_* \text{RHom}(L, f^! Q) \xrightarrow{c} f_* \text{RHom}(L, M),$$

where  $a$  (resp.  $c$ ) is defined by  $u$  (resp.  $v$ ), and  $b$  is the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6).

It follows easily from the definitions that the formation of  $\text{RHom}(u, v)$  is compatible with composition: for composable  $(u, v)$ ,  $(u', v')$  of type  $(2, 1)$  (resp.  $(3, 4)$ ), one has

$$\text{RHom}(u' u, v' v) = \text{RHom}(u', v') \text{RHom}(u, v). \quad (2.1.6)$$

## 2.2

Let  $X$  be an  $S$ -scheme. Recall that for  $L \in \text{ob } D^-(X)$ ,  $M \in \text{ob } D(X)$ ,  $N \in \text{ob } D^+(X)$ , one has the adjunction isomorphism (“dear to Cartan,” as Grothendieck says)

$$\text{Hom}(L \otimes^L M, N) = \text{Hom}(L, \text{RHom}(M, N)). \quad (2.2.1)$$

For  $E \in \text{ob } D^-(X)$ ,  $F \in \text{ob } D^+(X)$ , the identity arrow of  $\text{RHom}(E, F)$  defines via 2.2.1 an arrow

$$E \otimes^L \text{RHom}(E, F) \longrightarrow F, \quad (2.2.2)$$

called the *evaluation arrow*: it is derived from the arrow at the level of complexes

$$\text{Hom}^\bullet(E, F) \otimes E \longrightarrow F, \quad f \otimes x \longmapsto f(x).$$

Let, for  $i = 1, 2$ ,  $E_i, F_i \in \text{ob } D_{\text{ctf}}(X)$ . By adjunction (2.2.1), the arrow

$$E_1 \otimes^L E_2 \otimes \text{RHom}(E_1, F_1) \otimes \text{RHom}(E_2, F_2) \longrightarrow F_1 \otimes^L F_2,$$

composed of a symmetry isomorphism and of the tensor product of the evaluation arrows relative to  $(E_1, F_1)$  and  $(E_2, F_2)$ , supplies an arrow

$$\text{RHom}(E_1, F_1) \otimes \text{RHom}(E_2, F_2) \longrightarrow \text{RHom}(E_1 \otimes^L E_2, F_1 \otimes^L F_2). \quad (2.2.3)$$

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme,  $X = X_1 \times_S X_2$ ,  $p_i : X \rightarrow X_i$  the canonical projection,  $E_i, F_i \in \text{ob } D_{\text{ctf}}(X_i)$ ,  $E = E_1 \otimes_S^L E_2$ ,  $F = F_1 \otimes_S^L F_2$ . One has a canonical arrow

$$\text{RHom}(E_1, F_1) \otimes_S^L \text{RHom}(E_2, F_2) \longrightarrow \text{RHom}(E, F), \quad (2.2.4)$$

obtained by composing the tensor product of the arrows  $p_i^* \text{RHom}(E_i, F_i) \rightarrow \text{RHom}(p_i^* E_i, p_i^* F_i)$  and 2.2.3 for  $(p_i^* E_i, p_i^* F_i)$ .

### Proposition 2.3

The arrow 2.2.4 is an isomorphism.

By localization and dévissage, one reduces to  $E_i = f_{i!} A_{U_i}$  for  $f_i : U_i \rightarrow X_i$  étale. One has canonical isomorphisms (duality)

$$\text{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} f_{i*} f_i^* F_i, \quad \text{RHom}(E, F) \xrightarrow{\sim} f_* f^* F \quad (f = f_1 \times_S f_2). \quad (1)$$

Taking into account the Künneth formula 1.6.4, it suffices, to conclude, to verify that these isomorphisms make the square

$$\begin{array}{ccc} \text{RHom}(f_{1!} A_{U_1}, F_1) \otimes_S^L \text{RHom}(f_{2!} A_{U_2}, F_2) & \xrightarrow{2.2.3} & \text{RHom}(f_! A_U, F) \\ \sim \downarrow & & \downarrow \sim \\ f_{1*} f_1^* F_1 \otimes_S^L f_{2*} f_2^* F_2 & \xrightarrow{1.6.4} & f_* f^* F \end{array}$$

commute, where  $U = U_1 \times_S U_2$ . Since  $f_i$  is étale, the isomorphism (1) relative to  $f_i$  (resp.  $f$ ) is adjoint to the composite arrow

$$f_i^* \text{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} \text{RHom}(f_i^* f_{i!} A_{U_i}, f_i^* F_i) \longrightarrow f_i^* F_i, \quad (3)$$

where the second arrow is deduced from the adjunction  $A_{U_i} \rightarrow f_i^* f_{i!} A_{U_i}$  (resp. the analogous composite arrow with  $f$ ). The commutativity of (2) is equivalent to that of the square

$$\begin{array}{ccc} f_1^* \mathrm{RHom}(f_{1!} A_{U_1}, F_1) \otimes_S^L f_2^* \mathrm{RHom}(f_{2!} A_{U_2}, F_2) & \longrightarrow & f^* \mathrm{RHom}(f_! A_U, F) \\ \downarrow & & \downarrow \\ f_1^* F_1 \otimes_S^L f_2^* F_2 & \xrightarrow{\sim} & f^* F, \end{array}$$

where the vertical arrows are given by (3). But the commutativity of this square follows from the fact that 2.2.3 is functorial in  $E_i$  and reduces to the identity for  $E_i = A$ , whence the proposition.

## 2.4

Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism,

$$f = f_1 \times_S f_2 : X \rightarrow Y,$$

$E_i, F_i \in \mathrm{ob} D_{\mathrm{ctf}}(X_i)$ ,  $P_i, Q_i \in \mathrm{ob} D_{\mathrm{ctf}}(Y_i)$ ,

$$E = E_1 \otimes_S^L E_2, \quad F = F_1 \otimes_S^L F_2, \quad P = P_1 \otimes_S^L P_2, \quad Q = Q_1 \otimes_S^L Q_2.$$

Consider the diagram

$$\begin{array}{ccccc} (E_1, F_1) & & X_1 \longleftarrow X_1 \times_S Y_2 \longleftarrow X & & (E, F) \\ \downarrow f_1 & & \downarrow & \swarrow f & \downarrow \\ (P_1, Q_1) & & Y_1 \longleftarrow Y \longleftarrow Y_1 \times_S X_2 & & \\ & & \downarrow & \swarrow & \downarrow \\ S \longleftarrow Y_2 \longleftarrow X_2 & & f_2 & & (E_2, F_2). \end{array} \quad (2.4.0)$$

Give ourselves  $(u_1, v_1) \in \mathrm{Hom}_{f_1}^{(3,4)}((E_1, F_1), (P_1, Q_1))$ ,  $(u_2, v_2) \in \mathrm{Hom}_{f_2}^{(2,1)}((E_2, F_2), (P_2, Q_2))$  (notation of 2.1.2 and 1.4). By 1.8, one deduces from  $(u_i, v_i)$  a square

$$\begin{array}{ccc} (E, F) & \xleftarrow{(E_1 \otimes_S^L u_2, F_1 \otimes_S^L v_2)} & (E_1 \otimes_S^L P_2, F_1 \otimes_S^L Q_2) \\ \downarrow (u_1 \otimes_S^L E_2, v_1 \otimes_S^L F_2) & & \downarrow (u_1 \otimes_S^L P_2, v_1 \otimes_S^L Q_2) \\ (P_1 \otimes_S^L E_2, Q_1 \otimes_S^L F_2) & \xrightarrow{(P_1 \otimes_S^L u_2, Q_1 \otimes_S^L v_2)} & (P, Q), \end{array} \quad (2.4.1)$$

where the vertical arrows (resp. horizontal arrows) are of type  $(3, 4)$  (resp.  $(2, 1)$ ).

We may now state the main compatibility of this number, which will play an essential role in the verification of the Lefschetz–Verdier formula.

## Proposition 2.5

The diagram 2.5.1 below, where the square on the right is deduced from 2.4.1 by applying  $\mathrm{RHom}$  in the sense of 2.1.3 and 2.1.5, is commutative:

$$\begin{array}{ccccc}
 \mathrm{RHom}(E_1, F_1) \otimes_S^L \mathrm{RHom}(E_2, F_2) & \xrightarrow{2.2.4} & \mathrm{RHom}(E, F) & & \\
 \downarrow & & \searrow & & \\
 \mathrm{RHom}(u_1, v_1) \otimes_S^L \mathrm{RHom}(u_2, v_2) & & & & \mathrm{RHom}(E_1 \otimes_S^L P_2, F_1 \otimes_S^L Q_2) \\
 \downarrow & & & & \downarrow \\
 \mathrm{RHom}(P_1, Q_1) \otimes_S^L \mathrm{RHom}(P_2, Q_2) & \xleftarrow{2.2.4} & \mathrm{RHom}(P, Q) & \xrightarrow{\quad} & \mathrm{RHom}(P_1 \otimes_S^L E_2, Q_1 \otimes_S^L F_2)
 \end{array} \tag{2.5.1}$$

## 2.6

The proof of 2.5 will occupy the rest of no. 2. It follows from 1.6.3 that, if  $w_i : L_i \rightarrow M_i$  is an arrow of type 1 over  $f_i$ , the arrow  $w = w_1 \otimes_S^L w_2$  makes the diagram

$$\begin{array}{ccc}
 & L_1 \otimes_S^L L_2 & \\
 L_1 \otimes_S^L w_2 \swarrow & & \searrow w_1 \otimes_S^L L_2 \\
 L_1 \otimes_S^L M_2 & & M_1 \otimes_S^L L_2 \\
 w_1 \otimes_S^L M_2 \searrow & & \swarrow M_1 \otimes_S^L w_2 \\
 & M_1 \otimes_S^L M_2 &
 \end{array}$$

commute. Applying this remark to  $w_i = \mathrm{RHom}(u_i, v_i)$ , one sees that it suffices to prove the following special cases of 2.5.

### Corollary 2.6.1

Let  $f : X \rightarrow Y$  be an  $S$ -morphism,  $S'$  an  $S$ -scheme,  $f' : X' \rightarrow Y'$  the morphism deduced from  $f$  by the base change  $S' \rightarrow S$ ,  $E, F \in \mathrm{ob} D_{\mathrm{ctf}}(X)$ ,  $P, Q \in \mathrm{ob} D_{\mathrm{ctf}}(Y)$ ,  $P', Q' \in \mathrm{ob} D_{\mathrm{ctf}}(S')$ , and  $(u, v) \in \mathrm{Hom}_f^{(2,1)}((E, F), (P, Q))$  (resp.  $\mathrm{Hom}_f^{(3,4)}((E, F), (P, Q))$ ). Then one has a commutative square, whose vertical arrows are of type 1 over  $f'$ :

$$\begin{array}{ccc}
 \mathrm{RHom}(E, F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{2.2.4} & \mathrm{RHom}(E \otimes_S^L P', F \otimes_S^L Q') \\
 \mathrm{RHom}(u, v) \otimes_S^L \mathrm{RHom}(P', Q') \downarrow & & \downarrow \mathrm{RHom}(u \otimes_S^L P', v \otimes_S^L Q') \\
 \mathrm{RHom}(P, Q) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{2.2.4} & \mathrm{RHom}(P \otimes_S^L P', Q \otimes_S^L Q').
 \end{array} \tag{2.6.1.1}$$

First examine the case where  $(u, v)$  is of type (2, 1). By functoriality, one may suppose that  $E = f^*P$ ,  $F = f^*Q$ ,  $(u, v)$  being given by the identity of  $(f^*P, f^*Q)$ . The commutativity of 2.6.1.1 in the case

$P' = Q' = A_{S'}$  is evident; to establish it in the general case, it therefore suffices to verify that, for  $L, M \in \text{ob } D_{\text{ctf}}(Y)$ , the square

$$\begin{array}{ccc} f^* \text{RHom}(E, F) \otimes^L f^* \text{RHom}(L, M) & \xrightarrow{f^*(2.2.3)} & f^* \text{RHom}(E \otimes^L L, F \otimes^L M) \\ \downarrow 2.1.1 \otimes 2.1.1 & & \downarrow 2.1.1 \\ \text{RHom}(f^* E, f^* F) \otimes^L \text{RHom}(f^* L, f^* M) & \xrightarrow{2.2.3} & \text{RHom}(f^*(E \otimes^L L), f^*(F \otimes^L M)) \end{array}$$

is commutative. But one sees immediately that the two composites of this square are adjoint to the tensor product of the evaluation arrows

$$f^* E \otimes^L f^* \text{RHom}(E, F) \longrightarrow f^* F \quad \text{and} \quad f^* L \otimes^L f^* \text{RHom}(L, M) \longrightarrow f^* M,$$

which proves the assertion.

Now pass to the case where  $(u, v)$  is of type  $(3, 4)$ . By functoriality, one may suppose that  $P = f_! E$ ,  $F = f^! Q$ ,  $(u, v)$  being defined by  $(\text{Id}_P, \text{Id}_F)$ . One must prove the commutativity of the square

$$\begin{array}{ccc} f_* \text{RHom}(E, f^! Q) \otimes_S^L \text{RHom}(P', Q') & \xrightarrow{b} & f'_* \text{RHom}(E \otimes_S^L P', f^! Q \otimes_S^L Q') \\ \downarrow a \otimes \text{RHom}(P', Q') & & \downarrow a' \\ \text{RHom}(f_! E, Q) \otimes_S^L \text{RHom}(P', Q') & \xrightarrow{2.2.4} & \text{RHom}(f_! E \otimes_S^L P', Q \otimes_S^L Q'), \end{array}$$

where  $b$  is the composite of  $f'_*(2.2.4)$  and a base-change isomorphism,  $a$  is the duality isomorphism (SGA 4 XVIII 3.1.9.6), and  $a'$  is the composite of the duality isomorphism (relative to  $E \otimes_S^L P'$  and  $Q \otimes_S^L Q'$ ) and the base-change isomorphisms

$$f'_!(E \otimes_S^L P') \xrightarrow{\sim} f_! E \otimes_S^L P', \quad f^! Q \otimes_S^L Q' \xrightarrow{\sim} f'^!(Q \otimes_S^L Q').$$

Recall that  $a$  is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_* \text{RHom}(E, f^! Q) \longrightarrow \text{RHom}(f_! E, f_! f^! Q)$$

and of the arrow defined by the adjunction arrow  $f_! f^! Q \rightarrow Q$ . There is an analogous description of  $a'$ , as the composite of the quoted arrow

$$f'_* \text{RHom}(E \otimes_S^L P', f'^!(Q \otimes_S^L Q')) \longrightarrow \text{RHom}(f'_!(E \otimes_S^L P'), f'_! f'^!(Q \otimes_S^L Q')),$$

of the arrows defined by  $c$  and  $d$ , and of the arrow defined by the adjunction arrow

$$f'_! f'^!(Q \otimes_S^L Q') \longrightarrow Q \otimes_S^L Q'.$$

By definition,  $d$  is adjoint to the arrow

$$f'_!(f^! Q \otimes_S^L Q') \longrightarrow Q \otimes_S^L Q'$$

defined by the base-change isomorphism  $f'_!(f^! Q \otimes_S^L Q') \simeq f_! f^! Q \otimes_S^L Q'$  and the adjunction arrow  $f_! f^! Q \rightarrow Q$ ; in other words, the square

$$\begin{array}{ccc} f'_!(f^! Q \otimes_S^L Q') & \xrightarrow{f'_! d} & f'_! f'^!(Q \otimes_S^L Q') \\ \downarrow \text{base ch. } \sim & & \downarrow \text{adj} \\ f_! f^! Q \otimes_S^L Q' & \xrightarrow{\text{adj} \otimes 1} & Q \otimes_S^L Q' \end{array}$$

is commutative. It follows that one may also describe  $a'$  as the composite of the arrow above, of the arrow defined by  $c$ , and of those defined by the base-change isomorphism

$$f'_!(f^!Q \otimes_S^L Q') \xrightarrow{\sim} f_!f^!Q \otimes_S^L Q'$$

and the adjunction arrow  $f_!f^!Q \rightarrow Q$ . This decomposition of  $a'$  and that of  $a$  given above, together with the commutativity of the square

$$\begin{array}{ccc} \mathrm{RHom}(f_!E, f_!f^!Q) \otimes_S^L \mathrm{RHom}(P', Q') & \longrightarrow & \mathrm{RHom}(f_!E \otimes_S^L P', f_!f^!Q \otimes_S^L Q') \\ \downarrow & & \downarrow \\ \mathrm{RHom}(f_!E, Q) \otimes_S^L \mathrm{RHom}(P', Q') & \longrightarrow & \mathrm{RHom}(f_!E \otimes_S^L P', Q \otimes_S^L Q'), \end{array}$$

defined by 2.2.4 and the adjunction arrow  $f_!f^!Q \rightarrow Q$ , show that, to establish the commutativity of 2.6.2, it suffices to prove the following compatibility.

### Lemma 2.6.3

Let  $f, f', E, F, P', Q'$  be as in 2.6.1. Then one has a commutative square

$$\begin{array}{ccc} f_*\mathrm{RHom}(E, F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{(1)} & f'_*\mathrm{RHom}(E \otimes_S^L P', F \otimes_S^L Q') \\ (2) \downarrow & & \downarrow (4) \\ \mathrm{RHom}(f_!E, f_!F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{(3)} & \mathrm{RHom}(f_!E \otimes_S^L P', f_!F \otimes_S^L Q'), \end{array}$$

where (1) is the composite of the base-change isomorphism

$$f_*\mathrm{RHom}(E, F) \otimes_S^L - \xrightarrow{\sim} f'_*(\mathrm{RHom}(E, F) \otimes_S^L -)$$

and  $f'_*(2.2.4)$ , (2) is deduced from the canonical arrow

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_!E, f_!F)$$

(SGA 4 XVIII 3.1.9.5) by applying  $\otimes_S^L \mathrm{RHom}(P', Q')$ , (3) is given by 2.2.4, and finally (4) is the composite of the canonical arrow

$$f'_*\mathrm{RHom}(E \otimes_S^L P', F \otimes_S^L Q') \longrightarrow \mathrm{RHom}(f'_!(E \otimes_S^L P'), f'_!(F \otimes_S^L Q'))$$

(loc. cit.) and the arrows defined by the base-change isomorphisms

$$f'_!(E \otimes_S^L P') \xrightarrow{\sim} f_!E \otimes_S^L P', \quad f'_!(F \otimes_S^L Q') \xrightarrow{\sim} f_!F \otimes_S^L Q'.$$

Returning to the definition of the arrow (SGA 4 XVIII 3.1.9.5), choose a compactification of  $f$ <sup>8</sup>; one is then reduced to proving the lemma separately in the case where  $f$  is an open immersion and in the case where  $f$  is a proper morphism.

---

<sup>8</sup>The arrow quoted is defined with the aid of such a compactification, but it does not depend on it: this point, tacitly admitted in (loc. cit.), is easily verified.

a) *Case where  $f$  is an open immersion.* By adjunction (2.2.1), it suffices to see that, if one tensors the square 2.6.3 by  $f_!E \otimes_S^L P'$  and composes with the evaluation arrow 2.2.2

$$(f_!E \otimes_S^L P') \otimes^L \mathrm{RHom}(f_!E \otimes_S^L P', f_!F \otimes_S^L Q') \longrightarrow f_!F \otimes_S^L Q',$$

the square obtained is commutative. But the sums of this square are zero outside  $X'$ , so that it suffices to verify that the square induced on  $X'$  commutes, which is trivial.

b) *Case where  $f$  is a proper morphism.* Then  $f_! = f_*$ ,  $f'_! = f'_*$ . One sees easily that the composite of

$$f_* \mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_*E, f_*F)$$

with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E, f_*F) \xrightarrow{\sim} f_* \mathrm{RHom}(f^*f_*E, F)$$

is none other than the arrow deduced from the adjunction arrow  $f^*f_*E \rightarrow E$  by applying  $f_* \mathrm{RHom}(-, F)$ . Similarly, the composite of (4) with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E \otimes_S^L P', f_*F \otimes_S^L Q') \xrightarrow{\sim} f'_* \mathrm{RHom}(f^*f_*E \otimes_S^L P', F \otimes_S^L Q')$$

( $f_*F \otimes_S^L Q'$  being identified with  $f'_*(F \otimes_S^L Q')$  by the canonical isomorphism) is none other than the arrow deduced from the adjunction arrow  $f^*f_*E \rightarrow E$  by applying  $f'_* \mathrm{RHom}(- \otimes_S^L P', F \otimes_S^L Q')$ . Thus one is reduced to verifying the commutativity of the square

$$\begin{array}{ccc} \mathrm{RHom}(f_*E, f_*F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{(3)} & \mathrm{RHom}(f_*E \otimes_S^L P', f_*F \otimes_S^L Q') \\ \downarrow & & \downarrow \\ f_* \mathrm{RHom}(f^*f_*E, F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{(1')} & f'_* \mathrm{RHom}(f^*f_*E \otimes_S^L P', F \otimes_S^L Q'), \end{array}$$

where the vertical arrows are the trivial-duality isomorphisms and (1') is (1) with  $E$  replaced by  $f^*f_*E$ . But this is a special case of 2.6.1.1 in the case, already treated, where  $(u, v)$  is of type (2, 1) (take  $u : f^*f_*E \rightarrow f_*E$  given by the identity of  $f^*f_*E$ , and  $v : F \rightarrow f_*F$  given by the identity of  $f_*F$ ). We have therefore proved 2.6.3. As we have seen, the commutativity of 2.6.2 follows, which completes the proof of 2.6.1 and, finally, of 2.5.

## 4 Cohomological Correspondences

### 3.1

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme and  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ , and let

$$X = X_1 \times_S X_2, \quad p_i : X \longrightarrow X_i, \quad q_i : X_i \longrightarrow S$$

be the canonical projections:

$$\begin{array}{ccc} & X & \\ p_1 \swarrow & & \searrow p_2 \\ X_1 & & X_2 \\ q_1 \searrow & & \swarrow q_2 \\ & S & \end{array}$$

With the notation of 1.3, one has a canonical isomorphism (2.2.4)

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \text{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2).$$

Composing with the Künneth isomorphism

$$q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} p_2^! L_2 \quad (1.7.3),$$

one obtains an isomorphism

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \text{RHom}(p_1^* L_1, p_2^! L_2). \quad (3.1.1)$$

This isomorphism will play a fundamental role throughout. For  $X_i = S$ , 3.1.1 reduces to

$$L_1^\vee \overset{\mathbf{L}}{\otimes} L_2 \xrightarrow{\sim} \text{RHom}(L_1, L_2),$$

a special case of an isomorphism valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.7). We set

$$\text{RHom}(p_1^* L_1, p_2^! L_2) = \text{RHom}_S(L_1, L_2), \quad \text{Hom}(p_1^* L_1, p_2^! L_2) = \text{Hom}_S(L_1, L_2). \quad (3.1.2)$$

The elements of  $\text{Hom}_S(L_1, L_2)$  will be called *cohomological correspondences from  $L_1$  to  $L_2$* . Instead of “cohomological correspondence from  $A_{X_1}$  to  $A_{X_2}$ ”, we shall simply say *cohomological correspondence from  $X_1$  to  $X_2$*  (understood: with coefficients in  $A$ ).

If  $X_1$  is smooth, purely of relative dimension  $r_1$ , then the same is true of  $p_2$ , and by the duality theorem (SGA 4 XVIII 3.2.5) one has

$$p_2^! L_2 \xrightarrow{\sim} p_2^* L_2(r_1)[2r_1],$$

therefore

$$\text{RHom}_S(L_1, L_2) = \text{RHom}(p_1^* L_1, p_2^* L_2)(r_1)[2r_1], \quad \text{Hom}_S(L_1, L_2) = H^{2r_1}(X, \text{RHom}(p_1^* L_1, p_2^* L_2)(r_1)). \quad (3.1.3)$$

In particular, the group of cohomological correspondences from  $X_1$  to  $X_2$  is identified in this case with  $H^{2r_1}(X, A(r_1))$ .

### 3.2

The cohomological correspondences met in practice are associated with  $S$ -morphisms  $X_2 \rightarrow X_1$ , or more generally with cycles on  $X$ , or with schemes over  $X$ . Let

$$c : C \longrightarrow X$$

be a scheme over  $X$  (one sometimes says that  $c$  is a correspondence between  $X_1$  and  $X_2$ ), and put  $c_i = p_i c : C \rightarrow X_i$  for the projections. We shall call *cohomological correspondences from  $L_1$  to  $L_2$  with support in  $c$*  (or in  $C$ , if there is no risk of confusion) the elements of the group

$$H^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \quad (= \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \text{ if } c \text{ is proper}).$$

By the induction formula (SGA 4 XVIII 3.1.12.2), one has a canonical isomorphism

$$c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}(c_1^* L_1, c_2^! L_2), \quad (3.2.1)$$

whence, applying  $c_!$  and composing with the adjunction arrow  $c_! c^! \rightarrow \mathrm{Id}$ , canonical arrows

$$c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2), \quad (3.2.2)$$

$$H^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \longrightarrow \mathrm{Hom}_S(L_1, L_2). \quad (3.2.3)$$

Thus every correspondence from  $L_1$  to  $L_2$  with support in  $C$  canonically defines, by 3.2.3, a correspondence (without support) from  $L_1$  to  $L_2$ .

*Examples.* a) Suppose that  $c$  is proper and that  $c_2$  has finite tor-dimension and relative dimension  $\leq N$ . The trace morphism (SGA 4<sup>1/2</sup> Cycle 2.3.3)

$$\mathrm{Tr}_{c_2} : c_{2!} A_C \longrightarrow A_{X_2}(-N)[-2N]$$

defines by adjunction a cohomological correspondence from  $A_{X_1}$  to  $A_{X_2}(-N)[2N]$  with support in  $C$ , which will be called the *class of  $(c, N)$* , and will be denoted

$$\mathrm{cl}(c, N) \in H^{-2N}(C, c_2^! A_{X_2}(-N)). \quad (3.2.4)$$

The most important case is that in which  $N = 0$ , i.e.  $c_2$  is finite (and of finite tor-dimension); in that case  $\mathrm{cl}(c)$  is a cohomological correspondence from  $X_1$  to  $X_2$  with support in  $C$ . If  $X_1$  is smooth, purely of relative dimension  $r_1$ , then, by 3.1.3 and 3.2.1,

$$H^{-2N}(C, c_2^! A_{X_2}(-N)) = H^{2(r_1-N)}(C, c^! A_X(r_1 - N)). \quad (3.2.5)$$

If moreover  $c$  is a closed immersion, the second member rewrites as

$$H_C^{2(r_1-N)}(X, A_X(r_1 - N)),$$

and the class of  $(c, N)$ , considered as an element of this group, is precisely the cohomology class associated with  $c$  in the sense of (SGA 4<sup>1/2</sup> Cycle 2.3.1), as follows immediately from the definitions.

b) Let  $F : X_2 \rightarrow X_1$  be an  $S$ -morphism, and denote by  $c(F) : C(F) \rightarrow X$  its graph, i.e. the image of the section of  $p_2$  defined by  $F$  ("the set of points  $(Fx, x)$ "):  $c_2$  is an isomorphism and  $c_1 c_2^{-1} = F$ . One therefore has canonical isomorphisms

$$\mathrm{RHom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{RHom}(F^* L_1, L_2), \quad \mathrm{Hom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{Hom}(F^* L_1, L_2). \quad (3.2.6)$$

In other words, the cohomological correspondences from  $L_1$  to  $L_2$  with support in the graph of  $F$  are the “lifts” of  $F$  to a homomorphism  $F^*L_1 \rightarrow L_2$ . Here are two important examples of such correspondences:

(i)  $X_1 = X_2$ ,  $L_1 = L_2$ ,  $F = 1$  is the identity of  $X_1$ , whose graph  $\Delta$  is the diagonal. The correspondence with support in  $\Delta$  defined by the identity of  $L_1$  is called the *diagonal correspondence*. It is the element

$$1 \in \text{Hom}(L_1, L_1) = H^0(X, \Delta_X^! \text{RHom}_S(L_1, L_1))$$

(a group that is identified with  $H_{\Delta}^{2r}(X, \text{RHom}_S(L_1, L_1)(r))$  when  $X_1$  is smooth, purely of relative dimension  $r$ ).

(ii) Suppose that  $S$  is the spectrum of a finite field  $\mathbb{F}_q$ , that  $X_1 = X_2$ ,  $L_1 = L_2$ , and take for  $F : X_1 \rightarrow X_1$  the relative Frobenius endomorphism over  $\mathbb{F}_q$ , the identity on the underlying space and raising to the  $q$ -th power on  $\mathcal{O}_{X_1}$  (cf. XIV), and take as lift of  $F$  to  $L_1$  the canonical isomorphism

$$F^*L_1 \xrightarrow{\sim} L_1$$

(loc. cit. §2, Prop. 1) (more precisely, the  $a$ -th iterate of the isomorphism of (loc. cit.) if  $q = p^a$ ). The cohomological correspondence from  $L_1$  to  $L_1$  with support in the graph of  $F$  so defined is called the *Frobenius correspondence* (relative to  $\mathbb{F}_q$ ).

c) Certain cohomological correspondences between sheaves on curves, related to Hecke operators, have been studied by Langlands in ([13] §7).

d) Cohomological correspondences between constant sheaves (the most frequent ones in nature) furnish in a natural way, by “integration”, cohomological correspondences between complexes, as we shall now see.

### 3.3

Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism, and put

$$f = f_1 \times_S f_2 : X \rightarrow Y, \quad f'_1 = f_1 \times_S Y_2, \quad f'_2 = Y_1 \times_S f_2,$$

$$p'_1 = X_1 \times_S f_2, \quad p'_2 = f_1 \times_S X_2.$$

$$\begin{array}{ccc} X_1 \times_S Y_2 & \xleftarrow{p'_1} & X \\ & \searrow f & \downarrow p'_2 \\ Y & \xleftarrow{f'_2} & Y_1 \times_S X_2 \\ \parallel & & \\ Y_1 \times_S Y_2 & & \end{array} \quad (3.3.0)$$

Let  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ . There exists a unique isomorphism

$$f_* \text{RHom}_S(L_1, L_2) \xrightarrow{\sim} \text{RHom}_S(f'_1! L_1, f'_{2*} L_2) \quad (3.3.1)$$

whose inverse makes the square

$$\begin{array}{ccc}
D(f'_{1!}L_1) \otimes_S^{\mathbf{L}} f'_{2*}L_2 & \xrightarrow{3.1.1} & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \\
& & \downarrow \\
f_{1!}DL_1 \otimes_S^{\mathbf{L}} f'_{2*}L_2 & \xrightarrow{(2)} & f_*\mathrm{RHom}_S(L_1, L_2),
\end{array}$$

commutative, where (1) is deduced from the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6)

$$Df'_{1!}L_1 \xrightarrow{\sim} f_{1*}DL_1$$

by applying  $-\otimes_S^{\mathbf{L}} f'_{2*}L_2$ , and (2) is the composite of the Künneth isomorphism

$$f_{1*}DL_1 \otimes_S^{\mathbf{L}} f'_{2*}L_2 \xrightarrow{\sim} f_*(DL_1 \otimes_S^{\mathbf{L}} L_2)$$

and  $f_*(3.1.1)$ . Applying  $R\Gamma(Y, -)$  to 3.3.1, one obtains an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f'_{1!}L_1, f'_{2*}L_2). \quad (3.3.2)$$

For  $u \in \mathrm{Hom}_S(L_1, L_2)$ , the correspondence  $f_*u$  will be called the *direct image* of  $u$  by  $f$ . When  $Y_1 = Y_2 = S$ , we shall write  $u_*$  rather than  $f_*u$ . If  $Y_1 = Y_2 = S$  is the spectrum of a separably closed field, 3.3.2 reads

$$\mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)). \quad (3.3.3)$$

Thus cohomological correspondences “operate” on cohomology. The following statement will allow us to interpret this operation in several ways, and in particular to relate it to the usual definitions in situations such as a) or b) above.

**Proposition 3.4.** With the notation of 3.3.0, one has a commutative diagram of isomorphisms in  $D_{\mathrm{ctf}}(Y)$ :

$$\begin{array}{ccccc}
& & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) & & \\
& \swarrow (1) & \downarrow & \searrow (2) & \\
f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2) & & f_*\mathrm{RHom}_S(L_1, L_2) & & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2) \quad (3.4.1) \\
& \searrow f'_{1*}(3) & & \swarrow f'_{2*}(4) & \\
& & f_*\mathrm{RHom}_S(L_1, L_2) & & 
\end{array}$$

where (5) is the inverse of 3.3.1 and the arrows (1) to (4) are defined by the commutative diagrams below; in these diagrams “base ch.” denotes an arrow deduced from a base-change or Künneth isomorphism of one of the types considered in no. 1, “dual.” an arrow deduced from a duality

isomorphism of type 2.1.4 or (SGA 4 XVIII 3.1.9.6), and “triv.” a trivial isomorphism:

$$\begin{array}{c}
\mathrm{RHom}(f'_{1!}L_1 \otimes_S^{\mathbf{L}} A_{Y_2}, K_{Y_1} \otimes_S^{\mathbf{L}} f'_{2*}L_2) \xrightarrow{\text{base ch.}} \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \\
\downarrow \text{base ch.} \\
\mathrm{RHom}(f'_{1!}(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}), K_{Y_1} \otimes_S^{\mathbf{L}} f'_{2*}L_2) \\
\downarrow \text{dual.} \\
f'_{1*}\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}, f_1^!(K_{Y_1} \otimes_S^{\mathbf{L}} f'_{2*}L_2)) \\
\downarrow \text{base ch.} \\
f'_{1*}\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}, K_{X_1} \otimes_S^{\mathbf{L}} f_{2*}L_2) \xrightarrow{\text{base ch.}} f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2), \\
\\
\mathrm{RHom}(f_{1!}L_1 \otimes_S^{\mathbf{L}} A_{X_2}, K_{Y_1} \otimes_S^{\mathbf{L}} L_2) \xrightarrow{\text{base ch.}} \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\downarrow \text{base ch.} \\
\mathrm{RHom}(f_{1!}L_1 \otimes_S^{\mathbf{L}} A_{X_2}, f'_{2*}(K_{Y_1} \otimes_S^{\mathbf{L}} L_2)) \\
\downarrow \text{dual.} \\
f'_{2*}\mathrm{RHom}(f_2^*(f_{1!}L_1 \otimes_S^{\mathbf{L}} A_{X_2}), K_{Y_1} \otimes_S^{\mathbf{L}} L_2) \\
\downarrow \text{triv.} \\
f'_{2*}\mathrm{RHom}(f_{1!}L_1 \otimes_S^{\mathbf{L}} A_{X_2}, K_{Y_1} \otimes_S^{\mathbf{L}} L_2) \xrightarrow{\text{base ch.}} f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2), \\
\\
\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}, K_{X_1} \otimes_S^{\mathbf{L}} f_{2*}L_2) \xrightarrow{\text{base ch.}} \mathrm{RHom}_S(L_1, f_{2*}L_2) \\
\downarrow \text{base ch.} \\
\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}, p'_{1*}(K_{X_1} \otimes_S^{\mathbf{L}} L_2)) \\
\downarrow \text{dual.} \\
p'_{1*}\mathrm{RHom}(p_1^*(L_1 \otimes_S^{\mathbf{L}} A_{Y_2}), K_{X_1} \otimes_S^{\mathbf{L}} L_2) \\
\downarrow \text{triv.} \\
p'_{1*}\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{X_2}, K_{X_1} \otimes_S^{\mathbf{L}} L_2) \xrightarrow{\text{base ch.}} p'_{1*}\mathrm{RHom}_S(L_1, L_2),
\end{array}$$

$$\begin{array}{ccc}
\mathrm{RHom}(f_{1!}L_1 \otimes_S^{\mathbf{L}} A_{X_2}, K_{Y_1} \otimes_S^{\mathbf{L}} L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\downarrow \text{base ch.} & & \\
\mathrm{RHom}(p_{2!}'(L_1 \otimes_S^{\mathbf{L}} A_{X_2}), K_{Y_1} \otimes_S^{\mathbf{L}} L_2) & & \\
\downarrow \text{dual.} & & \\
p_{2*}'\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{X_2}, p_2'^!(K_{Y_1} \otimes_S^{\mathbf{L}} L_2)) & & \\
\downarrow \text{base ch.} & & \\
p_{2*}'\mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} A_{X_2}, K_{X_1} \otimes_S^{\mathbf{L}} L_2) & \xrightarrow{\text{base ch.}} & p_{2*}'\mathrm{RHom}_S(L_1, L_2).
\end{array}$$

This is only an explication of 2.5 in the particular case where  $(u_1, v_1) : (L_1, K_{X_1}) \rightarrow (f_{1!}L_1, K_{Y_1})$  is given by the identity of  $f_{1!}L_1$  and the canonical isomorphism  $f_1^!K_{Y_1} \xrightarrow{\sim} K_{X_1}$ , and where  $(u_2, v_2) : (A_{X_2}, L_2) \rightarrow (A_{Y_2}, f_{2*}L_2)$  is given by the canonical isomorphism  $A_{X_2} \xrightarrow{\sim} f_2^!A_{Y_2}$  and the identity of  $f_{2*}L_2$ .

**Corollary 3.5.** Assume that  $Y_1 = Y_2 = S$ . Let  $u \in \mathrm{Hom}_S(L_1, L_2)$ . The direct image  $f_*u \in \mathrm{Hom}(f_{1!}L_1, f_{2*}L_2)$  (3.3.2) can be obtained in the following two equivalent ways:

a) Identify  $u$ , by trivial duality, with an arrow

$$L_1 \longrightarrow p_{1*}p_2^!L_2;$$

compose with the base-change isomorphism

$$p_{1*}p_2^!L_2 \xrightarrow{\sim} f_1^!f_{2*}L_2;$$

$f_*u$  is the adjoint of this composite arrow.

b) Identify  $u$ , by adjunction, with an arrow

$$p_{2!}p_1^*L_1 \longrightarrow L_2;$$

compose with the base-change isomorphism

$$f_2^*f_{1!}L_1 \xrightarrow{\sim} p_{2!}p_1^*L_1;$$

$f_*u$  is the adjoint of this composite arrow.

Description a) (respectively b)) indeed corresponds to the path  $(f_{1*}(3) \circ (1))^{-1}$  (respectively  $f_{2*}(4) \circ (2)$ ) in 3.4.1.

### 3.6

In practice, it is mainly description 3.5 b) that is used. Let us spell it out at the level of cohomology classes, assuming, in order to lighten the notation, that  $S$  is the spectrum of a separably closed field. Let

$$u \in \mathrm{Hom}_S(L_1, L_2) = \mathrm{Hom}(p_1^*L_1, p_2^!L_2).$$

For  $x \in H_c^i(X_1, L_1)$ ,  $u_*(x) \in H^i(X_2, L_2)$  is defined as follows: consider the element  $p_1^*x \in H^i(X_2, p_{2!}p_1^*L_1)$  deduced from  $x$  by inverse image, apply  $u$  to it, i.e. form the product

$$u \cdot p_1^*x \in H^i(X_2, p_{2!}p_2^!L_2),$$

then apply the arrow

$$a : H^i(X_2, p_{2!}p_2^!L_2) \longrightarrow H^i(X_2, L_2)$$

defined by the adjunction arrow  $p_{2!}p_2^! \rightarrow \text{Id}$ ; in short:

$$u_*(x) = a(u \cdot p_1^*x). \quad (3.6.1)$$

Let  $c : C \rightarrow X$  be a proper morphism, and let  $u \in \text{Hom}(c_1^*L_1, c_2^!L_2)$ . Denote by

$$u_* \in \text{Hom}(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2))$$

the image of  $u$  under the composite of 3.2.3 and 3.3.3. It follows from 3.6.1 that, for  $x \in H_c^i(X_1, L_1)$ , one has

$$u_*(x) = a(u \cdot c_1^*x), \quad (3.6.2)$$

where  $c_1^*x$  is the inverse image of  $x$  in  $H^i(X_2, c_{2!}c_1^*L_1)$  and

$$a : H^i(X_2, c_{2!}c_2^!L_2) \longrightarrow H^i(X_2, L_2)$$

is the arrow deduced from the adjunction  $c_{2!}c_2^!L_2 \rightarrow \text{Id}$ . In the situation of Example 3.2 a), one thus finds

$$\text{cl}(c, N)_*(x) = \text{Tr}_{c_2}(c_1^*x), \quad (3.6.3)$$

where

$$\text{Tr}_{c_2} : H^i(X_2, Rc_{2!}A_C) \longrightarrow H^{i-2N}(X_2, A(-N))$$

is the arrow induced by the trace morphism relative to  $c_2$ . For  $N = 0$ , this formula agrees with (SGA 4<sup>1/2</sup> Cycle 3.6):  $\text{cl}(c)_*$  is the homomorphism denoted  $\gamma_{X, X_1}^*$  there. If  $X_1$  is smooth, purely of relative dimension  $r_1$ , and  $c$  is a closed immersion, it follows from 3.6.1 and from the remark made at the end of 3.2 a) that one also has

$$\text{cl}(c, N)_*(x) = \text{Tr}_{p_2}(\text{cl}(c) p_1^*x), \quad (3.6.4)$$

where  $\text{cl}(c) \in H_C^{2(r_1-N)}(X, A(r_1 - N))$  is the cohomology class of  $C$  and

$$\text{Tr}_{p_2} : H^*(X, Rp_{2!}A_X) \longrightarrow H^{*-2r_1}(X_2, A(-r_1))$$

is the arrow defined by the trace morphism relative to  $p_2$ . This recovers the usual definition of the homomorphism induced on cohomology by a correspondence ([12] 1.3), (SGA 4<sup>1/2</sup> Cycle 3.2).

### 3.6.5

In the situation of 3.2 b), it follows from 3.6.2 that, for  $u \in \text{Hom}(F^*L_1, L_2)$ ,

$$u_* : H_c^i(X_1, L_1) \longrightarrow H^i(X_2, L_2)$$

is the composite of the “inverse image” homomorphism

$$H_c^i(X_1, L_1) \longrightarrow H^i(X_2, F^*L_1)$$

and the homomorphism induced by  $u$  (more generally,  $f_*u : Rf_{1!}L_1 \rightarrow Rf_{2*}L_2$  is the composite of the canonical homomorphism

$$Rf_{1!}L_1 \longrightarrow Rf_{2*}F^*L_1$$

and of  $Rf_{2*}u$ ). In particular, the diagonal correspondence induces the canonical arrow

$$Rf_{1!}L_1 \longrightarrow Rf_{1*}L_1$$

(the identity if  $X_1$  is proper over  $S$ ).

### 3.7

Return to the situation of the beginning of 3.3, assuming in addition that a commutative square

$$\begin{array}{ccc} X & \xleftarrow{c} & C \\ f \uparrow & & \downarrow g \\ Y & \xleftarrow{d} & D \end{array} \quad (3.7.1)$$

is given, with  $c$  proper. Put  $c_i = p_i c$ ,  $d_i = q_i d$ , where  $p_i : X \rightarrow X_i$ ,  $q_i : Y \rightarrow Y_i$  are the projections. From 3.7.1 one deduces a commutative diagram whose inner square is cartesian:

$$\begin{array}{ccccc} & & C & & \\ & \swarrow c & \downarrow i & \searrow g & \\ X & \xleftarrow{c'} & C' & \xleftarrow{\quad} & D \\ & & \downarrow g' & & \\ Y & \xleftarrow{d} & D & & \end{array} \quad (3.7.2)$$

For  $K \in \text{ob } D_{\text{ctf}}(X)$ , there is a canonical arrow

$$f_*c_!c^!K \longrightarrow d_*d^!f_*K, \quad (3.7.3)$$

which is the composite of the arrow

$$f_*c_!c^!K = f_*c'_!(i_*i^!)c^!K \longrightarrow f_*c'_!c^!K = d_*g'_!c^!K$$

defined by the adjunction arrow  $i_*i^! \rightarrow \text{Id}$ , and the isomorphism

$$d_*g'_!c^!K \xrightarrow{\sim} d_*d^!f_*K$$

defined by the canonical isomorphism  $g'_!d^! \xrightarrow{\sim} d^!f_*$  (SGA 4 XVIII 3.1.12.3). Taking  $K = \text{RHom}_S(L_1, L_2)$ , and composing 3.7.3 with the isomorphism deduced from 3.3.1, one obtains a canonical arrow

$$f_*c_!c^!\text{RHom}_S(L_1, L_2) \longrightarrow d_*d^!\text{RHom}_S(f_{1!}L_1, f_{2*}L_2), \quad (3.7.4)$$

which, by 3.2.1, may be rewritten as

$$f_*c_!\text{RHom}(c_1^*L_1, c_2^!L_2) \longrightarrow d_*\text{RHom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \quad (3.7.5)$$

One deduces

$$f_* : \text{Hom}(c_1^*L_1, c_2^!L_2) \longrightarrow \text{Hom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \quad (3.7.6)$$

The arrow 3.7.5 (respectively 3.7.6) generalizes 3.3.1 (respectively 3.3.2), with which it coincides when  $c$  and  $d$  are the identity. Also note that, for  $c$  proper, 3.2.2 (respectively 3.2.3) is nothing other than 3.7.5 (respectively 3.7.6) when  $f_1$ ,  $f_2$ , and  $d$  are the identity arrows. When  $Y_1 = Y_2 = S = D$ , we shall abbreviate  $f_*u$  as  $u_*$ , as above.

## 5 Pairings of Correspondences. Lefschetz Formula

### 4.1

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme,  $X = X_1 \times_S X_2$ ,  $p_i : X \rightarrow X_i$  the projections,  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ , and  $K_{X_i}$  the dualizing complex (1.3). The canonical arrows

$$p_i^* \text{RHom}(L_i, K_{X_i}) \overset{\mathbf{L}}{\otimes} p_i^* L_i \longrightarrow p_i^* K_{X_i}, \quad (4.1.1)$$

composed from 2.1.1 and the evaluation arrow 2.2.2, give, by tensor product on  $X$  and composition with the Künneth isomorphism 1.7.6, a pairing

$$(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2) \overset{\mathbf{L}}{\otimes} (L_1 \overset{\mathbf{L}}{\otimes}_S DL_2) \longrightarrow K_X, \quad (4.1.2)$$

which may be rewritten, by 3.1.1 and 3.1.2, as

$$\text{RHom}_S(L_1, L_2) \overset{\mathbf{L}}{\otimes} \text{RHom}_S(L_2, L_1) \longrightarrow K_X. \quad (4.1.3)$$

One deduces a pairing, called cup product,

$$\langle \ , \ \rangle : \text{Hom}_S(L_1, L_2) \otimes \text{Hom}_S(L_2, L_1) \longrightarrow H^0(X, K_X). \quad (4.1.4)$$

One may also regard 4.1.2 as the external tensor product of the evaluation arrows

$$DL_i \overset{\mathbf{L}}{\otimes} L_i \longrightarrow K_{X_i}.$$

In view of the Künneth isomorphism 2.2.4, one thus sees that 4.1.2 is a “perfect duality”, i.e. identifies by the adjunction isomorphism 2.2.1 each of the complexes  $DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$  and  $L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$  with the dual of the other (with values in  $K_X$ ).

When  $X_1 = X_2 = S$ , 4.1.3 is a special case of a pairing valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.8), and, with the notation of (SGA 6 I 8.3), one has

$$\langle u, v \rangle = \text{Tr}(vu) = \text{Tr}(uv). \quad (4.1.5)$$

### 4.2

Let  $c : C \rightarrow X$ ,  $d : D \rightarrow X$ , and  $e = c \times_X d : E = C \times_X D \rightarrow X$ . For  $P, Q \in \text{ob } D_{\text{ctf}}(X)$ , one has a canonical arrow (which is not in general an isomorphism)

$$c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q \longrightarrow e^! (P \overset{\mathbf{L}}{\otimes} Q), \quad (4.2.1)$$

defined, in the same way as 1.7.3, as the adjoint of the composite arrow

$$e_!(c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q) \xrightarrow[(1)]{\sim} c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \xrightarrow{(2)} P \overset{\mathbf{L}}{\otimes} Q,$$

where (1) is the Künneth isomorphism (SGA 4 XVII 5.4.2.2) and (2) is the tensor product of the adjunction arrows. Applying  $e_!$  to 4.2.1 and composing with the Künneth isomorphism just cited, one obtains a canonical arrow

$$c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \longrightarrow e_! e^! (P \overset{\mathbf{L}}{\otimes} Q). \quad (4.2.2)$$

Similarly, composing the Künneth arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow e_*(c^! P \otimes_X^{\mathbf{L}} d^! Q)$$

with the arrow deduced from 4.2.1 by applying  $e_*$ , one obtains a canonical arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow e_* e^!(P \otimes^{\mathbf{L}} Q). \quad (4.2.3)$$

For  $P = DL_1 \otimes_S^{\mathbf{L}} L_2$ ,  $Q = L_1 \otimes_S^{\mathbf{L}} DL_2$ , one deduces from 4.2.2 and 4.2.3, by composition with 4.1.2 and the isomorphism  $e^! K_X = K_E$ , and with the identifications 3.1.1 and 3.2.1, pairings

$$c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_! \mathrm{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow e_! K_E, \quad (4.2.4)$$

$$c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_* \mathrm{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow e_* K_E, \quad (4.2.4')$$

$$\langle \ , \ \rangle : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) \longrightarrow H^0(E, K_E). \quad (4.2.5)$$

These pairings are compatible with étale localization: given a commutative square (not necessarily cartesian)

$$\begin{array}{ccc} & E' & \\ & \swarrow \quad \searrow & \\ C' & & D' \\ & \searrow \quad \swarrow & \\ & X' & \end{array} \quad \text{étale over} \quad \begin{array}{ccc} & E & \\ & \swarrow \quad \searrow & \\ C & & D \\ & \searrow \quad \swarrow & \\ & X & \end{array}$$

if  $c'$ ,  $d'$ ,  $e'$  denote the composites  $C' \rightarrow C \rightarrow X$ ,  $D' \rightarrow D \rightarrow X$ ,  $E' \rightarrow E \rightarrow X$ , one has a commutative square.

$$\begin{array}{ccc} \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) & \xrightarrow{4.2.5} & H^0(E, K_E) \\ \downarrow & & \downarrow \\ \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(d_2'^* L_2, d_1'^! L_1) & \longrightarrow & H^0(E', K_{E'}) \end{array} \quad (4.2.6)$$

where the vertical arrows are the restrictions, and the lower horizontal arrow is the composite of the arrow analogous to 4.2.5 and the restriction

$$H^0(E'', K_{E''}) \longrightarrow H^0(E', K_{E'}),$$

with  $E'' = C' \times_{X'} D'$  (there are analogous commutative squares with arrows of type 4.2.4, 4.2.4', which we leave to the reader to write down). If  $Z$  is étale over  $E$ , we shall denote, for

$$\begin{aligned} u &\in \mathrm{Hom}(c_1^* L_1, c_2^! L_2), & v &\in \mathrm{Hom}(d_2^* L_2, d_1^! L_1), \\ \langle u, v \rangle_Z &\in H^0(Z, K_Z) \end{aligned} \quad (4.2.7)$$

the image of  $\langle u, v \rangle$  in  $H^0(Z, K_Z)$  by restriction (in practice,  $Z$  will be a connected component of  $E$ ). If  $z$  is an isolated closed point of  $E$ , and  $\bar{z}$  a geometric point above  $z$ , it follows from 4.2.6 that

$$\langle u, v \rangle_z \in H^0(z, K_z) = A$$

depends only on the situation deduced by strict localization at the images of  $\bar{z}$  in  $C, D, X$ .

**Example 4.3.** Assume that  $X_i$  is smooth, purely of relative dimension  $r_i$ , and that  $c$  and  $d$  are closed immersions, with  $c_2$  (respectively  $d_1$ ) finite. For  $L_1 = A_{X_1}$ ,  $L_2 = A_{X_2}$ , we have, by 3.1.3,

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = A_X(r_1)[2r_1], \quad L_1 \otimes_S^{\mathbf{L}} DL_2 = A_X(r_2)[2r_2],$$

and 4.1.2 is identified with the product

$$A_X(r_1)[2r_1] \otimes_X^{\mathbf{L}} A_X(r_2)[2r_2] \longrightarrow A_X(r)[2r],$$

where  $r = r_1 + r_2$ , and hence, taking 3.2.1 into account, 4.2.5 is identified with the usual cup product

$$H_C^{2r_1}(X, A(r_1)) \otimes H_D^{2r_2}(X, A(r_2)) \longrightarrow H_{C \cap D}^{2r}(X, A(r)).$$

Take as cohomological correspondence from  $X_1$  to  $X_2$  (respectively from  $X_2$  to  $X_1$ ) the class of  $c$  (respectively  $d$ ) (3.2 a)). It follows from (SGA 4<sup>1/2</sup> Cycle 2.3.8) that, at an isolated point  $z$  of  $C \cap D$ ,  $\langle \text{cl}(c), \text{cl}(d) \rangle_z$  is nothing but the image in  $A$  of the intersection multiplicity of  $C$  and  $D$  at  $z$ .

We now give the main result of the exposé.

**Theorem 4.4.** Let, for  $i = 1, 2$ ,  $f_i : X_i \longrightarrow Y_i$  be a proper  $S$ -morphism,

$$f = f_1 \times_S f_2 : X \longrightarrow Y, \quad p_i : X \longrightarrow X_i, \quad q_i : Y \longrightarrow Y_i$$

the projections. Let moreover

$$\begin{array}{ccccc} & & C' & \xrightarrow{\quad} & C \\ & \swarrow c' & \downarrow & \swarrow c & \downarrow g \\ X & \xleftarrow{f'} & C'' & & \\ \downarrow f & & \downarrow d' & \xrightarrow{f''} & D \\ & \swarrow d' & D' & \xrightarrow{f''} & D \\ Y & \xleftarrow{d''} & D'' & & \end{array} \quad (4.4.0)$$

be a commutative cube with cartesian horizontal faces, with  $c', c'', d', d''$  proper. Put

$$c'_i = p_i c', \quad c''_i = p_i c'', \quad d'_i = q_i d', \quad d''_i = q_i d''.$$

Finally, let  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ . Then there is a commutative square

$$\begin{array}{ccc} f_* c'_* \text{RHom}(c'_1{}^* L_1, c'_2{}^! L_2) \otimes^{\mathbf{L}} f_* c''_* \text{RHom}(c''_2{}^* L_2, c''_1{}^! L_1) & \xrightarrow{(1)} & f_* c_* K_C \\ \downarrow (2) & & \downarrow (4) \\ d'_* \text{RHom}(d'_1{}^! M_1, d'_2{}^! M_2) \otimes^{\mathbf{L}} d''_* \text{RHom}(d''_2{}^! M_2, d''_1{}^! M_1) & \xrightarrow{(3)} & d_* K_D \end{array} \quad (4.4.1)$$

where  $M_i = f_{i*} L_i$ , (3) is given by 4.2.4, (1) is deduced from 4.2.4 by applying  $f_*$  and composing with the canonical arrow

$$f_* c'_*(-) \otimes^{\mathbf{L}} f_* c''*(-) \longrightarrow f_*(c'_*(-) \otimes^{\mathbf{L}} c''*(-)),$$

(2) is the tensor product of the arrows 3.7.5 relative to the faces containing  $f$ , and (4) is defined by the adjunction arrow  $g_*K_C = g_*g^!K_D \longrightarrow K_D$ .

*Proof.* Put

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = P, \quad L_1 \otimes_S^{\mathbf{L}} DL_2 = Q, \quad DM_1 \otimes_S^{\mathbf{L}} M_2 = E, \quad M_1 \otimes_S^{\mathbf{L}} DM_2 = F.$$

Consider the diagram

$$\begin{array}{ccccc}
f_*c'_!P \otimes^{\mathbf{L}} f_*c''_!Q & \xrightarrow{(1)} & f_*c'_!(P \otimes^{\mathbf{L}} Q) & \xrightarrow{(2)} & f_*c'_!K_X \\
(8) \downarrow & & (9) \downarrow & & \downarrow (10) \\
d'_*d'^!f_*P \otimes^{\mathbf{L}} d''_*d''^!f_*Q & \xrightarrow{(3)} & d_*d'^!(f_*P \otimes^{\mathbf{L}} f_*Q) & \xrightarrow{(4)} & d_*d'^!(P \otimes^{\mathbf{L}} Q) & \xrightarrow{(5)} & d_*d'^!K_X \\
(11) \downarrow & & (12) \downarrow & & \downarrow (13) \\
d'_*d'^!E \otimes^{\mathbf{L}} d''_*d''^!F & \xrightarrow{(6)} & d_*d'^!(E \otimes^{\mathbf{L}} F) & \xrightarrow{(7)} & d_*d'^!K_Y.
\end{array} \tag{4.4.2}$$

The arrows are as follows: (1) is deduced from 4.2.2 by applying  $f_*$ ; (2) is defined by 4.1.2; (3) is given by 4.2.2; (4) is given by the canonical arrow  $f_*P \otimes^{\mathbf{L}} f_*Q \longrightarrow f_*(P \otimes^{\mathbf{L}} Q)$ ; (5) by 4.1.2; (6) by 4.2.2; (7) by 4.1.2; (8) is the tensor product of arrows 3.7.3; (9) and (10) are defined by 3.7.3; (11) and (12) by the isomorphisms  $f_*P \longrightarrow E$ ,  $f_*Q \longrightarrow F$  (3.3.1); and (13) by the adjunction arrow  $f_*K_X \longrightarrow K_Y$ . By definition, 4.4.1 is identified, via 3.1.1 and 3.2.1, with the outer boundary of 4.4.2. It remains only to prove that the squares (A), (B), (C), (D) are commutative. The commutativity of (B) and (C) is clear by functoriality. It remains to prove that of (A) and (D).

a) Commutativity of (D). It is enough to prove that the square

$$\begin{array}{ccc}
f_*P \otimes^{\mathbf{L}} f_*Q & \longrightarrow & f_*K_X \\
\downarrow & & \downarrow \\
E \otimes^{\mathbf{L}} F & \longrightarrow & K_Y
\end{array} \tag{4.4.3}$$

(corresponding to (D) for  $d = \text{Id}$ ) commutes. Recall that the pairing

$$P \otimes^{\mathbf{L}} Q \longrightarrow K_X \quad (\text{respectively } E \otimes^{\mathbf{L}} F \longrightarrow K_Y)$$

is the external tensor product of the evaluation arrows

$$DL_i \otimes^{\mathbf{L}} L_i \longrightarrow K_{X_i} \quad (\text{respectively } DM_i \otimes^{\mathbf{L}} M_i \longrightarrow K_{Y_i}).$$

By the Künneth isomorphisms

$$f_{1*}DL_1 \otimes_S^{\mathbf{L}} f_{2*}L_2 \xrightarrow{\sim} f_*P, \quad f_{1*}L_1 \otimes_S^{\mathbf{L}} f_{2*}DL_2 \xrightarrow{\sim} f_*Q,$$

4.4.3 is therefore identified with the external tensor product of the squares

$$\begin{array}{ccc}
f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow{(1)} & f_{i*}K_{X_i} \\
(3) \downarrow & & \downarrow (4) \\
DM_i \otimes^{\mathbf{L}} M_i & \xrightarrow{(2)} & K_{Y_i},
\end{array} \tag{4.4.4}$$

where (2) is the evaluation arrow, (1) the composite of the canonical arrow

$$f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \longrightarrow f_{i*}(DL_i \otimes^{\mathbf{L}} L_i)$$

and the arrow defined by the evaluation arrow, (3) is the tensor product of the duality isomorphism  $f_{i*}DL_i \longrightarrow DM_i$  and the identity arrow of  $M_i$ , and (4) is the adjunction arrow. It is therefore enough to verify the commutativity of the squares 4.4.4. Since the above duality isomorphism is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i})$$

and the arrow deduced from the adjunction arrow

$$f_{i*}K_{X_i} \longrightarrow K_{Y_i},$$

it is enough to see that the following square commutes:

$$\begin{array}{ccc} f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow{(1)} & f_{i*}K_{X_i} \\ (3) \downarrow & & \downarrow \mathrm{Id} \\ \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow{(2)} & f_{i*}K_{X_i}, \end{array} \quad (4.4.5)$$

where (1) is the arrow (1) of 4.4.4, (2) the evaluation arrow, and (3) the tensor product by  $f_{i*}L_i$  of the canonical arrow just mentioned. More generally, for any arrow  $N_i \longrightarrow f_{i*}L_i$  in  $D_{\mathrm{ctf}}(X_i)$ , i.e. any arrow  $L_i \xrightarrow{u} N_i$  of type 1 over  $f_i$ , the triangle

$$\begin{array}{ccc} f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} N_i & & \\ \downarrow & \searrow & \\ \mathrm{RHom}(N_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} N_i & \longrightarrow & f_{i*}K_{X_i} \end{array}$$

deduced from 4.4.5 commutes: indeed, by functoriality, one may assume that  $L_i = f_i^*N_i$ , with  $u$  given by the identity of  $L_i$ ; the arrow

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(N_i, f_{i*}K_{X_i})$$

is then the inverse of the trivial duality isomorphism, and the commutativity of the triangle is verified immediately at the level of complexes, resolving  $N_i$  flatly and  $K_{X_i}$  injectively.

b) Commutativity of (A). Referring to the definition of 3.7.3, one sees that it is enough to verify this commutativity in each of the following cases:

- (i)  $X_i = Y_i$ ,  $f_i$  = the identity arrow of  $X_i$ ; (ii) the lateral faces of 4.4.0 are cartesian.

Place ourselves in case (i). The square (A) is the outer boundary of the diagram

$$\begin{array}{ccccc} c'_*(c'^!P \otimes_X^{\mathbf{L}} c''^!Q) & \longrightarrow & c_*(c'^!P \otimes_X^{\mathbf{L}} c''^!Q) & \longrightarrow & c_*c'^!(P \otimes Q) \\ \downarrow & & \downarrow & & \downarrow \\ d'_*d'^!P \otimes_X^{\mathbf{L}} d''_*d''^!Q & \longrightarrow & d_*(d'^!P \otimes_X^{\mathbf{L}} d''^!Q) & \longrightarrow & d_*d'^!(P \otimes Q), \end{array}$$

in which the horizontal isomorphisms of  $A_1$  are Künneth isomorphisms, the horizontal arrows of  $A_2$  are given by 4.2.1, and the middle vertical arrow is deduced by applying  $d_*$  to the arrow

$$g_*(c'^!P \otimes_X^{\mathbf{L}} c''^!Q) \longrightarrow d'^!P \otimes_X^{\mathbf{L}} d''^!Q$$

defined, via Künneth, by the external tensor product of the adjunction arrows

$$f'_*c'^!P \longrightarrow d'^!P, \quad f''_*c''^!Q \longrightarrow d''^!Q.$$

These adjunction arrows and the identity arrows of  $d'^!d'^!P$ ,  $d''^!d''^!Q$  define commutative triangles of arrows of type 3 (1.4)

$$\begin{array}{ccc} & c'^!P & \\ \swarrow & \downarrow & \\ d'^!d'^!P & \longleftarrow & d'^!P \end{array} \quad \begin{array}{ccc} & c''^!Q & \\ \swarrow & \downarrow & \\ d''^!d''^!Q & \longleftarrow & d''^!Q \end{array}$$

over

$$\begin{array}{ccc} & C' & \\ \swarrow c' & \downarrow f' & \\ X & \xleftarrow{d'} & D' \end{array} \quad \begin{array}{ccc} & C'' & \\ \swarrow c'' & \downarrow f'' & \\ X & \xleftarrow{d''} & D'' \end{array}.$$

By external tensor product (over  $X$ ), these triangles give a commutative triangle

$$\begin{array}{ccc} & c'^!P \otimes_X^{\mathbf{L}} c''^!Q & \\ \swarrow & \downarrow & \\ d'^!d'^!P \otimes^{\mathbf{L}} d''^!d''^!Q & \longleftarrow & d'^!P \otimes_X^{\mathbf{L}} d''^!Q \end{array} \quad \text{over} \quad \begin{array}{ccc} & C & \\ \swarrow c & \downarrow g & \\ X & \xleftarrow{d} & D \end{array}.$$

The commutativity of this triangle is equivalent to that of the square deduced from  $A_1$  by inverting the horizontal arrows. On the other hand, there are commutative triangles of arrows of type 3

$$\begin{array}{ccc} & c'^!P & \\ \swarrow & \downarrow & \\ P & \xleftarrow{\quad} & d'^!P \end{array} \quad \begin{array}{ccc} & c''^!Q & \\ \swarrow & \downarrow & \\ Q & \xleftarrow{\quad} & d''^!Q \end{array}$$

defined by the identity arrows of  $d'^!P$ ,  $c'^!P$ ,  $d''^!Q$ ,  $c''^!Q$ . By external tensor product, one obtains a commutative triangle

$$\begin{array}{ccc} & c'^!P \otimes_X^{\mathbf{L}} c''^!Q & \\ \swarrow & \downarrow & \\ P \otimes Q & \xleftarrow{\quad} & d'^!P \otimes_X^{\mathbf{L}} d''^!Q \end{array}$$

which can be interpreted as a morphism of arrows of type 3 over  $g : C \rightarrow D$ :

$$\begin{array}{ccc} c'^! P \otimes_X^{\mathbf{L}} c''^! Q & \longrightarrow & c'^!(P \otimes^{\mathbf{L}} Q) \\ \downarrow & & \downarrow \\ d'^! P \otimes_X^{\mathbf{L}} d''^! Q & \longrightarrow & d'^!(P \otimes Q), \end{array}$$

or again as a commutative square over  $D$ :

$$\begin{array}{ccc} g_*(c'^! P \otimes_X^{\mathbf{L}} c''^! Q) & \longrightarrow & g_* c'^!(P \otimes^{\mathbf{L}} Q) \\ \downarrow & & \downarrow \\ d'^! P \otimes_X^{\mathbf{L}} d''^! Q & \longrightarrow & d'^!(P \otimes Q). \end{array}$$

But the square deduced from the latter by applying  $d_*$  is none other than  $A_2$ , whence the commutativity of (A) in case (i).

It remains to treat case (ii). Since the faces of 4.4.0 containing  $f$  are cartesian, the arrows

$$f'_* c'^! P \rightarrow d'^! f_* P, \quad f''_* c''^! Q \rightarrow d''^! f_* Q$$

(3.7.3) are isomorphisms. Their inverses define arrows of type 1

$$c'^! P \rightarrow d'^! f_* P, \quad c''^! Q \rightarrow d''^! f_* Q$$

over  $f', f''$ , whence, by direct image, commutative squares of arrows of type 1

$$\begin{array}{ccc} c'_* c'^! P & \longleftarrow & c'^! P \\ \downarrow & & \downarrow \\ d'_* d'^! P & \longleftarrow & d'^! f_* P \end{array} \quad \begin{array}{ccc} c''_* c''^! Q & \longleftarrow & c''^! Q \\ \downarrow & & \downarrow \\ d''_* d''^! Q & \longleftarrow & d''^! f_* Q \end{array}$$

over the faces of 4.4.0 containing  $f$ . Taking the tensor product, one obtains a commutative square in  $D_{\text{ctf}}(Y)$

$$\begin{array}{ccc} d'_* d'^! f_* P \otimes_Y^{\mathbf{L}} d''_* d''^! f_* Q & \longrightarrow & d_*(d'^! f_* P \otimes_Y^{\mathbf{L}} d''^! f_* Q) \\ \downarrow & & \downarrow \\ f_* c'_* c'^! P \otimes^{\mathbf{L}} f_* c''_* c''^! Q & \longrightarrow & d_* g_*(c'^! P \otimes_X^{\mathbf{L}} c''^! Q), \end{array} \tag{4.4.6}$$

where the left vertical arrow is the inverse of (8), the horizontal arrows are given by Künneth, and the right vertical arrow by the external tensor product over  $Y$  of the arrows

$$f'_* c'^! P \rightarrow d'^! f_* P, \quad f''_* c''^! Q \rightarrow d''^! f_* Q.$$

The same tensor product defines the left vertical arrow of the square

$$\begin{array}{ccc} d'^! f_* P \otimes_Y^{\mathbf{L}} d''^! f_* Q & \longrightarrow & d'^! f_*(P \otimes^{\mathbf{L}} Q) \\ \downarrow & & \downarrow \\ g_*(c'^! P \otimes_X^{\mathbf{L}} c''^! Q) & \longrightarrow & g_* c'^!(P \otimes Q), \end{array} \tag{4.4.7}$$

where the right vertical arrow is 3.7.3, and the horizontal arrows are defined by 4.2.1. The outer boundary of the diagram composed of 4.4.6 and  $d_*(4.4.7)$  is the square deduced from (A) by inverting the vertical arrows. It is therefore enough to prove that 4.4.7 commutes, in other words that the arrows 4.2.1

$$c'^! P \otimes_X^{\mathbf{L}} c''^! Q \longrightarrow c'^!(P \otimes^{\mathbf{L}} Q), \quad d'^! f_* P \otimes_Y^{\mathbf{L}} d''^! f_* Q \longrightarrow d'^! f_*(P \otimes^{\mathbf{L}} Q)$$

define a morphism from

$$c'^! P \otimes_X^{\mathbf{L}} c''^! Q \longrightarrow d'^! f_* P \otimes_Y^{\mathbf{L}} d''^! f_* Q$$

to

$$c'^!(P \otimes Q) \longrightarrow d'^! f_*(P \otimes Q),$$

viewed as arrows of type 1 over  $g$ . By adjunction, we are reduced to verifying that the tensor products of adjunction arrows

$$c'_* c'^! P \otimes_X^{\mathbf{L}} c''_* c''^! Q \longrightarrow P \otimes^{\mathbf{L}} Q, \quad d'_* d'^! f_* P \otimes_Y^{\mathbf{L}} d''_* d''^! f_* Q \longrightarrow f_* P \otimes^{\mathbf{L}} f_* Q$$

define a morphism, between arrows of type 1 over  $f$ , from the tensor product of the arrows

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P, \quad c''_* c''^! Q \longrightarrow d''_* d''^! f_* Q$$

to the tensor product of the arrows

$$P \longrightarrow f_* P, \quad Q \longrightarrow f_* Q.$$

For this it is enough to verify that the adjunction arrows

$$c'_* c'^! P \longrightarrow P, \quad d'_* d'^! f_* P \longrightarrow f_* P$$

define a morphism from

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P$$

to

$$P \longrightarrow f_* P,$$

and likewise for the pair  $(c'', d'')$ . It is enough to consider the case of  $(c', d')$ , the statement being the same in the two cases. We have to see that the triangle

$$\begin{array}{ccc} & f_* c'_* c'^! P & \\ & \swarrow & \downarrow \sim \\ f_* P & \longleftarrow & d'_* d'^! f_* P \end{array}$$

where the horizontal and oblique arrows are given by adjunction and the vertical arrow by the isomorphism 3.7.3, is commutative. Taking into account the definition of 3.7.3 (SGA 4 XVIII 3.1.12.3) as the transpose of a base-change isomorphism, we are reduced, by applying the functor  $\mathrm{Hom}(R, -)$  ( $R \in \mathrm{ob} D_{\mathrm{ctf}}(Y)$ ), to verifying the commutativity of the square

$$\begin{array}{ccc} c'_* c'^* f^* R & \xrightarrow{\mathrm{Id}} & c'_* f'^* d'^* R \\ \uparrow & & \uparrow \sim \\ f^* R & \longrightarrow & f^* d'_* d'^* R \end{array}$$

where the left vertical arrow (respectively the lower horizontal arrow) is defined by the adjunction arrow  $1 \rightarrow c'_*c'^*$  (respectively  $1 \rightarrow d'_*d'^*$ ), and the right vertical arrow by the base-change isomorphism

$$f^*d'_* \xrightarrow{\sim} c'_*f'^*.$$

But this square commutes by the very definition of the base-change arrow. The commutativity of (A) is therefore established, and this completes the proof of the theorem.

**Corollary 4.5.** Under the hypotheses of 4.4, there is a commutative square

$$\begin{array}{ccc} \mathrm{Hom}(c'_1{}^*L_1, c_2'^!L_2) \otimes \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1) & \xrightarrow{4.2.5} & \mathrm{H}^0(C, K_C) \\ \downarrow 3.7.6 \otimes 3.7.6 & & \downarrow g_* \\ \mathrm{Hom}(d_1''^*f_{1*}L_1, d_2'^!f_{2*}L_2) \otimes \mathrm{Hom}(d_2''^*f_{2*}L_2, d_1''^!f_{1*}L_1) & \xrightarrow{4.2.5} & \mathrm{H}^0(D, K_D), \end{array}$$

where  $g_*$  is defined by the adjunction arrow

$$g_*K_C = g_*g^!K_D \rightarrow K_D.$$

In other words, for

$$u \in \mathrm{Hom}(c'_1{}^*L_1, c_2'^!L_2), \quad v \in \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1),$$

one has

$$\langle f_*u, f_*v \rangle = g_*\langle u, v \rangle. \quad (4.5.1)$$

For every proper  $S$ -scheme  $t : T \rightarrow S$ , denote by

$$\int_T : \mathrm{H}^0(T, K_T) \rightarrow \mathrm{H}^0(S, K_S) = \mathrm{H}^0(S, A) \quad (4.6)$$

the arrow defined by the adjunction arrow

$$t_!K_T \rightarrow A.$$

**Corollary 4.7 – Lefschetz–Verdier formula.** Under the hypotheses of 4.4, suppose that  $Y_1 = Y_2 = S = D' = D''$ . For

$$u \in \mathrm{Hom}(c'_1{}^*L_1, c_2'^!L_2), \quad v \in \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1),$$

one has

$$\langle u_*, v_* \rangle = \int_C \langle u, v \rangle, \quad (4.7.1)$$

where

$$u_* \in \mathrm{Hom}(f_{1*}L_1, f_{2*}L_2), \quad v_* \in \mathrm{Hom}(f_{2*}L_2, f_{1*}L_1)$$

are the homomorphisms deduced from  $u, v$  by direct image over  $S$  (3.7.6), and

$$\langle u_*, v_* \rangle = \mathrm{Tr}(u_*v_*) = \mathrm{Tr}(v_*u_*)$$

is the cup product, in the sense of (SGA 6 I 8.3), of the homomorphisms of perfect complexes  $u_*, v_*$  (4.1.5).

Taking 3.6.5 into account, in particular one has:

**Corollary 4.8.** Let  $t : T \rightarrow S$  be a proper  $S$ -scheme,  $\Delta \subset T \times_S T$  the diagonal,  $F : T \rightarrow T$  an  $S$ -endomorphism of  $T$ ,  $T^F = (\text{graph of } F) \times_{(T \times_S T)} \Delta$  the scheme of fixed points of  $F$ ,  $L \in \text{ob } D_{\text{ctf}}(T)$ ,  $u \in \text{Hom}(F^*L, L)$  a correspondence from  $L$  to  $L$  with support in the graph of  $F$ , and  $1 \in \text{Hom}(L, L)$  the identity correspondence, with support in  $\Delta$  (3.2 b)). Then

$$\text{Tr}(u_*) = \int_{T^F} \langle u, 1 \rangle,$$

where  $u_*$  is the endomorphism of  $t_*L$  obtained by composing  $t_*L \rightarrow t_*F^*L$  (inverse image) with  $t_*u : t_*F^*L \rightarrow t_*L$ .

From the remarks of 3.2 a) and 3.6 one also deduces:

**Corollary 4.9.** Under the hypotheses of 4.7, assume that  $X_i$  is smooth over  $S$ , purely of relative dimension  $r_i$ , and that  $C'$  (respectively  $C''$ ) is a closed subscheme of  $X$  finite and of finite tor-dimension over  $X_2$  (respectively  $X_1$ ). Let

$$\text{cl}(C') \in H_{C'}^{2r_1}(X, A(r_1)), \quad \text{cl}(C'') \in H_{C''}^{2r_2}(X, A(r_2))$$

be the cohomology classes of  $C'$  and  $C''$ , and let

$$\text{cl}(C')_* \in \text{Hom}(Rf_{1*}A, Rf_{2*}A), \quad \text{cl}(C'')_* \in \text{Hom}(Rf_{2*}A, Rf_{1*}A)$$

be the corresponding homomorphisms (3.2 a), 3.7.6). Then

$$\langle \text{cl}(C')_*, \text{cl}(C'')_* \rangle = \int_{C' \cap C''} \text{cl}(C) \text{cl}(D),$$

where

$$\text{cl}(C) \text{cl}(D) \in H_{C' \cap C''}^{2(r_1+r_2)}(X, A(r_1+r_2)) = H^0(C' \cap C'', K_{C' \cap C''})$$

is the usual cup product.

Taking 4.3 into account, in particular one has:

**Corollary 4.10.** Under the hypotheses of 4.9, suppose that  $C' \cap C''$  is finite. For  $z \in C' \cap C''$ , denote by  $(C' \cdot C'')_z$  the image in  $A$  of the intersection multiplicity of  $C'$  and  $C''$  at  $z$ . Then

$$\langle \text{cl}(C')_*, \text{cl}(C'')_* \rangle = \sum_{z \in C' \cap C''} (C' \cdot C'')_z.$$

**4.11.** Under the hypotheses of 4.9, one has in general

$$\langle \text{cl}(C')_*, \text{cl}(C'')_* \rangle = \sum_{Z \in \pi_0(C' \cap C'')} (\text{cl}(C') \text{cl}(C''))_Z, \quad (4.11.1)$$

where  $\pi_0(C' \cap C'')$  denotes the set of connected components of  $C' \cap C''$ , and  $(\text{cl}(C') \text{cl}(C''))_Z$  the restriction of  $\text{cl}(C') \text{cl}(C'')$  to  $H^0(Z, K_Z)$ . When  $Z$  is a component of dimension  $\geq 1$ , the calculation of  $(\text{cl}(C') \text{cl}(C''))_Z$  presents problems, because one is in an “excess” situation. If  $Z$  is smooth, purely

of dimension  $d$ , if  $C'$  and  $C''$  are smooth in a neighbourhood of  $Z$ , and if  $X$  is projective, one can show that

$$\int_Z (\text{cl}(C') \text{cl}(C''))_Z = \int_Z c_d(N_Z), \quad (4.11.2)$$

where  $N_Z$  is the locally free sheaf of rank  $d$  defined by the exact sequence of normal bundles

$$0 \longrightarrow N_{Z/C'} \oplus N_{Z/C''} \longrightarrow N_{Z/X} \longrightarrow N_Z \longrightarrow 0,$$

and  $c_d(N_Z) \in H^{2d}(Z, A(d))$  denotes its  $d$ -th Chern class (SGA 5 VII) (use (SGA 6 XIV 6.3) to reduce to a calculation of intersection number in K-theory, and apply (SGA 6 VII 2.5), cf. ([10] 4.1); the projectivity hypothesis is no doubt unnecessary).

Let  $T$  be a smooth projective  $S$ -scheme; take as correspondence  $C'$  (respectively  $C''$ ) the graph of an  $S$ -endomorphism  $F$  (respectively the diagonal), and suppose that the scheme of fixed points  $T^F = C' \cap C''$  is smooth. The sheaf  $N_{(T^F)}$  above is nothing other than the tangent sheaf to  $T^F$ , so that 4.11.2 implies, by the Gauss–Bonnet formula (SGA 5 VII 4.9), the analogue of a formula well known to topologists:

$$\text{Tr}(F, R\Gamma(T, A)) = \text{EP}(T_s^F), \quad (4.11.3)$$

where  $\text{EP}(T_s^F)$  denotes the Euler–Poincaré characteristic of  $T_s^F$ , for  $s$  a geometric point above  $S$  (loc. cit.).

The calculation of  $(\text{cl}(C') \text{cl}(C''))_Z$  for a non-smooth component  $Z$  of dimension  $\geq 1$  has not been treated in general, at least to the knowledge of the editor.

**4.12.** In the situation of 4.7, let  $z$  be an isolated point of  $C$ , with projection  $z_i$  on  $X_i$ ,  $a'$  (respectively  $a''$ ) on  $C'$  (respectively  $C''$ ), and let  $\bar{z}$  be a geometric point above  $z$ , with image  $\bar{z}_i$  above  $z_i$ . Assume that  $c'_2$  (respectively  $c''_1$ ) is quasi-finite and flat at  $a'$  (respectively  $a''$ ), that  $L_i$  is perfect in a neighbourhood of  $\bar{z}_i$ , and that  $u$  (respectively  $v$ ) is given in a neighbourhood of  $a'$  (respectively  $a''$ ) by

$$\text{Tr}_{c'_2} : c'_2! c'_2^* L_2 \longrightarrow L_2 \quad \text{and} \quad s \in \text{Hom}(c'_1^* L_1, c'_2^* L_2)$$

(respectively

$$\text{Tr}_{c''_1} : c''_1! c''_1^* L_1 \longrightarrow L_1 \quad \text{and} \quad t \in \text{Hom}(c''_2^* L_2, c''_1^* L_1)).$$

Let  $s_{\bar{z}} \in \text{Hom}(L_{1\bar{z}_1}, L_{2\bar{z}_2})$  (respectively  $t_{\bar{z}} \in \text{Hom}(L_{2\bar{z}_2}, L_{1\bar{z}_1})$ ) be the homomorphism deduced from  $s$  (respectively  $t$ ) by passage to the fibres. Let again, as in 3.2 a),  $\text{cl}(c')$  (respectively  $\text{cl}(c'')$ ) denote the correspondence from  $X_1$  to  $X_2$  (respectively from  $X_2$  to  $X_1$ ) defined, in a neighbourhood of  $a'$  (respectively  $a''$ ), by  $c'_2$  (respectively  $c''_1$ ), and let  $(C' \cdot C'')_z$  be the corresponding local term  $\langle \text{cl}(c'), \text{cl}(c'') \rangle_z$  (equal to the image in  $A$  of the intersection multiplicity of  $C'$  and  $C''$  at  $z$  when  $C'$  and  $C''$  are closed subschemes of  $X$ , with  $X_1$  and  $X_2$  smooth). Then:

$$\langle u, v \rangle_z = (C' \cdot C'')_z \langle s_{\bar{z}}, t_{\bar{z}} \rangle. \quad (4.12.1)$$

This follows easily from 4.2.6.

**4.13.** If  $T$  is a scheme over  $S$ , denote by  $DF(T)$  the filtered derived category of complexes of  $A_T$ -modules with finite filtration ([9] V 1), and by  $DF_{\text{ctf}}(T)$  the full subcategory formed by the  $L$  such that  $\text{gr}(L) \in \text{ob } D_{\text{ctf}}(T)$ . The kind of compatibility of filtered derived functors with passage to associated graded objects implies that  $DF_{\text{ctf}}$  is stable under the six operations  $f^*$ ,  $f_*$ ,  $f_!$ ,  $f^!$ ,  $\otimes^L$ ,  $\text{RHom}$ . The definitions and results of the preceding numbers are valid, *mutatis mutandis*,

in  $DF_{\text{ctf}}$ . The pairing 4.1.2 is compatible with passage to associated graded objects ( $K_X$  being regarded as endowed with the trivial filtration), and it follows that, with the notation of 4.2.5, for  $L_i \in \text{ob } DF_{\text{ctf}}(X_i)$ ,

$$u \in \text{Hom}_{DF}(c_1^* L_1, c_2^! L_2), \quad v \in \text{Hom}_{DF}(d_2^* L_2, d_1^! L_1),$$

one has the following equality in  $H^0(E, K_E)$ :

$$\langle u, v \rangle = \sum_i \langle \text{gr}^i u, \text{gr}^i v \rangle. \quad (4.13.1)$$

As Deligne observes, this remark applies in particular to the situation considered by Langlands in ([13] §7). The correspondences that he considers in (loc. cit. 7.11) indeed preserve, near the fixed points, finite filtrations on the sheaves with associated graded objects inducing, on the strict localization at the image of a fixed point, either a constant sheaf or a constant sheaf on the complement of the closed point extended by zero. The calculation of the local terms is easily reduced, by 4.13, to 4.12.1 and to the case of sheaves concentrated on the closed points that are the images of the fixed points, a case treated directly. Applying 4.7, one obtains the Lefschetz formula required in ([13] 7.11). We point out that the general statement (loc. cit. 7.12) is true; the proof, more delicate, will be given in III B.

## 5. Complements

### 5.1. Dual of a correspondence

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme, let  $X = X_1 \times_S X_2$ , and let  $p_i : X \rightarrow X_i$  be the projections. Let  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ . Every cohomological correspondence  $u$  from  $L_1$  to  $L_2$  gives, by applying  $D_X$  and identifying  $Dp_2^!$  with  $p_2^* D$  and  $Dp_1^*$  with  $p_1^! D$ , a correspondence  $Du$  from  $DL_2$  to  $DL_1$ , called the *dual* of  $u$ . In other words,  $D$  defines a homomorphism

$$D : \text{Hom}(p_1^* L_1, p_2^! L_2) \longrightarrow \text{Hom}(p_2^* DL_2, p_1^! DL_1). \quad (5.1.1)$$

One checks that 5.1.1 is obtained, by applying  $H^0(X, -)$ , from the isomorphism

$$D : \text{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \text{RHom}(p_2^* DL_2, p_1^! DL_1) \quad (5.1.2)$$

which makes the square

$$\begin{array}{ccc} p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 & \xrightarrow{3.1.1} & \text{RHom}(p_1^* L_1, p_2^! L_2) \\ \downarrow & & \downarrow \\ p_2^* DDL_2 \otimes^{\mathbf{L}} p_1^* DL_1 & \xrightarrow{3.1.1} & \text{RHom}(p_2^* DL_2, p_1^! DL_1), \end{array} \quad (5.1.3)$$

commutative. Here the left vertical arrow is the composite of the symmetry isomorphism and the one defined by the biduality isomorphism  $L_2 \rightarrow DDL_2$ . It follows easily from the definitions that the following square is commutative:

$$\begin{array}{ccc} \text{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}} \text{RHom}_S(L_2, L_1) & \xrightarrow{4.1.3} & K_X \\ \downarrow D \otimes D & & \downarrow \text{Id} \\ \text{RHom}_S(DL_2, DL_1) \otimes^{\mathbf{L}} \text{RHom}_S(DL_1, DL_2) & \xrightarrow{4.1.3} & K_X. \end{array} \quad (5.1.4)$$

Let  $c : C \rightarrow X$  be an  $S$ -morphism and put  $c_i = p_i c$ . Applying  $c_* c^\dagger$  to 5.1.2, and using (3.2.1), one deduces an isomorphism

$$D : \mathrm{Hom}(c_1^* L_1, c_2^\dagger L_2) \xrightarrow{\sim} \mathrm{Hom}(c_2^* D L_2, c_1^\dagger D L_1), \quad (5.1.5)$$

which is checked to identify with the one obtained by applying the functor  $D_X$  and the identifications of  $D c_1^*$  with  $c_1^\dagger D$  and  $D c_2^\dagger$  with  $c_2^* D$ . If  $c' : C' \rightarrow X$  and  $c'' : C'' \rightarrow X$  are  $S$ -morphisms, it follows from 5.1.4 that, for

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^\dagger L_2), \quad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^\dagger L_1),$$

one has

$$\langle u, v \rangle = \langle Du, Dv \rangle, \quad (5.1.6)$$

with the notation of 4.2.5.

On the other hand, one checks that, in the situation of 3.3, the following square is commutative:

$$\begin{array}{ccc} f_* \mathrm{RHom}_S(L_1, L_2) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_1^\dagger L_1, f_2^* L_2) \\ f_* D \downarrow & & \downarrow D \\ f_* \mathrm{RHom}_S(D L_2, D L_1) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_2^\dagger D L_2, f_1^* D L_1), \end{array} \quad (5.1.7)$$

and consequently, in the situation of 3.7.6, one has

$$f_* D u = D f_* u. \quad (5.1.8)$$

## 5.2. Composition of correspondences

Let, for  $i = 1, 2, 3$ ,  $X_i$  be an  $S$ -scheme, let  $X_{ij} = X_i \times_S X_j$ ,  $X = X_1 \times_S X_2 \times_S X_3$ , and let  $p_i : X \rightarrow X_i$ ,  $p_{ij} : X \rightarrow X_{ij}$  be the projections. Let  $L_i \in \mathrm{ob} D_{\mathrm{ctf}}(X_i)$ . One has  $X = X_{12} \times_{X_2} X_{23}$ , and

$$(D L_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (D L_2 \otimes_S^{\mathbf{L}} L_3) = p_1^* D L_1 \otimes^{\mathbf{L}} p_2^* L_2 \otimes^{\mathbf{L}} p_2^* D L_2 \otimes^{\mathbf{L}} p_3^* L_3.$$

The evaluation arrow  $L_2 \otimes_S^{\mathbf{L}} D L_2 \rightarrow K_{X_2}$  therefore defines a pairing

$$(D L_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (D L_2 \otimes_S^{\mathbf{L}} L_3) \longrightarrow p_{13}^\dagger (D L_1 \otimes_S^{\mathbf{L}} L_3), \quad (5.2.1)$$

the second member being identified with  $(D L_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} K_{X_2}$  by Künneth (1.7.3). Taking 3.1.1 and 3.1.2 into account, one may rewrite 5.2.1 in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow p_{13}^\dagger \mathrm{RHom}_S(L_1, L_3) = \mathrm{RHom}(p_1^* L_1, p_3^\dagger L_3), \quad (5.2.2)$$

the identification with the second member being given by the induction isomorphism (SGA 4 XVIII 3.1.12.2). Hence one obtains a pairing, called *composition of correspondences*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_3) \longrightarrow \mathrm{Hom}(p_1^* L_1, p_3^\dagger L_3), \quad (5.2.3)$$

which will be denoted  $(u, v) \mapsto vu$ . One checks that, if  $p_{ij,i} : X_{ij} \rightarrow X_i$  and  $p_{ij,j} : X_{ij} \rightarrow X_j$  denote the projections, 5.2.2 can also be obtained by composing the arrow

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow \mathrm{RHom}(p_{12}^* p_{12,1}^\dagger L_1, p_{12}^\dagger p_{12,2}^* L_2) \otimes \mathrm{RHom}(p_{23}^\dagger p_{23,2}^* L_2, p_{23}^\dagger p_{23,3}^\dagger L_3)$$

(the tensor product of the canonical arrows  $p_{12}^* \mathrm{RHom}_S(L_1, L_2) \rightarrow \mathrm{RHom}(p_{12}^* L_1, p_{12}^! L_2)$  and  $p_{23}^* \mathrm{RHom}_S(L_2, L_3) \rightarrow \mathrm{RHom}(p_{23}^! L_2, p_{23}^! L_3)$ ), the arrow defined by the base-change arrow  $p_{12}^! p_{12,2}^* L_2 \rightarrow p_{23}^* p_{23,2}^* L_2$ , and a usual composition arrow on the  $\mathrm{RHom}$ 's.

Let  $c' : C' \rightarrow X_{12}$  and  $c'' : C'' \rightarrow X_{23}$  be correspondences, and let  $c : C = C' \times_{X_2} C'' \rightarrow X$  be the “composite” correspondence. From 5.2.2 one deduces, or else defines directly by a composition analogous to the one just described, a pairing

$$\mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}(c_2''^* L_2, c_3^! L_3) \longrightarrow \mathrm{RHom}(c_1^* L_1, c_3^! L_3), \quad (5.2.4)$$

hence a “composition” pairing  $(u, v) \mapsto vu$ :

$$\mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3^! L_3) \longrightarrow \mathrm{Hom}(c_1^* L_1, c_3^! L_3). \quad (5.2.5)$$

One checks that composition of correspondences is compatible with direct images in the following sense. Let, for  $1 \leq i \leq 3$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism, with  $f_2$  proper, and let  $c', c'', c$  be morphisms as above, with  $c'$  and  $c''$  proper. Let  $d' : D' \rightarrow Y_{12}$  and  $d'' : D'' \rightarrow Y_{23}$  be proper morphisms,  $d : D = D' \times_{Y_2} D'' \rightarrow Y$ , and suppose given the commutative squares

$$\begin{array}{ccc} X_{12} & \longleftarrow & C' \\ f_{12} \uparrow & & \downarrow \\ Y_{12} & \longleftarrow & D' \end{array} \quad \begin{array}{ccc} X_{23} & \longleftarrow & C'' \\ f_{23} \uparrow & & \downarrow \\ Y_{23} & \longleftarrow & D'' \end{array}$$

Then one has a commutative square

$$\begin{array}{ccc} \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3^! L_3) & \xrightarrow{5.2.5} & \mathrm{Hom}(c_1^* L_1, c_3^! L_3) \\ f_{12*} \otimes f_{23*} \downarrow & & \downarrow f_{13*} \\ \mathrm{Hom}(d_1^* M_1, d_2^! M_2) \otimes \mathrm{Hom}(d_2''^* M_2, d_3^! M_3) & \xrightarrow{5.2.5} & \mathrm{Hom}(d_1^* M_1, d_3^! M_3), \end{array} \quad (5.2.6)$$

where  $M_i = f_{i!} L_i$ .

Finally, suppose that  $X_1 = X_3$  and  $L_1 = L_3$ . The fundamental pairing 4.1.2 may be interpreted as the composite of the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* L_2 \otimes^{\mathbf{L}} q_2^* DL_2 \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \quad (5.2.7)$$

defined by  $L_2 \otimes^{\mathbf{L}} DL_2 \rightarrow K_{X_2}$ , or, equivalently, induced by 5.2.1 on  $X_{12}$  via the diagonal  $X_{12} \rightarrow X_1 \times_S X_2 \times_S X_1$  ( $q_i : X_{12} \rightarrow X_i$  denoting the projection), and the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow K_{X_{12}} \quad (5.2.8)$$

defined by  $DL_1 \otimes^{\mathbf{L}} L_1 \rightarrow K_{X_1}$ . It follows that, for  $u \in \mathrm{Hom}_S(L_1, L_2)$  and  $v \in \mathrm{Hom}_S(L_2, L_1)$ , one has, in  $H^0(X_1 \times_S X_2, K_{X_1 \times_S X_2})$ ,

$$\langle u, v \rangle = \langle vu, 1 \rangle, \quad (5.2.9)$$

where 1 denotes the identity correspondence from  $L_1$  to  $L_1$  on  $X_1 \times_S X_1$ , supported on the diagonal (3.2 b)), and the cup product in the second member is taken in the sense of 4.2.5, taking account of the fact that the fibred product  $X \times_{(X_1 \times_S X_1)} X_1$  (via  $p_{13} : X \rightarrow X_{13} = X_1 \times_S X_1$  and the

diagonal  $X_1 \rightarrow X_1 \times_S X_1$ ) identifies with  $X_1 \times_S X_2$ . Likewise, more generally, if  $c' : C' \rightarrow X_{12}$  and  $c'' : C'' \rightarrow X_{21}$  are correspondences, if  $c : C = C' \times_{X_{12}} C'' \rightarrow X_1 \times_S X_2 \times_S X_1$  is the composite correspondence, and if  $d : C' \times_{X_{12}} C'' = D \rightarrow X_{12}$  is the intersection correspondence, then, for  $u \in \text{Hom}(c_1^* L_1, c_2'^! L_2)$  and  $v \in \text{Hom}(c_2''^* L_2, c_1''^! L_1)$ , one has the following equality in  $H^0(D, K_D)$ :

$$\langle u, v \rangle = \langle vu, 1 \rangle, \quad (5.2.10)$$

where 1 denotes, as above, the diagonal correspondence, and  $D$  is identified with the fibred product of  $C \rightarrow X_1 \times_S X_1$  and the diagonal  $X_1 \rightarrow X_1 \times_S X_1$ . Formulas 5.2.9 and 5.2.10 generalize a well-known formula in the case of constant coefficients ([12] 1.3.6 (ii) c)).

### 5.3. External tensor product of correspondences

Let, for  $1 \leq i \leq 4$ ,  $X_i$  be an  $S$ -scheme, let  $X = X_1 \times_S X_2 \times_S X_3 \times_S X_4$ ,  $X_{ij} = X_i \times_S X_j$ , and let  $L_i \in \text{ob } D_{\text{ctf}}(X_i)$ . The Künneth isomorphism (1.7.6, 2.2.4)

$$DL_1 \otimes_S^{\mathbf{L}} DL_3 \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3)$$

defines an isomorphism on  $X$

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_S^{\mathbf{L}} (DL_3 \otimes_S^{\mathbf{L}} L_4) \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} (L_2 \otimes_S^{\mathbf{L}} L_4), \quad (5.3.1)$$

which, by 3.1.1, may be rewritten in the form

$$\text{RHom}_S(L_1, L_2) \otimes_S^{\mathbf{L}} \text{RHom}_S(L_3, L_4) \xrightarrow{\sim} \text{RHom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4). \quad (5.3.2)$$

Hence a pairing, called the *external tensor product*,

$$\text{Hom}_S(L_1, L_2) \otimes \text{Hom}_S(L_3, L_4) \longrightarrow \text{Hom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4), \quad (5.3.3)$$

which will be denoted  $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$ . The pairing 5.3.2 may also be interpreted as the composite of the external tensor product

$$\text{RHom}(E_1, E_2) \otimes_S^{\mathbf{L}} \text{RHom}(E_3, E_4) \xrightarrow{\sim} \text{RHom}(E_1 \otimes_S^{\mathbf{L}} E_3, E_2 \otimes_S^{\mathbf{L}} E_4)$$

(where  $E_1 = p_{12}^* p_{12,1}^* L_1$ , etc., with the evident notation) and of the isomorphism defined by Künneth

$$E_2 \otimes_S^{\mathbf{L}} E_4 \xrightarrow{\sim} p_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4) \quad (1.7.3).$$

Let  $c' : C' \rightarrow X_{12}$  and  $c'' : C'' \rightarrow X_{34}$  be correspondences, and let  $c : C = C' \times_S C'' \rightarrow X = X_{13} \times_S X_{24}$  be the product correspondence. From 5.3.2 one deduces, or else defines analogously, a pairing

$$\text{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes_S^{\mathbf{L}} \text{RHom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \text{RHom}(c_{13}^* (L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4)), \quad (5.3.4)$$

hence an external tensor product pairing,  $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$ ,

$$\text{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \text{Hom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \text{Hom}(c_{13}^* (L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4)). \quad (5.3.5)$$

One checks that these pairings are compatible with direct images, in the following sense. Let, for  $1 \leq i \leq 4$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism, let  $c'$ ,  $c''$ ,  $c$  be as above, with  $c'$  and  $c''$  proper, let

$d' : D' \rightarrow Y_{12}$  and  $d'' : D'' \rightarrow Y_{34}$  be proper morphisms, let  $d : D = D' \times_S D'' \rightarrow Y = Y_{12} \times_S Y_{34}$ , and suppose given the commutative squares

$$\begin{array}{ccc} X_{12} & \longleftarrow & C' \\ f_{12} \uparrow & & \downarrow \\ Y_{12} & \longleftarrow & D' \end{array} \quad \begin{array}{ccc} X_{34} & \longleftarrow & C'' \\ f_{34} \uparrow & & \downarrow \\ Y_{34} & \longleftarrow & D'' \end{array}$$

Then one has a commutative square

$$\begin{array}{ccc} \mathrm{Hom}(c'_1{}^* L_1, c'_2{}^! L_2) \otimes \mathrm{Hom}(c''_3{}^* L_3, c''_4{}^! L_4) & \xrightarrow{5.3.5} & \mathrm{Hom}(c_{13}^* L_{13}, c_{24}^! L_{24}) \\ f_{12*} \otimes f_{34*} \downarrow & & \downarrow f_{13,24*} \\ \mathrm{Hom}(d'_1{}^* M_1, d'_2{}^! M_2) \otimes \mathrm{Hom}(d''_3{}^* M_3, d''_4{}^! M_4) & \xrightarrow{5.3.5} & \mathrm{Hom}(d_{13}^* M_{13}, d_{24}^! M_{24}), \end{array} \quad (5.3.6)$$

where  $M_i = f_{i!} L_i$  if  $i = 1, 3$  and  $f_{i*} L_i$  otherwise,  $L_{ij} = L_i \otimes_S^{\mathbf{L}} L_j$ , and  $M_{ij} = M_i \otimes_S^{\mathbf{L}} M_j$  (there is also an analogous square with the  $\mathrm{RHom}$ 's).

Suppose that  $X_3 = X_2$ ,  $X_4 = X_1$ ,  $L_3 = L_2$ , and  $L_4 = L_1$ . A reasoning analogous to that of the last paragraph of 5.2 shows that, for  $u \in \mathrm{Hom}_S(L_1, L_2)$  and  $v \in \mathrm{Hom}_S(L_2, L_1)$ , one has

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \quad (5.3.7)$$

where 1 denotes the diagonal correspondence from  $L_{12}$  to  $L_{12}$  on  $X_{12}$ , and the cup product in the second member is taken in the sense of 4.2.5. More generally, if  $c' : C' \rightarrow X_{12}$  and  $c'' : C'' \rightarrow X_{21}$  are correspondences, their “intersection”  $C = C' \times_{X_{12}} C''$  identifies with the fibred product of  $C' \times_S C'' \rightarrow X = X_{12} \times_S X_{21}$  and the diagonal  $X_{12} \rightarrow X$ , and, for  $u \in \mathrm{Hom}(c'_1{}^* L_1, c'_2{}^! L_2)$  and  $v \in \mathrm{Hom}(c''_2{}^* L_2, c''_1{}^! L_1)$ , one has the following equality in  $H^0(C, K_C)$ :

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \quad (5.3.8)$$

where 1 denotes, as above, the diagonal correspondence. Thus there are two “diagonal” procedures (5.2.10, 5.3.8) for the calculation of the “Verdier local term”  $\langle u, v \rangle$ .

## 6. Appendix. Lefschetz formula for coherent sheaves

In this number,  $S$  denotes a noetherian scheme. The  $S$ -schemes considered will be assumed separated and of finite type. Recall that, by Nagata, every  $S$ -morphism between such schemes is compactifiable, i.e. factors as an open immersion followed by a proper morphism. For every  $S$ -scheme  $X$ , we shall write  $D(X)$  for the derived category of the category of  $\mathcal{O}_X$ -modules.

### 6.1

Recall ([8] App.) that for every  $S$ -morphism  $f : X \rightarrow Y$  one has a functor

$$f^! : D^+(Y) \longrightarrow D^+(X)$$

(and transitivity isomorphisms  $(gf)^! = f^!g^!$  for a composite  $gf$ , with a cocycle condition). If  $f$  is proper,  $f^!$  is right “adjoint” to the functor  $f_* : D^+(X)_{\text{coh}} \rightarrow D^+(Y)_{\text{coh}}$  (where the subscript “coh” denotes the full subcategory formed by complexes with coherent cohomology), this adjunction being made precise by an isomorphism

$$f_* \text{RHom}(E, f^! F) \xrightarrow{\sim} \text{RHom}(f_* E, F) \quad (6.1.1)$$

for  $E \in \text{ob } D(X)_{\text{coh}}$ ,  $F \in \text{ob } D^+(Y)$ , defined analogously to the isomorphism (SGA 4 XVIII 3.1.9.6). The considerations of ([8] App.5), together with the duality theorem for projective space, show that, if  $f$  is smooth, purely of dimension  $d$ , there is a canonical isomorphism

$$f^! L \xleftarrow{\sim} f^* L \otimes \Omega_{X/Y}^d[d]; \quad (6.1.2)$$

and, if moreover  $f$  is proper, the arrow

$$\text{Tr}_f : f_*(f^* L \otimes \Omega_{X/Y}^d[d]) \longrightarrow L \quad (6.1.3)$$

deduced, by 6.1.2, from the adjunction arrow  $f_* f^! \rightarrow I$ , coincides with the trace morphism ([8] VI).

## 6.2

Contrary to what happens in the case of discrete coefficients, there is no reasonable subcategory of  $D$  here that is stable under the “six operations”  $(f^*, f_*, f^!, f_!, \otimes^{\mathbf{L}}, \text{RHom})$ . The most natural finiteness notion seems to be relative perfection (SGA 6 III 4). If  $X$  is an  $S$ -scheme and  $L \in \text{ob } D(X)$ , recall that  $L$  is said to be perfect relative to  $S$  if  $L$  has finite tor-dimension relative to  $S$  and coherent cohomology. We shall denote by  $D(X)_{\text{parf}/S}$  the full subcategory of  $D(X)$  formed by the complexes perfect relative to  $S$ . It is not stable in general either under  $\otimes^{\mathbf{L}}$  or under  $\text{RHom}$ . However, if  $E$  is perfect in the absolute sense and  $F$  is perfect relative to  $S$ , then  $\text{RHom}(E, F)$  is perfect relative to  $S$  (SGA 6 III 4.9). On the other hand, let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme and  $L_i \in \text{ob } D^-(X_i)$ . We shall put, as in 1.5,

$$L_1 \otimes_S^{\mathbf{L}} L_2 = Lp_1^* L_1 \otimes^{\mathbf{L}} Lp_2^* L_2,$$

where  $p_i : X_1 \times_S X_2 \rightarrow X_i$  is the projection. Considering this object is reasonable only if  $X_1$  and  $X_2$  are tor-independent over  $S$ , i.e. (SGA 6 III 1.5) if  $\text{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0$  for  $i \neq 0$ . Suppose this is the case: then, if  $L_1$  and  $L_2$  are perfect relative to  $S$ , so is  $L_1 \otimes_S^{\mathbf{L}} L_2$  (SGA 6 III 4.7).

Let  $f : X \rightarrow Y$  be an  $S$ -morphism. If  $f$  is proper, then  $f_* D(X)_{\text{parf}/S} \subset D(Y)_{\text{parf}/S}$  (SGA 6 III 4.8). If  $f$  has finite tor-dimension, then  $f^* D(Y)_{\text{parf}/S} \subset D(X)_{\text{parf}/S}$  (SGA 6 III 4.5), and  $f^! D(Y)_{\text{parf}/S} \subset D(X)_{\text{parf}/S}$  (SGA 6 III 4.9).

For every  $S$ -scheme  $a : X \rightarrow S$ , we put

$$K_X = a^! \mathcal{O}_S, \quad (6.2.1)$$

and write  $D_X$  (or simply  $D$ ) for the functor  $\text{RHom}(-, K_X)$ . If  $X$  has finite tor-dimension over  $S$ , then  $K_X$  is perfect relative to  $S$  and, for every  $L \in \text{ob } D(X)_{\text{parf}/S}$ , one has  $DL \in \text{ob } D(X)_{\text{parf}/S}$ , and the canonical arrow  $L \rightarrow DDL$  is an isomorphism (SGA 6 III 4.9).

Let  $f : X \rightarrow Y$  be an  $S$ -morphism,  $L \in \text{ob } D(X)$ , and  $M \in \text{ob } D(Y)$ . If  $f^* M$  is defined (for example if  $M \in \text{ob } D^-(Y)$ , or if  $f$  has finite tor-dimension), put

$$\text{Hom}_f^{(1)}(L, M) = \text{Hom}(f^* M, L), \quad \text{Hom}_f^{(2)}(L, M) = \text{Hom}(L, f^* M).$$

If  $f^!M$  is defined (for example if  $M \in \text{ob } D^+(Y)$ , or if  $f$  is smooth, or finite of finite tor-dimension, or a composite of such morphisms, since in these cases  $f^!$  extends, by 6.1.1 and 6.1.2, to a functor defined on the whole category  $D(Y)$ ), put

$$\text{Hom}_f^{(3)}(L, M) = \text{Hom}(L, f^!M), \quad \text{Hom}_f^{(4)}(L, M) = \text{Hom}(f^!M, L).$$

The elements of  $\text{Hom}_f^{(i)}(L, M)$  will be called, as in 1.4, *arrows of type  $i$  from  $L$  to  $M$  over  $f$*  (or  $f$ -arrows of type  $i$  from  $L$  to  $M$ ). We leave to the reader the task of making explicit the composition of arrows of type  $i$ , for fixed  $i$ , and the corresponding fibred and cofibred categories. We recall only the partial adjunction formulas

$$\text{Hom}_f^{(1)}(L, M) = \text{Hom}(M, f_*L) \quad (M \in \text{ob } D^-(Y), L \in \text{ob } D^+(X)),$$

$$\text{Hom}_f^{(3)}(L, M) = \text{Hom}(f_*L, M) \quad (f \text{ proper}, L \in \text{ob } D(X)_{\text{coh}}, M \in \text{ob } D^+(Y)).$$

### 6.3

In the situation of 1.6, suppose that  $X_1$  and  $X_2$  are tor-independent over  $S$ , and likewise  $Y_1$  and  $Y_2$ . Then, for  $L_i \in \text{ob } D(X_i)_{\text{parf}/S}$ , the Künneth arrow 1.6.4 is defined (use the fact that finite relative tor-dimension is preserved by direct image (SGA 6 III 3.7.1)) and is an isomorphism (to verify this, one may assume  $S$  and the  $Y_i$  affine, and, for a Čech calculation, reduce to also assuming  $X_1$  and  $X_2$  affine; the conclusion is then a trivial consequence of the tor-independence hypotheses). On the other hand, for  $M_i \in \text{ob } D(Y_i)_{\text{parf}/S}$ , with  $f_1^!M_1$  or  $f_2^!M_2$  in  $D^b$ , the Künneth arrow 1.7.3 is defined<sup>9</sup>, and is an isomorphism when  $f_1$  and  $f_2$  have finite tor-dimension, as is easily checked with the help of the Künneth isomorphism of type 1.6.4 just indicated, conjugated with 6.1.2, and with 6.1.1 in the case where  $f_i$  is a closed immersion. In particular, if  $X_1$  and  $X_2$  have finite tor-dimension over  $S$  and are tor-independent, 1.7.6 is an isomorphism.

One defines  $\text{RHom}(u, v)$  as in 2.1 for  $(u, v)$  of type  $(2, 1)$  (respectively  $(3, 4)$ , with  $f$  proper in order to avoid the pro-objects caused by an  $f_!$ ). Under suitable degree hypotheses (see for example ([9] V 2.3)), the evaluation arrows (2.2.2) and tensor-product arrows (2.2.3, 2.2.4) are defined, and the arrows 2.2.3 and 2.2.4 (provided they are defined) are isomorphisms as soon as  $E_1$  and  $E_2$  are perfect in the absolute sense.

In the situation of 2.4, suppose that the pairs  $(X_1, X_2)$ ,  $(X_1, Y_2)$ ,  $(Y_1, X_2)$ ,  $(Y_1, Y_2)$  are tor-independent over  $S$ , that  $f_1$  is proper, that  $E_i, F_i, P_i, Q_i$ ,  $\text{RHom}(E_i, F_i)$ ,  $\text{RHom}(P_i, Q_i)$  are perfect relative to  $S$ , and that the Künneth arrows

$$f_1^!Q_1 \otimes_S^{\mathbf{L}} Q_2 \rightarrow (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} Q_2), \quad f_1^!Q_1 \otimes_S^{\mathbf{L}} F_2 \rightarrow (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} F_2)$$

are isomorphisms. Then 2.4.1 and the diagram of 2.5 are defined, and the latter is commutative (however, the horizontal arrows are not necessarily isomorphisms). The verification is the same as for 2.5.

### 6.4

Let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme, let  $X = X_1 \times_S X_2$ , and let  $p_i : X \rightarrow X_i$ ,  $q_i : X_i \rightarrow S$  be the canonical projections. Suppose that  $X_1$  and  $X_2$  are tor-independent and that  $q_1$  has finite

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<sup>9</sup>In the proper or lissifiable case, the definition of the arrow is immediate; the general case can be reduced to the lissifiable case by cohomological descent.

tor-dimension. Let  $L_i \in \text{ob } D(X_i)_{\text{parf}/S}$ . By the analogue of 2.2.4, one has a canonical arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \text{RHom}(L_1 \otimes^{\mathbf{L}} \mathcal{O}_{X_2}, q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2), \quad (6.4.1)$$

and hence, by composing with the Künneth isomorphism (6.3)  $q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2 \xrightarrow{\sim} p_2^! L_2$ , an arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \text{RHom}(p_1^* L_1, p_2^! L_2). \quad (6.4.2)$$

The arrow 6.4.1, and therefore also 6.4.2, is an isomorphism, as one sees by reducing to the case where  $q_1$  is a closed immersion.

As in 3.1.2, we shall put

$$\text{RHom}(p_1^* L_1, p_2^! L_2) = \text{RHom}_S(L_1, L_2), \quad \text{Hom}(p_1^* L_1, p_2^! L_2) = \text{Hom}_S(L_1, L_2), \quad (6.4.3)$$

and call the elements of  $\text{Hom}_S(L_1, L_2)$  *cohomological correspondences from  $L_1$  to  $L_2$* . If  $c : C \rightarrow X$  is a correspondence (proper in practice), then by the induction formula ([8] III 8.8)<sup>10</sup>

$$c^! \text{RHom}(p_1^* L_1, p_2^! L_2) = \text{RHom}(c_1^* L_1, c_2^! L_2), \quad (6.4.4)$$

and the elements of

$$\text{Hom}(c_1^* L_1, c_2^! L_2) = H^0(X, c_* c^! \text{RHom}(p_1^* L_1, p_2^! L_2))$$

will be called *cohomological correspondences from  $L_1$  to  $L_2$  with support in  $C$* . We shall see below examples analogous to those of 3.2.

## 6.5

Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be a proper  $S$ -morphism, and let  $f = f_1 \times_S f_2 : X \rightarrow Y$ . Suppose that the  $X_i$  and  $Y_i$  have finite tor-dimension over  $S$ , and that the squares of 2.4.0 are tor-independent. Let  $L_i \in \text{ob } D(X_i)_{\text{parf}/S}$ . As in 3.3.1, one defines an isomorphism

$$f_* \text{RHom}_S(L_1, L_2) \xrightarrow{\sim} \text{RHom}_S(f_{1*} L_1, f_{2*} L_2), \quad (6.5.1)$$

hence, by applying  $R\Gamma(Y, -)$ , an isomorphism

$$f_* : \text{Hom}_S(L_1, L_2) \xrightarrow{\sim} \text{Hom}_S(f_{1*} L_1, f_{2*} L_2), \quad (6.5.2)$$

which we shall simply denote by  $u \mapsto u_*$  when  $Y_1 = Y_2 = S$ . There is a commutative diagram analogous to 3.4.1, as follows from the analogue of 2.5 indicated in 6.3. One deduces from it a corollary analogous to 6.5 and, in the case where  $Y_1 = Y_2 = S$  is the spectrum of a field, a description, analogous to that of 3.6, of the operations  $u_*$  on cohomology.

Given a commutative square 3.7.1, with  $c$  proper, one has, for  $K \in \text{ob } D^+(C)$ , a canonical arrow

$$f_* c_* c^! K \longrightarrow d_* d^! f_* K, \quad (6.5.3)$$

defined in the same way as 3.7.3 (the isomorphism  $g'_* c^! \xrightarrow{\sim} d^! f_*$  is also valid in the present context: definition of the arrow by transposition of a base-change arrow, and verification that it is an isomorphism by reduction to the cases where  $d$  is smooth and where  $d$  is a closed immersion). Taking  $K = \text{RHom}_S(L_1, L_2)$ , one deduces from 6.5.3 and 6.5.1 canonical arrows

$$f_* c_* \text{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow d_* \text{RHom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \quad (6.5.4)$$

$$f_* : \text{Hom}(c_1^* L_1, c_2^! L_2) \longrightarrow \text{Hom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \quad (6.5.5)$$

analogous to 3.7.5 and 3.7.6.

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<sup>10</sup>The “lissifiability” hypothesis of (loc. cit.) is unnecessary.

## 6.6

Let, for  $i = 1, 2$ ,  $X_i$  be  $S$ -schemes of finite tor-dimension and tor-independent over  $S$ , and let  $L_i \in \text{ob } D(X_i)_{\text{parf}/S}$ . The external tensor product of the evaluation arrows  $DL_i \otimes^{\mathbf{L}} L_i \rightarrow K_{X_i}$  defines, as in 4.1, “cup-product” pairings analogous to 4.1.2, 4.1.3, and 4.1.4.

Let  $c : C \rightarrow X$  and  $d : D \rightarrow X$  be proper morphisms, let  $e = c \times_X d : E \rightarrow X$ , and let  $P, Q \in \text{ob } D(X)_{\text{parf}/S}$ . By 6.1.1, one has

$$c_* c^! P = \text{RHom}(Rc_* \mathcal{O}_C, P)$$

(and an analogous isomorphism with  $d$ ). Assuming  $c$  and  $d$  of finite tor-dimension at the points above  $e(E)$ , one deduces, by tensor product, a canonical arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow \text{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, P \otimes^{\mathbf{L}} Q). \quad (6.6.1)$$

When  $c$  and  $d$  are tor-independent, the second member is rewritten, by Künneth and 6.1.1, as  $e_* e^! (P \otimes^{\mathbf{L}} Q)$ , and 6.6.1 takes the form 4.2.3. For  $P = DL_1 \otimes_S^{\mathbf{L}} L_2$ ,  $Q = L_1 \otimes_S^{\mathbf{L}} DL_2$ , one deduces from 6.6.1 and the pairings indicated above pairings analogous to 4.2.4 and 4.2.5:

$$c_* \text{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_* \text{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow \text{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X), \quad (6.6.2)$$

$$\langle \ , \ \rangle : \text{Hom}(c_1^* L_1, c_2^! L_2) \otimes \text{Hom}(d_2^* L_2, d_1^! L_1) \longrightarrow \text{Hom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X). \quad (6.6.3)$$

Just like the pairings of 4.2, these are compatible with étale localization. When  $c$  and  $d$  are tor-independent, the second member of 6.6.2 (respectively 6.6.3) identifies canonically with  $e_* K_E$  (respectively  $H^0(E, K_E)$ ).

Theorem 4.4 has the following analogue:

**Theorem 6.7.** Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be a proper  $S$ -morphism, let  $f = f_1 \times_S f_2 : X \rightarrow Y$ , and let  $p_i : X \rightarrow X_i$ ,  $q_i : Y \rightarrow Y_i$  be the projections. Suppose given, on the other hand, a cube 4.4.0 with cartesian horizontal faces, with  $c', c'', d', d''$  proper, and with  $c', c''$  (respectively  $d', d''$ ) of finite tor-dimension at the points above  $c(C)$  (respectively  $d(D)$ ). Put  $c'_i = p_i c'$ ,  $c''_i = p_i c''$ ,  $d'_i = q_i d'$ , and  $d''_i = q_i d''$ . Finally, let  $L_i \in \text{ob } D(X_i)_{\text{parf}/S}$ . Assume that the pairs  $(X_1, X_2)$ ,  $(X_1, Y_2)$ ,  $(Y_1, X_2)$ ,  $(Y_1, Y_2)$  are tor-independent over  $S$ , and that  $X_i$  and  $Y_i$  have finite tor-dimension over  $S$ . Then one has a commutative square

$$\begin{array}{ccc} f_* c'_* \text{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes f_* c''_* \text{RHom}(c_2''^* L_2, c_1''^! L_1) & \xrightarrow{(1)} & f_* \text{RHom}(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''}, K_X) \\ \downarrow (2) & & \downarrow (4) \\ d'_* \text{RHom}(d_1'^* M_1, d_2'^! M_2) \otimes d''_* \text{RHom}(d_2''^* M_2, d_1''^! M_1) & \xrightarrow{(3)} & \text{RHom}(Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''}, K_Y), \end{array} \quad (6.7.1)$$

where  $M_i = f_{i!} L_i$ , (3) is given by 6.6.2, (1) is deduced from 6.6.2 by application of  $f_*$  and composition with the canonical arrow  $f_* c'_* \otimes f_* c''_* \rightarrow f_*(c'_* \otimes c''_*)$ , (2) is the tensor product of the arrows 6.5.4 relative to the faces containing  $f$ , and (4) is the composite of the duality isomorphism 6.1.1 and of the arrow defined by the canonical arrow

$$Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''} \longrightarrow f_*(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''})$$

which is adjoint to the tensor product of the base-change arrows  $f^* Rd'_* \mathcal{O}_{D'} \rightarrow Rc'_* \mathcal{O}_{C'}$ ,  $f^* Rd''_* \mathcal{O}_{D''} \rightarrow Rc''_* \mathcal{O}_{C''}$ .

The proof is essentially the same as that of 4.4; we leave to the reader the small modifications to be made.

When  $c'$  and  $c''$  are tor-independent, and likewise  $d'$  and  $d''$ , the arrow (4) of 6.7.1 is rewritten in the form  $f_*c_*K_C \rightarrow d_*K_D$ ; it is simply defined by the adjunction arrow  $g_*K_C = g_*g^!K_D \rightarrow K_D$  (as is arrow (4) of 4.4.1). However, it would not be natural to consider only this case, since that would exclude correspondences with excess intersection.

From 6.7 one deduces formulas analogous to 4.5.1 and 4.7.1: under the hypotheses of 6.7, one has, for  $u \in \text{Hom}(c_1^*L_1, c_2^!L_2)$  and  $v \in \text{Hom}(c_2''^*L_2, c_1''^!L_1)$ ,

$$\langle f_*u, f_*v \rangle = f_*\langle u, v \rangle, \quad (6.7.2)$$

where  $f_*u$  (respectively  $f_*v$ ) denotes the direct image correspondence of  $u$  (respectively  $v$ ) in the sense of 6.5.5, and  $f_*$  in the second member is the homomorphism

$$\text{Hom}(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''}, K_X) \longrightarrow \text{Hom}(Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''}, K_Y) \quad (6.7.3)$$

which is the composite of the duality isomorphism 6.1.1 and the arrow induced by the canonical arrow

$$Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''} \longrightarrow f_*(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''})$$

cited in 6.7. When  $Y_1 = Y_2 = S = D' = D''$ , 6.7.2 is the *Lefschetz-Verdier formula*.

As an illustration, we shall show that this formula implies the Woods Hole formula [1].

## 6.8

For this purpose we shall need a few reminders on the “Hodge” cohomology class associated with a regularly embedded subscheme of a smooth scheme and on Grothendieck residues ([8] III 9). Let  $f : X \rightarrow S$  be a smooth morphism purely of relative dimension  $N$ , and let  $i : Y \hookrightarrow X$  be a regular immersion of codimension  $d$ , i.e. ((SGA IV 16.9.2), (SGA 6 VII 1)) a closed subscheme defined locally by a regular sequence of  $d$  equations, or, in equivalent terminology ((8 III 1), (SGA 6 VIII 1)), a closed immersion, locally a complete intersection, of codimension  $d$ . Put  $g = fi$ . Then one associates with  $Y$  an element

$$\text{cl}_{X/S}^d(Y) \in \text{Ext}_{\mathcal{O}_X}^d(\mathcal{O}_Y, \Omega_{X/S}^d), \quad (6.8.1)$$

called the *cohomology class associated with  $Y$* , defined as follows. By the “purity” theorem ([8] III 7.2), one has

$$\text{Ext}_{\mathcal{O}_X}^i(\mathcal{O}_Y, \Omega_{X/S}^d) = \begin{cases} 0, & i \neq d, \\ \text{Hom}(\wedge^d N_{Y/X}, \Omega_{X/S}^d \otimes \mathcal{O}_Y), & i = d, \end{cases} \quad (6.8.2)$$

where  $N_{Y/X} = I/I^2$  is the conormal sheaf ( $I = \text{Ker } \mathcal{O}_X \rightarrow \mathcal{O}_Y$ ). Therefore

$$\text{Ext}_{\mathcal{O}_X}^d(\mathcal{O}_Y, \Omega_{X/S}^d) = H^0(Y, \underline{\text{Hom}}(\wedge^d N_{Y/X}, \Omega_{X/S}^d \otimes \mathcal{O}_Y)). \quad (6.8.3)$$

The differential  $d_{X/S} : \mathcal{O}_X \rightarrow \Omega_{X/S}^1$  induces an  $\mathcal{O}_Y$ -linear map

$$d_{X/S} \otimes 1 : N_{Y/X} \longrightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y.$$

The  $d$ -th exterior power of this map is a global section, on  $Y$ , of  $\Lambda^d N_{Y/X}^\vee \otimes \Omega_{X/S}^d$ ; its image in  $\text{Ext}_{\mathcal{O}_X}^d(\mathcal{O}_Y, \Omega_{X/S}^d)$  under the isomorphism 6.8.3 is, by definition, the class 6.8.1.

Assume  $g$  finite and flat (hence  $d = N$ ). Let  $\omega \in H^0(Y, \Lambda^N N_{Y/X}^\vee \otimes \Omega_{X/S}^N)$ . By 6.8.2 and 6.1, one may view  $\omega$  as a homomorphism

$$\mathcal{O}_Y \longrightarrow g^! \mathcal{O}_S (= \Lambda^N N_{Y/X}^\vee \otimes \Omega_{X/S}^N),$$

and therefore, by adjunction, as a homomorphism  $g_* \mathcal{O}_Y \rightarrow \mathcal{O}_S$ . The residue (relative to  $g$ ) of  $\omega$  is defined as the section of  $\mathcal{O}_S$  which is the image of 1 under this homomorphism, and is denoted  $\text{Res}_{Y/S}(\omega)$  (cf. ([8] III 9)). Let, on the other hand,  $i_* \omega \in H^N(X, \Omega_{X/S}^N)$  denote the image of  $\omega$  by the composite of the isomorphism 6.8.3 and the canonical arrow

$$\text{Ext}_{\mathcal{O}_X}^N(\mathcal{O}_Y, \Omega_{X/S}^N) \longrightarrow H^N(X, \Omega_{X/S}^N).$$

When  $f$  is proper, it follows from transitivity for trace morphisms (or adjunction arrows)  $i_* i^! \rightarrow 1$ ,  $f_* f^! \rightarrow 1$ , that

$$\text{Res}_{Y/S}(\omega) = \text{Tr}_f(i_* \omega), \quad (6.8.4)$$

where  $\text{Tr}_f : H^N(X, \Omega_{X/S}^N) \rightarrow H^0(S, \mathcal{O}_S)$  is defined by the adjunction arrow  $f_* f^! \mathcal{O}_S = Rf_* \Omega_{X/S}^N[N] \rightarrow \mathcal{O}_S$  (6.1). On the other hand, without any properness hypothesis, one has the important compatibility ([8] III 9 R6)

$$\text{Res}_{Y/S}(y \, \text{cl}_{X/S}^N(Y)) = \text{Tr}_{Y/S}(y), \quad (6.8.5)$$

for  $y \in H^0(Y, \mathcal{O}_Y)$ , where  $\text{Tr}_{Y/S} : g_* \mathcal{O}_Y \rightarrow \mathcal{O}_S$  denotes the trace in the classical sense, i.e.  $\text{Tr}_{Y/S}(y)$  is the trace of the endomorphism of  $g_* \mathcal{O}_Y$  defined by multiplication by  $y$  (this compatibility is not proved in loc. cit.; the case  $N = 1$  is treated in [Raynaud, Anneaux locaux henséliens, VII 1, Lecture Notes in Math. n° 169, Springer Verlag, 1970]). In other words,  $\text{Tr}_{Y/S}$  corresponds, by adjunction, to the homomorphism  $\mathcal{O}_Y \rightarrow g^! \mathcal{O}_S$  defined by multiplication by  $\text{cl}_{X/S}^N(Y)$ . It follows that, for  $F \in \text{ob } D(S)_{\text{coh}}$ , the arrow

$$g^* F \longrightarrow g^! F = g^* F \otimes^{\mathbf{L}} g^! \mathcal{O}_S \quad (6.8.6)$$

given by multiplication by  $\text{cl}_{X/S}^N(Y)$  corresponds by adjunction to the arrow

$$g_* g^* F = F \otimes^{\mathbf{L}} g_* \mathcal{O}_Y \longrightarrow F \quad (6.8.7)$$

defined by  $\text{Tr}_{Y/S} : g_* \mathcal{O}_Y \rightarrow \mathcal{O}_S$ .

## 6.9

Let  $X$  (respectively  $Y$ ) be a proper smooth  $S$ -scheme of relative dimension  $m$  (respectively  $n$ ),  $Z = X \times_S Y$ , and let  $a : A \hookrightarrow Z$ ,  $b : B \hookrightarrow Z$  be regular immersions such that  $a_2 : A \rightarrow Y$ ,  $b_1 : B \rightarrow X$  are finite and flat (write  $p_1 : Z \rightarrow X$ ,  $p_2 : Z \rightarrow Y$  for the projections, and  $a_i = p_i a$ ,  $b_i = p_i b$ ). Let  $c : C = A \times_Z B \rightarrow Z$ ,  $L \in \text{ob } D(X)_{\text{parf}}$ ,  $M \in \text{ob } D(Y)_{\text{parf}}$  (the subscript *parf* means “perfect in the absolute sense”, which is also equivalent to being perfect rel. to  $S$ , since  $X$  and  $Y$  are smooth). Assume that  $C$  is finite and flat over  $S$ . The hypotheses made therefore imply that  $c$  is a regular immersion and that  $a$  and  $b$  are tor-independent.

One has  $p_1^* \Omega_{X/S}^1 = \Omega_{Z/Y}^1$ ,  $p_2^* \Omega_{Y/S}^1 = \Omega_{Z/X}^1$ , and a decomposition

$$\Omega_{Z/S}^1 = \Omega_{Z/Y}^1 \oplus \Omega_{Z/X}^1,$$

whence, for every integer  $r$ , a decomposition

$$\Omega_{Z/S}^r = \bigoplus_{i+j=r} \Omega_{Z/Y}^i \otimes \Omega_{Z/X}^j. \quad (6.9.0)$$

On the other hand, by 6.1.2, one has  $p_2^! M = p_2^* M \otimes \Omega_{Z/Y}^m[m]$ , hence

$$H^0(Z, a_* a^! \mathrm{RHom}(p_1^* L, p_2^! M)) = \mathrm{Ext}_{\mathcal{O}_Z}^m(a_1^* L, a_2^* M \otimes \Omega_{Z/Y}^m),$$

whence, by 6.4.4 and 6.8.2, an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) = H^0(A, \mathrm{RHom}(a_1^* L, a_2^* M) \otimes \Lambda^m N_{A/Z}^\vee \otimes \Omega_{Z/Y}^m). \quad (6.9.1)$$

Similarly one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) = H^0(B, \mathrm{RHom}(b_2^* M, b_1^* L) \otimes \Lambda^n N_{B/Z}^\vee \otimes \Omega_{Z/X}^n). \quad (6.9.2)$$

Let  $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$ ,  $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$ , and write

$$\langle u, v \rangle \in H^0(C, \mathcal{O}_C) \quad (6.9.3)$$

for the section  $\mathrm{Tr}(u_C v_C) = \mathrm{Tr}(v_C u_C)$ , where  $u_C \in \mathrm{Hom}(c_1^* L, c_2^* M)$  and  $v_C \in \mathrm{Hom}(c_2^* M, c_1^* L)$  are induced by  $u$  and  $v$ . Consider the cohomological correspondence  $u^! \in \mathrm{Hom}(a_1^* L, a_2^! M)$  (respectively  $v^! \in \mathrm{Hom}(b_2^* M, b_1^! L)$ ) deduced from  $u$  (respectively  $v$ ) by composition with the trace homomorphism 6.8.7  $a_{2*} a_2^* M \rightarrow M$  (respectively  $b_{1*} b_1^* L \rightarrow L$ ), or equivalently (6.8.6) by product with  $\mathrm{cl}_{Z/Y}(A)$  (respectively  $\mathrm{cl}_{Z/X}(B)$ ). Note in passing that  $\mathrm{cl}_{Z/Y}(A)$  (respectively  $\mathrm{cl}_{Z/X}(B)$ ) is the component in  $H^0(A, \Lambda^m N_{A/Z}^\vee \otimes \Omega_{Z/Y}^m)$  (respectively  $H^0(B, \Lambda^n N_{B/Z}^\vee \otimes \Omega_{Z/X}^n)$ ) of  $\mathrm{cl}_{Z/S}(A)$  (respectively  $\mathrm{cl}_{Z/S}(B)$ ) defined by the projection  $\Omega_{Z/S}^m \rightarrow \Omega_{Z/Y}^m$  (respectively  $\Omega_{Z/S}^n \rightarrow \Omega_{Z/X}^n$ ) of 6.9.0. One checks easily that the cup-product

$$\langle u^!, v^! \rangle \in \mathrm{Ext}_{\mathcal{O}_Z}^{m+n}(\mathcal{O}_C, \Omega_{Z/S}^{m+n}) = H^0(C, \Lambda^{m+n} N_{C/Z}^\vee \otimes \Omega_{Z/S}^{m+n})$$

defined in 6.6.3 is given by

$$\langle u^!, v^! \rangle = \langle u, v \rangle \mathrm{cl}_{Z/Y}(A) \mathrm{cl}_{Z/X}(B). \quad (6.9.4)$$

On the other hand one checks that the homomorphism 6.7.3

$$\mathrm{Ext}_{\mathcal{O}_Z}^{m+n}(\mathcal{O}_C, \Omega_{Z/S}^{m+n}) \longrightarrow H^0(S, \mathcal{O}_S)$$

is the composite of the canonical arrow  $\mathrm{Ext}_{\mathcal{O}_Z}^{m+n}(\mathcal{O}_C, \Omega_{Z/S}^{m+n}) \rightarrow H^{m+n}(Z, \Omega_{Z/S}^{m+n})$  and the trace morphism  $\mathrm{Tr}_{Z/S} : H^{m+n}(Z, \Omega_{Z/S}^{m+n}) \rightarrow H^0(S, \mathcal{O}_S)$ . The Lefschetz-Verdier formula 6.7.2 therefore gives the relation

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Tr}_{Z/S}(\langle u, v \rangle \mathrm{cl}_{Z/Y}(A) \mathrm{cl}_{Z/X}(B)). \quad (6.9.5)$$

Thanks to 6.8.4 one may rewrite the second member as a residue; thus, in summary, one has obtained:

**Theorem 6.10.** Under the hypotheses of the first paragraph of 6.9, let  $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$ ,  $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$ , and consider the cohomological correspondence  $u^!$  (respectively  $v^!$ ) from  $L$  to  $M$  (respectively from  $M$  to  $L$ ), with support in  $A$  (respectively  $B$ ), deduced from  $u$  (respectively  $v$ ) by composition with the trace  $a_{2*} a_2^* M \rightarrow M$  (respectively  $b_{1*} b_1^* L \rightarrow L$ ) (6.8.7), and the homomorphism  $(u^!)_* : f_* L \rightarrow g_* M$  (respectively  $(v^!)_* : g_* M \rightarrow f_* L$ ) deduced from  $u^!$  (respectively  $v^!$ ) by direct

image (6.5.5) ( $f : X \rightarrow S$ ,  $g : Y \rightarrow S$  denoting the projections). Then, with the notation of 6.8, 6.9.3, one has

$$\langle (u^!)_*, (v^!)_* \rangle = \text{Res}_{C/S}(\langle u, v \rangle \text{cl}_{Z/Y}(A) \text{cl}_{Z/X}(B)). \quad (6.10.1)$$

**Remarks 6.11.** a) As in 3.6, using the description 3.5 b) of the direct image of a correspondence, one checks that, if  $h : Z \rightarrow S$  is the projection,  $(u^!)_*$  (respectively  $(v^!)_*$ ) is the composite of the arrows

$$\begin{aligned} f_* L &\xrightarrow{(1)} (ha)_* a_1^* L \xrightarrow{(ha)_* u} (ha)_* a_2^* M = g_* a_{2*} a_2^* M \xrightarrow{(2)} g_* M, \\ (\text{respectively } g_* M &\xrightarrow{(1)} (hb)_* b_2^* M \xrightarrow{(hb)_* v} (hb)_* b_1^* L = f_* b_{1*} b_1^* L \xrightarrow{(2)} f_* L), \end{aligned}$$

where (1) is the usual functoriality arrow (defined by the adjunction arrow  $1 \rightarrow a_{1*} a_1^*$  (respectively  $1 \rightarrow b_{2*} b_2^*$ )) and (2) is given by the trace  $a_{2*} a_2^* M \rightarrow M$  (respectively  $b_{1*} b_1^* L \rightarrow L$ ) (6.8.7).

b) Suppose that  $S$  is the spectrum of a field. The intersection  $C = A \times_{(X \times_S Y)} B$  is then formed of isolated points. At each point  $z$  of  $C$ , choose, near the images of  $z$ , regular sequences  $(s_1, \dots, s_m)$ ,  $(t_1, \dots, t_n)$  of equations of  $A, B$ . Then 6.10.1 is written, with the notation of the residue symbols of ([8] III 9), as

$$\langle (u^!)_*, (v^!)_* \rangle = \sum_{z \in C} \text{Res}_z \left( \langle u, v \rangle \prod_{1 \leq i \leq m} \frac{d_{Z/Y} s_i}{s_i} \prod_{1 \leq j \leq n} \frac{d_{Z/X} t_j}{t_j} \right), \quad (6.11.1)$$

where  $d_{Z/Y} s_i$  (respectively  $d_{Z/X} t_j$ ) denotes the component of type  $(1, 0)$  (respectively  $(0, 1)$ ) of  $d_{Z/S} s_i$  (respectively  $d_{Z/S} t_j$ ) in the decomposition 6.9.0.

**Corollary 6.12.** (Woods-Hole formula [1]). Assume that  $S$  is the spectrum of a field. Let  $X$  be a proper smooth  $S$ -scheme,  $F$  an  $S$ -endomorphism of  $X$ ,  $L$  an object of  $D(X)_{\text{parf}}$  (i.e. a perfect complex on  $X$ ), and  $u \in \text{Hom}(F^* L, L)$ . Assume that the scheme of fixed points  $X^F$  is finite and consists of transverse points (i.e. points where the graph of  $F$  meets the diagonal transversely). Then

$$\sum_i (-1)^i \text{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \frac{\text{Tr}(u_x)}{\det(1 - dF_x)},$$

where  $(F, u)$  is the endomorphism of  $H^i(X, L)$  which is the composite of  $H^i(X, L) \rightarrow H^i(X, F^* L)$  (inverse image) and  $H^i(X, u)$ ,  $\text{Tr}(u_x)$  denotes the value at  $x$  of the endomorphism induced by  $u$  on the fibre of  $L$  at  $x$ , and  $\det(1 - dF_x)$  is the value at  $x$  of the determinant of the map  $(1 - \text{map induced by } F \text{ on } \Omega_{X,x}^1)$ .

*Proof.* Apply 6.10, taking account of 6.11 a), b). Put  $Z = X \times_S X$ . Let  $x \in X^F$ . Near  $x$ , choose a system of local coordinates  $(x_1, \dots, x_n)$  (defining an étale morphism into  $\mathbb{A}_S^n$ ), put  $F_i = x_i F$ , and write  $(x_1, \dots, x_n, y_1, \dots, y_n)$  for the corresponding local coordinates on  $Z$  near  $z = (x, x)$  ( $x_i = x_i \otimes 1$ ,  $y_i = 1 \otimes x_i$ ). Then, near  $z$ ,  $(x_1 - y_1, \dots, x_n - y_n)$  (respectively  $(y_1 - F_1, \dots, y_n - F_n)$ ) is a regular system of equations for the diagonal  $A$  (respectively for the graph  $B$  of  $F$ ). The contribution at  $z$  of the second member of 6.11.1 is

$$\langle u, 1 \rangle_z = \text{Res}_z \left( \text{Tr}(u) \frac{dx_1 \cdots dx_n dy_1 \cdots dy_n}{(x_1 - y_1) \cdots (x_n - y_n)(y_1 - F_1) \cdots (y_n - F_n)} \right).$$

Observe that in the numerator one may replace  $dy_i$  by

$$dy_i - dF_i = dy_i - \sum_{1 \leq j \leq n} (dF_i/dx_j) dx_j.$$

Applying then formula ([8] III 9 R3) to the situation  $(B \hookrightarrow Z \rightarrow S)$ ,  $(s_1, \dots, s_p) = (y_1 - F_1, \dots, y_n - F_n)$ ,  $(t_1, \dots, t_n) = (x_1 - y_1, \dots, x_n - y_n)$ ,  $\omega = (\text{Tr}(u)dx_1 \cdots dx_n)$ , one obtains

$$\langle u, 1 \rangle_z = \text{Res}_x \left( \text{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right). \quad (*)$$

Since  $x$  is transverse, the  $x_i - F_i$  form a regular system of parameters of  $\mathcal{O}_{X,x}$ , and hence one may write, near  $x$ ,

$$x_i = \sum_j (x_j - F_j) c_{ij},$$

where  $(c_{ij})$  is a matrix of sections of  $\mathcal{O}_X$  invertible at  $x$ , whose determinant has value at  $x$  equal to  $\det(1 - dF_x)^{-1}$ . Thanks to ([8] III 9 R1), one therefore deduces from (\*) that

$$\langle u, 1 \rangle_z = \text{Res}_x \left( \text{Tr}(u) \det(c_{ij}) \frac{dx_1 \cdots dx_n}{x_1 \cdots x_n} \right),$$

whence, by ([8] III 9 R6),

$$\langle u, 1 \rangle_z = \text{Tr}(u(x)) \det(c_{ij}(x)) = \text{Tr}(u(x)) \det(1 - dF_x)^{-1}.$$

The corollary therefore follows from 6.11.1.

**Remark 6.12.1.** If  $X^F$  is assumed only finite, the preceding calculation shows that one has

$$\sum_i (-1)^i \text{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \text{Res}_x \left( \text{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right),$$

where  $(x_i)$  is a system of local coordinates at  $x$ , and the  $F_i$  are the coordinates of  $F$  (near  $x$ ) in this system.

The reader will convince himself, moreover, that the preceding formula is still valid if  $X$  is proper and is assumed only to be smooth at the points of  $X^F$ .

## 6.13

Assume that  $S$  is the spectrum of an algebraic closure of the finite field  $\mathbb{F}_q$  and that  $X/S$  is obtained by extension of scalars from a proper smooth scheme  $X_0/\mathbb{F}_q$ . Take for  $F$  the Frobenius endomorphism of  $X/S$  “defined by raising coordinates to the  $q$ -th power”. One has  $X^F = X_0(\mathbb{F}_q)$ , and  $X^F$  consists of transverse points since  $dF = 0$ . If  $L$  is a locally free sheaf of finite type on  $X$  and  $u \in \text{Hom}(F^*L, L)$ , the Woods-Hole formula is therefore written as

$$\sum_i (-1)^i \text{Tr}((F, u), H^i(X, L)) = \sum_{x \in X_0(\mathbb{F}_q)} \text{Tr}(u_x, L_x). \quad (6.13.1)$$

In particular, for  $L = \mathcal{O}_X$ , and  $u$  the canonical isomorphism  $F^*\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_X$ , one finds:

$$\sum_i (-1)^i \text{Tr}(F, H^i(X, \mathcal{O}_X)) = \text{Card}(X_0(\mathbb{F}_q)) \pmod{p}. \quad (6.13.2)$$

Artin-Schreier theory makes it possible to rewrite the first member in the form  $\sum_i (-1)^i \text{Tr}(F, H^i(X, \mathbb{F}_p))$ , where  $H^*(X, \mathbb{F}_p)$  denotes the étale cohomology of  $X$  with values in the constant sheaf  $\mathbb{F}_p$ . By a

refinement of this argument, Deligne, in (SGA 4<sup>1/2</sup> Fonctions L), deduced from 6.13.1 the following generalization of 6.13.2:

**Theorem 6.13.3 (Deligne).** Let  $X_0$  be a separated scheme of finite type over  $\mathbb{F}_q$ , let  $X$  be the scheme deduced from  $X_0$  by extension of scalars to an algebraic closure  $k$  of  $\mathbb{F}_q$ , and let  $F$  be the Frobenius endomorphism of  $X/k$ . Then, for every  $L_0 \in \text{ob } D_{\text{ctf}}^b(X_0, \mathbb{F}_p)$ , if  $L$  denotes the inverse image of  $L_0$  on  $X$ , one has

$$\sum_i (-1)^i \text{Tr}(F, H_c^i(X, L)) = \sum_{x \in X^F} \text{Tr}(F, L_x).$$

Let us point out that the particular case of 6.13.3 corresponding to  $L_0 = \mathbb{F}_p$  was established by Katz in (SGA 7 XXII) by a completely different method, based on Dwork's calculation of the zeta function of a hypersurface. Moreover, 6.13.2 in the proper smooth case is also an immediate consequence of the Lefschetz formula in crystalline cohomology ([5] VII 3.1.9). In fact, crystalline cohomology gives the more precise information that each factor  $\det(1 - Ft, H^i(X_0/W)) \in W[t]$  of the zeta function is congruent mod  $p$  to  $\det(1 - Ft, H^i(X_0, \mathcal{O}_{X_0}))$  (at least if  $H^*(X_0/W)$  is torsion-free). Katz showed [11] that, using estimates of B. Mazur [14], one could obtain higher congruences (involving the  $H^i(X_0, \Omega_{X_0}^j)$  and the Cartier operator) for smooth hypersurfaces (or complete intersections) for which the Hodge numbers (in the interesting dimension) are assumed to vanish up to a certain level.

On the other hand, if  $X_0$  is a proper scheme over  $\mathbb{F}_q$ , geometrically integral, and such that  $H^i(X_0, \mathcal{O}_{X_0}) = 0$  for  $i > 0$ , the Lefschetz-Verdier formula 6.7.2, applied to  $X = X_0 \otimes k$  and the Frobenius endomorphism of  $X$ , immediately implies that  $X_0$  has at least one rational point over  $\mathbb{F}_q$  (since  $\text{Tr}(F, R\Gamma(X, \mathcal{O}_X)) = 1$ ). Serre asks whether this conclusion is still true when one replaces  $\mathbb{F}_q$  by a field satisfying  $(C_1)$  ([17] II 3), for example  $\mathbb{C}(t)$  (Tsen), or  $\mathbb{C}((t))$  (Lang) (see loc. cit. for other examples); indeed, he observes that if  $X_0$  is a complete intersection of multidegree  $(m_1, \dots, m_r)$  in  $\mathbb{P}^n$ , the condition of triviality of cohomology is equivalent ([16] no. 78) to the condition  $m_1 + \dots + m_r < n + 1$ , which is also the condition that occurs when one writes condition  $(C_1)$ .

## 6.14

The Woods Hole formula has many other applications, for which we refer to [2] and to Beauville's exposé [4].

## 6.15

The editor has not examined the case of components of fixed points of dimension  $\geq 1$ , but it is not impossible that one could deduce from 6.7.2 (probably conjugated with Riemann-Roch) Lefschetz-Riemann-Roch formulae in the style of those of Donovan [7], Iversen [10], Nielsen [15] (at least with values in the base field). Nor is it impossible, on the other hand, that the formula 6.7.2, applied over the dual numbers, might provide residue formulae for vector fields analogous to those of Bott [6], [3].

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# Exposé III B

## Calculations of Local Terms

by Luc Illusie

### Introduction

This exposé, written in January 1977, does not correspond to any oral exposé of the seminar. It comprises two independent parts. In the first, we compute the Lefschetz–Verdier local terms for cohomological correspondences between smooth curves over an algebraically closed field, supported on correspondences satisfying certain general-position hypotheses. The formula that we obtain gives as corollaries Verdier’s formula [6] for transversal endomorphisms (which applies in particular to the case of the Frobenius endomorphism), and the formula stated by Langlands in ([4] 7.12). The method of proof is parallel to that followed by Artin–Verdier in [6]. One reduces to proving the vanishing of the local term when the given complexes of sheaves have zero fibre at the images of the fixed point under consideration, and in fact one proves a stronger property, namely that a certain restriction map is zero (2.5). To establish this property, one first reduces to the tamely ramified case by covers of the same degree of the two given curves (so as not to destroy the general-position hypotheses), and then treats this case directly by means of Abhyankar’s lemma and the relative purity theorem. The second part of this exposé, of a much more technical nature, is inspired by the method used by Grothendieck to establish the Lefschetz formula for certain cohomological correspondences on curves (see XII and (SGA 4<sup>1/2</sup> Rapport)). In no. 5, for complexes of modules on a topos, we develop a theory of non-commutative traces which generalizes the commutative theory of (SGA 6 I) and, in the case of a punctual topos, recovers the non-commutative theory developed by Grothendieck in the oral seminar (this was written up in Exposé XI, unfortunately lost). We apply the techniques of no. 5 to define, in no. 6, Lefschetz–Verdier local terms for cohomological correspondences between complexes of modules over rings that are not necessarily commutative. These local terms behave like non-commutative traces and make it possible, in particular when one takes as base ring the algebra of a group, to divide canonically the ordinary local terms attached to equivariant correspondences. As an application, we prove Grothendieck’s divisibility conjecture of (XII 4.5), in a more general form. We also show that the local terms defined by Grothendieck in the Lefschetz formula of XII are indeed the Lefschetz–Verdier local terms, and we generalize the formula of (loc. cit.).

I thank P. Deligne for the useful suggestions he made to me during many discussions; I am particularly grateful to him for having pointed out, and helped me correct, an error in an early proof of Theorem 1.2.

## Summary

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## I. Correspondences in General Position between Curves

In this part,  $S$  denotes the spectrum of an algebraically closed field  $k$  of characteristic  $p$ , and  $\Lambda$  a noetherian commutative ring killed by an integer prime to  $p$ . For every scheme  $X$  over  $S$  we denote by  $D(X)$  the derived category of sheaves of  $\Lambda$ -modules on  $X$ , and by  $D_{\text{ctf}}(X)$  the full subcategory of  $D(X)$  formed by complexes of finite tor-dimension and constructible cohomology.

### 1. Statement of the theorem and corollaries

#### 1.1

Let  $X$  (resp.  $Y$ ) be the strict localization, at a closed point, of a smooth separated  $S$ -curve, let  $A$  (resp.  $B$ ) be a strictly local  $S$ -scheme, and let

$$a : A \longrightarrow X \times_S Y, \quad b : B \longrightarrow X \times_S Y$$

be  $S$ -morphisms. We denote by  $p_1 : X \times_S Y \rightarrow X$ ,  $p_2 : X \times_S Y \rightarrow Y$  the projections, put  $a_i = p_i a$ ,  $b_i = p_i b$ , and denote by  $x$  (resp.  $y$ ) the closed point of  $X$  (resp.  $Y$ ). We assume that  $a_2$  and  $b_1$  are finite and surjective, and that  $A \times_{(X \times_S Y)} B$  is reduced to the closed point  $z$  whose image in  $X \times_S Y$  is  $(x, y)$ .

Let  $L \in \text{ob } D_{\text{ctf}}(X)$ ,  $M \in \text{ob } D_{\text{ctf}}(Y)$ ,  $u \in \text{Hom}(a_1^* L, a_2^! M)$ , and  $v \in \text{Hom}(b_2^* M, b_1^! L)$ . Denote by

$$u_z \in \text{Hom}(L_x, M_y) \quad (\text{resp. } v_z \in \text{Hom}(M_y, L_x))$$

the composite homomorphism

$$L_x = R\Gamma(X, L) \xrightarrow{a_1^*} R\Gamma(A, a_1^* L) \xrightarrow{R\Gamma(A, u)} R\Gamma(A, a_2^! M) \xrightarrow{a_{2*}} R\Gamma(Y, M) = M_y,$$

where  $a_{2*}$  is defined by the adjunction map  $a_{2*} a_2^! M \rightarrow M$  (resp. the analogous composite homomorphism defined by means of  $(b, v)$ ). Since  $L_x$  and  $M_y$  are perfect complexes of  $\Lambda$ -modules, one can form the cup-product (SGA 6 I 8.3)

$$\langle u_z, v_z \rangle = \text{Tr}(u_z v_z) = \text{Tr}(v_z u_z) \in \Lambda. \quad (1.1.1)$$

Put  $\Lambda_X(1)[2] = K_X$ ,  $\Lambda_Y(1)[2] = K_Y$ . By (I 5) (or (SGA 4<sup>1/2</sup> Duality 1, or Finitude 4.1)) the functor  $D_X = \text{RHom}(-, K_X)$  (resp.  $D_Y = \text{RHom}(-, K_Y)$ ) on  $D_{\text{ctf}}(X)$  (resp.  $D_{\text{ctf}}(Y)$ ) is dualizing, i.e. for

every  $E \in \text{ob } D_{\text{ctf}}(X)$  (resp.  $E \in \text{ob } D_{\text{ctf}}(Y)$ ), one has  $E \xrightarrow{\sim} D_X D_X E$  (resp.  $E \xrightarrow{\sim} D_Y D_Y E$ ). We shall sometimes write  $D$  instead of  $D_X, D_Y$ .

On the other hand, denote by  $p_i^!$  the functor  $p_i^*(-)(1)[2]$ . We have trivially

$$K_X \otimes_S^{\mathbf{L}} M \xrightarrow{\sim} p_2^! M, \quad L \otimes_S^{\mathbf{L}} K_Y \xrightarrow{\sim} p_1^! L,$$

and it follows from (III 2.3), by passage to the limit, that the Künneth maps analogous to (III 3.1.1)

$$DL \otimes_S^{\mathbf{L}} M \longrightarrow \text{RHom}(p_1^* L, p_2^! M), \quad L \otimes_S^{\mathbf{L}} DM \longrightarrow \text{RHom}(p_2^* M, p_1^! L)$$

are isomorphisms. Since  $a_2$  is finite,  $a$  is finite, and hence the functor  $a^!$  is defined; one verifies easily that one has the transitivity isomorphism  $a^! p_2^! = a_2^!$ . By the induction formula for  $a$  (SGA 4 XVIII 3.1.12.2), one therefore has a canonical isomorphism analogous to (III 3.2.1)

$$a^! \text{RHom}(p_1^* L, p_2^! M) \xrightarrow{\sim} \text{RHom}(a_1^* L, a_2^! M),$$

whence an isomorphism

$$\text{Hom}(a_1^* L, a_2^! M) \xrightarrow{\sim} H^0(X \times_S Y, a_* a^! (DL \otimes_S^{\mathbf{L}} M)). \quad (*)$$

Likewise one has an isomorphism

$$\text{Hom}(b_2^* M, b_1^! L) \xrightarrow{\sim} H^0(X \times_S Y, b_* b^! (L \otimes_S^{\mathbf{L}} DM)). \quad (**)$$

As in (III 4.1.2), one defines a natural pairing

$$(DL \otimes_S^{\mathbf{L}} M) \otimes^{\mathbf{L}} (L \otimes_S^{\mathbf{L}} DM) \longrightarrow K_X \otimes_S^{\mathbf{L}} K_Y = \Lambda_{X \times_S Y}(2)[4] \stackrel{\text{dfn}}{=} K_{X \times_S Y}.$$

Composing with the pairing analogous to (III 4.2.3)

$$a_* a^! P \otimes^{\mathbf{L}} b_* b^! Q \longrightarrow c_* c^! (P \otimes^{\mathbf{L}} Q),$$

where  $P = DL \otimes_S^{\mathbf{L}} M$ ,  $Q = L \otimes_S^{\mathbf{L}} DM$ , and  $c : Z = A \times_{(X \times_S Y)} B \rightarrow X \times_S Y$  is the canonical projection, and taking into account the isomorphisms (\*), (\*\*), one deduces a pairing analogous to (III 4.2.5)

$$\langle \ , \ \rangle : \text{Hom}(a_1^* L, a_2^! M) \otimes \text{Hom}(b_2^* M, b_1^! L) \longrightarrow H^0(X \times_S Y, c_* c^! K_{X \times_S Y}).$$

One verifies easily, by passage to the limit, that  $c^! K_{X \times_S Y} = \Lambda_z$ , so that  $H^0(X \times_S Y, c_* c^! K_{X \times_S Y}) = \Lambda$ . Thus one has a “cup-product”

$$\langle u, v \rangle \in \Lambda. \quad (1.1.2)$$

### 1.1.3

Let  $X'$  (resp.  $Y'$ ) be a smooth  $S$ -curve,  $A'$  (resp.  $B'$ ) a separated  $S$ -curve,

$$a' : A' \longrightarrow X' \times_S Y', \quad b' : B' \longrightarrow X' \times_S Y'$$

finite  $S$ -morphisms,  $x'$  (resp.  $y'$ ) a closed point of  $X'$  (resp.  $Y'$ ),  $z'$  an isolated closed point of  $A' \times_{(X' \times_S Y')} B'$ , with image  $s'$  (resp.  $t'$ ) in  $A'$  (resp.  $B'$ ), and such that  $a'_2(= p_2 a')$  (resp.  $b'_1(= p_1 b')$ ) is quasi-finite at  $s'$  (resp.  $t'$ ),

$$L' \in \text{ob } D_{\text{ctf}}(X'), \quad M' \in \text{ob } D_{\text{ctf}}(Y'), \quad u' \in \text{Hom}(a_1'^* L', a_2'^! M'), \quad v' \in \text{Hom}(b_2'^* M', b_1'^! L').$$

One can then consider the local term (III 4.2.7)  $\langle u', v' \rangle_{z'} \in \Lambda$ . On the other hand, if  $X, Y, A, B$  are the strict localizations of  $X', Y', A', B'$  at  $x', y', s', t'$ , if  $a : A \rightarrow X \times_S Y$ ,  $b : B \rightarrow X \times_S Y$  are the morphisms deduced from  $a', b'$ , if  $L = L'|X$ ,  $M = M'|Y$ , and if  $u \in \text{Hom}(a_1^* L, a_2^! M)$ ,  $v \in \text{Hom}(b_2^* M, b_1^! L)$  are the maps deduced from  $u', v'$ , then the data  $(X, Y, a, b, u, v)$  satisfy the hypotheses at the beginning of 1.1, and hence a cup-product 1.1.2  $\langle u, v \rangle \in \Lambda$  is defined. It follows from the compatibility of the formation of local terms (III 4.2.7) with localization (III 4.2.6) that  $\langle u', v' \rangle_{z'} = \langle u, v \rangle$ . It is also easy to see that any data  $(X, Y, a, b, u, v)$  satisfying the hypotheses at the beginning of 1.1 are induced by data  $(X', Y', a', b', x', y', z', s', t', u', v')$  as above, where one may also assume, if desired, that  $A' \times_{(X' \times_S Y')} B'$  is reduced to  $z'$ .

In general the cup-products 1.1.1 and 1.1.2 are unequal. Nevertheless one has the following theorem, which is the main result of this first part:

**Theorem 1.2.** Under the hypotheses at the beginning of 1.1, assume moreover that  $A$  and  $B$  are in general position, i.e. that the intersection of the tangent cones at  $(x, y)$  to the images of  $A$  and  $B$  in  $X \times_S Y$  is reduced to  $(x, y)$ , and that likewise  $A$  and  $X \times \{y\}$  are in general position, as are  $\{x\} \times Y$  and  $B$ . Then

$$\langle u_z, v_z \rangle = \langle u, v \rangle. \quad (1.2.1)$$

By the Lefschetz–Verdier formula (III 4.7), taking 1.1.3 into account, one deduces:

**Corollary 1.3.** Let  $F$  be an endomorphism of a proper smooth curve  $X/S$  such that the fixed-point scheme  $X^F$  is finite and formed of points where the graph of  $F$  is transverse to the diagonal. Let  $L \in \text{ob } D_{\text{ctf}}(X)$ , and  $u \in \text{Hom}(F^* L, L)$ . Then

$$\text{Tr}((F, u), R\Gamma(X, L)) = \sum_{x \in X^F} \text{Tr}(u_x, L_x),$$

where the trace on both sides is taken in the sense of (SGA 6 I 8).

For  $L$  concentrated in degree 0, 1.3 is the result of Artin–Verdier ([6] 4.1). By a passage to the limit explained in XV, one deduces from 1.3 an analogous result in  $\ell$ -adic cohomology ([6] 1.1), (XV §2 no. 3 prop. 2):

**Corollary 1.3.1.** With  $X$  and  $F$  as in 1.3, let  $L$  be a constructible  $\mathbb{Q}_\ell$ -sheaf on  $X$  ( $\ell$  prime to  $p$ ), and let  $u \in \text{Hom}(F^* L, L)$ . Then:

$$\sum_i (-1)^i \text{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \text{Tr}(u_x, L_x).$$

The hypotheses of 1.3 and 1.3.1 apply in particular to the case where  $(F, u)$  is a Frobenius correspondence (XV 2), whence Grothendieck’s trace formula ([2] 5.1), (SGA 4<sup>1/2</sup> Rapport 3.2).

## 1.4

With the notation of 1.1, assume that  $b_2$  and  $a_1$  are finite and surjective (instead of  $a_2$  and  $b_1$ ), and consider the dual correspondences (III 5.1)

$$Du \in \text{Hom}(a_2^* DM, a_1^! DL), \quad Dv \in \text{Hom}(b_1^* DL, b_2^! DM),$$

and the homomorphisms  $(Du)_z \in \text{Hom}((DM)_y, (DM)_x)$ ,  $(Dv)_z \in \text{Hom}((DL)_x, (DM)_y)$ . Assume that  $(A, \{x\} \times Y)$  is in general position, as are  $(X \times \{y\}, B)$  and  $(A, B)$ . Applying 1.2 to  $Du, Dv$ , one obtains

$$\langle Du, Dv \rangle = \langle (Du)_z, (Dv)_z \rangle,$$

whence, by (III 5.1.6) and (III 4.2.6),

$$\langle u, v \rangle = \langle (Du)_z, (Dv)_z \rangle. \quad (1.4.1)$$

From this remark and from 1.2 one deduces, by the Lefschetz–Verdier formula (III 4.7):

**Corollary 1.5.** Let  $X, Y$  be proper smooth curves over  $S$ ,

$$a : A \longrightarrow X \times_S Y, \quad b : B \longrightarrow X \times_S Y$$

correspondences where  $A$  and  $B$  are proper curves over  $S$ , let  $L \in \text{ob } D_{\text{ctf}}(X)$ ,  $M \in \text{ob } D_{\text{ctf}}(Y)$ ,  $u \in \text{Hom}(a_1^* L, a_2^! M)$ , and  $v \in \text{Hom}(b_2^* M, b_1^! L)$ . Assume that  $C = A \times_{(X \times_S Y)} B$  is finite over  $S$  and of the form  $C = C_1 \amalg C_2$ , with the following properties: for  $z \in C_1$ , the morphisms deduced from  $a_2, b_1$  by strict localization at the images of  $z$  are finite, and, if  $(x, y)$  denotes the image of  $z$  in  $X \times_S Y$ , each of the pairs  $(A, B)$ ,  $(A, X \times \{y\})$ ,  $(\{x\} \times Y, B)$  is in general position at  $(x, y)$ ; for  $z \in C_2$ , the morphisms deduced from  $a_1, b_2$  by strict localization at the images of  $z$  are finite, and each of the pairs  $(A, B)$ ,  $(A, \{x\} \times Y)$ ,  $(B, X \times \{y\})$  is in general position at  $(x, y)$  (“general position” meaning that the intersection of the tangent cones at  $(x, y)$  to the images of the curves of the pair under consideration is reduced to  $(x, y)$ ). Then, if  $u_* \in \text{Hom}(R\Gamma(X, L), R\Gamma(Y, M))$ ,  $v_* \in \text{Hom}(R\Gamma(Y, M), R\Gamma(X, L))$  are the morphisms defined by  $u, v$  (III 3.7.6), one has, with the notation of 1.1.1, (III 5.1.5),

$$\langle u_*, v_* \rangle = \sum_{z \in C_1} \langle u_z, v_z \rangle + \sum_{z \in C_2} \langle (Du)_z, (Dv)_z \rangle. \quad (1.5.1)$$

## 1.6

Assuming the formalism of  $\ell$ -adic cohomology, one deduces from 1.5, by the passage-to-the-limit technique of XIV, that, under the hypotheses of 1.5 on  $X, Y, a, b$ , for  $L \in \text{ob } D^b(X, \mathbb{Q}_\ell)$  ( $\ell$  prime  $\neq p$ ),  $M \in \text{ob } D^b(Y, \mathbb{Q}_\ell)$ ,  $u \in \text{Hom}(a_1^* L, a_2^! M)$ ,  $v \in \text{Hom}(b_2^* M, b_1^! L)$ , one has the analogue of 1.5.1 in  $\mathbb{Q}_\ell$ . This formula contains, as a special case, Proposition 7.12 of Langlands [4]: apply the formula with  $X = Y$ ,  $M$  the sheaf  $F$  considered by Langlands;  $L = \alpha M$  ( $\alpha \in \mathbb{Q}_\ell^*$ ),  $A$  the diagonal of  $X \times X$ ,  $b$  the correspondence  $\Phi$  of (loc. cit.),  $u$  the correspondence supported on  $A$  given by  $\alpha^{-1} : L \rightarrow M$ ,  $v$  the correspondence  $\Phi$  of (loc. cit.); with the notation of (loc. cit.),  $C_1$  (resp.  $C_2$ ) is the set of fixed points where  $a < d$  (resp.  $a > d$ ).

## 1.7

Under the hypotheses of 1.2, take for  $u$  (resp.  $v$ ) the correspondence  $\text{cl}(A)$  (resp.  $\text{cl}(B)$ ) defined by  $\text{Tr}_{a_2} : a_{2*} \Lambda \rightarrow \Lambda$  (resp.  $\text{Tr}_{b_1} : b_{1*} \Lambda \rightarrow \Lambda$ ). Formula 1.2.1 becomes

$$\deg(a_2) \deg(b_1) = \langle \text{cl}(A), \text{cl}(B) \rangle, \quad (1.7.1)$$

where  $\deg(a_2)$  (resp.  $\deg(b_1)$ ) denotes the generic degree of  $a_2$  (resp.  $b_1$ ). Taking into account the hypotheses of general position, the left hand side of 1.7.1 is none other than the intersection multiplicity at  $(x, y)$  of the cycles  $a_*A$ ,  $b_*B$ , the direct images of  $A$  and  $B$ . It is not difficult to deduce 1.7.1 directly from the compatibility of the formation of the cohomology class associated with a cycle with intersections (SGA 4<sup>1/2</sup> Cycle 2.3.8).

## Exposé III B: Calculations of Local Terms

### 2. Reduction to a vanishing theorem

#### 2.1

Under the hypotheses of 1.1, let  $s$  (resp.  $t$ ) denote the image of  $z$  in  $A$  (resp.  $B$ ), and let  $i : U = X - \{x\} \hookrightarrow X$  and  $j : V = Y - \{y\} \hookrightarrow Y$  be the canonical inclusions. Since  $a_2$  and  $b_1$  are finite, one has  $a_2^{-1}(y) = s$ ,  $b_1^{-1}(x) = t$ , hence

$$a_1^{-1}(U) \subset a_2^{-1}(V), \quad b_2^{-1}(V) \subset b_1^{-1}(U),$$

and there are commutative diagrams

$$\begin{array}{ccc} U & \xleftarrow{a'_1} & a_1^{-1}(U) \xrightarrow{a'_2} V \\ & & \downarrow j \\ X & \xleftarrow{a_1} & A \xrightarrow{a_2} Y \end{array} \quad \begin{array}{ccc} U & \xleftarrow{b'_1} & b_2^{-1}(V) \xrightarrow{b'_2} V \\ & & \downarrow j \\ X & \xleftarrow{b_1} & B \xrightarrow{b_2} Y, \end{array}$$

where the squares corresponding to  $U \leftarrow a_1^{-1}(U) \rightarrow A \rightarrow X$  and to  $b_2^{-1}(V) \rightarrow V \rightarrow Y$  are cartesian.

**Lemma 2.2.** To prove 1.2, it is enough to consider the following two cases:

$$\text{a) } L_x = 0, \quad M_y = 0; \quad \text{b) } L|U = 0, \quad M|V = 0.$$

Consider the decreasing filtrations on  $L$  and  $M$  defined by the subcomplexes

$$F^n L = \begin{cases} L, & n \leq 0, \\ i_! i^* L, & n = 1, \\ 0, & n > 1, \end{cases} \quad F^n M = \begin{cases} M, & n \leq 0, \\ j_! j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The complex  $a_{2*} a_1^* L$  (resp.  $b_{1*} b_2^* M$ ) is then filtered by

$$F^n a_{2*} a_1^* L = a_{2*} a_1^* F^n L.$$

We have  $F^n a_{2*} a_1^* L = a_{2*} a_1^* L$  for  $n \leq 0$ , and 0 for  $n > 1$ ; moreover, since  $a_2$  is finite and the square is cartesian,

$$F^1 a_{2*} a_1^* L = j_!(a'_2)_!(a'_1)^* i^* L.$$

Similarly,

$$F^n b_{1*} b_2^* M = \begin{cases} b_{1*} b_2^* M, & n \leq 0, \\ i_!(b'_1)_!(b'_2)^* j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

We may suppose that  $u$  (resp.  $v$ ) is defined by a homomorphism of complexes

$$u : a_{2*}a_1^*L \longrightarrow M \quad (\text{resp. } v : b_{1*}b_2^*M \longrightarrow L).$$

The preceding formulas show that the composite

$$F^1a_{2*}a_1^*L \longrightarrow a_{2*}a_1^*L \xrightarrow{u} M$$

(resp.  $F^1b_{1*}b_2^*M \rightarrow b_{1*}b_2^*M \xrightarrow{v} L$ ) factors uniquely through  $F^1M$  (resp.  $F^1L$ ); in other words,  $u$  (resp.  $v$ ) is a homomorphism of filtered complexes. Thus  $u$  (resp.  $v$ ) underlies a filtered correspondence, in the sense of (III 4.13), which we again denote by

$$u \in \text{Hom}_{DF}(a_1^*L, a_2^!M), \quad v \in \text{Hom}_{DF}(b_2^*M, b_1^!L).$$

By additivity of cup-products ([3] V 3.8.7), (III 4.13.1), one has

$$\langle u_z, v_z \rangle = \langle \text{gr}^0 u_z, \text{gr}^0 v_z \rangle + \langle \text{gr}^1 u_z, \text{gr}^1 v_z \rangle,$$

and

$$\langle u, v \rangle = \langle \text{gr}^0 u, \text{gr}^0 v \rangle + \langle \text{gr}^1 u, \text{gr}^1 v \rangle.$$

The lemma follows, since  $\text{gr}^1 L = i_! i^* L$  (resp.  $\text{gr}^1 M = j_! j^* M$ ) has zero fibre at  $x$  (resp.  $y$ ), whereas  $\text{gr}^0 L$  (resp.  $\text{gr}^0 M$ ) is concentrated at  $x$  (resp.  $y$ ).

**Lemma 2.3.** Under the hypotheses of 1.1, suppose that  $L|U = 0$  and  $M|V = 0$ . Then

$$\langle u_z, v_z \rangle = \langle u, v \rangle.$$

The idea of the proof is that then  $u$  and  $v$  are direct images of correspondences between complexes concentrated at the closed points, and that the formula to be proved merely expresses the Lefschetz formula for the projections  $x \rightarrow S$ ,  $y \rightarrow S$ . In view of 1.1.3, the data of 1.1 may be replaced by the following data:  $X$  (resp.  $Y$ ) is a smooth  $S$ -curve endowed with a closed point  $x$  (resp.  $y$ ),  $A$  (resp.  $B$ ) is a separated  $S$ -curve,  $a : A \rightarrow X \times_S Y$ ,  $b : B \rightarrow X \times_S Y$  are finite  $S$ -morphisms such that  $A \times_{X \times_S Y} B$  is reduced to a closed point  $z$  above  $(x, y)$ ,

$$u \in \text{Hom}(a_1^*L, a_2^!M), \quad v \in \text{Hom}(b_2^*M, b_1^!L),$$

with

$$L = h_{1*}L', \quad M = h_{2*}M', \quad L' \in \text{ob } D_{\text{ctf}}(\{x\}), \quad M' \in \text{ob } D_{\text{ctf}}(\{y\}),$$

where  $h_1 : \{x\} \rightarrow X$  and  $h_2 : \{y\} \rightarrow Y$  are the canonical inclusions. We must show that

$$\langle u_z, v_z \rangle = \langle u, v \rangle_z,$$

where

$$u_z \in \text{Hom}(\text{R}\Gamma(X, L), \text{R}\Gamma(Y, M)), \quad v_z \in \text{Hom}(\text{R}\Gamma(Y, M), \text{R}\Gamma(X, L))$$

are defined as in 1.1, and  $\langle u, v \rangle_z$  is the value at  $z$  of the cup-product (III 4.2.7). We have  $\text{R}\Gamma(X, L) = \text{R}\Gamma_c(X, L) = L_x$ ,  $\text{R}\Gamma(Y, M) = \text{R}\Gamma_c(Y, M) = M_y$ , and  $u_z$  (resp.  $v_z$ ) is nothing other than the direct image  $u_*$  (resp.  $v_*$ ) of  $u$  (resp.  $v$ ) on  $S$  in the sense of (III 3.7.6), as is shown by the calculation procedure (III 3.5 b)).

On the other hand, if  $h = h_1 \times_S h_2 : (x, y) \rightarrow X \times_S Y$  is the inclusion, and if  $a' : A' \rightarrow (x, y)$ ,  $b' : B' \rightarrow (x, y)$  denote the fibres of  $a, b$  at  $(x, y)$ , each reduced to a point, then there are canonical isomorphisms

$$\begin{aligned} h_* a'_* a'^! (DL_x \otimes_{\Lambda}^L M_y) &\xrightarrow{\sim} a_* a^! (DL \otimes_{\Lambda}^L M), \\ h_* b'_* b'^! (L_x \otimes_{\Lambda}^L DM_y) &\xrightarrow{\sim} b_* b^! (L \otimes_{\Lambda}^L DM), \end{aligned}$$

and the arrows

$$\begin{aligned} h_* : \text{Hom}(a_1'^* L_x, a_2'^! M_y) &\rightarrow \text{Hom}(a_1^* L, a_2^! M), \\ h_* : \text{Hom}(b_2^* M_y, b_1'^! L_x) &\rightarrow \text{Hom}(b_2^* M, b_1^! L) \end{aligned}$$

(III 3.7.6) are isomorphisms. The correspondences  $u$  and  $v$  are therefore written uniquely as

$$u = h_* u', \quad v = h_* v',$$

with

$$u' \in \text{Hom}(a_1'^* L_x, a_2'^! M_y), \quad v' \in \text{Hom}(b_2^* M_y, b_1'^! L_x).$$

By the Lefschetz formula (III 4.5.1) for  $(h_1, h_2, A' \rightarrow A, B' \rightarrow B)$ , one has

$$\langle u, v \rangle_z = \langle u', v' \rangle_z.$$

On the other hand, by the Lefschetz formula for  $\{x\} \rightarrow S, \{y\} \rightarrow S$ , one has

$$\langle u_*, v_* \rangle = \langle u', v' \rangle_z.$$

In view of the preceding remark, the conclusion follows by comparing these two equalities.

**Lemma 2.4.** To prove 1.2, one may suppose that  $a$  and  $b$  are closed immersions, and it is enough to prove that, if  $L_x = 0$  and  $M_y = 0$ , the restriction map

$$H_A^0(X \times_S Y, DL \otimes_{\Lambda}^L M) \longrightarrow H_z^0(B, b^*(DL \otimes_{\Lambda}^L M)) \quad (2.4.1)$$

is zero.

Put  $X \times_S Y = Z$ . The closed subschemes  $a' : A' \rightarrow Z, b' : B' \rightarrow Z$ , images of  $a$  and  $b$ , satisfy the same hypotheses as  $a$  and  $b$ . Let  $u'$  (resp.  $v'$ ) be the correspondence supported on  $A'$  (resp.  $B'$ ) which is the direct image of  $u$  (resp.  $v$ ) by the arrow deduced from the adjunction arrow  $f_* f^! \rightarrow 1$  (resp.  $g_* g^! \rightarrow 1$ ), where  $f : A \rightarrow A'$  (resp.  $g : B \rightarrow B'$ ) is the projection. It is immediate that  $u_z = u'_z$  and  $v_z = v'_z$ . On the other hand, it follows easily from the definition of the cup-product 1.1.2 that

$$\langle u, v \rangle = \langle u', v' \rangle,$$

hence the first assertion.

By 2.2 and 2.3, one may suppose that  $L_x = 0$  and  $M_y = 0$ , and the conclusion of 1.2 then becomes  $\langle u, v \rangle = 0$ . Let  $P, Q \in \text{ob } D_{\text{ctf}}^b(Z)$ . The cup-product

$$H_A^0(Z, P) \otimes H_B^0(Z, Q) \longrightarrow H_z^0(Z, P \otimes_{\Lambda}^L Q)$$

decomposes as

$$H_A^0(Z, P) \otimes H_B^0(Z, Q) \xrightarrow{b^* \otimes 1} H_z^0(B, b^* P) \otimes H_B^0(Z, Q) \longrightarrow H_z^0(Z, P \otimes_{\Lambda}^L Q),$$

where  $b^*$  is the restriction and the second arrow, a variant of the product considered in (SGA 4<sup>1/2</sup> Cycle I 2.1), is derived from the evident product at the level of sections of sheaves. This assertion is easily checked, for example by means of the flasque resolutions of (SGA 4 XVII 4.2). If  $a' : \{z\} \rightarrow B$ ,  $b' : \{z\} \rightarrow A$ ,  $c : \{z\} \rightarrow Z$  are the inclusions, one may also invoke the standard decomposition of the Künneth arrow (III 4.2.1)

$$a'^! P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q \longrightarrow c^! (P \otimes_{\Lambda}^{\mathbb{L}} Q)$$

into a base-change arrow

$$a'^! P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q \longrightarrow a'^! b^* P \otimes_{\Lambda}^{\mathbb{L}} a^* b'^! Q,$$

followed by projection arrows

$$a'^! b^* P \otimes_{\Lambda}^{\mathbb{L}} a^* b'^! Q \longrightarrow a'^! (b^* P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q) \longrightarrow a'^! b'^! (P \otimes_{\Lambda}^{\mathbb{L}} Q).$$

Taking  $P = DL \otimes_{\Lambda}^{\mathbb{L}} M'$ ,  $Q = L \otimes_{\Lambda}^{\mathbb{L}} DM$ , and returning to the definition of  $\langle u, v \rangle$  (III 4.2.5), one sees that it is enough to prove that  $b^* u = 0$ , whence the conclusion.

In view of 2.4, 1.2 will therefore follow from the following more precise result.

**Proposition 2.5.** Under the hypotheses of 1.2, suppose that  $a$  and  $b$  are closed immersions and that  $L_x = 0$ ,  $M_y = 0$ . Let  $T$  be the strict localization of  $X \times_S Y$  at  $z = (x, y)$ , and identify  $A$ ,  $B$  with closed subschemes of  $T$ . Then the restriction arrows

$$b^* : R\Gamma_A(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T) \longrightarrow R\Gamma_z(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)),$$

$$a^* : R\Gamma_B(T, (L \otimes_{\Lambda}^{\mathbb{L}} DM)|T) \longrightarrow R\Gamma_z(A, a^*(L \otimes_{\Lambda}^{\mathbb{L}} DM))$$

are zero.

## 2.6

The proof of 2.5 will occupy the rest of number 2 and numbers 3 and 4. By symmetry of the statement, it is enough to prove the assertion concerning  $b^*$ . We shall rewrite it in a form which will be more convenient. First note that

$$R\Gamma(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T) = (DL \otimes_{\Lambda}^{\mathbb{L}} M)_z = (DL)_x \otimes_{\Lambda}^{\mathbb{L}} M_y = 0,$$

because  $M_y = 0$ , and similarly

$$R\Gamma(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)) = (DL \otimes_{\Lambda}^{\mathbb{L}} M)_z = 0.$$

Consequently, the relative cohomology triangles give isomorphisms

$$R\Gamma(T - A, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T - A) \xrightarrow{\sim} R\Gamma_A(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T)[1],$$

$$R\Gamma(B - z, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)|B - z) \xrightarrow{\sim} R\Gamma_z(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M))[1],$$

under which the arrow  $b^*$  of 2.5, shifted by 1, is identified with the restriction

$$b^* : R\Gamma(T - A, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T - A) \longrightarrow R\Gamma(B - z, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|B - z). \quad (2.6.1)$$

Since  $L_x = 0$ ,  $M_y = 0$ , one may write

$$L = i_! P, \quad M = j_! Q, \quad (2.6.2)$$

where  $i : X - x \rightarrow X$ ,  $j : Y - y \rightarrow Y$  are the inclusions, and  $P \in \text{ob } D_{\text{ctf}}(X - x)$ ,  $Q \in \text{ob } D_{\text{ctf}}(Y - y)$ . By the duality isomorphism (SGA 4 XVIII 3.19.6), together with passage to the limit, one has

$$DL = Ri_* DP = Ri_* P,$$

hence

$$DL \otimes_{\Lambda}^{\mathbb{L}} M = Ri_* P \otimes_{\Lambda}^{\mathbb{L}} j_! Q.$$

Put  $X \times_S Y = Z$ , and identify  $X \times \{y\}$  and  $\{x\} \times Y$  with closed subschemes of  $Z$  (resp.  $T$ ), which we shall simply denote by  $X$  and  $Y$ . Let

$$f' : Z - Y \hookrightarrow Z, \quad g' : Z - (X \cup Y) \hookrightarrow Z - Y,$$

$$f : T - Z \hookrightarrow T, \quad g : T - (X \cup Y) \hookrightarrow T - Y$$

be the inclusions. The Künneth formula (III 1.6.4) gives, by passage to the limit,

$$DL \otimes_{\Lambda}^{\mathbb{L}} M = Rf'_* g'_! (P \otimes_{\Lambda}^{\mathbb{L}} Q),$$

and hence

$$(DL \otimes_{\Lambda}^{\mathbb{L}} M)|_T = Rf_* g_! (P \otimes_{\Lambda}^{\mathbb{L}} Q)|_{T - (X \cup Y)}. \quad (2.6.3)$$

With this identification, 2.6.1 becomes

$$b^* : R\Gamma(T - (Y \cup A), g_!(E)|_{T - (Y \cup A)}) \longrightarrow R\Gamma(B - z, g_!(E)|_{B - z}), \quad (2.6.4)$$

where

$$E = (P \otimes_{\Lambda}^{\mathbb{L}} Q)|_{T - (X \cup Y)}.$$

In the next two numbers, we shall show that the arrow 2.6.4 is zero: first we shall reduce to the case where the cohomology sheaves of  $E$  have tame ramification, more precisely where the inertia action occurs through an  $\ell$ -group, for a prime number  $\ell \neq p$ , and then we shall treat that case directly by means of a purity theorem.

### 3. Reduction to the tame case

#### 3.1

Recall that the base ring  $\Lambda$  is annihilated by an integer prime to  $p$ , hence is a direct product of noetherian rings  $\Lambda_i$ , each  $\Lambda_i$  being annihilated by a power of a prime number  $\ell_i \neq p$ . For every scheme  $S'$  over  $S$ , the derived category  $D(S', \Lambda)$  is the product of the categories  $D(S', \Lambda_i)$ , so that, in order to prove the vanishing of 2.6.4, we may suppose that  $\Lambda$  is annihilated by a power of a prime number  $\ell \neq p$ . We may also suppose that  $P$  (resp.  $Q$ ) is bounded and has locally free components of finite type.

### 3.2

Put  $X - x = U$ ,  $Y - y = V$ . Let  $U''/U$  be a finite Galois covering with group  $G$  which trivializes the components of  $P$ , let  $H$  be an  $\ell$ -Sylow subgroup of  $G$ , let  $U' = U''/H$  be the quotient covering, and let  $X'$  be the normalized strictly local trait of  $X$  in  $U'$ . Then  $X'/X$  is a finite connected covering of degree  $d_1$  prime to  $\ell$ , and the representation of  $\pi_1(U')$  defined by the components of  $P' = P|U'$  factors through a finite  $\ell$ -group. Proceeding similarly with  $Q$ , one finds a finite connected covering  $Y'/Y$  of degree  $d_2$  prime to  $\ell$ , inducing on  $V$  an étale covering  $V'$  such that the representation of  $\pi_1(V')$  defined by the components of  $Q' = Q|V'$  factors through a finite  $\ell$ -group. Let

$$d = d_1 e_1 = d_2 e_2 = \text{lcm}(d_1, d_2).$$

After adjoining an  $e_1$ -th root (resp. an  $e_2$ -th root) of a uniformizer of  $X'$  (resp.  $Y'$ ), we may suppose that  $X'$  (resp.  $Y'$ ) has the same properties as above, and moreover that  $d_1 = d_2 = d$ .

Let  $x'$  (resp.  $y'$ ) be the closed point of  $X'$  (resp.  $Y'$ ),  $T'$  the strict localization of  $X' \times_S Y'$  at  $(x', y')$ ,  $z'$  the closed point of  $T'$ ,  $A'$  (resp.  $B'$ ) the inverse image of  $A$  (resp.  $B$ ) in  $T'$ , and

$$E' = (P \overset{\text{L}}{\otimes}_{\Lambda} Q)|T' - (X' \cup Y') = (P' \overset{\text{L}}{\otimes}_{\Lambda} Q')|T' - (X' \cup Y').$$

We then have a commutative square of restriction arrows

$$\begin{array}{ccc} \text{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) & \xrightarrow{b^*} & \text{R}\Gamma(B - z, g_!(E)|B - z) \\ \downarrow & & \downarrow (1) \\ \text{R}\Gamma(T' - (Y' \cup A'), g'_!(E')|T' - (Y' \cup A')) & \xrightarrow{b'^*} & \text{R}\Gamma(B' - z', g'_!(E')|B' - z'), \end{array}$$

where  $b' : B' \rightarrow T'$ ,  $g' : T' - (X' \cup Y') \rightarrow T' - Y'$  are the inclusions. The covering  $T'/T$  is finite, of degree  $d^2$ , and étale outside  $X \cup Y$ , hence induces an étale covering of degree  $d^2$  from  $B' - z'$  onto  $B - z$ . The composite of (1) and the trace morphism

$$\text{R}\Gamma(B' - z', g'_!(E')|B' - z') \longrightarrow \text{R}\Gamma(B - z, g_!(E)|B - z)$$

is therefore multiplication by  $d^2$ , which is an isomorphism, since  $\ell$  does not divide  $d$ . To prove that the arrow 2.6.4 is zero, it is therefore enough to prove that  $b'^* = 0$ . We are thus reduced to the case where the representation of  $\pi_1(U)$  (resp.  $\pi_1(V)$ ) defined by the components of  $P$  (resp.  $Q$ ) factors through a finite  $\ell$ -group, provided one ensures that  $A'$ ,  $B'$  satisfy the general-position hypotheses of 1.2. This will follow from the following lemma.

**Lemma 3.3.** Under the hypotheses of 1.2,  $a$  and  $b$  being closed immersions, let  $X'/X$ ,  $Y'/Y$  be finite coverings of strictly local  $S$ -traits, of the same degree  $d$ . Let  $A'$  (resp.  $B'$ ) denote the inverse image of  $A$  (resp.  $B$ ) in  $X' \times_S Y'$ . Then the pairs  $(A', B')$ ,  $(A', X')$ ,  $(Y', B')$  are each in general position in the sense of 1.2.

The proof of 3.3 will be given after a few preliminaries.

**Lemma 3.4.** Let  $X \xleftarrow{s_1} W \xrightarrow{s_2} Y$  be finite morphisms of  $S$ -traits, let  $s = (s_1, s_2) : W \rightarrow X \times_S Y$ , and let  $u, v, w$  be uniformizers of  $X, Y, W$  respectively, so that

$$s_1^* u = a w^m \pmod{w^{m+1}}, \quad s_2^* v = b w^n \pmod{w^{n+1}},$$

with  $a, b \in k^*$ . Then: (i) if  $m < n$ ,  $s(W)$  is tangent to  $X$  at  $z = (x, y)$ ; (ii) if  $m = n$ ,  $s(W)$  is tangent to the subscheme of  $X \times_S Y$  with equation

$$ap_2^*(v) - bp_1^*(u) = 0.$$

Let  $Z = X \times_S Y$ , and let  $W'$  be the reduced closed subscheme which is the image of  $s$ . In the tangent space to  $Z$  at  $z$ , the slope, with respect to the tangent to  $X$ , of the tangent to  $W'$  is the value at the origin of the function induced on  $W'$  by  $p_2^*(v)/p_1^*(u)$ ; this is also the value at the origin of the function induced by  $p_2^*(v)/p_1^*(u)$  on  $W$ , and is therefore 0 (resp.  $b/a$ ) in case (i) (resp. (ii)).

In the situation of 3.4, we shall define the slope of  $W$  with respect to  $((X, u), (Y, v))$  as the element of  $k \cup \{\infty\}$  equal to 0 if  $m < n$ , to  $\infty$  if  $m > n$ , and to  $b/a$  if  $m = n$ . This element is independent of the choice of  $w$ .

**Lemma 3.5.** Let  $f : X' \rightarrow X$ ,  $g : Y' \rightarrow Y$  be finite morphisms of  $S$ -traits with the same ramification index  $d$ , and let  $u, v, u', v'$  be uniformizers of  $X, Y, X', Y'$  respectively. Then there exists  $C \in k^*$  such that, for every commutative diagram of finite morphisms of  $S$ -traits

$$\begin{array}{ccccc} X' & \xleftarrow{s'_1} & W' & \xrightarrow{s'_2} & Y' \\ & & \downarrow h & & \downarrow g \\ X & \xleftarrow{s_1} & W & \xrightarrow{s_2} & Y, \end{array}$$

one has  $\lambda = C\lambda'^d$ , where  $\lambda$  (resp.  $\lambda'$ ) is the slope of  $W$  (resp.  $W'$ ) with respect to  $((X, u), (Y, v))$  (resp.  $((X', u'), (Y', v'))$ ).

By hypothesis,

$$f^*u = \alpha u'^d \pmod{u'^{d+1}}, \quad g^*v = \beta v'^d \pmod{v'^{d+1}},$$

with  $\alpha, \beta \in k^*$ . We show that  $C = \beta/\alpha$  has the required properties. Let  $w$  (resp.  $w'$ ) be a uniformizer of  $W$  (resp.  $W'$ ). We have

$$\begin{aligned} s_1^*u &= aw^m \pmod{w^{m+1}}, & s_2^*v &= bw^n \pmod{w^{n+1}}, \\ s_1'^*u' &= a'w'^{m'} \pmod{w'^{m'+1}}, & s_2'^*v' &= b'w'^{n'} \pmod{w'^{n'+1}}, \\ h^*w &= \gamma w'^r \pmod{w'^{r+1}}, \end{aligned}$$

with  $a, b, a', b', \gamma \in k^*$ . Commutativity of the squares in the diagram implies

$$m'd = rm, \quad n'd = rn,$$

hence  $m'/n' = m/n$ . Suppose first that  $m = n$ . Then  $\lambda = b/a$ ,  $\lambda' = b'/a'$ . If  $N = m'd = rm = n'd = rn$ , the preceding congruences give

$$(fs_1')^*u = \alpha a'^d w'^N \pmod{w'^{N+1}}, \quad (s_1h)^*u = a\gamma^m w'^N \pmod{w'^{N+1}},$$

whence  $\alpha a'^d = a\gamma^m$ . Similarly  $\beta b'^d = b\gamma^n$ , whence  $\lambda = C\lambda'^d$ , for  $\lambda \neq 0, \infty$ . But, since  $m'/n' = m/n$ ,  $\lambda = 0$  (resp.  $\infty$ ) is equivalent to  $\lambda' = 0$  (resp.  $\infty$ ), which completes the proof.

We now prove 3.3. Let  $Z = X \times_S Y$ ,  $Z' = X' \times_S Y'$ , let  $\tilde{A}$  (resp.  $\tilde{B}$ ) be the normalization of  $A$  (resp.  $B$ ), and let  $\tilde{A}'$  (resp.  $\tilde{B}'$ ) be the normalization of  $A'$  (resp.  $B'$ ). Since  $\tilde{A}'$  (resp.  $\tilde{B}'$ ) is also the normalization of  $\tilde{A} \times_Z Z'$  (resp.  $\tilde{B} \times_Z Z'$ ), we have commutative squares

$$\begin{array}{ccccc} \tilde{A}' & \longrightarrow & Z' & \longleftarrow & \tilde{B}' \\ \downarrow & & & & \downarrow \\ \tilde{A} & \longrightarrow & Z & \longleftarrow & \tilde{B}. \end{array}$$

Let  $(x_i)_{1 \leq i \leq m}$  (resp.  $(y_j)_{1 \leq j \leq n}$ ) be the points of  $\tilde{A}$  (resp.  $\tilde{B}$ ) above  $x$  (resp.  $y$ ), and let  $(x'_i)_{1 \leq i \leq m'}$  (resp.  $(y'_j)_{1 \leq j \leq n'}$ ) be the points of  $\tilde{A}'$  (resp.  $\tilde{B}'$ ) above  $x'$  (resp.  $y'$ ). Since  $A$  and  $B$  are strictly local, we have decompositions

$$\tilde{A} = \coprod_{1 \leq i \leq m} A_i, \quad \tilde{B} = \coprod_{1 \leq j \leq n} B_j, \quad \tilde{A}' = \coprod_{1 \leq i \leq m'} A'_i, \quad \tilde{B}' = \coprod_{1 \leq j \leq n'} B'_j,$$

where  $A_i, B_j, A'_i, B'_j$  are strictly local traits with respective closed points  $x_i, y_j, x'_i, y'_j$ . Choose uniformizers  $u, v, u', v'$  of  $X, Y, X', Y'$  respectively; let  $\lambda_i$  (resp.  $\mu_j$ ) be the slope of  $A_i$  (resp.  $B_j$ ) with respect to  $((X, u), (Y, v))$ , and let  $\lambda'_i$  (resp.  $\mu'_j$ ) be that of  $A'_i$  (resp.  $B'_j$ ) with respect to  $((X', u'), (Y', v'))$ . The general-position hypotheses on  $A$  and  $B$  are expressed by

$$\lambda_i \neq 0 \text{ for all } i, \quad \mu_j \neq \infty \text{ for all } j, \quad \lambda_i \neq \mu_j \text{ for all } i, j, \quad (1)$$

and we have to show that

$$\lambda'_i \neq 0 \text{ for all } i, \quad \mu'_j \neq \infty \text{ for all } j, \quad \lambda'_i \neq \mu'_j \text{ for all } i, j. \quad (2)$$

Let  $i' \in [1, m']$ ,  $j' \in [1, n']$ , and let  $x_i$  (resp.  $y_j$ ) be the image of  $x'_{i'}$  (resp.  $y'_{j'}$ ). The preceding diagram provides commutative squares of finite morphisms of  $S$ -traits

$$\begin{array}{ccc} X' \longleftarrow A'_{i'} & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X \longleftarrow A_i & \longrightarrow & Y \end{array} \quad \begin{array}{ccc} X' \longleftarrow B'_{j'} & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X \longleftarrow B_j & \longrightarrow & Y. \end{array}$$

By 3.5, there exists  $C \in k^*$  such that

$$\lambda_i = C \lambda'_{i'}{}^d, \quad \mu_j = C \mu'_{j'}{}^d.$$

Thus (1) implies (2), which completes the proof of 3.3.

## 4. End of the proof of 1.2

### 4.1

To complete the proof of 2.5, hence of 1.2, it remains to establish the vanishing of 2.6.4 under the following additional hypotheses:  $\Lambda$  is annihilated by a power of a prime number  $\ell \neq p$ , and

$P$  and  $Q$  are bounded, with locally free components of finite type, such that the corresponding representations of the fundamental groups factor through finite  $\ell$ -groups. We have

$$g_!(E)|T - (Y \cup A) = g_{1!}(E_1),$$

where  $g_1 : T - (X \cup Y \cup A) \hookrightarrow T - (Y \cup A)$  is the inclusion, and

$$E_1 = (P \otimes_{\Lambda}^L Q)|T - (X \cup Y \cup A).$$

Now  $E_1$  is a complex with bounded, constructible, locally constant cohomology such that the corresponding representation of  $\pi_1(T - (X \cup Y \cup A))$  factors through an  $\ell$ -group. The assertion to be proved will therefore follow from the following more general statement.

**Proposition 4.2.** Let  $T$  be the strict localization at a closed point of a smooth  $S$ -scheme of dimension 2, let  $t$  be the closed point of  $T$ , and let  $A, B, X$  be closed subschemes of dimension 1 of  $T$ . Let  $i : T - (A \cup X) \hookrightarrow T - A$  be the inclusion. Suppose that  $X$  is regular, and that  $A$  is in general position with respect to  $B$  and  $X$ ; that is, the intersection of the tangent cones at  $t$  to  $A$  and  $B \cup X$  is reduced to  $t$ . Let  $E \in \text{ob } D^b(T - (A \cup X))$  be such that  $H^*(E)$  is constructible and locally constant, and that the corresponding representation of  $\pi_1(T - (A \cup X))$  factors through an  $\ell$ -group. Then the restriction arrow

$$R\Gamma(T - A, i_!(E)) \longrightarrow R\Gamma(B - t, i_!(E)|B - t)$$

is zero.

The proof will rely on the following lemma.

**Lemma 4.3.** Let  $Z$  be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to  $\ell$ , and let  $X, Y$  be regular divisors meeting transversely at a point  $z$ , giving a diagram of inclusions

$$\begin{array}{ccc} Z - (X \cup Y) & \xrightarrow{i'} & Z - Y \\ j \downarrow & & \downarrow j \\ Z - X & \xrightarrow{i} & Z \\ m \downarrow & & \downarrow m \\ Y - z & \xrightarrow{i_1} & Y. \end{array}$$

Assume the purity hypothesis

$$(P) \quad R^q j_*(\mathbb{Z}/\ell) = 0 \quad \text{for } q > 1.$$

Let  $L \in \text{ob } D^b(Z - (X \cup Y), \Lambda)$  have constructible locally constant cohomology, with the corresponding representation of  $\pi_1(Z - (X \cup Y))$  factoring through an  $\ell$ -group. Then:

a) There is a canonical isomorphism

$$i_{1!} m'^* Rj_* i'_! L \xrightarrow{\sim} m^* Rj_*(i'_! L),$$

and the complex  $m'^* Rj_* i'_! L$  has constructible, locally constant cohomology.

- b) Suppose that  $Y$  is connected, and let  $\bar{s}$  be a geometric point of  $Y$  localized at a closed point  $s \neq z$ . Then the restriction arrow

$$R\Gamma(Y, m^* Rj_*(i'_! L)) \longrightarrow (Rj_*(i'_! L))_{\bar{s}} \quad (4.3.1)$$

is zero.

- c) Under the hypotheses of b), let  $C$  be a closed subscheme of dimension 1 of  $Z$  such that  $Y \cap C = s$ , and let  $\tilde{C}$  be the strict localization of  $C$  at  $\bar{s}$ . Then the restriction arrow

$$R\Gamma(Z - Y, i'_! L) \longrightarrow R\Gamma(\tilde{C} - \bar{s}, i'_!(L)|_{\tilde{C} - \bar{s}}) \quad (4.3.2)$$

is zero.

The trivial isomorphism

$$i^* Rj_* i'_! L \xrightarrow{\sim} Rj_*(i'^* i'_! L) = Rj_* L$$

gives an isomorphism

$$m'^* Rj_* i'_! L \xrightarrow{\sim} i^* m^* Rj_* i'_! L,$$

hence, by adjunction, an arrow

$$i_{1!} m'^* Rj_* i'_! L \longrightarrow m^* Rj_*(i'_! L),$$

which induces an isomorphism outside  $z$ . We show that it is an isomorphism, that is, that if  $\bar{z}$  is a geometric point above  $z$ , then

$$(Rj_*(i'_! L))_{\bar{z}} = 0.$$

For this, one may replace  $\Lambda$  by  $\mathbb{Z}/\ell^\nu$ , where  $\ell^\nu$  annihilates  $\Lambda$ , and by dévissage and passage to the limit reduce to the case where  $L$  is a locally constant  $\mathbb{Z}/\ell^\nu$ -module of finite type corresponding to a representation of  $\pi_1$  through an  $\ell$ -group. Since the only finite abelian group of  $\ell$ -torsion which is simple under the action of an  $\ell$ -group is  $\mathbb{Z}/\ell$  with trivial action, a further dévissage reduces to the case where  $L$  is the constant sheaf  $\mathbb{Z}/\ell$ .

Let  $\tilde{Z}$  be the strict localization of  $Z$  at  $\bar{z}$ ,  $\tilde{X} = X \times_Z \tilde{Z}$ ,  $\tilde{Y} = Y \times_Z \tilde{Z}$ , and let  $i' : \tilde{Z} - (\tilde{X} \cup \tilde{Y}) \hookrightarrow \tilde{Z} - \tilde{Y}$  be the inclusion. One has

$$(Rj_*(i'_!(\mathbb{Z}/\ell)))_{\bar{z}} = R\Gamma(\tilde{Z} - \tilde{Y}, i'_!(\mathbb{Z}/\ell)),$$

so it remains to prove that the restriction arrow

$$H^n(\tilde{Z} - \tilde{Y}, \mathbb{Z}/\ell) \longrightarrow H^n(\tilde{X} - z, \mathbb{Z}/\ell)$$

is an isomorphism for all  $n$ . For  $n \leq 1$  this is a consequence of Abhyankar's lemma, and for  $n > 1$  both sides vanish by the purity hypothesis. The first assertion of a) is therefore proved. We prove the second. By dévissage, we may suppose  $L$  concentrated in degree 0. The hypothesis (P) implies, by dévissage, that  $R^q j_* L = 0$  for  $q > 2$ . It remains to prove constructibility and local constancy of  $m'^* R^q j_* i'_! L$  for  $q \leq 1$ , which is standard. For constructibility, resolving  $L$  on the right by sheaves of the form  $p_* p^* L$ , where  $p : Z' \rightarrow Z$  is a finite covering tamely ramified along  $Y$ , reduces to the case where  $L$  is constant, of value  $L_0$ : then

$$m'^* Rj_* i'_! L = (L_0)_{Y-z}, \quad m'^* R^1 j_* i'_! L = (L_0)_{Y-z}(-1)$$

by Abhyankar. For local constancy, one has to see that the specialization arrows are isomorphisms. One may, for this, replace  $\Lambda$  by  $\mathbb{Z}/\ell^\nu$  as above, and then reduce by dévissage to the case where  $L$  is the constant sheaf  $\mathbb{Z}/\ell$ ; the conclusion follows from Abhyankar's lemma.

We have therefore proved a). We prove b). Let  $\bar{\eta}$  be a geometric generic point of  $Y$ . Choose specialization arrows

$$a : \bar{\eta} \rightarrow Y(\bar{z}), \quad b : \bar{\eta} \rightarrow Y(\bar{s}),$$

where  $Y(\bar{z})$  (resp.  $Y(\bar{s})$ ) is the strict localization of  $Y$  at  $\bar{z}$  (resp.  $\bar{s}$ ). Put

$$M = m^* Rj_*(i'_! L).$$

The commutative square

$$\begin{array}{ccc} Y & \longrightarrow & Y(\bar{s}) \\ \uparrow & & \uparrow b \\ Y(\bar{z}) & \xrightarrow{a} & \bar{\eta} \end{array}$$

induces a commutative square

$$\begin{array}{ccc} R\Gamma(Y, M) & \xrightarrow{4.3.1} & M_{\bar{s}} \\ \downarrow & & \downarrow b^* \\ M_{\bar{z}} & \xrightarrow{a^*} & M_{\bar{\eta}}. \end{array}$$

Since  $M|_{Y-z}$  has locally constant cohomology,  $b^*$  is an isomorphism. On the other hand, since  $M = i_{1!} i_1^* M$ , one has  $M_{\bar{z}} = 0$ , hence the arrow 4.3.1 is zero. It remains to prove c). We have  $\tilde{C} = C \times_Z \tilde{Z}$ , where  $\tilde{Z}$  denotes, as above, the strict localization of  $Z$  at  $\bar{z}$ . Put  $\tilde{Y} = Y \times_Z \tilde{Z}$ . Then there is a commutative diagram of restriction arrows

$$\begin{array}{ccc} R\Gamma(Z - Y, i'_! L) = R\Gamma(Z, Rj_*(i'_! L)) & \xrightarrow{4.3.2} & R\Gamma(\tilde{C} - \bar{s}, i'_! L)|_{\tilde{C} - \bar{s}} \\ \downarrow & & \uparrow \\ R\Gamma(Y, m^* Rj_*(i'_! L)) & \xrightarrow{4.3.1} & (Rj_*(i'_! L))_{\bar{s}} = R\Gamma(\tilde{Z} - \tilde{Y}, (i'_! L)|_{\tilde{Z} - \tilde{Y}}) \end{array}$$

The vanishing of 4.3.2 therefore follows from b), completing the proof of 4.3.

We prove 4.2. Let  $f : T' \rightarrow T$  be the blowing-up of the maximal ideal of  $T$ , let  $D = f^{-1}(z) (\simeq \mathbb{P}_S^1)$  be the exceptional divisor, and let  $A', B', X'$  be the respective strict transforms of  $A, B, X$ . Then  $T'$  is regular, connected, of dimension 2,  $X'$  is a regular divisor meeting  $D$  transversely at one point  $z'$ , and the general-position hypotheses on  $A, B, X$  mean that  $A'$  is disjoint from  $B' \cup X'$ . Let  $E'$  be the inverse image of  $E$  on  $T' - (A' \cup D \cup X')$ , and let

$$i' : T' - (A' \cup D \cup X') \hookrightarrow T' - (A' \cup D)$$

be the inclusion. We must prove that the restriction arrow

$$R\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow R\Gamma(B' \cup D, i'_!(E')|_{B' - (B' \cup D)}) \quad (*)$$

is zero. But  $B'$  is a disjoint union of strictly local schemes  $B'_j$  ( $1 \leq j \leq n$ ), of dimension 1, such that for every  $j$ ,  $B'_j \cap D$  is reduced to a point  $b_j$ . Applying 4.3 c) to  $(Z = T' - A', X = X', Y =$

$D - (D \cap A'), L = E', C = B'_j$ ), the hypothesis (P) being verified by virtue of the relative purity theorem (SGA 4 XVI 3), since  $(Z, Y)$  is a filtered projective limit of smooth pairs of codimension 1 over  $S$ , with affine étale transition morphisms, one obtains that the restriction arrow

$$\mathrm{R}\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B'_j - b_j, i'_!(E')|_{B'_j - b_j})$$

is zero. Therefore  $(*)$  is zero, which completes the proof of 4.2, hence of 2.5 and 1.2.

### 5.9. Traces and extension of scalars

Let  $A \rightarrow B$  be a morphism of flat  $K$ -algebras. It induces a morphism of  $K$ -algebras  $A^e \rightarrow B^e$ , hence an extension-of-scalars functor

$$- \otimes_{A^e}^{\mathbf{L}} B^e : D(A^e) \longrightarrow D(B^e).$$

For  $E, F$  as in 5.3.5, if  $F \otimes_{A^e}^{\mathbf{L}} B^e \in \mathrm{ob} D^b(B^e)$  and  $\mathrm{RHom}_B(B \otimes_A^{\mathbf{L}} E, F \otimes_{A^e}^{\mathbf{L}} B^e) \in \mathrm{ob} D^b((B^e)^\circ)$ , there is a commutative square in  $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_A(E, F) \otimes_{A^e}^{\mathbf{L}} E & \xrightarrow{(1)} & F \otimes_{A^e}^{\mathbf{L}} A \\ (3) \downarrow & & \downarrow (4) \\ \mathrm{RHom}_B(B \otimes_A^{\mathbf{L}} E, F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_{B^e}^{\mathbf{L}} (B \otimes_A^{\mathbf{L}} E) & \xrightarrow{(2)} & (F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_{B^e}^{\mathbf{L}} B, \end{array} \quad (5.9.1)$$

where the arrows (1) and (2) are defined by 5.3.6, (4) by  $A \rightarrow B$ , and (3) by the canonical arrow  $F \rightarrow F \otimes_{A^e}^{\mathbf{L}} B^e$ . The verification is immediate. Similarly, for  $E, F, G$  as in 5.4.2, if  $F \otimes_{A^e}^{\mathbf{L}} B^e \in \mathrm{ob} D^+(B^e)$  and  $(F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_B^{\mathbf{L}} (B \otimes_A^{\mathbf{L}} G) \in \mathrm{ob} D^+(B)$ , one has a commutative square in  $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_A(E, F) \otimes_A^{\mathbf{L}} G & \xrightarrow{(1)} & \mathrm{RHom}_A(E, F \otimes_A^{\mathbf{L}} G) \\ (3) \downarrow & & \downarrow (4) \\ \mathrm{RHom}_B(B \otimes_A^{\mathbf{L}} E, F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_B^{\mathbf{L}} (B \otimes_A^{\mathbf{L}} G) & \xrightarrow{(2)} & \mathrm{RHom}_B(B \otimes_A^{\mathbf{L}} E, (F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_B^{\mathbf{L}} (B \otimes_A^{\mathbf{L}} G)), \end{array} \quad (5.9.2)$$

where (1) and (2) are defined by 5.4.2, and (3) and (4) by  $F \rightarrow F \otimes_{A^e}^{\mathbf{L}} B^e$ . From this one deduces, for  $E, F$  as in 5.6.1, with  $F \otimes_{A^e}^{\mathbf{L}} B^e \in \mathrm{ob} D^b(B^e)$ , a commutative square in  $D(K)$ , analogous to the preceding ones, in which the vertical arrows are defined by 5.6.1:

$$\begin{array}{ccc} \mathrm{RHom}_A(E, F \otimes_A^{\mathbf{L}} E) & \xrightarrow{\mathrm{Tr}_A} & F \otimes_{A^e}^{\mathbf{L}} A \\ \downarrow & & \downarrow \\ \mathrm{RHom}_B(B \otimes_A^{\mathbf{L}} E, (F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_B^{\mathbf{L}} (B \otimes_A^{\mathbf{L}} E)) & \xrightarrow{\mathrm{Tr}_B} & (F \otimes_{A^e}^{\mathbf{L}} B^e) \otimes_{B^e}^{\mathbf{L}} B. \end{array} \quad (5.9.3)$$

Finally, if one is given  $F' \in \text{ob } D^b(B^e)$  such that  $\text{RHom}_A(E, F') \in \text{ob } D^b((B^\circ))$ , and  $f : F \overset{\text{L}}{\otimes}_{A^e} B^e \rightarrow F'$ , then there is a compatibility slightly more general than 5.9.1, expressed by an analogous commutative square, with  $F'$  in place of  $F \overset{\text{L}}{\otimes}_{A^e} B^e$  in the lower row and with the vertical arrows defined by  $f$ . One has similarly commutative squares analogous to 5.9.2 and 5.9.3 with  $F'$  in place of  $F \overset{\text{L}}{\otimes}_{A^e} B^e$ . Taking in particular  $F = A$  and, for  $f : A \overset{\text{L}}{\otimes}_{A^e} B^e \rightarrow B$ , the arrow defined by  $A \rightarrow B$ , one obtains a commutative square

$$\begin{array}{ccc} \text{RHom}_A(E, E) & \xrightarrow{\text{Tr}_A} & A \overset{\text{L}}{\otimes}_{A^e} A \\ \downarrow & & \downarrow \\ \text{RHom}_B(B \overset{\text{L}}{\otimes}_A E, B \overset{\text{L}}{\otimes}_A E) & \xrightarrow{\text{Tr}_B} & B \overset{\text{L}}{\otimes}_{B^e} B, \end{array} \quad (5.9.4)$$

where the left vertical arrow, respectively the right vertical arrow, is given by extension of scalars, respectively is the canonical arrow deduced from  $A \rightarrow B$ .

### 5.10. Traces and restriction of scalars

Let  $A \rightarrow B$  be a morphism of flat  $K$ -algebras such that  $B$ , as a left  $A$ -module, is flat and of finite presentation, i.e. locally a direct factor of a free  $A$ -module of finite type. This is the case for example if  $A \rightarrow B$  is the morphism  $K[H] \rightarrow K[G]$  defined by an injective morphism of discrete groups  $H \hookrightarrow G$  of finite index. The scalar-restriction homomorphism

$$\text{Hom}_B(B, B) \longrightarrow \text{Hom}_A(B, B),$$

where  $B$  is considered as a left module, identifies, by the canonical isomorphisms

$$B \xrightarrow{\sim} \text{Hom}_B(B, B), \quad \text{Hom}_A(B, A) \otimes_A B \xrightarrow{\sim} \text{Hom}_A(B, B),$$

with an arrow of  $B$ -bimodules

$$B \longrightarrow \text{Hom}_A(B, A) \otimes_A B. \quad (5.10.1)$$

The composite of 5.10.1 with the evaluation arrow  $\text{Hom}_A(B, A) \otimes_A B \rightarrow A_{\natural}$  (5.3.3) is the homomorphism  $f \mapsto \text{Tr}_A(f)$  ( $= \text{Tr}_A(d_f)$ , where  $d_f$  is right multiplication by  $f$ ); hence, by the formula  $\text{Tr}_A(f_g) = \text{Tr}_A(gf)$  (5.8.5), it factors through  $B_{\natural}$  as an arrow again denoted  $\text{Tr}_{B/A} : B_{\natural} \rightarrow A_{\natural}$ . Thus there is a commutative square

$$\begin{array}{ccc} B & \xrightarrow{\text{Tr}_B} & B_{\natural} \\ \text{5.10.1} \downarrow & & \downarrow \text{Tr}_{B/A} \\ \text{Hom}_A(B, A) \otimes_A B & \xrightarrow{\text{ev}} & A_{\natural}. \end{array} \quad (5.10.2)$$

In the situation  $({}_B E, {}^B P_A)$ , one has a homomorphism of right  $B$ -modules

$$\text{Hom}_B(E, P) \otimes_A B \longrightarrow \text{Hom}_B(E, P \otimes_A B), \quad u \otimes b \longmapsto (x \longmapsto \nu(x) \otimes b), \quad (5.10.3)$$

which is an isomorphism, as is checked immediately: it is enough to see that the underlying  $K$ -linear homomorphism is an isomorphism, and this follows from the fact that  $B$  is locally a direct factor of a free  $A$ -module of finite type. This isomorphism derives to an isomorphism of  $D(B)$

$$\mathrm{RHom}_B(E, P) \otimes_A^{\mathrm{L}} B \xrightarrow{\sim} \mathrm{RHom}_B(E, P \otimes_A^{\mathrm{L}} B), \quad (5.10.4)$$

for  $E \in \mathrm{ob} D^-(B)$ ,  $P \in \mathrm{ob} D^+(B \otimes_K A^\circ)$ : resolve  $E$  by modules of the form  $B \otimes_A L$ , where  $L$  is flat, and  $P$  by injective  $(B \otimes_K A^\circ)$ -modules, which are necessarily injective as  $B$ -modules, since  $A$  is flat over  $K$ ; this allows one to remove the  $R$  on the left, and the resolution chosen for  $E$  also allows one to remove it on the right.

For  $E \in \mathrm{ob} D^-(B)$ ,  $M \in \mathrm{ob} D^+(K)$ ,  $F \in \mathrm{ob} D^-(B)$ , define an arrow in  $D(K)$

$$\mathrm{RHom}_B(E, B \otimes_K^{\mathrm{L}} M) \otimes_B^{\mathrm{L}} F \longrightarrow \mathrm{RHom}_A(E, A \otimes_K^{\mathrm{L}} M) \otimes_A^{\mathrm{L}} F \quad (5.10.5)$$

as the composite of the arrows

$$\begin{aligned} & \mathrm{RHom}_B(E, B \otimes_K^{\mathrm{L}} M) \otimes_B^{\mathrm{L}} F \xrightarrow{(1)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A) \otimes_A^{\mathrm{L}} (B \otimes_K^{\mathrm{L}} M)) \otimes_B^{\mathrm{L}} F \\ & \xrightarrow{(2)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A) \otimes_K^{\mathrm{L}} M) \otimes_A^{\mathrm{L}} B \otimes_B^{\mathrm{L}} F \\ & \xrightarrow{(3)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A \otimes_K^{\mathrm{L}} M)) \otimes_A^{\mathrm{L}} F \\ & \xrightarrow{(4)} \mathrm{RHom}_A(E, A \otimes_K^{\mathrm{L}} M) \otimes_A^{\mathrm{L}} F, \end{aligned}$$

where (1) is defined by 5.10.1, (2) by the inverse isomorphism of 5.10.4 with  $P = \mathrm{Hom}_A(B, A) \otimes_K^{\mathrm{L}} M$ , (3) by the canonical isomorphism

$$\mathrm{Hom}_A(B, A) \otimes_K^{\mathrm{L}} M \xrightarrow{\sim} \mathrm{Hom}_A(B, A \otimes_K^{\mathrm{L}} M)$$

(5.4.2), and (4) by the adjunction isomorphism 5.2.2.

**Proposition 5.10.6.** Suppose that  $M \otimes_K^{\mathrm{L}} F \in \mathrm{ob} D^+(B)$ . Then there is a commutative square in  $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_B(E, B \otimes_K^{\mathrm{L}} M) \otimes_B^{\mathrm{L}} F & \xrightarrow{5.4.2} & \mathrm{RHom}_B(E, M \otimes_K^{\mathrm{L}} F) \\ \downarrow 5.10.5 & & \downarrow \mathrm{res} \\ \mathrm{RHom}_A(E, A \otimes_K^{\mathrm{L}} M) \otimes_A^{\mathrm{L}} F & \xrightarrow{5.4.2} & \mathrm{RHom}_A(E, M \otimes_K^{\mathrm{L}} F), \end{array} \quad (5.10.7)$$

where the right vertical arrow is restriction of scalars.

One has to verify that there is a commutative square

$$\begin{array}{ccc}
E \overset{\mathbb{L}}{\otimes}_K \mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_B F & \longrightarrow & M \overset{\mathbb{L}}{\otimes}_K F \\
\downarrow & & \downarrow \text{res} \\
E \overset{\mathbb{L}}{\otimes}_K \mathrm{RHom}_A(E, A \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_A F & \longrightarrow & M \overset{\mathbb{L}}{\otimes}_K F,
\end{array}$$

where the horizontal arrows are given by the evaluations (5.3.6). This square is deduced, by applying  $-\overset{\mathbb{L}}{\otimes}_B F$ , from the analogous square with  $F = B$ , so that one may suppose  $F = B$ . The commutativity of 5.10.7 in this case is equivalent to that of the triangle

$$\begin{array}{ccc}
\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) & & \\
(1) \downarrow & \searrow (2) & \\
\mathrm{RHom}_B(E, \mathrm{Hom}_A(B, B \overset{\mathbb{L}}{\otimes}_K M)) & \xrightarrow{(3)} & \mathrm{RHom}_A(E, B \overset{\mathbb{L}}{\otimes}_K M),
\end{array}$$

where (1) is defined by 5.10.1, (2) is restriction of scalars, and (3) is the adjunction isomorphism.

But the arrow  $B \overset{\mathbb{L}}{\otimes}_K M \rightarrow \mathrm{Hom}_A(B, B \overset{\mathbb{L}}{\otimes}_K M)$  given by 5.10.1 is nothing other than the adjunction arrow corresponding to the functors restriction of scalars and  $\mathrm{Hom}_A(B, -)$ ; the commutativity follows at once.

Let  $E \in \mathrm{ob} D^-(B)$ ,  $M \in \mathrm{ob} D^b(K)$  be such that  $\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \in \mathrm{ob} D^b(B^\circ)$ . Then, by 5.3.5, there is an evaluation arrow

$$\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_{B^\circ} E \longrightarrow M \overset{\mathbb{L}}{\otimes}_K (B \overset{\mathbb{L}}{\otimes}_{B^\circ} B),$$

which, by composition with the canonical arrow  $B \overset{\mathbb{L}}{\otimes}_{B^\circ} B \rightarrow B_{\natural}$ , gives an arrow

$$\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_B E \longrightarrow M \overset{\mathbb{L}}{\otimes}_K B_{\natural}. \quad (5.10.8)$$

**Proposition 5.10.9.** Suppose that  $A$  is of finite presentation and flat as a  $K$ -module, and that  $A_{\natural}$ ,  $B_{\natural}$  are flat over  $K$ . Let  $E \in \mathrm{ob} D^-(B)$ ,  $M \in \mathrm{ob} D^b(K)$  be such that  $\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \in \mathrm{ob} D^b(B^\circ)$ . Then  $\mathrm{RHom}_A(E, A \overset{\mathbb{L}}{\otimes}_K M) \in \mathrm{ob} D^b(A^\circ)$ , and there is a commutative square

$$\begin{array}{ccc}
\mathrm{RHom}_B(E, B \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_B E & \xrightarrow{5.10.8} & M \overset{\mathbb{L}}{\otimes}_K B_{\natural} \\
5.10.5 \downarrow & & \downarrow M \overset{\mathbb{L}}{\otimes} \mathrm{Tr}_{B/A} \\
\mathrm{RHom}_A(E, A \overset{\mathbb{L}}{\otimes}_K M) \overset{\mathbb{L}}{\otimes}_A E & \xrightarrow{5.10.8} & M \overset{\mathbb{L}}{\otimes}_K A_{\natural}.
\end{array} \quad (5.10.10)$$

The additional hypotheses on  $A$  and  $B$  (verified, for example, if  $A$  and  $B$  are  $K$ -algebras of finite groups) are no doubt useless, and one should have a commutative square analogous to 5.10.10 with  $A_{\natural}$  (resp.  $B_{\natural}$ ) replaced by  $A \overset{\mathbb{L}}{\otimes}_{A^\circ} A$  (resp.  $B \overset{\mathbb{L}}{\otimes}_{B^\circ} B$ ).

The first assertion of 5.10.9 is trivial. We prove the second. We may suppose that  $M$  has injective components and that  $E$  has components of the form  $B \otimes_K L$ , with  $L$  flat. Then

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E = \mathrm{Hom}_B^\bullet(E, B \otimes_K M) \otimes_B E,$$

$$\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E = \mathrm{Hom}_A^\bullet(E, A \otimes_K M) \otimes_A E,$$

and the arrows in 5.10.10 are defined at the level of complexes. Thus we are reduced to verifying that, if  $L, M$  are  $K$ -modules and  $E$  a left  $B$ -module, there is a commutative square

$$\begin{array}{ccc} \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M) \otimes_B (B \otimes_K L) & \xrightarrow{5.3.3} & M \otimes_K B_{\natural} \\ \downarrow (1) & & \downarrow M \otimes_K \mathrm{Tr}_{B/A} \\ \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M) \otimes_A (B \otimes_K L) & \xrightarrow{5.3.3} & M \otimes_K A_{\natural}, \end{array}$$

where the arrow (1) is defined by sending  $B \otimes_K M$  into  $\mathrm{Hom}_A(B, A) \otimes_A B \otimes_K M$  by 5.10.1 and composing with an isomorphism of the type 5.10.3 and an adjunction isomorphism. Since  $A$  is flat and of finite presentation as a  $K$ -module, the tensor-product arrows

$$B \otimes_K \mathrm{Hom}_K(L, M) = \mathrm{Hom}_B(B, B) \otimes_K \mathrm{Hom}_K(L, M) \rightarrow \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M),$$

$$\mathrm{Hom}_A(B, A) \otimes_K \mathrm{Hom}_K(L, M) \rightarrow \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M)$$

are isomorphisms, by which (1) identifies with the tensor product over  $K$  of 5.10.1 by  $\mathrm{Hom}_K(L, M) \otimes_K L$ . On the other hand, the upper horizontal arrow, respectively the lower horizontal arrow, of the square identifies under these isomorphisms with the tensor product over  $K$  of  $\mathrm{Tr}_B : B \rightarrow B_{\natural}$  (resp.  $\mathrm{ev} : \mathrm{Hom}_A(B, A) \otimes_A B \rightarrow A_{\natural}$ ) by the evaluation arrow  $\mathrm{Hom}_K(L, M) \otimes_K L \rightarrow M$ . The commutativity therefore follows from that of 5.10.2, completing the proof of 5.10.9.

**Corollary 5.10.11.** Under the hypotheses of 5.10.9, for  $E \in \mathrm{ob} D(B)_{\mathrm{parf}}$ ,  $M \in \mathrm{ob} D^b(K)$ , there is a commutative square in  $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow{\mathrm{Tr}_B} & M \overset{\mathrm{L}}{\otimes}_K B_{\natural} \\ \downarrow & & \downarrow M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A} \\ \mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow{\mathrm{Tr}_A} & M \overset{\mathrm{L}}{\otimes}_K A_{\natural}, \end{array} \quad (5.10.12)$$

where the left vertical arrow is restriction of scalars.

It is enough indeed to conjugate 5.10.6 and 5.10.9.

### 5.11. Application to finite groups: divisibility properties

Let  $G$  be a finite group, with neutral element denoted  $e$ , and put  $K[G] = B$ . The  $K$ -module  $B_{\natural}$  is free with basis the set  $G_{\natural}$  of conjugacy classes of  $G$ ; every section  $s$  of  $B_{\natural}$  is written uniquely

$$s = \sum_{x \in G_{\natural}} s(x)x.$$

**Proposition 5.11.1.** Cf. SGA 4<sup>1/2</sup>, Rapport 4.5. Let  $E \in \text{ob } D(B)_{\text{parf}}$ ,  $M \in \text{ob } D^b(K)$ . For  $u \in \text{Hom}_B(E, M \overset{\text{L}}{\otimes}_K E)$ , one has, in  $H^0(T, M)$ ,

$$\text{Tr}_K(u) = |G| \text{Tr}_B(u)(e). \quad (5.11.1)$$

It is enough to apply 5.10.11 with  $A = K$ , noting that  $\text{Tr}_{B/K} : B_{\natural} \rightarrow K$  is given by

$$\text{Tr}_{B/K}(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases} \quad (5.11.2)$$

**Proposition 5.11.3.** Under the hypotheses of 5.11.1, let  $g \in G$ , let  $Z_g$  be the centralizer of  $g$  in  $G$ , and put  $A = K[Z_g]$ . Let  $u \in \text{Hom}_B(E, M \overset{\text{L}}{\otimes}_K E)$ , and let  $gu \in \text{Hom}_A(E, M \overset{\text{L}}{\otimes}_K E)$  be the composite of the image of  $u$  in  $\text{Hom}_A(E, M \overset{\text{L}}{\otimes}_K E)$  with left multiplication by  $g$ . Then, in  $H^0(T, M)$ ,

$$\text{Tr}_A(gu)(e) = \text{Tr}_B(u)(g^{-1}). \quad (5.11.4)$$

We shall show more generally that there is a commutative square

$$\begin{array}{ccc} \text{RHom}_B(E, M \overset{\text{L}}{\otimes}_K E) & \xrightarrow{\text{Tr}_B} & M \overset{\text{L}}{\otimes}_K B_{\natural} \xrightarrow{x \mapsto x(g^{-1})} M \\ \downarrow & & \nearrow x \mapsto x(e) \\ \text{RHom}_A(E, M \overset{\text{L}}{\otimes}_K E) & \xrightarrow{\text{Tr}_A} & M \overset{\text{L}}{\otimes}_K A_{\natural} \end{array}$$

where the left vertical arrow is the composite of restriction of scalars and left multiplication by  $g$ . This square is the boundary of the diagram obtained by inserting  $\text{Tr}_{B/A}$  and the left multiplication  $s_g$  by  $g$ ; the central square commutes by 5.10.11, and the lower one by the definition of  $\text{Tr}_A$ . It remains only to prove that, for  $h \in G$ ,

$$\text{Tr}_A(x \mapsto gxh)(e) = \text{the value at } g^{-1} \text{ of the image of } h \text{ in } B_{\natural},$$

i.e.

$$\text{Tr}_A(x \mapsto gxh)(e) = \begin{cases} 1, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases} \quad (*)$$

For this one may replace  $T$  by the punctual topos and  $K$  by  $\mathbb{Z}$ , and it is then enough to prove (\*) after multiplication by  $|Z_g|$ . In view of 5.11.1, we are therefore reduced to showing that

$$\text{Tr}_{\mathbb{Z}}(x \mapsto gxh) = \begin{cases} |Z_g|, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases}$$

But this follows from the fact that the left hand side is the number of fixed points of the map  $x \mapsto gxh$  from  $G$  to  $G$ , whence the commutativity of 5.11.4 and the conclusion of 5.11.3.

**Corollary 5.11.5.** Under the hypotheses of 5.11.3, one has

$$\text{Tr}_K(gu) = |Z_g| \text{Tr}_B(u)(g^{-1}) = |Z_g| \text{Tr}_A(gu)(e). \quad (5.11.5)$$

### 5.12. Traces and external tensor products

Let, for  $i = 1, 2$ ,  $A_i$  be a flat  $K$ -algebra, and put  $A = A_1 \otimes_K A_2$ . Let  $E_i \in \text{ob } D^-(A_i)$ ,  $F_i \in \text{ob } D^b(A_i^e)$  be such that  $\text{RHom}_{A_i}(E_i, F_i) \in \text{ob } D^b(A_i^\circ)$ , let  $F = F_1 \overset{\text{L}}{\otimes}_K F_2 \in \text{ob } D^b(A^e)$ , and suppose  $\text{RHom}_A(E, F) \in \text{ob } D^b(A^\circ)$ , where  $E = E_1 \overset{\text{L}}{\otimes}_K E_2$ . Then there is a commutative square

$$\begin{array}{ccc} (\text{RHom}_{A_1}(E_1, F_1) \overset{\text{L}}{\otimes}_{A_1} E_1) \overset{\text{L}}{\otimes}_K (\text{RHom}_{A_2}(E_2, F_2) \overset{\text{L}}{\otimes}_{A_2} E_2) & \xrightarrow{(1)} & (F_1 \overset{\text{L}}{\otimes}_{A_1^e} A_1) \overset{\text{L}}{\otimes}_K (F_2 \overset{\text{L}}{\otimes}_{A_2^e} A_2) \\ \downarrow (2) & & \downarrow (3) \\ \text{RHom}_A(E, F) \overset{\text{L}}{\otimes}_A E & \xrightarrow{\hspace{10em}} & F \overset{\text{L}}{\otimes}_{A^e} A, \end{array} \quad (5.12.1)$$

where (1) is the tensor product of the arrows 5.3.5, (2) is defined by the tensor-product arrow

$$\text{RHom}_{A_1}(E_1, F_1) \overset{\text{L}}{\otimes}_K \text{RHom}_{A_2}(E_2, F_2) \longrightarrow \text{RHom}_A(E, F), \quad (5.12.2)$$

and (3) is the evident canonical isomorphism. The verification is essentially trivial. One checks in the same way that, for  $E_i \in \text{ob } D^-(A_i)$ ,  $F_i \in \text{ob } D^+(A_i^e)$ ,  $G_i \in \text{ob } D^-(A_i)$ , with  $F_i \overset{\text{L}}{\otimes}_{A_i} G_i \in \text{ob } D^b(A_i)$ ,  $\text{RHom}_{A_i}(E_i, F_i) \in \text{ob } D^b(A_i^\circ)$ ,  $\text{RHom}_{A_i}(E_i, F_i \overset{\text{L}}{\otimes}_{A_i} G_i) \in \text{ob } D^b(K)$ ,  $F \overset{\text{L}}{\otimes}_A G \in \text{ob } D^b(A)$ , where  $G = G_1 \overset{\text{L}}{\otimes}_K G_2$ , there is a commutative square

$$\begin{array}{ccc} (\text{RHom}_{A_1}(E_1, F_1) \overset{\text{L}}{\otimes}_{A_1} G_1) \overset{\text{L}}{\otimes}_K (\text{RHom}_{A_2}(E_2, F_2) \overset{\text{L}}{\otimes}_{A_2} G_2) & \xrightarrow{(1)} & \text{RHom}_{A_1}(E_1, F_1 \overset{\text{L}}{\otimes}_{A_1} G_1) \overset{\text{L}}{\otimes}_K \text{RHom}_{A_2}(E_2, F_2 \overset{\text{L}}{\otimes}_{A_2} G_2) \\ \downarrow (2) & & \downarrow (3) \\ \text{RHom}_A(E, F) \overset{\text{L}}{\otimes}_A G & \xrightarrow{\hspace{10em} 5.4.2 \hspace{10em}} & \text{RHom}_A(E, F \overset{\text{L}}{\otimes}_A G), \end{array} \quad (5.12.3)$$

where (2) and (3) are tensor-product arrows of the type 5.12.2, and (1) is the tensor product of the arrows 5.4.2. From the commutativity of the squares 5.12.1 and 5.12.3 it follows that, for  $E_i \in \text{ob } D(A_i)_{\text{parf}}$ ,  $F_i \in \text{ob } D^b(A_i^e)$  such that  $F = F_1 \overset{\text{L}}{\otimes}_K F_2 \in \text{ob } D^b(A^e)$ , there is a commutative square

$$\begin{array}{ccc} \text{RHom}_{A_1}(E_1, F_1 \overset{\text{L}}{\otimes}_{A_1} E_1) \overset{\text{L}}{\otimes}_K \text{RHom}_{A_2}(E_2, F_2 \overset{\text{L}}{\otimes}_{A_2} E_2) & \xrightarrow{(1)} & (F_1 \overset{\text{L}}{\otimes}_{A_1^e} A_1) \overset{\text{L}}{\otimes}_K (F_2 \overset{\text{L}}{\otimes}_{A_2^e} A_2) \\ \downarrow (2) & & \downarrow (3) \\ \text{RHom}_A(E, F \overset{\text{L}}{\otimes}_A E) & \xrightarrow{\hspace{10em} 5.6.1 \hspace{10em}} & F \overset{\text{L}}{\otimes}_{A^e} A, \end{array} \quad (5.12.4)$$

where (1) is the tensor product of the arrows  $\text{Tr}_{A_i}$  (5.6.1), (2) is defined by 5.12.2, and (3) is the canonical isomorphism appearing in 5.12.1. Consequently, for  $u_i \in \text{Hom}_{A_i}(E_i, F_i \overset{\text{L}}{\otimes}_{A_i} E_i)$ , one has

$$\text{Tr}_A(u_1 \overset{\text{L}}{\otimes}_K u_2) = \text{Tr}_{A_1}(u_1) \text{Tr}_{A_2}(u_2), \quad (5.12.5)$$

the product on the right being defined by the canonical arrow

$$H^0(T, F_1 \overset{\text{L}}{\otimes}_{A_1^e} A_1) \otimes H^0(T, F_2 \overset{\text{L}}{\otimes}_{A_2^e} A_2) \longrightarrow H^0(T, F \overset{\text{L}}{\otimes}_{A^e} A).$$

### 5.13. Traces of semilinear endomorphisms

Let  $A$  be a  $K$ -algebra and  $\varphi$  an endomorphism of  $A$ . If  $E, F$  are left  $A$ -modules, recall that an additive homomorphism  $u : E \rightarrow F$  is called  $\varphi$ - $A$ -linear if  $u(ax) = \varphi(a)u(x)$  for all  $a \in A, x \in E$ , i.e. if  $u$  is an  $A$ -linear homomorphism from  $E$  to the  $A$ -module  ${}_{(\varphi)}F$  deduced from  $F$  by restriction of scalars along  $\varphi$ . If  $F$  is an  $A$ -bimodule and  $G$  a left  $A$ -module, there is an evident isomorphism

$${}_{(\varphi)}F \otimes_A G \xrightarrow{\sim} {}_{(\varphi)}(F \otimes_A G),$$

which derives trivially to

$${}_{(\varphi)}F \overset{\mathbf{L}}{\otimes}_A G \xrightarrow{\sim} {}_{(\varphi)}(F \overset{\mathbf{L}}{\otimes}_A G).$$

Assuming  $A$  flat over  $K$ , one deduces from 5.4.2 the arrow

$$\mathrm{RHom}_A(E, {}_{(\varphi)}F) \overset{\mathbf{L}}{\otimes}_A G \longrightarrow \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathbf{L}}{\otimes}_A G)) \quad (5.13.1)$$

for  $E \in \mathrm{ob} D^-(A), F \in \mathrm{ob} D^+(A^e), G \in \mathrm{ob} D^-(A)$ , with  $F \overset{\mathbf{L}}{\otimes}_A G \in \mathrm{ob} D^+(A)$ . For  $E \in \mathrm{ob} D(A)_{\mathrm{parf}}$ , 5.13.1 is an isomorphism, which allows one to define, by 5.3.5, a trace arrow

$$\mathrm{Tr}_{A,\varphi} : \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathbf{L}}{\otimes}_A E)) \longrightarrow {}_{(\varphi)}F \overset{\mathbf{L}}{\otimes}_{A^e} A. \quad (5.13.2)$$

In particular, for  $F = A$ , one obtains

$$\mathrm{Tr}_{A,\varphi} : \mathrm{Hom}_A(E, {}_{(\varphi)}E) \longrightarrow H^0(T, A_{\natural,\varphi}), \quad (5.13.3)$$

where  $A_{\natural,\varphi} = {}_{(\varphi)}A \otimes_{A^e} A$  is the quotient  $K$ -module of  $A$  by the submodule locally generated by local sections of the form  $ab - \varphi(b)a$ . In the case where  $A = K[G]$ ,  $G$  is a discrete group, and  $\varphi$  is defined by an endomorphism of  $G$ ,  $A_{\natural,\varphi}$  is the free  $K$ -module with basis  $G_{\natural,\varphi}$ , where  $G_{\natural,\varphi}$  is the quotient of  $G$  by the action  $(g, x) \mapsto \varphi(g)xg^{-1}$ ; the orbits are sometimes called  $\varphi$ -conjugacy classes of  $G$ . The traces 5.13.2 coincide, for  $T$  punctual, with those considered by Grothendieck and discussed in Exposé XII.

In the situation of 5.10, if  $\varphi, \psi$  are respective endomorphisms of  $A, B$ , compatible with  $A \rightarrow B$ , there is a unique arrow

$$\mathrm{Tr}_A : B_{\natural,\psi} \longrightarrow A_{\natural,\varphi}$$

making the square, analogous to 5.10.2,

$$\begin{array}{ccc} {}_{(\psi)}B = \mathrm{Hom}_B(B, {}_{(\psi)}B) & \xrightarrow{\mathrm{Tr}_{B,\psi}} & B_{\natural,\psi} \\ \downarrow & & \downarrow \mathrm{Tr}_A \\ \mathrm{Hom}_A(B, {}_{(\varphi)}A) \otimes_A B & \xrightarrow{\mathrm{ev}} & A_{\natural,\varphi} \end{array} \quad (5.13.4)$$

commute, where  $\mathrm{ev}$  is the evaluation arrow, and the left vertical arrow is restriction of scalars. One defines an arrow analogous to 5.10.5, with  $A \otimes_K M$  (resp.  $B \otimes_K M$ ) replaced by  ${}_{(\varphi)}A \otimes_K M$  (resp.  ${}_{(\psi)}B \otimes_K M$ ), and there are compatibilities analogous to 5.10.6, 5.10.9, 5.10.11, which we leave to the reader to state.

Let  $G$  be a finite group and  $\varphi$  an endomorphism of  $G$ . For  $g \in G$ , let  $Z_{g,\varphi}$  be the  $\varphi$ -centralizer of  $g$  in  $G$ , i.e. the set of  $x \in G$  such that  $\varphi(x)gx^{-1} = g$ . Put  $A = K[G]$ , and let  $E \in \mathrm{ob} D(A)_{\mathrm{parf}}$ ,

$M \in \text{ob } D^b(K)$ . The preceding compatibilities easily imply that, for  $u \in \text{Hom}_A(E, {}_{(\varphi)}(M \overset{\text{L}}{\otimes}_K E))$  and  $g \in G$ ,

$$\text{Tr}_K(gu) = |Z_{g^{-1}, \varphi}| \text{Tr}_{A, \varphi}(u)(g^{-1}), \quad (5.13.5)$$

the details of the verification being left to the reader.

## 6. Equivariant Correspondences and Divisibility of Local Terms

### 6.1. Notation

In this number,  $S$  denotes the spectrum of a field of characteristic  $p$ , and  $\Lambda$  a noetherian commutative ring annihilated by an integer not divisible by  $p$ . The  $S$ -schemes considered will be assumed separated and of finite type. The  $\Lambda$ -algebras considered will be associative, unital, and of finite type, but not necessarily commutative. Most often, and in particular for everything concerning “non-commutative Lefschetz–Verdier local terms,” they will be assumed flat; in applications, they will be algebras of finite groups.

If  $X$  is an  $S$ -scheme and  $A$  a  $\Lambda$ -algebra, we write  $D({}_AX)$  (resp.  $D(X_A)$ ) for the derived category of the category of sheaves of left (resp. right)  $A$ -modules on  $X$ . For  $A$  commutative, we shall also write  $D(X, A)$  for  $D({}_AX)$ , and  $D(X) = D(X, \Lambda)$ . We denote by  $D_c(-)$  (resp.  $D_{\text{ctf}}(-)$ ) the full subcategory formed by complexes with constructible cohomology (resp. finite tor-dimension and constructible cohomology). If  $A, B$  are  $\Lambda$ -algebras, we write  $D({}_AX_B)$  for the derived category of sheaves of  $(A, B)$ -bimodules, left over  $A$  and right over  $B$ . Finally, we use the notations  $A^o, A^e, A_{\natural}$  of 5.1.

### 6.2

The formalism of (III 1, 2, 3) extends without difficulty to the “non-commutative” case. We briefly indicate the main points of this generalization.

Let  $X$  be an  $S$ -scheme and let  $A, B$  be  $\Lambda$ -algebras. If  $L \in \text{ob } D_{\text{ctf}}(X_A)$  and  $M \in \text{ob } D({}_AX_B)$ , with  $M$  of finite tor-dimension and constructible cohomology as an object of  $D(X_B)$ , then

$$L \overset{\text{L}}{\otimes}_A M \in \text{ob } D_{\text{ctf}}(X_B)$$

as is immediate. On the other hand, it follows from (SGA 4<sup>1/2</sup> Finiteness 1.6) that, for  $M$  as above and  $L \in \text{ob } D_{\text{ctf}}({}_AX)$ , one has

$$\text{RHom}_A(L, M) \in \text{ob } D_{\text{ctf}}(X_B).$$

It also follows from loc. cit. that, if  $f$  is an  $S$ -morphism,  $D_{\text{ctf}}({}_A-)$  is stable under the operations  $(f^*, f_*, f_!, f^!)$ , the case of  $f^*$  being trivial.

Let, for  $i = 1, 2$ ,  $f_i : X_i \rightarrow Y_i$  be an  $S$ -morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \longrightarrow Y = Y_1 \times_S Y_2,$$

and let  $A$  be a  $\Lambda$ -algebra. For  $L_1 \in \text{ob } D^-(X_{1A})$ ,  $L_2 \in \text{ob } D^-(X_{2A}^o)$ , there is a Künneth arrow in  $D^-(Y)$

$$f_{1!} L_1 \overset{\text{L}}{\otimes}_{S,A} f_{2!} L_2 \longrightarrow f_*(L_1 \overset{\text{L}}{\otimes}_{S,A} L_2), \quad (6.2.1)$$

where

$$L_1 \otimes_{S,A}^L L_2 \stackrel{\text{dfn}}{=} p_1^* L_1 \otimes_A^L p_2^* L_2.$$

As in (III 1.6), one deduces that it is an isomorphism. If  $B, C$  are  $\Lambda$ -algebras, with  $B$  or  $C$  assumed flat over  $\Lambda$  in order to define

$$\otimes_A^L : D^-(X_{BA}) \times D^-(X_{AC}) \longrightarrow D^-(X_{BC}),$$

there is an isomorphism analogous to 6.2.1 in  $D^-(Y_{BC})$ , for  $L_1 \in \text{ob } D^-(X_{B_1A})$ ,  $L_2 \in \text{ob } D^-(X_{A_2B})$ . One checks similarly that, for  $M_1 \in \text{ob } D^-(Y_{1A})$ ,  $M_2 \in \text{ob } D^-(Y_{2A}^o)$ , there is a Künneth isomorphism in  $D^-(X)$

$$f^! M_1 \otimes_{S,A}^L f^! M_2 \longrightarrow f^! (M_1 \otimes_{S,A}^L M_2), \quad (6.2.2)$$

and an analogous isomorphism in  $D^-({}_B X_C)$  for  $M_1 \in \text{ob } D^-(Y_{B_1A})$ ,  $M_2 \in \text{ob } D^-(Y_{A_2C})$ .

Finally, let, for  $i = 1, 2$ ,  $A_i$  be a  $\Lambda$ -algebra,  $A_2$  being flat,  $E_1 \in \text{ob } D_{\text{ctf}}(A_1 X_1)$ ,  $F_2 \in \text{ob } D^+(X_{A_2})$ ,  $F_1 \in \text{ob } D(A_1 X_{1A_2})$ , with  $F_1$  of finite tor-dimension and constructible cohomology as an object of  $D(X_{1A_2})$ . Then

$$\text{RHom}_{A_1}(E_1, F_1) \in \text{ob } D_{\text{ctf}}(X_{1A_2}),$$

as seen above, and there is a canonical arrow in  $D^+(X)$

$$\text{RHom}_{A_1}(E_1, F_1) \otimes_{S,A_2}^L F_2 \longrightarrow \text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \otimes_{S,A_2}^L F_2), \quad (6.2.3)$$

defined by composing the “inverse-image” arrow (III 2.1.1)

$$p_1^* \text{RHom}_{A_1}(E_1, F_1) \otimes_{A_2}^L p_2^* F_2 \longrightarrow \text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \otimes_{A_2}^L p_2^* F_2$$

with the arrow analogous to 5.4.2

$$\text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \otimes_{A_2}^L p_2^* F_2 \longrightarrow \text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \otimes_{A_2}^L p_2^* F_2),$$

adjoint, by 5.2.3, to the arrow

$$p_1^* E_1 \otimes^L \text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \otimes_{A_2}^L p_2^* F_2 \longrightarrow p_1^* F_1 \otimes_{A_2}^L p_2^* F_2$$

defined by the evaluation arrow 5.3.6. One deduces from 6.2.1, as in (III 2.3), that 6.2.3 is an isomorphism. In particular, for  $A_2 = \Lambda$ ,  $F_2 = \Lambda_{X_2}$ , 6.2.3 gives an isomorphism

$$p_1^* \text{RHom}_{A_1}(E_1, F_1) \xrightarrow{\sim} \text{RHom}_{A_1}(p_1^* E_1, p_1^* F_1). \quad (6.2.4)$$

### 6.3

If  $t : T \rightarrow S$  is an  $S$ -scheme, and  $A$  is a  $\Lambda$ -algebra, we put

$$t_A^! = K A_T \quad (\in \text{ob } D^+({}_A T_A)). \quad (6.3.1)$$

For  $A = \Lambda$ , we write  $K_T$  instead of  $K \Lambda_T$ . It follows from 6.2.2 that there is a canonical isomorphism in  $D^+({}_A T_A)$

$$K_T \otimes_{\Lambda}^L A \xrightarrow{\sim} K A_T. \quad (6.3.2)$$

We denote by  $D_A$  the functor

$$\mathrm{RHom}_A(-, KA_T).$$

It sends  $D_{\mathrm{ctf}}({}_AT)$  into  $D_{\mathrm{ctf}}(T_A)$  and conversely, and the proof of (SGA 4<sup>1/2</sup> Finiteness Theorem 4) shows that, for  $L \in \mathrm{ob} D_{\mathrm{ctf}}({}_AT)$ , the canonical arrow

$$L \longrightarrow D_AD_AL$$

is an isomorphism. For  $A = \Lambda$ , we write  $D$  instead of  $D_A$ .

## 6.4

Let  $A$  be a flat  $\Lambda$ -algebra, and let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme,  $X = X_1 \times_S X_2$ ,  $p_i : X \rightarrow X_i$  the projections, and  $L_i \in \mathrm{ob} D_{\mathrm{ctf}}({}_AX_i)$ . By 6.2.3 applied to  $A_1 = A_2 = A$ ,  $E_1 = L_1$ ,  $F_2 = L_2$ ,  $E_2 = KA_{X_1}$ , there is an isomorphism

$$D_AL_1 \otimes_{S,A}^L L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^*L_1, KA_{X_1} \otimes_{S,A}^L L_2),$$

whence, by composing with the Künneth isomorphism 6.2.2

$$KA_{X_1} \otimes_{S,A}^L L_2 \xrightarrow{\sim} p_2^!L_2,$$

an isomorphism

$$D_AL_1 \otimes_{S,A}^L L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^*L_1, p_2^!L_2), \quad (6.4.1)$$

which generalizes (III 3.1.1).

Let  $c : C \rightarrow X$  be an  $S$ -morphism, and put  $c_i = p_i c$ . By the induction formula (SGA 4 XVIII 3.1.12.2), there is a canonical isomorphism, analogous to (III 3.2.1),

$$c^! \mathrm{RHom}_A(p_1^*L_1, p_2^!L_2) \xrightarrow{\sim} \mathrm{RHom}_A(c_1^*L_1, c_2^!L_2), \quad (6.4.2)$$

and hence, by composing with 6.4.1, an isomorphism

$$H^0(X, c_* c^!(D_AL_1 \otimes_{S,A}^L L_2)) \xrightarrow{\sim} \mathrm{Hom}_A(c_1^*L_1, c_2^!L_2). \quad (6.4.3)$$

The elements of  $\mathrm{Hom}_A(c_1^*L_1, c_2^!L_2)$  will be called cohomological correspondences from  $L_1$  to  $L_2$  with support in  $c$ . The terminology differs slightly from that of loc. cit.; the two terminologies coincide when  $c$  is proper, which is the case in practice.

Given  $S$ -morphisms  $f_i : X_i \rightarrow Y_i$  ( $i = 1, 2$ ) and a commutative square (III 3.7.1) with  $c$  proper, one defines, as in loc. cit., direct-image arrows analogous to (III 3.7.5), (III 3.7.6), for  $L_i \in \mathrm{ob} D_{\mathrm{ctf}}({}_AX_i)$ .

## 6.5

Let  $A$  be a flat  $\Lambda$ -algebra,  $X$  an  $S$ -scheme,  $X^2 = X \times_S X$ ,  $p_1, p_2 : X^2 \rightarrow X$  the canonical projections,  $\delta : X \rightarrow X^2$  the diagonal, and

$$c : C \longrightarrow X^2$$

an  $S$ -morphism. Put  $c_i = p_i c$ , and let  $X^c = C \times_{X^2} X$  be the scheme of fixed points of  $c$ , giving a cartesian square

$$\begin{array}{ccc} X^c & \xrightarrow{i^c} & X \\ \delta^c \downarrow & & \downarrow \delta \\ C & \xrightarrow{c} & X^2. \end{array} \quad (6.5.1)$$

Put

$$KA_{X, L_{\mathfrak{h}}} = K_X \overset{\mathbf{L}}{\otimes}_{\Lambda} (A \overset{\mathbf{L}}{\otimes}_{A^e} A), \quad KA_{X, \mathfrak{h}} = K_X \overset{\mathbf{L}}{\otimes}_{\Lambda} A_{\mathfrak{h}}. \quad (6.5.2)$$

Let  $L \in \text{ob } D_{\text{ctf}}({}_A X)$ . We define a “local trace” homomorphism

$$\text{Tr}_A : \delta^* c_* \text{RHom}_A(p_1^* L, p_2^! L) \longrightarrow KA_{X, L_{\mathfrak{h}}}, \quad (6.5.3)$$

by composing the isomorphism

$$\delta^* \text{RHom}_A(p_1^* L, p_2^! L) \xrightarrow{\sim} \delta^* (D_A L \overset{\mathbf{L}}{\otimes}_{S, A} L) \quad (6.5.4)$$

deduced from the inverse of 6.4.1 by applying  $\delta^*$ , the trivial isomorphism

$$\delta^* (D_A L \overset{\mathbf{L}}{\otimes}_{S, A} L) = D_A L \overset{\mathbf{L}}{\otimes}_A L, \quad (6.5.5)$$

and the evaluation arrow 5.3.5

$$D_A L \overset{\mathbf{L}}{\otimes}_A L \longrightarrow KA_{X, L_{\mathfrak{h}}}. \quad (6.5.6)$$

It follows at once from the definition that, for  $X = S$ , 6.5.3 coincides with the arrow

$$\text{Tr}_A : \text{RHom}_A(L, L) \longrightarrow A \overset{\mathbf{L}}{\otimes}_{A^e} A$$

of 5.6.1. Composing the arrow deduced from 6.5.3 by applying  $i^c c^!$  with the arrow

$$\delta^* c_*^c \text{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i_*^c i^{c!} \delta^* \text{RHom}_A(p_1^* L, p_2^! L) \quad (6.5.7)$$

deduced from 6.4.2 and the base-change arrow

$$\delta^* c_* c^! \longrightarrow i_*^c i^{c!} \delta^*,$$

one obtains an arrow

$$\text{Tr}_A : \delta^* c_* \text{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i_*^c i^{c!} KA_{X, L_{\mathfrak{h}}} = i_*^c KA_{X^c, L_{\mathfrak{h}}}, \quad (6.5.8)$$

and hence the arrow

$$\text{Tr}_A : \text{Hom}_A(c_1^* L, c_2^! L) \longrightarrow H^0(X^c, KA_{X^c, L_{\mathfrak{h}}}). \quad (6.5.9)$$

We again write  $\text{Tr}_A$  for the composites of 6.5.3, 6.5.8, 6.5.9 with the arrows deduced from the canonical arrow

$$KA_{X, L_{\mathfrak{h}}} \longrightarrow KA_{X, \mathfrak{h}}.$$

One checks easily that, for  $A = \Lambda$ , one has, for  $u \in \text{Hom}(c_1^* L, c_2^! L)$ ,

$$\text{Tr}(u) = \langle u, 1 \rangle, \quad (6.5.10)$$

with the notation of (III 4.2.5), where 1 denotes the diagonal correspondence. More precisely, the arrow

$$c_* \text{RHom}(c_1^* L, c_2^! L) \longrightarrow \delta_* i_*^c K_{X^c}$$

adjoint to 6.5.8 is the one defined by (III 4.2.4') with  $X_1 = X_2$ ,  $L_1 = L_2$ ,  $d = \delta$ , and the diagonal correspondence  $1 \in \text{Hom}(L, L)$ .

## 6.6

Let  $A$  be a flat  $\Lambda$ -algebra, and let, for  $i = 1, 2$ ,  $X_i$  be an  $S$ -scheme,  $X = X_1 \times_S X_2$ , and  $L_i \in \text{ob } D_{\text{ctf}}(AX_i)$ . We define a pairing

$$\text{RHom}_A(p_1^*L_1, p_2^!L_2) \overset{\text{L}}{\otimes}_{\Lambda} \text{RHom}_A(p_2^*L_2, p_1^!L_1) \longrightarrow KA_{X, L_{\natural}} \quad (6.6.1)$$

as follows. One identifies the left hand side with

$$\text{RHom}_A(p_1^*L_1, p_1^*KA_{X_1}) \overset{\text{L}}{\otimes} p_2^*L_2 \overset{\text{L}}{\otimes} \text{RHom}_A(p_2^*L_2, p_2^*KA_{X_2}) \overset{\text{L}}{\otimes} p_1^*L_1$$

by the isomorphisms 6.4.1, then with

$$A \overset{\text{L}}{\otimes}_{A^e} \text{RHom}_A(\ ) \overset{\text{L}}{\otimes}_{\Lambda} p_2^*L_2 \overset{\text{L}}{\otimes}_{\Lambda} \text{RHom}_A(\ ) \overset{\text{L}}{\otimes}_{\Lambda} p_1^*L_1 \overset{\text{L}}{\otimes}_{A^e} A$$

by 5.3.1. This object is sent to

$$A \overset{\text{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\text{L}}{\otimes}_{\Lambda} KA_{X_2}) \overset{\text{L}}{\otimes}_{A^e} A = R$$

by the tensor product of the evaluation arrows 5.3.6. Finally  $R$  is identified with

$$A \overset{\text{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\text{L}}{\otimes}_{\Lambda} KA_{X_2})$$

by 5.8.2, and then with  $KA_{X, L_{\natural}}$  by 6.5.2 and (III 1.7.6).

If  $c : C \rightarrow X$ ,  $d : D \rightarrow X$  are correspondences, and

$$e = c \times_X d : E \rightarrow X,$$

one deduces from 6.6.1, as in (III 4.2), a pairing

$$c_* \text{RHom}_A(c_1^*L_1, c_2^!L_2) \overset{\text{L}}{\otimes}_{\Lambda} d_* \text{RHom}_A(d_2^*L_2, d_1^!L_1) \longrightarrow e_* KA_{E, L_{\natural}}, \quad (6.6.2)$$

and hence a cup-product

$$\langle \ , \ \rangle_A : \text{Hom}_A(c_1^*L_1, c_2^!L_2) \otimes \text{Hom}_A(d_2^*L_2, d_1^!L_1) \longrightarrow H^0(E, KA_{E, L_{\natural}}). \quad (6.6.3)$$

It is not difficult to extend to the non-commutative case the notion of composition of correspondences studied in (III 5.2), and one checks that, for

$$u \in \text{Hom}_A(c_1^*L_1, c_2^!L_2), \quad v \in \text{Hom}_A(d_2^*L_2, d_1^!L_1),$$

one has the formula analogous to (III 5.2.9)

$$\langle u, v \rangle_A = \text{Tr}_A(\bar{v}u) = \text{Tr}_A(u\bar{v}). \quad (6.6.4)$$

The Lefschetz formula (III 4.4) generalizes without difficulty in this context. Given proper  $S$ -morphisms  $f_i : X_i \rightarrow Y_i$  ( $i = 1, 2$ ) and a commutative cube (III 4.4.0), for  $L_i \in \text{ob } D_{\text{ctf}}(AX_i)$  one has a commutative square analogous to (III 4.4.1), with  $K_C$  (resp.  $K_D$ ) replaced by  $KA_{C, L_{\natural}}$  (resp.  $KA_{D, L_{\natural}}$ ), and a formula analogous to (III 4.5.1), in  $H^0(D, KA_{D, L_{\natural}})$ :

$$\langle f_*u, f_*v \rangle_A = g_*\langle u, v \rangle_A. \quad (6.6.5)$$

The arrow

$$f_*c_*KA_{C, L_{\natural}} \longrightarrow d_*KA_{D, L_{\natural}}$$

is deduced from arrow (4) of (III 4.4.1) by tensoring with

$$A \overset{\text{L}}{\otimes}_{A^e} A.$$

## 6.7

Let  $X$  and  $c$  be as in 6.5, and let  $A \rightarrow B$  be a morphism of flat  $\Lambda$ -algebras. With respect to extension and restriction of scalars, the arrows  $\mathrm{Tr}_A$  (6.5.3, 6.5.8, 6.5.9) behave like the arrows  $\mathrm{Tr}_A$  of 5.6.1, 5.6.2. The point is that, if  $f$  is an  $S$ -morphism, the functors  $(f^*, f_*, f_!, f^!)$  commute with extension and restriction of scalars; consequently, the study of the behavior of  $\mathrm{Tr}_A$  (6.5.3) with respect to extension and restriction of scalars reduces to the behavior of the “tensor-product” arrow

$$\mathrm{RHom}_A(p_1^*L, p_1^*KA_X) \overset{\mathrm{L}}{\otimes}_A p_2^*L \longrightarrow \mathrm{RHom}_A(p_1^*L, p_1^*KA_X \overset{\mathrm{L}}{\otimes}_A p_2^*L)$$

and of the “evaluation” arrow

$$\mathrm{RHom}_A(L, KA_X) \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X, L_{\mathfrak{h}}}.$$

Thus the commutativity of the squares 5.9.1, 5.9.2 implies the commutativity of the square

$$\begin{array}{ccc} \delta^*c_*\mathrm{RHom}_A(c_1^*L, c_2^!L) & \xrightarrow{\mathrm{Tr}_A} & i_*^c KA_{X^c, L_{\mathfrak{h}}} \\ \downarrow & & \downarrow \\ \delta^*c_*\mathrm{RHom}_B(c_1^*(B \overset{\mathrm{L}}{\otimes}_A L), c_2^!(B \overset{\mathrm{L}}{\otimes}_A L)) & \xrightarrow{\mathrm{Tr}_B} & i_*^c KB_{X^c, L_{\mathfrak{h}}}, \end{array} \quad (6.7.1)$$

where the left vertical arrow is extension of scalars and the right vertical arrow is defined by the canonical arrow

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A \longrightarrow B \overset{\mathrm{L}}{\otimes}_{B^e} B.$$

It follows that, for  $u \in \mathrm{Hom}_A(c_1^*L, c_2^!L)$ ,

$$\mathrm{Tr}_B(B \overset{\mathrm{L}}{\otimes}_A u) = \varphi \mathrm{Tr}_A(u), \quad (6.7.2)$$

where

$$\varphi : H^0(X^c, KA_{X^c, L_{\mathfrak{h}}}) \longrightarrow H^0(X^c, KB_{X^c, L_{\mathfrak{h}}})$$

is the canonical map.

Assume  $A$  is finite type and flat as a  $\Lambda$ -module,  $B$  finite type and flat as a left  $A$ -module, and  $A_{\mathfrak{h}}$ ,  $B_{\mathfrak{h}}$  flat over  $\Lambda$ . There is a map  $\mathrm{Tr}_{B/A} : B_{\mathfrak{h}} \rightarrow A_{\mathfrak{h}}$ ; we shall again denote by

$$\mathrm{Tr}_{B/A} : KB_{X, \mathfrak{h}} \longrightarrow KA_{X, \mathfrak{h}} \quad (6.7.3)$$

the arrow deduced from it by applying  $K_X \overset{\mathrm{L}}{\otimes} -$  (6.5.2). It follows from 5.10.6 and 5.10.9 that, for  $L \in \mathrm{ob} D_{\mathrm{ctf}}(BX)$ , there is a commutative square

$$\begin{array}{ccc} \delta^*c_*\mathrm{RHom}_B(c_1^*L, c_2^!L) & \xrightarrow{\mathrm{Tr}_B} & i_*^c KB_{X^c, \mathfrak{h}} \\ \downarrow & & \downarrow \mathrm{Tr}_{B/A} \\ \delta^*c_*\mathrm{RHom}_A(c_1^*L, c_2^!L) & \xrightarrow{\mathrm{Tr}_A} & i_*^c KA_{X^c, \mathfrak{h}}, \end{array} \quad (6.7.4)$$

where the left vertical arrow is restriction of scalars. Consequently, for  $u \in \mathrm{Hom}_B(c_1^*L, c_2^!L)$ , one has, in  $H^0(X^c, KA_{X^c, \mathfrak{h}})$ ,

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_{B/A}(\mathrm{Tr}_B(u)). \quad (6.7.5)$$

## 6.8

Let  $A, B$  be flat  $\Lambda$ -algebras,  $X, Y$   $S$ -schemes,  $Z = X \times_S Y$ , and let  $c : C \rightarrow X^2$ ,  $d : D \rightarrow Y^2$  be correspondences. Let

$$e = c \times_S d : E = C \times_S D \longrightarrow Z^2$$

be their product. Then

$$Z^e = X^c \times_S Y^d.$$

Put  $R = A \otimes_\Lambda B$ . As in (III 5.3), one defines the external tensor product

$$\mathrm{Hom}_A(c_1^* L, c_2^! L) \otimes \mathrm{Hom}_B(d_1^* M, d_2^! M) \xrightarrow{\otimes_S^L} \mathrm{Hom}_R(e_1^*(L \otimes_S^L M), e_2^!(L \otimes_S^L M)), \quad (6.8.1)$$

and one deduces easily from the commutativity of the squares 5.12.1, 5.12.3 that, for

$$u \in \mathrm{Hom}_A(c_1^* L, c_2^! L), \quad v \in \mathrm{Hom}_B(d_1^* M, d_2^! M),$$

one has, in  $H^0(Z^e, KR_{Z^e, L_{\mathfrak{h}}})$ ,

$$\mathrm{Tr}_R(u \otimes_S^L v) = \mathrm{Tr}_A(u) \mathrm{Tr}_B(v), \quad (6.8.2)$$

the product on the right being defined by the canonical arrow

$$H^0(X^c, KA_{X^c, L_{\mathfrak{h}}}) \otimes H^0(Y^d, KB_{Y^d, L_{\mathfrak{h}}}) \longrightarrow H^0(Z^e, KR_{Z^e, L_{\mathfrak{h}}}), \quad (6.8.3)$$

coming from the isomorphism

$$KA_{X^c, L_{\mathfrak{h}}} \otimes_S^L KB_{Y^d, L_{\mathfrak{h}}} \longrightarrow K(A \otimes B)_{Z^e, L_{\mathfrak{h}}}$$

given by (III 1.7.6) and arrow (4) of 5.9.1.

## 6.9

Let  $G$  be a finite group. If  $X$  is a  $G$ - $S$ -scheme, that is, an  $S$ -scheme endowed with an action of  $G$  by  $S$ -automorphisms, and  $A$  is a  $\Lambda$ -algebra, we denote by  $D(G, {}_A X)$  the derived category of the category of sheaves of left  $G$ - $A$ -modules on  $X$ . For the notions of  $G$ -sheaf of sets, rings, modules, and so on, on  $X$  for the étale topology, we refer the reader to (X 1). We denote by the subscript  $c$  (resp.  $\mathrm{ctf}$ ) the full subcategory formed by complexes of  $G$ - $A$ -modules which, as complexes of  $A$ -modules, have constructible cohomology (resp. finite tor-dimension and constructible cohomology).

Similarly, if  $B$  is a  $\Lambda$ -algebra, we denote by  $D(G, {}_A X_B)$  the derived category of sheaves of  $G$ -( $A, B$ )-bimodules on  $X$ . If  $G$  acts trivially on  $X$ , a left  $G$ - $A$ -module on  $X$  is just a left  $A[G]$ -module on  $X$  without operators, hence

$$D(G, {}_A X) = D({}_A[G] X).$$

Since  $G$  is finite, one also has

$$D_c({}_A X) = D_c({}_A[G] X),$$

but only an inclusion

$$D_{\mathrm{ctf}}({}_A[G] X) \subset D_{\mathrm{ctf}}(G, {}_A X).$$

The duality formalism of (SGA 4 XVII, XVIII) extends without difficulty to  $G$ - $S$ -schemes. Since the functor forgetting the action of  $G$  is faithful, it is enough to extend naturally the definition of

the “six operations” ( $\overset{L}{\otimes}, \text{RHom}, f^*, f_*, f_!, f^!$ ) and the canonical arrows considered in loc. cit. (base change, Künneth, duality, etc.). This extension presents no difficulty: for the definition of the functors  $f_!, f^!$ , it is enough to use equivariant compactifications. We leave to the reader the task of paraphrasing the definitions and results 6.2 to 6.8 in the setting of  $G$ - $S$ -schemes.

If, for  $i = 1, 2$ ,  $X_i$  is a  $G$ - $S$ -scheme,  $c : C \rightarrow X = X_1 \times_S X_2$  is a  $G$ - $S$ -morphism, and  $L_i \in \text{ob } D_{\text{ctf}}(G, {}_A X_i)$ , the elements of

$$\text{Hom}_A(c_1^* L_1, c_2^! L_2),$$

where  $\text{Hom}_A$  is taken in the category  $D(G, {}_A -)$ , will be called equivariant cohomological correspondences from  $L_1$  to  $L_2$  with support in  $c$ . Note that, when  $G$  acts trivially on  $X_i$  ( $i = 1, 2$ ) and  $L_i \in \text{ob } D_{\text{ctf}}(A[G]X_i)$ , the isomorphism 6.4.1 in  $D(G, X)$  refines to an isomorphism

$$D_{A[G]} L_1 \overset{L}{\otimes}_{S, A[G]} L_2 \xrightarrow{\sim} \text{RHom}_{A[G]}(p_1^* L_1, p_2^! L_2).$$

It follows that, in a  $G$ -equivariant situation of type 6.5, with trivial action of  $G$  on the spaces, and with  $L \in \text{ob } D_{\text{ctf}}(A[G]X) \subset D_{\text{ctf}}(G, {}_A X)$ , one may, for

$$u \in \text{Hom}_{A[G]}(c_1^* L, c_2^! L),$$

define a local term

$$\text{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, L_{\mathfrak{h}}}),$$

finer, in the evident sense, than the local term

$$\text{Tr}_A(u) \in H^0(G, X^c, KA_{X^c, L_{\mathfrak{h}}})$$

obtained by using only finite tor-dimension of  $L$  with respect to  $A$ .

## 6.10

Let  $X, c$  be as in 6.5, and let  $L \in \text{ob } D_{\text{ctf}}(\Lambda[G]X)$ . For

$$u \in \text{Hom}_{\Lambda[G]}(c_1^* L, c_2^! L),$$

6.5.9 gives a local trace

$$\text{Tr}_{\Lambda[G]}(u) \in H^0(X^c, K\Lambda[G]_{X^c, \mathfrak{h}}) = H^0(X^c, K_{X^c} \otimes \Lambda[G]_{\mathfrak{h}}), \quad (6.10.1)$$

where  $G_{\mathfrak{h}}$  is the set of conjugacy classes of  $G$ . For every  $x \in G_{\mathfrak{h}}$ , one may consider the “value of  $\text{Tr}_{\Lambda[G]}(u)$  at  $x$ ,”

$$\text{Tr}_{\Lambda[G]}(u)(x) \in H^0(X^c, K_{X^c}),$$

the image of  $\text{Tr}_{\Lambda[G]}(u)$  under the projection

$$H^0(X^c, K_{X^c} \otimes \Lambda[G]_{\mathfrak{h}}) \longrightarrow H^0(X^c, K_{X^c})$$

defined by  $x$ . It follows from 6.7.5 and 5.11.2 that

$$\text{Tr}_{\Lambda}(u) = |G| \text{Tr}_{\Lambda[G]}(u)(e), \quad (6.10.2)$$

where  $e$  denotes the identity element of  $G$ .

More generally, let  $g \in G$ , let  $Z_g$  be the centralizer of  $g$ , and let

$$gu \in \mathrm{Hom}_{\Lambda[Z_g]}(c_1^*L, c_2^!L)$$

be the correspondence obtained by composing  $u$  with left multiplication by  $g : c_1^*L \rightarrow c_1^*L$ . We have  $L \in \mathrm{ob} D_{\mathrm{ctf}}(\Lambda[Z_g]X)$ , and hence 6.10.2 implies

$$\mathrm{Tr}_{\Lambda}(gu) = |Z_g| \mathrm{Tr}_{\Lambda[Z_g]}(gu)(e). \quad (6.10.3)$$

By the same argument as in 5.11.3, one deduces from 6.7.4 the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gu)(e) = \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}), \quad (6.10.4)$$

which allows 6.10.3 to be rewritten, if desired, in the form

$$\mathrm{Tr}_{\Lambda}(gu) = |Z_g| \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}). \quad (6.10.5)$$

## 6.11

Let  $f : X \rightarrow Y$  be a proper  $G$ - $S$ -morphism, and let

$$\begin{array}{ccc} C & \xrightarrow{c} & X \times_S X \\ \downarrow & & \downarrow f \times_S f \\ D & \xrightarrow{d} & Y \times_S Y \end{array} \quad (6.11.1)$$

be a commutative equivariant square of  $G$ - $S$ -schemes, with  $c$  and  $d$  proper. Assume that  $G$  acts trivially on  $Y$  and  $D$ . We denote by

$$f^{c/d} \quad (\text{or } f^c) : X^c \longrightarrow Y^d \quad (6.11.2)$$

the morphism induced by 6.11.1 on the schemes of fixed points of  $c$  and  $d$ , with the notation of 6.5.1. Let  $L \in \mathrm{ob} D_{\mathrm{ctf}}(G, {}_A X)$ , and let

$$u \in \mathrm{Hom}(c_1^*L, c_2^!L)$$

be a  $G$ -equivariant correspondence with support in  $c$ . By direct image, it gives a  $G$ -equivariant correspondence

$$f_*u \in \mathrm{Hom}_{\Lambda[G]}(d_1^*f_*L, d_2^!f_*L).$$

For  $g \in G$ , let  $Z_g$  denote the centralizer of  $g$ , and let

$$gu \in \mathrm{Hom}((gc)_1^*L, c_2^!L) \quad (6.11.3)$$

be the  $Z_g$ -equivariant correspondence with support in

$$gc \stackrel{\mathrm{dfn}}{=} (gc_1 = c_1g, c_2) : C \rightarrow X \times_S X, \quad (6.11.4)$$

defined as the composite of the arrows

$$g^*(c_1^*L) \xrightarrow{\alpha_g} c_1^*L \xrightarrow{u} c_2^!L,$$

where  $\alpha_g$  is the arrow giving the action of  $G$  on  $c_1^*L$ . It is clear that

$$f_*(gu) = g(f_*u). \quad (6.11.5)$$

**Proposition 6.12.** Under the hypotheses of 6.11, assume that

$$f_*L \in \text{ob } D_{\text{ctf}}(\Lambda_{[G]}Y).$$

Then, for every  $g \in G$ ,

$$f_*^{gc} \text{Tr}_\Lambda(gu) = |Z_g| \text{Tr}_{\Lambda[G]}(f_*u)(g^{-1}), \quad (6.12.1)$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the Gysin morphism defined by the adjunction arrow

$$f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \longrightarrow K_{Y^d}.$$

Indeed, by 6.10.5 applied to  $f_*u$ , and by 6.11.5, one has

$$\text{Tr}_\Lambda(f_*(gu)) = |Z_g| \text{Tr}_{\Lambda[G]}(f_*u)(g^{-1}),$$

and 6.12.1 therefore follows from the Lefschetz formula (III 4.5.1).

### 6.13

Let  $\ell$  be a prime number distinct from  $p$ . For every integer  $n \geq 1$ , put

$$\Lambda_n = \mathbb{Z}/\ell^n.$$

Under the hypotheses of 6.11, let  $L = (L_n)$  be a projective system of objects of

$$D_{\text{ctf}}(G, X, \Lambda_n)$$

such that

$$L_m \overset{\text{L}}{\otimes}_{\Lambda_m} \Lambda_n \xrightarrow{\sim} L_n \quad (m \geq n),$$

and let

$$u \in \text{Hom}(c_1^*L, c_2^!L)$$

be a projective system of equivariant correspondences  $u_n \in \text{Hom}(c_1^*L_n, c_2^!L_n)$  such that, for  $m \geq n$ ,

$$u_m \overset{\text{L}}{\otimes}_{\Lambda_m} \Lambda_n = u_n$$

modulo the isomorphism  $L_m \overset{\text{L}}{\otimes}_{\Lambda_m} \Lambda_n = L_n$ . By compatibility of local traces with extension of scalars, the local terms, for  $g \in G$ ,

$$\text{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\text{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) \stackrel{\text{dfn}}{=} \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms

$$f_*^{gc} \operatorname{Tr}_{\Lambda_n}(gu_n) = \operatorname{Tr}_{\Lambda_n}(gf_*u_n)$$

an element

$$f_* \operatorname{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(Y^d, K_{\ell, Y^d}) \stackrel{\text{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_{n, Y^d}).$$

Assume moreover that

$$f_* M_n \in \operatorname{ob} D_{\text{ctf}}(\Lambda_n[G]Y)$$

for all  $n$ . Then the local traces

$$\operatorname{Tr}_{\Lambda_n[G]}(f_*u_n)$$

similarly define an element

$$\operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*u) \in H^0(Y^d, K\mathbb{Z}_\ell[G]_{Y^d, \mathfrak{h}}) \stackrel{\text{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_n[G]_{Y^d, \mathfrak{h}}).$$

It follows from 6.12 that

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(gu) = |Z_g| \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}). \quad (6.13.1)$$

## 6.14

Let  $X$  be a  $G$ - $S$ -scheme, let  $j : U \rightarrow X$  be an equivariant open immersion, and let

$$c : C \rightarrow X \times_S X$$

be a morphism of  $G$ - $S$ -schemes such that  $c_2 = (p_2c) : C \rightarrow X$  is quasi-finite and of finite tor-dimension. Suppose that

$$c_1^{-1}(U) \subset c_2^{-1}(U).$$

Then there are commutative squares

$$\begin{array}{ccccccc} U & \longleftarrow & c_1^{-1}(U) & \longleftarrow & c_2^{-1}(U) & \longrightarrow & U \\ \downarrow & & \downarrow & & \downarrow & & \\ X & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X & & \\ \uparrow & & \uparrow & & \uparrow & & \\ X-U & \longleftarrow & c_2^{-1}(X-U) & \longrightarrow & X-U & & \end{array} \quad (6.14.1)$$

The trace morphism

$$\operatorname{Tr}_{c_2} : c_{2!}\Lambda_C \longrightarrow \Lambda_X$$

then defines a filtered  $G$ -equivariant correspondence

$$\underline{c} \in \operatorname{Hom}(c_1^*\Lambda_X, c_2^!\Lambda_X), \quad (6.14.2)$$

where  $\Lambda_X$  is endowed with the filtration defined by the submodule  $i_!\Lambda_U$ , that is,

$$F^n \Lambda_X = \Lambda_X \quad (n \leq 0), \quad F^1 \Lambda_X = i_!\Lambda_U, \quad F^n \Lambda_X = 0 \quad (n \geq 1).$$

The associated graded consists of two  $G$ -equivariant correspondences

$$\operatorname{gr}^0 \underline{c} = \underline{c}_{X-U, X} \in \operatorname{Hom}(c_1^*\Lambda_{X-U, X}, c_2^!\Lambda_{X-U, X}),$$

$$\mathrm{gr}^1 \underline{c} = \underline{c}_{U,X} \in \mathrm{Hom}(c_1^* \Lambda_{U,X}, c_2^! \Lambda_{X-U,X}), \quad (6.14.3)$$

where we have put

$$\Lambda_{X-U,X} = j_* \Lambda_{X-U}, \quad \Lambda_{U,X} = i_! \Lambda_U.$$

We have

$$\underline{c}_{X-U,U} = j_* \underline{c}_{X-U}, \quad (6.14.4)$$

where

$$\underline{c}_{X-U} \in \mathrm{Hom}(c_1^* \Lambda_{X-U}, c_2^! \Lambda_{X-U})$$

is the correspondence defined by  $\mathrm{Tr}_{c_2, X-U}$ . We write  $\underline{c}$ ,  $\underline{c}_{U,X}$ , and so on, when we want to specify the base ring.

Forgetting the action of  $G$ , there are local terms

$$\mathrm{Tr}_\Lambda(\underline{c}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{U,X}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \quad (6.14.5)$$

which, by additivity of traces (III 4.13.1), satisfy the relation

$$\mathrm{Tr}_\Lambda(\underline{c}) = \mathrm{Tr}_\Lambda(\underline{c}_{U,X}) + \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}). \quad (6.14.6)$$

Moreover, by 6.14.4,

$$\mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c \mathrm{Tr}_\Lambda(\underline{c}_{X-U}), \quad (6.14.7)$$

where

$$j_*^c : H^0((X-U)^c, K_{(X-U)^c}) \longrightarrow H^0(X^c, K_{X^c})$$

is the direct-image morphism defined by  $j^c : (X-U)^c \rightarrow X^c$ . Recall (III 4.3) that, if  $X$  is smooth,  $c$  a closed immersion, and  $X^c$  finite,  $\mathrm{Tr}_\Lambda(\underline{c})$  is the image in  $\Lambda$  of the intersection multiplicity  $(C, X)$  of  $C$  with the diagonal. If, moreover,  $X-U$  is smooth,  $\mathrm{Tr}_\Lambda(\underline{c}_{X-U})$  is the image in  $\Lambda$  of  $(c_2^{-1}(X-U), X-U)$ . Note also that, if  $X-U$  is proper over  $S$  and  $c|X-U$  is the diagonal, then the Lefschetz formula (III 4.7) implies that

$$\int_{X-U} \mathrm{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler–Poincaré characteristic of  $X-U$  with values in  $\Lambda$ .

## 6.15

Let there be a  $G$ -equivariant diagram 6.11.1, with  $f$ ,  $c$ ,  $d$  proper, and an equivariant open immersion  $U \rightarrow X$ . Suppose that  $c_2$  is quasi-finite and of finite tor-dimension, that

$$c_1^{-1}(U) \subset c_2^{-1}(U),$$

that  $G$  acts trivially on  $C$  and  $D$ , and that

$$f|U : U \longrightarrow f(U) = V$$

is an étale Galois covering with group  $G$ . This last hypothesis implies that

$$Rf_*(\Lambda_{U,X}) \in \mathrm{ob} D_{\mathrm{ctf}}(\Lambda[G]Y) :$$

indeed  $Rf_*(\Lambda_{U,X})$  is concentrated in degree 0 and is extension by zero of a sheaf on  $V$  locally isomorphic to  $\Lambda[G]$ . Hence one may apply 6.12 to  $L = \Lambda_{U,X}$ ,  $u = \underline{c}_{U,X}$ , and, in view of 6.14.6 and 6.14.7, one obtains:

**Proposition 6.16.** Under the hypotheses of 6.15, for every  $g \in G$  one has

$$f_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}) - (fj)_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}_{X-U}) = |Z_g| \operatorname{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1}), \quad (6.16.1)$$

where  $j : X - U \rightarrow X$  is the inclusion, and  $Z_g$  denotes the centralizer of  $g$ .

Applying 6.16 to the projective system of correspondences  $\underline{c}_\ell = (\underline{c}_{\Lambda_n})$ , where  $\Lambda_n = \mathbb{Z}/\ell^n$  with  $\ell$  prime distinct from  $p$ , and taking into account that 6.14.6 and 6.14.7 give, after passage to the limit and application of  $f_*^{gc}$ ,

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) = f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,U,X}) + (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}),$$

one deduces, by 6.13:

**Corollary 6.17.** Under the hypotheses of 6.15, for every  $g \in G$  one has

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) - (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}) = |Z_g| \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*\underline{c}_{\ell,U,X})(g^{-1}), \quad (6.17.1)$$

with the notation of 6.16.

In view of the compatibility of local traces with localization and of the remarks made in 6.14, one sees that 6.17 gives an affirmative solution, in a more general formulation, to Grothendieck's divisibility conjecture (XII 4.5), in the case where the endomorphism  $\varphi$  of  $G$  considered in loc. cit. is the identity.

In fact, by following 5.13, it is not difficult to give divisibility statements analogous to 6.12, 6.13, 6.16, 6.17, in the case where  $c$  is not  $G$ -equivariant, but where  $c_2$  is and  $c_1$  satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point  $x$  of  $G$ ,  $\varphi$  being a given endomorphism of  $G$ . The divisibility formulas are then written as above, with  $Z_g$  replaced by the  $\varphi$ -centralizer of  $g^{-1}$  (5.13.5).

## 6.18

Put

$$N_G = \sum_{g \in G} g \in \Lambda[G]. \quad (6.18.1)$$

Let  $X$  be an  $S$ -scheme. For  $L \in \operatorname{ob} D(\Lambda[G]X)$ , left multiplication by  $N_G$  induces an arrow in  $D(X)$

$$N_G : \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} L \longrightarrow \operatorname{RHom}_{\Lambda[G]}(\Lambda, L), \quad (6.18.2)$$

where  $\Lambda$  is regarded as a  $\Lambda[G]$ -algebra by the natural augmentation sending every  $g \in G$  to 1.

**Lemma 6.18.3.** If  $L \in \operatorname{ob} D_{\operatorname{ctf}}(\Lambda[G]X)$ , the arrow 6.18.2 is an isomorphism.

By dévissage, one reduces to the case where  $L$  is perfect, then to  $L = \Lambda[G]$ , and the assertion follows from the fact that  $N_G$  (6.18.1) is a basis of the module of  $G$ -invariants of the left  $\Lambda[G]$ -module  $\Lambda[G]$ .

**Proposition 6.19.** In the situation of 6.11, let  $M \in \text{ob } D_{\text{ctf}}(Y)$ ,

$$v \in \text{Hom}(d_1^* M, d_2^! M),$$

and let  $a : M \rightarrow f_* L$  be an arrow in  $D(\Lambda[G]Y)$ , where  $M$  is regarded as an object of  $D(\Lambda[G]Y)$  by the augmentation  $\Lambda[G] \rightarrow \Lambda$ , such that the square

$$\begin{array}{ccc} d_1^* M & \xrightarrow{v} & d_2^! M \\ \downarrow & & \downarrow \\ d_1^* f_* L & \xrightarrow{f_* u} & d_2^! f_* L \end{array}$$

is commutative. Suppose that  $f_* M \in \text{ob } D_{\text{ctf}}(\Lambda[G]Y)$  and that  $a$  induces an isomorphism

$$M \xrightarrow{\sim} \text{RHom}_{\Lambda[G]}(\Lambda, f_* L).$$

Then

$$\text{Tr}_{\Lambda}(v) = \sum_{g \in G_{\mathfrak{h}}} \text{Tr}_{\Lambda[G]}(f_* u)(g^{-1}). \quad (6.19.1)$$

By 6.18.3,  $a$ , composed with  $N_G^{-1}$ , gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{\text{L}}{\otimes}_{\Lambda[G]} f_* L,$$

under which  $v$  is identified with

$$\Lambda \overset{\text{L}}{\otimes}_{\Lambda[G]} f_* u.$$

By 6.7.2,  $\text{Tr}_{\Lambda}(v)$  is therefore the image of  $\text{Tr}_{\Lambda[G]}(f_* u)$  under the arrow induced by the augmentation

$$\Lambda[G] \longrightarrow \Lambda.$$

This augmentation, by definition, sends every conjugacy class to 1, whence the conclusion.

## 6.20

Let, for  $i = 1, 2$ ,  $X_i$  be a  $G$ - $S$ -scheme, and let

$$X = X_1 \times_S X_2, \quad c : C \rightarrow X$$

be a  $G$ - $S$ -morphism such that  $c_2$  is quasi-finite and of finite tor-dimension. Let  $L_i \in \text{ob } D^+(G, {}_A X_i)$ . The trace morphism

$$\text{Tr}_{c_2} : c_{2!} c_2^! L_2 \longrightarrow L_2$$

defines by adjunction an arrow in  $D(G, X)$ , again denoted

$$\text{Tr}_{c_2} : c_2^! L_2 \longrightarrow c_2'^! L_2, \quad (6.20.1)$$

which associates to every  $G$ -equivariant morphism

$$u \in \mathrm{Hom}(c_1^* L_1, c_2^* L_2)$$

a  $G$ -equivariant correspondence

$$u^! = \mathrm{Tr}_{c_2} \circ u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2). \quad (6.20.2)$$

**Proposition 6.21.** Let  $f : X \rightarrow Y$  be a proper  $G$ -morphism, and let 6.11.1 be an equivariant commutative square, with trivial action of  $G$  on  $Y$  and  $D$ . Suppose  $c, d$  are proper,  $c_2, d_2$  quasi-finite and of finite tor-dimension, and the square

$$\begin{array}{ccc} C & \xrightarrow{c_2} & X \\ h \downarrow & & \downarrow f \\ D & \xrightarrow{d_2} & Y \end{array}$$

is cartesian and tor-independent (SGA 6 III 1.5). Let  $M \in \mathrm{ob} D_{\mathrm{ctf}}(Y)$ ,

$$v \in \mathrm{Hom}(d_1^* M, d_2^* M),$$

hence a  $G$ -equivariant homomorphism

$$h^* v \in \mathrm{Hom}(c_1^* f^* M, c_2^* f^* M),$$

and  $G$ -equivariant correspondences

$$(h^* v)^! \in \mathrm{Hom}(c_1^* f^* M, c_2^! f^* M), \quad f_*((h^* v)^!) \in \mathrm{Hom}(d_1^*(f_* f^* M), d_2^! f_* f^* M).$$

Then there is a commutative square of arrows in  $D(\Lambda[G]Y)$ , where  $M$  is regarded as an object of  $D(\Lambda[G]Y)$  by the trivial action,

$$\begin{array}{ccc} d_1^* M & \xrightarrow{v^!} & d_2^! M \\ \downarrow & & \downarrow \\ d_1^* f_* f^* M & \xrightarrow{f_*((h^* v)^!)} & d_2^! f_* f^* M, \end{array}$$

where the vertical arrows are given by the adjunction arrow  $M \rightarrow f_* f^* M$ .

Put  $h^* v = u$ ,  $f^* M = L$ , and consider the diagram

$$\begin{array}{ccccccc} d_1^* M & \xrightarrow{v} & d_2^* M & \xrightarrow{\mathrm{Tr}_{d_2}} & d_2^! M & & \\ \downarrow & & \downarrow & & \downarrow & & \\ d_1^* f_* L & \longrightarrow & h_* c_1^* L & \xrightarrow{h_* \mathrm{Tr}_{c_2}} & h_* c_2^! L & \longrightarrow & d_2^! f_* L. \end{array}$$

The arrows (1) and (3) are given by the adjunction arrow  $M \rightarrow f_* L$ , (2) by the adjunction arrow  $d_2^* M \rightarrow h_* h^*(d_2^* M) = h_* c_2^* L$ , (4) by the base-change arrow, and (5) is the canonical isomorphism

(SGA 4 XVIII 3.1.12.3). By definition,  $v^!$  is the composite of the upper row. On the other hand, it follows from the interpretation of the direct image of a correspondence (III 3.5 b), 3.7) that  $f_*u^!$  is the composite of the lower row. It remains only to prove commutativity of the squares (A) and (B). The commutativity of (A) follows immediately from the definition of the base-change arrow. As for (B), it follows from the fact that, in view of the tor-independence hypothesis,  $\mathrm{Tr}_{d_2}$  is compatible with base change by  $f$  (SGA 4<sup>1/2</sup> Cycle 2.3.3).

Applying 6.19, one obtains:

**Corollary 6.22.** Under the hypotheses of 6.21, assume that

$$f_*f^*M \in \mathrm{ob} D_{\mathrm{ctf}}(\Lambda[G]Y),$$

and that the adjunction arrow

$$M \rightarrow f_*f^*M$$

induces an isomorphism

$$M \xrightarrow{\sim} \mathrm{RHom}_{\Lambda[G]}(\Lambda, f_*f^*M).$$

Then

$$\mathrm{Tr}_{\Lambda}(v^!) = \sum_{g \in G_{\mathfrak{t}}} \mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}). \quad (6.22.1)$$

**Proposition 6.23.** Under the hypotheses of 6.21, suppose moreover that there is an open immersion

$$i : V \rightarrow Y$$

such that

$$d_1^{-1}(V) = d_2^{-1}(V),$$

and that  $f$  induces a principal Galois covering with group  $G$ ,

$$f_V : U = f^{-1}(V) \rightarrow V.$$

Suppose also that  $M$  is of the form  $M = i_!E$ , where  $E \in \mathrm{ob} D(V)$ , and that there is an isomorphism

$$\alpha : f^*E \xrightarrow{\sim} \Lambda_X \overset{\mathrm{L}}{\otimes}_{\Lambda} E_0$$

in  $D(G, X)$ , with  $E_0 \in \mathrm{ob} D(\Lambda[G])$ , perfect over  $\Lambda$ . Let

$$v \in \mathrm{Hom}(d_1^*M, d_2^*M)$$

be as in 6.21, and let

$$v_0 \in \mathrm{Hom}_{\Lambda[G]}(E_0, E_0)$$

be the homomorphism defined, thanks to  $\alpha$ , by  $f^*v|_{c_1^{-1}(U)}$ . Then, if  $\underline{c}_{U,X}$  denotes the correspondence defined in 6.14.3, with the notation 6.20.2 one has

$$\mathrm{Tr}_{\Lambda}(v^!) = \sum_{g \in G_{\mathfrak{t}}} \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1}) \mathrm{Tr}_{\Lambda}(gv_0). \quad (6.23.1)$$

Note that, by the central character of  $\mathrm{Tr}_{\Lambda}$ , the right hand side does not depend on the choice of  $\alpha$ .

By the projection formula

$$f_* \Lambda_Y \otimes^{\mathbf{L}} M \xrightarrow{\sim} f_* f^* M$$

and 6.15,  $M$  satisfies the hypotheses of 6.22. On the other hand,  $\alpha$  defines an isomorphism

$$f^* M \xrightarrow{\sim} \Lambda_{U,X} \otimes E_0$$

under which  $(f^* v)^!$  is identified with the external tensor product  $\mathcal{C}_{U,X} \otimes v_0$ . By the projection formula,  $f_* f^* M$  is therefore identified with  $f_*(\Lambda_{U,X}) \otimes E_0$ , and  $f_*((f^* v)^!)$  with the external tensor product  $f_* \mathcal{C}_{U,X} \otimes v_0$ . For  $g \in G$ , the  $Z_g$ -equivariant correspondence  $gf_*((f^* v)^!)$  is therefore identified with

$$gf_* \mathcal{C}_{U,X} \otimes gv_0.$$

Since, by 6.10.4,

$$\mathrm{Tr}_{\Lambda[G]}(f_*((f^* v)^!))(g^{-1}) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*((f^* v)^!))(e),$$

it is enough, in view of 6.22.1, to prove the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_* \mathcal{C}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[G]}(f_* \mathcal{C}_{U,X})(g^{-1}) \mathrm{Tr}_{\Lambda}(gv_0),$$

or, equivalently by 6.10.4,

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_* \mathcal{C}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_* \mathcal{C}_{U,X})(e) \mathrm{Tr}_{\Lambda}(gv_0). \quad (6.23.2)$$

Replacing  $G$  by  $Z_g$  if necessary, we may suppose that  $g = e$ . Returning to the definition of the traces, one sees that it is enough to establish the following lemma.

**Lemma 6.23.3.** (cf. SGA 4<sup>1/2</sup> Report 4.7). Let  $P$  be a  $G_V$ -module locally a direct factor of a free module of finite type, let  $P' = i_! P$ , and let  $Q$  be a complex of  $\Lambda[G]$ -modules, perfect over  $\Lambda$ . Then  $P \otimes_{\Lambda}^{\mathbf{L}} Q$ , considered as an object of  $D(\Lambda[G]V)$  by the diagonal action of  $G$ , is perfect over  $\Lambda[G]$ , and there is a commutative square

$$\begin{array}{ccc} \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^* P', p_2^! P') \otimes^{\mathbf{L}} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \longrightarrow & K_Y \\ \downarrow & & \parallel \\ \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^* P' \otimes^{\mathbf{L}} Q, p_2^! (P' \otimes^{\mathbf{L}} Q)) & \longrightarrow & K_Y, \end{array}$$

where  $\delta : Y \rightarrow Y \times_S Y$  is the diagonal, the upper horizontal arrow is the tensor product of the arrows

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^* P', p_2^! P') \xrightarrow{6.5.3} K\Lambda[G]_{Y, \natural} \xrightarrow{x \mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q, Q) \xrightarrow{\mathrm{Tr}_{\Lambda}} \Lambda,$$

the vertical arrow is deduced from the external tensor product by restriction along the diagonal

$$\Lambda[G] \rightarrow \Lambda[G] \otimes \Lambda[G],$$

and the lower horizontal arrow is the composite of 6.5.3 and of  $x \mapsto x(e)$ .

The first assertion is evident, for one reduces to  $P = \Lambda[G]_V$ , and  $P \overset{\mathbb{L}}{\otimes}_{\Lambda} Q$  is then isomorphic to the tensor product of  $P$  by  $Q$  considered with the trivial action of  $G$ . For the second, first note that, by 6.5.4 and 6.5.5, one has

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^* P', p_2^! P') \xrightarrow{\sim} D_{\Lambda[G]}(i_! P) \overset{\mathbb{L}}{\otimes}_{\Lambda[G]} i_! P \xrightarrow{\sim} i_!(D_{\Lambda[G]} P \overset{\mathbb{L}}{\otimes}_{\Lambda[G]} P),$$

so that we are reduced to checking that the square induced by the preceding square on  $V$  commutes. We may therefore suppose that  $V = Y$ . The square then reads

$$\begin{array}{ccc} \mathrm{RHom}_{\Lambda[G]}(P, K_Y \overset{\mathbb{L}}{\otimes} P) \overset{\mathbb{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \longrightarrow & K_Y \\ \downarrow & & \parallel \\ \mathrm{RHom}_{\Lambda[G]}(P \overset{\mathbb{L}}{\otimes} Q, K_Y \overset{\mathbb{L}}{\otimes} P \overset{\mathbb{L}}{\otimes} Q) & \longrightarrow & K_Y, \end{array}$$

where the vertical arrow is the external tensor product, the upper horizontal arrow is given by

$$\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_{\Lambda},$$

and the lower horizontal arrow by  $\mathrm{Tr}_{\Lambda[G]}(-)(e)$ . By descent, one reduces to  $P = \Lambda[G]_Y$ , and, by the initial remark, one may then suppose that  $G$  acts trivially on  $Q$ . The assertion is then an immediate consequence of 5.12.5.

## 6.24

If  $\Lambda$  is annihilated by a power of a prime number  $\ell \neq p$ , the right hand side of 6.23.1 may be rewritten in the form

$$\sum_{g \in G_{\mathfrak{h}}} \mathrm{Tr}_{\mathbb{Z}_{\ell}[G]}(f_{*\mathcal{C}_{\ell,U,X}})(g^{-1}) \mathrm{Tr}_{\Lambda}(gv_0),$$

with the notation of 6.17. Formula 6.17.1 shows that the local terms defined by Grothendieck in (XII 6.2) coincide with Verdier's local terms, at least when the endomorphism  $\varphi$  of  $G$  considered in loc. cit. is the identity; but, as already observed, this restriction is not serious. If  $Y$  is proper over  $S$ , then by conjugating 6.23.1 with the Lefschetz formula for  $Y \rightarrow S$  (III 4.7), one obtains a formula generalizing (XII 6.3), and a fortiori the Lefschetz formula for Frobenius proved in (SGA 4<sup>1/2</sup> Report) and Part I of the present exposé.

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SGA 4<sup>1/2</sup>: *Cohomologie étale*, by P. Deligne, Lecture Notes in Mathematics 569, Springer-Verlag, 1977.

In SGA 4<sup>1/2</sup>:

Finiteness Theorem: finiteness theorems in  $\ell$ -adic cohomology;

Cycle: the cohomology class associated with a cycle;

Report: report on the trace formula.

## Exposé V. $J$ -adic Projective Systems

by J.-P. Jouanolou

Throughout this exposé, the letter  $A$  denotes a ring with unit. The present exposé is intended to develop a formalism which will be used in the following exposé for the study of  $\ell$ -adic cohomology.

### 1. Generalities on abelian $A$ -categories

**Definition 1.1.** An  $A$ -category (resp. an abelian  $A$ -category) is a pair  $(\mathcal{C}, \rho)$  formed by an additive (resp. abelian) category  $\mathcal{C}$  and a homomorphism of rings

$$\rho : A \longrightarrow \text{End}(\text{id}_{\mathcal{C}})$$

from  $A$  to the ring of endomorphisms of the identity functor from  $\mathcal{C}$  to  $\mathcal{C}$ .

Given two  $A$ -categories (resp. abelian  $A$ -categories)  $(\mathcal{C}, \rho)$  and  $(\mathcal{C}', \rho')$ , a morphism of  $A$ -categories (resp. of abelian  $A$ -categories) from  $(\mathcal{C}, \rho)$  to  $(\mathcal{C}', \rho')$  is an additive (resp. exact) functor  $F : \mathcal{C} \rightarrow \mathcal{C}'$  such that, for every object  $M$  of  $\mathcal{C}$  and every element  $a$  of  $A$ , the equality below is satisfied:

$$F(\rho(a)_M) = \rho'(a)_{F(M)}.$$

#### Example 1.1.1

Let  $\mathcal{C}$  be an additive (resp. abelian) category. Recall that the category of left  $(A, \mathcal{C})$ -modules, denoted  $(A, \mathcal{C})\text{-mod}$ , is the category described as follows:

- (i) Its objects are pairs  $(X, \alpha)$  consisting of an object  $X$  of  $\mathcal{C}$  and a homomorphism of rings

$$\alpha : A \longrightarrow \text{Hom}_{\mathcal{C}}(X, X).$$

- (ii) A morphism  $(X, \alpha) \rightarrow (Y, \beta)$  is a  $\mathcal{C}$ -morphism  $u : X \rightarrow Y$  such that, for every element  $a$  of  $A$ , the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ \alpha(a) \downarrow & & \downarrow \beta(a) \\ X & \xrightarrow{u} & Y. \end{array}$$

The category  $(A, \mathcal{C})\text{-mod}$  is evidently an  $A^\natural$ -category (resp. an abelian  $A^\natural$ -category), where  $A^\natural$  is the centre of  $A$ . Moreover, it has the following properties:

- (a) The functor  $(A, \mathcal{C})\text{-mod} \rightarrow \mathcal{C}$  obtained by forgetting the  $A$ -module structure commutes with inductive and projective limits.
- (b) Let  $I$  be a category. In order that the  $I$ -indexed inductive (resp. projective) limits of  $(A, \mathcal{C})\text{-mod}$  be representable, it is enough that this be so for the  $I$ -indexed inductive (resp. projective) limits of  $\mathcal{C}$ .

## 1.2. External tensor product

Let  $(\mathcal{C}, \rho)$  be an abelian  $A$ -category. If  $M$  is a right  $A$ -module and  $X$  an object of  $\mathcal{C}$ , the external tensor product of  $X$  by  $M$ , denoted  $M \otimes_A X$ , is the covariant functor from  $\mathcal{C}$  to the category of sets, defined on objects  $Y$  of  $\mathcal{C}$  by the equality below and in the evident way on arrows:

$$(M \otimes_A X)(Y) = \text{Hom}_A(M, \text{Hom}_{\mathcal{C}}(X, Y)),$$

where  $\text{Hom}_{\mathcal{C}}(X, Y)$  is regarded as a right  $A$ -module via the action of  $A$  on  $X$ . Interpreting  $M \otimes_A X$  and  $Y$  as objects of the category  $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}, \text{Ens})$ , the preceding equality takes the more familiar form

$$\text{Hom}_{\widehat{\mathcal{C}}}(M \otimes_A X, Y) = \text{Hom}_A(M, \text{Hom}_{\mathcal{C}}(X, Y)).$$

If the functor  $M \otimes_A X$  is representable, a representative will be denoted and named in the same way.

If  $u : M \rightarrow M'$  and  $v : X \rightarrow X'$  are two morphisms, respectively in  $(A\text{-mod})$  and in  $\mathcal{C}$ , one defines in the usual way a  $\mathcal{C}$ -morphism

$$u \otimes_A v : M \otimes_A X \longrightarrow M' \otimes_A X',$$

which allows us to speak of the external tensor-product bifunctor, covariant in both variables. Its properties are summarized in the following proposition.

**Proposition 1.2.1.** The tensor-product bifunctor is additive and commutes with inductive limits.

The proof, classical in the case of modules over a ring, is immediate.

**Proposition 1.2.2 (Representability criteria).**

- (a) For every object  $X$  of  $\mathcal{C}$ ,  $A \otimes_A X$  is represented by  $X$ .

- (b) For every finitely presented  $A$ -module  $M$  and every object  $X$  of  $\mathcal{C}$ , the functor  $M \otimes_A X$  is representable.
- (c) If the category  $\mathcal{C}$  has arbitrary direct sums, the functor  $M \otimes_A X$  is representable for every pair  $(M, X)$ .

*Proof.* Assertion (a) is clear. The other two assertions then follow from the lemma below, whose proof is immediate.

**Lemma 1.2.3.** Let  $M \rightarrow N \rightarrow P \rightarrow 0$  be an exact sequence of  $A$ -modules and let  $X$  be an object of  $\mathcal{C}$ . If the functors  $M \otimes_A X$  and  $N \otimes_A X$  are represented by objects  $Y$  and  $Z$  of  $\mathcal{C}$ , then the functor  $P \otimes_A X$  is represented by the cokernel of the morphism  $Y \rightarrow Z$  corresponding to  $\alpha \otimes_A \text{id}_X$ .

#### Remark 1.2.4

Let  $\mathfrak{G}$  be an ideal of finite type of  $A$ . For every object  $X$  of  $\mathcal{C}$ , the functor  $(A/\mathfrak{G}) \otimes_A X$  is representable, and the arrow

$$p_X : X \longrightarrow (A/\mathfrak{G}) \otimes_A X$$

coming from the canonical epimorphism from  $A$  onto  $A/\mathfrak{G}$  and from part (a) of 1.2.2 is an epimorphism. The same statement remains valid when the ideal  $\mathfrak{G}$  is no longer assumed to be of finite type, but  $\mathcal{C}$  has arbitrary direct sums. Put  $\mathfrak{G}X = \text{Ker}(p_X)$ . When the category has filtered inductive limits, it is clear that

$$\mathfrak{G}X = \sum_{a \in \mathfrak{G}} aX.$$

If  $u : X \rightarrow Y$  is an epimorphism,  $u$  maps  $\mathfrak{G}X$  epimorphically onto  $\mathfrak{G}Y$ . Finally, for an object  $X$  to be annihilated by every element of  $\mathfrak{G}$ , it is necessary and sufficient that  $\mathfrak{G}X = 0$ . Then every morphism from an object  $Z$  to  $X$  factors, by means of  $p_Z$ , through a morphism from  $(A/\mathfrak{G}) \otimes_A Z$  to  $X$ .

## 2. The Mittag-Leffler-Artin-Rees condition

Throughout this paragraph, the letter  $\mathcal{C}$  denotes an abelian category and we put

$$P = \text{Hom}(\mathbb{N}^\circ, \mathcal{C}),$$

the category of projective systems indexed by the ordered set  $\mathbb{N}$  of positive integers and with values in  $\mathcal{C}$ .

### 2.1. The Mittag-Leffler-Artin-Rees condition

Let  $X = (X_n, u_n)_{n \geq 0}$  be an object of the category  $P$ . If  $r$  denotes an integer  $\geq 0$ , we agree to write  $X[r]$  for the projective system  $(X_{n+r}, u_{n+r})_{n \geq 0}$ . A morphism  $h : X \rightarrow Y$  of  $P$  defines in the evident way a morphism denoted  $h[r]$  from  $X[r]$  to  $Y[r]$ , functorially. If  $r$  and  $s$  are two integers satisfying  $s \geq r \geq 0$ , the transition morphisms  $X_{n+s} \rightarrow X_{n+r}$  ( $n \geq 0$ ) define a morphism

$$w_{sr} : X[s] \longrightarrow X[r],$$

and it is clear that, if  $t$  is a third integer with  $t \geq s \geq r$ , the equality below is satisfied:

$$w_{sr} \circ w_{ts} = w_{tr}.$$

In other words, we have thus associated to  $X$  an object  $\overline{X}$  of the category  $\text{Hom}(\mathbb{N}^\circ, P)$ . Moreover, if  $h : X \rightarrow Y$  is a morphism of  $P$ , there corresponds to it in the evident way a morphism  $\overline{h}$  from  $\overline{X}$  to  $\overline{Y}$ , functorially. Finally, if

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

is an exact sequence of  $P$ , the corresponding sequence below in  $\overline{P} = \text{Hom}(\mathbb{N}^\circ, P)$  is exact:

$$0 \longrightarrow \overline{X} \xrightarrow{\overline{u}} \overline{Y} \xrightarrow{\overline{v}} \overline{Z} \longrightarrow 0.$$

**Definition 2.1.1.** Let an object of the category  $P$  be given. It is said to satisfy the Mittag-Leffler-Artin-Rees condition (abbreviated MLAR) if there exists an integer  $r$  such that, for every integer  $s \geq r$ , the equality below is satisfied:

$$\text{Im}(X[s] \xrightarrow{w_{s0}} X) = \text{Im}(X[r] \xrightarrow{w_{r0}} X).$$

Equivalently, for every integer  $n \geq 0$ , the images of  $X_{n+r}$  and  $X_{n+s}$  in  $X_n$  by the transition morphisms are the same. Finally, a third interpretation is that the associated projective system  $\overline{X}$  satisfies the Mittag-Leffler condition (abbreviated ML) (EGA 0<sub>III</sub> 13.1.2). The following proposition is deduced from this by (EGA 0<sub>III</sub> 13.2.1).

**Proposition 2.1.2.** Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of  $P$ .

- (i) If  $Y$  satisfies (MLAR), then  $Z$  satisfies (MLAR).
- (ii) If  $X$  and  $Z$  satisfy (MLAR), then  $Y$  satisfies (MLAR).

## 2.2. AR-null projective systems

**Definition 2.2.1.** Let  $X$  be an object of the category  $P$ . It is called AR-null if there exists an integer  $r \geq 0$  such that the canonical morphism  $X[r] \rightarrow X$  is zero.

It is clear in particular that an AR-null projective system satisfies the condition (MLAR).

**Proposition 2.2.2.** Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of  $P$ . The following assertions are equivalent:

- (i)  $Y$  is AR-null;
- (ii)  $X$  and  $Z$  are AR-null.

In other words, the full subcategory  $P_0$  of  $P$ , whose objects are the AR-null projective systems, is thick in  $P$ .

*Proof.* It is clear that (i) implies (ii). Conversely, let  $r$  and  $t$  be integers satisfying the condition of Definition 2.2.1 for  $X$  and  $Z$  respectively. One sees immediately that the integer  $s + t$  satisfies the conditions of Definition 2.2.1 for  $Y$ .

### 2.3. Reminders on the calculus of fractions and applications

Let  $\mathcal{E}$  be a category and  $S$  a subset of  $\text{Fl}(\mathcal{E})$ . A category of fractions of  $\mathcal{E}$  with respect to  $S$  is a pair  $(p, \mathcal{E}(S^{-1}))$  consisting of a category  $\mathcal{E}(S^{-1})$  and a functor  $p : \mathcal{E} \rightarrow \mathcal{E}(S^{-1})$  satisfying the following universal property: for a functor  $u$  from  $\mathcal{E}$  to a category  $\mathcal{F}$  to factor through  $p$ , it is necessary and sufficient that it make all arrows belonging to  $S$  invertible; the factorization is then unique. The uniqueness of such a pair up to isomorphism is clear. We refer the reader to Exposé I of the Séminaire Homotopique de Strasbourg for the proof of existence, and also for most of the results stated without proof in what follows in this paragraph. See also P. Gabriel and M. Zisman, *Calculus of fractions and homotopy theory*, Erg. der Math., 35, Springer-Verlag, 1967.

Certain properties of the multiplicative part  $S$  ensure a simple description of the category  $\mathcal{E}(S^{-1})$ .

**Definition 2.3.1.** A subset  $S$  of  $\text{Fl}(\mathcal{E})$  is said to admit a right calculus of fractions if the following conditions hold:

- (a) The identity arrows of  $\mathcal{E}$  belong to  $S$ .
- (b) If  $u : X \rightarrow Y$  and  $v : Y \rightarrow Z$  belong to  $S$ , the composite  $v \circ u$  belongs to  $S$ .
- (c) Every diagram of type  $(D_1)$  below, in which  $t$  belongs to  $S$ , can be completed to a diagram of type  $(D_2)$ , in which  $s$  belongs to  $S$ :

$$(D_1) \quad X \xrightarrow{u} Y \xleftarrow{t} Y' \quad (D_2) \quad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ s \downarrow & & \downarrow t \\ X & \xrightarrow{u} & Y \end{array}$$

- (d) For every commutative diagram

$$X \begin{array}{c} \xrightarrow{f} \\ \rightrightarrows \\ \xrightarrow{g} \end{array} Y \xrightarrow{s} Z,$$

in which  $s$  belongs to  $S$ , there exists a commutative diagram

$$T \xrightarrow{t} X \begin{array}{c} \xrightarrow{f} \\ \rightrightarrows \\ \xrightarrow{g} \end{array} Y,$$

in which  $t$  belongs to  $S$ .

Dually one defines the following notion.

**Definition 2.3.2.** A subset  $S$  of  $\text{Fl}(\mathcal{E})$  is said to admit a left calculus of fractions if the following conditions hold:

- (a') The identity arrows of  $\mathcal{E}$  belong to  $S$ .
- (b') The composite of two arrows belonging to  $S$  belongs to  $S$ .
- (c') Every diagram of type  $(D_1)'$  below, in which  $s$  belongs to  $S$ , can be completed to a diagram of type  $(D_2)'$ , in which  $t$  belongs to  $S$ :

$$(D_1)' \quad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ s \downarrow & & \\ X & & \end{array} \quad (D_2)' \quad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ s \downarrow & & \downarrow t \\ X & \xrightarrow{u} & Y. \end{array}$$

- (d') For every commutative diagram  $\Delta$  of the form below, in which  $s$  belongs to  $S$ , there exists a morphism  $t$  with source  $Y$  and belonging to  $S$  such that  $t \circ f = t \circ g$ :

$$(\Delta) \quad X' \xrightarrow{s} X \rightrightarrows[Y]{f, g} Y.$$

### 2.3.3

Suppose that  $S$  is a subset of  $\text{Fl}(\mathcal{E})$  admitting a right calculus of fractions. It follows from part (d) of 2.3.1 that, for every object  $X$  of  $\mathcal{E}$ , the category  $(S/X)^\circ$ , opposite to the category of arrows over  $X$  belonging to  $S$ , is filtering (SGA 4 I 2.7). Consider then the category  $\mathcal{E}_S$  described as follows:

- the objects of  $\mathcal{E}_S$  are the objects of  $\mathcal{E}$ ;
- if  $X$  and  $Y$  are two objects of  $\mathcal{E}$ ,

$$\text{Hom}_{\mathcal{E}_S}(X, Y) = \varinjlim_{(S/X)^\circ} \text{Hom}_{\mathcal{E}}(\text{source}(\cdot), Y).$$

If  $P_S$  denotes the evident functor from  $\mathcal{E}$  to  $\mathcal{E}_S$ , the pair  $(P_S, \mathcal{E}_S)$  is a solution of the universal problem described at the beginning of the paragraph. In this case we agree to write  $\mathcal{E}(S^{-1})$  for this particular solution.

### 2.3.4

Let now  $S$  be a subset of  $\text{Fl}(\mathcal{E})$  admitting a left calculus of fractions. Then the functor  $P_S : \mathcal{E} \rightarrow \mathcal{E}(S^{-1})$  commutes with finite inductive limits. In particular, if  $\mathcal{E}$  is additive,  $\mathcal{E}(S^{-1})$  is additive and  $P_S$  is an additive functor.

### 2.3.5

If  $\mathcal{E}$  is an abelian category and if  $S$  is a subset of  $\text{Fl}(\mathcal{E})$  admitting both a right calculus of fractions and a left calculus of fractions, the category  $\mathcal{E}(S^{-1})$  is abelian and the functor  $P_S$  is exact. Moreover, the pair  $(P_S, \mathcal{E}(S^{-1}))$  is a quotient of  $\mathcal{E}$  by the thick abelian subcategory  $P_S^{-1}(0)$ .

#### 2.4. The category $P_{AR} = \text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$

Returning to the notation of 2.1, let  $AR$  be the set of canonical arrows (cf. 2.1)

$$v_{rX} : X[r] \longrightarrow X,$$

where  $X$  runs through the set of objects of  $P$  and  $r$  through the set of integers  $\geq 0$ .

**Proposition 2.4.1.** The subset  $AR$  of  $\text{Fl}(P)$  admits both a right calculus of fractions and a left calculus of fractions.

*Proof.* The parts (a)  $\Rightarrow$  (a') and (b)  $\Rightarrow$  (b') of 2.3.1 and 2.3.2 are clear.

We prove (c). Let  $(D_1)$  be the commutative diagram below, in which  $v_{rY}$  denotes the canonical arrow from  $Y[r]$  to  $Y$ :

$$\begin{array}{ccc} Y[r] & & \\ v_{rY} \downarrow & & \\ X & \longrightarrow & Y. \end{array}$$

The diagram  $(D_2)$  below is commutative, and  $v_{rX}$ , the canonical morphism from  $X[r]$  to  $X$ , belongs to  $AR$ , whence the assertion:

$$\begin{array}{ccc} X[r] & \xrightarrow{u[r]} & Y[r] \\ v_{rX} \downarrow & & \downarrow v_{rY} \\ X & \xrightarrow{u} & Y. \end{array}$$

We prove (d). Let a commutative diagram be given:

$$X \xrightleftharpoons[g]{f} Z[r] \xrightarrow{v_{rZ}} Z.$$

Applying translation by  $r$ , one obtains a commutative diagram

$$X[r] \xrightleftharpoons[g[r]]{f[r]} Z[2r] \xrightarrow{v_{rZ}[r]} Z[r].$$

It is clear that the following diagram is commutative, so that  $f \circ v_{rX} = g \circ v_{rX}$ :

$$\begin{array}{ccc} X[r] & \xrightleftharpoons[g[r]]{f[r]} & Z[2r] \\ v_{rX} \downarrow & & \downarrow v_{rZ}[r] \\ X & \xrightleftharpoons[g]{f} & Z[r]. \end{array}$$

We prove (c'). First observe that, if  $Y$  denotes an object of  $P$  and  $r$  a positive integer, there exists an object  $T$  of  $P$  such that  $Y = T[r]$  and such that, for every object  $X$  of  $P$ , the homomorphism given by translation by  $r$

$$\text{Hom}_P(X, T) \longrightarrow \text{Hom}_P(X[r], Y)$$

is a bijection. The projective system  $(T_n, w_n)$  defined as follows plainly works:

$$T_n = 0 \quad \text{for } n \geq r - 1,$$

and  $T_n = Y_{n-r}$  otherwise, with  $w_n = 0$  for  $n \geq r - 1$  and  $w_n = u_{n-r}$  otherwise.

This said, suppose a diagram of the following type is given, and show that it can be completed as indicated in the statement of (c'):

$$\begin{array}{ccc} X[r] & \xrightarrow{u'} & Y' \\ v_{rX} \downarrow & & \\ X & & \end{array}$$

For this, let  $Y$  be a projective system such that  $Y' = Y[r]$  and such that the canonical map

$$\text{Hom}_P(X, Y) \longrightarrow \text{Hom}_P(X[r], Y')$$

is a bijection. Then it is clear that the following diagram is commutative, where  $u$  denotes the unique morphism from  $X$  to  $Y$  such that  $u' = u[r]$ :

$$\begin{array}{ccc} X[r] & \xrightarrow{u'} & Y' = Y[r] \\ v_{rX} \downarrow & & \downarrow v_{rY} \\ X & \xrightarrow{u} & Y. \end{array}$$

We prove (d'). Consider the commutative diagram below, and show that there exists a morphism  $w$  from  $Y$  such that  $w \circ f = w \circ g$ :

$$X[r] \xrightarrow{v_{rX}} X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} Y.$$

It is first of all clear that the following diagram is commutative:

$$\begin{array}{ccc} X[r] & \begin{array}{c} \xrightarrow{f[r]} \\ \xrightarrow{g[r]} \end{array} & Y[r] \\ v_{rX} \downarrow & & \downarrow v_{rY} \\ X & \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} & Y. \end{array}$$

In particular  $v_{rY} \circ f[r] = v_{rY} \circ g[r]$ . Let now  $T$  be the system associated to  $Y$  and  $r$  as at the beginning of the proof of (c'), so that in particular  $Y = T[r]$ . The translates by  $r$  of the morphisms  $v_{rT} \circ f$  and  $v_{rT} \circ g$  are respectively equal to  $v_{rY} \circ f$  and  $v_{rY} \circ g[r]$ , hence are equal. It follows from the fact that translation by  $r$  is injective on morphisms from  $X$  to  $T$  that the equality below holds, whence the assertion:

$$v_{rT} \circ f = v_{rT} \circ g.$$

**Definition 2.4.2.** Let  $\mathcal{C}$  be an abelian category. The category of projective systems of  $\mathcal{C}$  up to translation, denoted  $\text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ , is the category described as follows:

- (i) The objects of  $\text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$  are the objects of  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ .
- (ii) If  $X$  and  $Y$  are two objects of  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ , a morphism from  $X$  to  $Y$  in the category  $\text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$  is an element of the set

$$\text{Hom}_{AR}(X, Y) = \varinjlim_r \text{Hom}(X[r], Y),$$

where, for every pair  $(r, s)$  of integers with  $s \geq r$ , the map

$$\text{Hom}(X[r], Y) \longrightarrow \text{Hom}(X[s], Y)$$

comes from the canonical morphism

$$X[s] \longrightarrow X[r].$$

The composition of morphisms is defined in the evident way.

In what follows, the category  $\mathcal{C}$  being fixed, we put, to simplify notation,

$$P = \text{Hom}(\mathbb{N}^\circ, \mathcal{C}), \quad P_{AR} = \text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C}).$$

### 2.4.3

We have seen (2.4.1) that the subset  $AR$  of  $\text{Fl}(P)$  admits a right calculus of fractions. This allows us (2.3.3) to consider the category

$$P'_{AR} = P(AR^{-1}),$$

which, recall, has the same objects as  $P$ . Let  $X$  and  $Y$  be two objects of  $P$ , and denote by  $\text{Hom}'_{AR}(X, Y)$  the set of isomorphisms from  $X$  to  $Y$  in  $P'_{AR}$ . An element of  $\text{Hom}_{AR}(X, Y)$  defines in the evident way an element of  $\text{Hom}'_{AR}(X, Y)$ . More precisely, one thus defines a functor

$$m : P_{AR} \longrightarrow P'_{AR}.$$

**Proposition 2.4.3.** The functor  $m$  is an isomorphism of categories.

*Proof.* We must see that, for every pair  $(X, Y)$  of objects of  $P$ , the corresponding map

$$m_{XY} : \text{Hom}_{AR}(X, Y) \longrightarrow \text{Hom}'_{AR}(X, Y)$$

is a bijection. It is plainly a surjection. For injectivity, one is immediately reduced to seeing that, for every commutative diagram of the type below, there exists an integer  $t$  greater than or equal to  $r$  and  $s$  such that the composite of  $\rho$  and the canonical morphism from  $X[t]$  to  $X[s]$  is the canonical morphism from  $X[t]$  to  $X[r]$ :

$$\begin{array}{ccc} & & X[s] \\ & \swarrow \rho & \\ X[r] & & \\ \downarrow v_{rX} & & \\ X & & \end{array}$$

But this follows from commutativity of the diagram below, in which the letter  $c$  denotes all canonical morphisms involved:

$$\begin{array}{ccc}
 X[r+s] & \xrightarrow{\rho(s)} & X[2s] \\
 \downarrow c & \swarrow c & \downarrow c \\
 X[r] & \xleftarrow{\rho} & X[s]
 \end{array}$$

**Proposition 2.4.4.**

- (i) The category  $P_{AR}$  is abelian.
- (ii) The functor  $p_{AR} : P \rightarrow P_{AR}$  is exact.
- (iii) For an object  $X$  of  $P$  to be AR-null, it is necessary and sufficient that  $p_{AR}(X) = 0$ .
- (iv) The pair  $(p_{AR}, P_{AR})$  is a quotient of the abelian category  $P$  by the thick abelian subcategory of AR-null objects.

*Proof.* If (iii) is admitted, the other assertions are consequences of (2.3.5). It follows from (2.3.4) that, for an object  $X$  of  $P$  to be AR-null, it is necessary and sufficient that  $p_{AR}(\text{id}_X) = 0$ , whence (iii).

**Corollary 2.4.5.** Let  $u : X \rightarrow Y$  be a morphism in the category  $P$ . The following assertions are equivalent:

- (i)  $p_{AR}(u)$  is an isomorphism.
- (ii) The kernel and cokernel of  $u$  are AR-null.

A morphism  $u : X \rightarrow Y$  in the category  $P$  of projective systems, satisfying the equivalent conditions (i) and (ii), will be called an AR-isomorphism.

**Definition 2.4.6.** Two objects  $X$  and  $Y$  of the category  $P$  are called AR-isomorphic if  $p_{AR}(X)$  and  $p_{AR}(Y)$  are isomorphic in the category  $P_{AR}$ . If  $X$  and  $Y$  are two AR-isomorphic objects of the category  $P$ , there need not exist in general an AR-isomorphism  $u$  from  $X$  to  $Y$ .

## 2.5. Characterization of projective systems satisfying the Mittag-Leffler-Artin-Rees condition

**Proposition 2.5.1.** The following assertions for an object  $X$  of  $P$  are equivalent:

- (i)  $X$  satisfies (MLAR) (2.1.1).
- (ii) There exists a strict projective system  $Y$  and an AR-isomorphism  $u : Y \rightarrow X$  (2.4.5).

- (iii) There exists a strict projective system  $Y$  such that  $X$  and  $Y$  are isomorphic in the category  $P_{AR}$ .
- (iv)  $X$  satisfies the Mittag-Leffler condition and, denoting by  $X'$  the projective system of universal images of  $X$ , the projective system  $X/X'$  is AR-null.

*Proof.* We shall prove the chain of implications

$$(i) \Rightarrow (iv) \Rightarrow (iii) \Rightarrow (ii) \Rightarrow (i).$$

$(i) \Rightarrow (iv)$ . There exists an integer  $m$  such that the canonical morphism  $w_{mX}$  from  $X[m]$  to  $X$  factors through the canonical inclusion  $i$  of  $X'$  in  $X$ . It follows that  $p_{AR}(i)$  is a direct epimorphism and therefore, since  $p_{AR}$  is exact, that  $p_{AR}(X/X') = 0$ . We conclude by (2.4.4) (iii).

$(iv) \Rightarrow (iii)$ . Condition (iv) means, with the preceding notation, that  $p_{AR}(i)$  is an epimorphism. But because  $p_{AR}$  is exact, it is also a monomorphism, hence an isomorphism. For  $Y$  one takes the projective system  $X'$ .

$(iii) \Rightarrow (ii)$ . The  $P_{AR}$ -isomorphism from  $Y$  to  $X$  whose existence is assumed is the image under  $p_{AR}$  of a  $P$ -morphism  $Y[r] \rightarrow X$ , where  $r$  denotes a positive or zero integer. The conclusion follows from the fact that  $Y[r]$  is strict.

$(ii) \Rightarrow (i)$ . We may suppose  $Y$  contained in  $X$  and  $u$  the canonical injection. Let  $T$  be the cokernel of  $u$ . The hypothesis means that  $T$  is AR-null, hence there exists an integer  $r$  such that, for every  $s \geq r$ , the canonical morphism from  $T[s]$  to  $T$  is zero. It follows that, for  $s \geq r$ , the image of  $X[s]$  in  $X$  is contained in  $Y$ ; since it also plainly contains the image, equal to  $Y$ , of  $Y[s]$  in  $Y$ , it is equal to  $Y$ . Thus  $Y$  is the image of  $X[s]$  for  $s$  sufficiently large, whence  $X$  satisfies (MLAR), and  $Y$  is the projective system of universal images of  $X$ .

**Corollary 2.5.2.** Let  $u : Y \rightarrow X$  be an AR-isomorphism, with  $Y$  strict. Under these conditions,  $X$  satisfies (MLAR) and  $u(Y)$  is the projective system of universal images of  $X$ .

### 3. $J$ -adic and AR- $J$ -adic projective systems

Let  $A$  be a commutative ring with unit, let  $J$  be an ideal of  $A$ , and let  $(\mathcal{C}, \rho)$  be an abelian  $A$ -category. We suppose either that  $J$  is of finite type, or that  $\mathcal{C}$  has infinite direct sums. The notation is the same as in the preceding paragraph. In particular, the symbols  $P$  and  $P_{AR}$  denote respectively the categories  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$  and  $\text{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ .

#### 3.1. $J$ -adic projective systems

**Definition 3.1.1.** Let  $X$  be an object of  $P$ . It is called  $J$ -adic if it satisfies the following conditions:

- (i) for every integer  $n \geq 0$ ,  $J^{n+1}X_n = 0$ ;
- (ii) for every pair  $(m, n)$  of integers with  $m \geq n \geq 0$ , the morphism

$$A/J^{n+1} \otimes_A X_m \longrightarrow X_n$$

deduced from the transition morphism  $X_m \rightarrow X_n$  is an isomorphism.

We shall write  $J\text{-ad}(\mathcal{C})$  for the full subcategory of  $P$  generated by the  $J$ -adic projective systems.

**Remark 3.1.2**

Let  $I$  and  $J$  be two ideals of  $A$  and suppose either that  $I$  and  $J$  are of finite type, or that  $\mathcal{C}$  has infinite direct sums. Then one can show that, if  $I$  and  $J$  define the same topology on  $A$ , the categories  $I\text{-ad}(\mathcal{C})$  and  $J\text{-ad}(\mathcal{C})$  are equivalent.

**Proposition 3.1.3 (Stability under exact sequences).** Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of  $P$ .

- (i) If  $X$  is strict and  $Y$  is  $J$ -adic, then  $Z$  is  $J$ -adic.
- (ii) If  $Y$  is strict and  $Z$  is  $J$ -adic, then  $X$  is strict.
- (iii) If  $X$  and  $Z$  are  $J$ -adic, then  $Y$  is  $J$ -adic.

*Proof.* If  $m \geq n$ , it is clear that the diagram  $(T)$  below, in which the vertical arrows denote the transition morphisms, is commutative and exact:

$$\begin{array}{ccccccc} A/J^{n+1} \otimes_A X_m & \xrightarrow{\text{id} \otimes u_m} & A/J^{n+1} \otimes_A Y_m & \xrightarrow{\text{id} \otimes v_m} & A/J^{n+1} \otimes_A Z_m & \longrightarrow & 0 \\ f_{nm} \downarrow & & g_{nm} \downarrow & & h_{nm} \downarrow & & \\ 0 & \longrightarrow & X_n & \longrightarrow & Y_n & \longrightarrow & Z_n \longrightarrow 0. \end{array}$$

In case (i),  $f_{nm}$  is an epimorphism and  $g_{nm}$  an isomorphism. In case (ii),  $g_{nm}$  is an epimorphism and  $h_{nm}$  an isomorphism. In case (iii),  $f_{nm}$  and  $h_{nm}$  are isomorphisms. In all three cases, the snake lemma applied to the diagram  $(T)$  gives the conclusion.

### 3.2. AR- $J$ -adic projective systems

**Proposition 3.2.1.** The following assertions, for an object  $X$  of  $P$  such that  $J^{n+1}X_n = 0$  for every  $n \geq 0$ , are equivalent:

- (i) There exists a  $J$ -adic projective system  $Y$  and an AR-isomorphism  $u : Y \rightarrow X$  (2.4.5).
- (ii) There exists a  $J$ -adic projective system  $Y$ , isomorphic to  $X$  in the category  $P_{AR}$ .

*Proof.* We have to show that (ii) implies (i). A  $P_{AR}$ -isomorphism from  $Y$  to  $X$  is represented, for some integer  $r$ , by an AR-isomorphism  $v : Y[r] \rightarrow X$ . Since  $J^{n+1}X_n = 0$  for every  $n \geq 0$  and  $Y$  is  $J$ -adic,  $v$  is the composite of a  $P$ -morphism  $u : Y \rightarrow X$  and the canonical morphism from  $Y[r]$  to  $Y$ . It is then clear that  $u$  is an AR-isomorphism, whence the assertion.

**Definition 3.2.2.** A projective system indexed by  $\mathbb{N}$  with values in  $\mathcal{C}$  is called AR- $J$ -adic if:

- (a)  $J^{n+1}X_n = 0$  for every  $n \geq 0$ ;
- (b) it satisfies the equivalent conditions (i) and (ii) of Proposition 3.2.1.

**Proposition 3.2.3.** Let  $X$  be a projective system indexed by  $\mathbb{N}$  with values in  $\mathcal{C}$ . Suppose that  $J^{n+1}X_n = 0$  for every  $n \geq 0$ . Then, in order that  $X$  be AR- $J$ -adic, it is necessary and sufficient that it satisfy (MLAR) and that, denoting by  $X'$  its projective system of universal images, there exist an integer  $r \geq 0$  such that, for every pair  $(m, n)$  of integers with  $m \geq n + r$ , the transition morphism below is an isomorphism:

$$X'_m / J^{n+1}X'_m \longrightarrow X'_{n+r} / J^{n+1}X'_{n+r}.$$

*Proof.*

- (a) The conditions of the statement are sufficient. If  $X$  satisfies (MLAR), it is AR-isomorphic to its projective system of universal images  $X'$ . Moreover, the hypothesis made on  $X'$  means that the projective system

$$(X'_{n+r} / J^{n+1}X'_{n+r})_{n \geq 0},$$

with the evident transition morphisms, is  $J$ -adic. It is clear that this projective system is AR-isomorphic to  $X'$ .

- (b) The conditions of the statement are necessary. If  $X$  is AR- $J$ -adic, it is AR-isomorphic to a strict projective system, hence satisfies (MLAR). The projective system  $X'$  of universal images of  $X$  is AR-isomorphic to  $X$ , hence is AR- $J$ -adic. To prove the second point, it remains to see that, if  $X$  is a strict AR- $J$ -adic projective system, there exists an integer  $r$  such that the projective system  $\ell_r(X)$  defined by

$$\ell_r(X)_n = X_{n+r} / J^{n+1}X_{n+r}$$

is  $J$ -adic. For this, let  $u : Y \rightarrow X$  be an AR-isomorphism, with  $Y$   $J$ -adic. Since the kernel  $T$  of  $u$  is AR-null, let  $r$  be an integer such that the canonical morphism  $T[r] \rightarrow T$  is zero. It is immediate that the canonical morphism

$$\ell_r(u) : \ell_r(Y) \longrightarrow \ell_r(X)$$

is an isomorphism, whence the assertion.

**Proposition 3.2.4 (Stability under exact sequences).** Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of  $P$ , the projective systems  $X$ ,  $Y$ , and  $Z$  being such that

$$J^{n+1}X_n = J^{n+1}Y_n = J^{n+1}Z_n = 0 \quad (n \geq 0).$$

- (i) If  $X$  satisfies (MLAR) and  $Y$  is AR- $J$ -adic, then  $Z$  is AR- $J$ -adic.
- (ii) If  $Y$  satisfies (MLAR) and  $Z$  is AR- $J$ -adic, then  $X$  satisfies (MLAR).

*Proof.* We use freely, for every projective system  $X$  and every integer  $r$ , the notation  $\ell_r(X)$  already made explicit.

Proof of (i). If  $X'$  and  $Y'$  denote the projective systems of universal images of  $X$  and  $Y$ , then  $u$  maps  $X'$  into  $Y'$ , whence the evident exact commutative diagram below, in which the arrows are evident:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X' & \longrightarrow & Y' & \longrightarrow & \tilde{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & 0. \end{array}$$

Since the category  $P_{AR}$  is abelian and the functor  $P \rightarrow P_{AR}$  is exact, it follows from this diagram, and from the fact that the inclusions of  $X'$  and  $Y'$  into  $X$  and  $Y$  respectively are AR-isomorphisms (2.5.1), that the morphism  $\tilde{Z} \rightarrow Z$  is an AR-isomorphism.

This allows us to reduce to the case where  $X$  and  $Y$ , and therefore  $Z$ , are strict. Then there exists an integer  $r$  such that the projective system  $\ell_r(Y)$  is  $J$ -adic. The functor  $\ell_r$  is plainly right exact, whence an exact sequence

$$\ell_r(X) \xrightarrow{\ell_r(u)} \ell_r(Y) \xrightarrow{\ell_r(v)} \ell_r(Z) \longrightarrow 0.$$

After replacing  $\ell_r(X)$  by its image under  $\ell_r(u)$ , we are under the hypotheses for applying (3.1.3 (i)), which shows that  $\ell_r(Z)$  is  $J$ -adic, whence the assertion.

(ii) Let  $Y'$  and  $Z'$  be the projective systems of universal images of  $Y$  and  $Z$ . Consider the evident exact commutative diagram below, in which the two vertical arrows on the right are the canonical inclusions:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \tilde{X} & \longrightarrow & Y' & \longrightarrow & Z' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \longrightarrow & 0. \end{array}$$

It shows that the morphism  $\tilde{X} \rightarrow X$  is an AR-isomorphism, which allows us to reduce to the case where  $Y$  and  $Z$  are strict. In this case, since  $Z$  is strict and AR- $J$ -adic, it follows from 3.2.3 that there exists an integer  $r$  such that  $\ell_r(Z)$  is  $J$ -adic. Considering the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \tilde{X} & \longrightarrow & \ell_r(Y) & \longrightarrow & \ell_r(Z) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \longrightarrow & 0, \end{array}$$

in which, as before,  $\tilde{X} \rightarrow X$  is an AR-isomorphism, allows us to reduce to the case where  $Y$  is strict and  $Z$  is  $J$ -adic. Then (3.1.3 (ii)) gives the conclusion.

## 4. Filtrations and graduations

### 4.1. Generalities

#### 4.1.1

Let  $\mathcal{C}$  be an abelian category and let  $X = (X_n)_{n \in \mathbb{Z}}$  be a projective system indexed by  $\mathbb{Z}$  with values in  $\mathcal{C}$ . Suppose that

$$X_i = 0 \quad (i < 0).$$

For every element  $p$  of  $\mathbb{Z}$ , define a subobject  $F^p X$  of  $X$  in the category  $P_1 = \text{Hom}(\mathbb{Z}^\circ, \mathcal{C})$  by setting

$$(F^p X)_n = F^p X_n = \begin{cases} \text{Ker}(X_n \rightarrow X_{p-1}), & n \geq p-1, \\ 0, & n < p-1, \end{cases}$$

with the transition morphisms defined in the evident way. Thus one obtains a decreasing filtration  $(F^p X)_{p \in \mathbb{Z}}$  of  $X$  satisfying

$$\begin{aligned} F^p X &= X \quad \text{if } p \leq 0, \\ (F^p X)_{p-1} &= 0 \quad \text{for all } p. \end{aligned}$$

Moreover, the snake lemma applied to the exact commutative diagram  $(D)$  below proves that, as soon as  $n \geq p$ , the canonical morphism

$$F^p X_n / F^{p+1} X_n \longrightarrow \text{Im}(X_n \rightarrow X_p) \cap \text{Ker}(X_p \rightarrow X_{p-1})$$

is an isomorphism:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \text{Ker}(X_n \rightarrow X_p) & \longrightarrow & X_n & \longrightarrow & \text{Im}(X_n \rightarrow X_p) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & 0 & \longrightarrow & X_{p-1} & \xrightarrow{\sim} & X_{p-1} & \longrightarrow & 0. \end{array}$$

If now  $X$  is a projective system indexed by  $\mathbb{N}$  with values in  $\mathcal{C}$ , we extend it to a projective system indexed by  $\mathbb{Z}$  by putting  $X_n = 0$  for  $n < 0$ , with the evident transition morphisms, which allows the preceding constructions to be applied to it. This allows us also to consider the category

$$P = \text{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

as a full subcategory of  $P_1$ .

**Definition 4.1.2.** Let  $X$  be a projective system indexed by  $\mathbb{Z}$ . We say that  $X$  is essentially constant if there exists an integer  $p$  such that, for every pair  $(r, s)$  of integers greater than or equal to  $p$  and satisfying  $s \geq r$ , the transition morphism

$$X_s \longrightarrow X_r$$

is an isomorphism. It is clear that it is enough to require this when  $r = p$ .

**Proposition 4.1.3 (The property (MLAR) and associated gradeds).** The following assertions for an object  $X$  of the category  $P$  are equivalent:

- (i)  $X$  satisfies (MLAR).
- (ii) The projective system with values in the category of objects of  $\mathcal{C}$  graded by  $\mathbb{N}$ ,

$$r \longmapsto (\text{gr}^\alpha(X_{\alpha+r}))_{\alpha \in \mathbb{N}},$$

with the evident transition morphisms, is essentially constant.

(iii) The projective system with values in the category of objects of  $\mathcal{C}$  graded by  $\mathbb{N}$ ,

$$n \longmapsto (\mathrm{gr}^p(X_n))_{p \in \mathbb{N}},$$

with the evident transition morphisms, satisfies (MLAR).

*Proof.* (i)  $\Rightarrow$  (ii) follows from the expression of  $F^p X_n / F^{p+1} X_n$  given in 4.1.1.

(ii)  $\Rightarrow$  (iii). Let  $r$  be an integer such that, for every  $s \geq r$  and every  $p$ , the canonical morphism

$$\mathrm{gr}^p(X_{p+s}) \longrightarrow \mathrm{gr}^p(X_{p+r})$$

is an isomorphism. We shall see that, for every pair  $(m, n)$  of integers with  $m \geq n + r$ , the equality below is satisfied for every integer  $p$ :

$$\mathrm{Im}(\mathrm{gr}^p(X_m) \rightarrow \mathrm{gr}^p(X_n)) = \mathrm{Im}(\mathrm{gr}^p(X_{n+r}) \rightarrow \mathrm{gr}^p(X_n)).$$

Indeed, if  $p \geq n + 1$ , then  $\mathrm{gr}^p(X_n) = 0$ ; if  $p \leq n$ , the chain of inequalities  $m \geq n + r \geq p + r$  shows that the canonical morphism

$$\mathrm{gr}^p(X_m) \longrightarrow \mathrm{gr}^p(X_{p+r})$$

is an isomorphism, whence the assertion.

(iii)  $\Rightarrow$  (i). Denote by  $Y$  the projective system of (iii), and let  $r$  be an integer such that

$$\mathrm{Im}(Y[s] \rightarrow Y) = \mathrm{Im}(Y[r] \rightarrow Y) \quad (s \geq r).$$

If  $n$  and  $\ell$  are two integers, with  $\ell \geq n$ , the filtration induced on  $\mathrm{Im}(X_\ell \rightarrow X_n)$  by that of  $X_n$  is also the quotient filtration of that of  $X_\ell$ , and consequently, endowing  $\mathrm{Im}(X_\ell \rightarrow X_n)$  with this filtration, the equality below is satisfied:

$$\mathrm{gr}^p(\mathrm{Im}(X_\ell \rightarrow X_n)) = \mathrm{Im}(\mathrm{gr}^p(X_\ell) \rightarrow \mathrm{gr}^p(X_n)).$$

In particular, condition (iii) is expressed by saying that, as soon as  $\ell \geq n + r$ , the canonical monomorphism

$$j : \mathrm{Im}(X_\ell \rightarrow X_n) \longrightarrow \mathrm{Im}(X_{n+r} \rightarrow X_n)$$

induces an isomorphism on associated graded. Since the gradings are finite, it follows that  $j$  itself is an isomorphism, whence (i).

#### 4.1.4

If  $X$  is an object of  $\mathcal{P}$  satisfying (ML), it follows from the expression of the associated graded recalled in (4.1.1) that, for every integer  $p$ , the projective system

$$\mathrm{gr}^p(X) = (\mathrm{gr}^p(X_n))_{n \in \mathbb{N}}$$

is essentially constant.

Its projective limit, which exists without any hypothesis on the abelian category  $\mathcal{C}$ , will be called the  $p$ -th component of the strict associated graded of  $X$ , and will be denoted

$$\mathrm{gr}^p(X).$$

## 4.2. Projective systems adapted to $J$ and the Artin-Rees lemma

We suppose given a commutative ring  $A$ , an ideal  $J$  of  $A$ , and an abelian  $A$ -category  $\mathcal{C}$ .

**Definition 4.2.1.** An object  $X$  of the category  $\mathcal{P} = \text{Hom}(\mathbb{N}^\circ, \mathcal{C})$  is said to be adapted to  $J$  if its filtration  $(F^p X)_{p \in \mathbb{N}}$  as a projective system (4.1.1) is compatible with the  $J$ -adic filtration of  $X$ , i.e. if for every pair  $(p, q)$  of non-negative integers the inclusion below holds:

$$J^p F^q(X) \subset F^{p+q}(X).$$

**Lemma 4.2.2.** Let  $X$  be an object of the category  $\mathcal{P}$ . The following assertions are equivalent:

- (i)  $X$  is adapted to  $J$ .
- (ii) For every pair  $(n, p)$  of integers with  $p \leq n + 1$ , one has

$$J^p F^{n-p+1}(X_n) = 0.$$

*Proof.* (i)  $\Rightarrow$  (ii): then

$$J^p F^{n-p+1}(X_n) \subset F^{n+1}(X_n) = 0.$$

(ii)  $\Rightarrow$  (i): if  $n < p + q$ ,  $F^{p+q}(X_n) = 0$  by definition. Moreover

$$J^p F^q(X_n)$$

is contained in  $J^{n+1-q} F^q(X_n)$ , and the latter is zero by hypothesis. If  $n > p + q$ , the image of  $J^p F^q(X_n)$  in  $X_{p+q-1}$  under the transition morphism is contained in  $J^p F^q(X_{p+q-1})$ , which is zero by hypothesis. The inclusion

$$J^p F^q(X_n) \subset F^{p+q}(X_n)$$

therefore follows by definition of the filtration of a projective system.

In particular, it follows from (4.2.2) that every projective subsystem of a projective system adapted to  $J$  is adapted to  $J$ .

**Examples.** a) A  $J$ -adic projective system is adapted to  $J$ . It follows that every projective subsystem of a  $J$ -adic projective system is adapted to  $J$ .

b) If a projective system  $X$  satisfies condition (ML) and is adapted to  $J$ , then its projective system of universal images is also adapted to  $J$ .

### 4.2.3. The associated graded of a projective system adapted to $J$

Let  $X$  be a projective system adapted to  $J$ . Since the filtration of  $X$  is compatible with  $J$ , it is clear that the graded object

$$(\text{gr}^p(X))_{p \in \mathbb{N}}$$

is canonically endowed with the structure of a graded  $\mathrm{gr}_J(A)$ -module. In particular, by passage to the limit, one deduces that when  $X$  satisfies condition (ML), the graded object

$$\mathrm{grs}(X) = (\mathrm{grs}^p(X))_{p \in \mathbb{N}}$$

is canonically endowed with the structure of a graded  $\mathrm{gr}_J(A)$ -module.

Let us briefly spell out how, when moreover  $X$  is strict, one defines the corresponding morphisms:

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q}).$$

Since the projective system  $X$  is strict, we know (4.1.1 and 4.1.2) that the following canonical morphism is an isomorphism:

$$\varphi : F^q(X_{p+q})/F^{q+1}(X_{p+q}) \longrightarrow F^q(X_q).$$

Moreover, the fact that the filtration of  $X$  is adapted to  $J$  gives a morphism

$$J^p \otimes_A F^q(X_{p+q}) \longrightarrow F^{p+q}(X_{p+q})$$

which factors, since

$$J^{p+1}F^q(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0, \quad J^pF^{q+1}(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0,$$

through

$$J^p/J^{p+1} \otimes_A (F^q(X_{p+q})/F^{q+1}(X_{p+q}))$$

as a morphism

$$\beta : (J^p/J^{p+1}) \otimes_A (F^q(X_{p+q})/F^{q+1}(X_{p+q})) \longrightarrow F^{p+q}(X_{p+q}).$$

The morphism  $\omega_{pq}$  is then given by the following expression:

$$\omega_{pq} = \beta \circ (\mathrm{id}_{J^p/J^{p+1}} \otimes_A \varphi)^{-1}.$$

**Lemma 4.2.4.** Let  $X$  be a  $J$ -adic projective system. The canonical morphism

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X),$$

coming from the graded  $\mathrm{gr}_J(A)$ -module structure of  $\mathrm{grs}(X)$ , is an epimorphism.

*Proof.* This follows immediately from the description of the morphisms  $\omega_{pq}$  given in (4.2.3).

**Proposition 4.2.5.** Characterization of strict AR- $J$ -adic projective systems by means of their associated gradeds. Let  $X$  be a strict projective system indexed by  $\mathbb{N}$ , with values in  $\mathcal{C}$ . Suppose that for every integer  $n$ , one has

$$J^{n+1}X_n = 0.$$

The following assertions are equivalent:

- (i)  $X$  is AR- $J$ -adic.

(ii) There exists an integer  $r > 0$  such that, for every integer  $n$ , the following inclusion holds:

$$F^n(X_n) \subset J^{n-r}X_n.$$

(iii) There exists an integer  $r$  such that, for every integer  $m$ , the following canonical morphism is an isomorphism whenever  $n \geq m + r$ :

$$\mathrm{gr}^n(J^m X) \longrightarrow \mathrm{gr}^n(X).$$

In the statement of (ii), we agree to set  $J^{n-r}X_n = X_n$  for  $n < r$ .

*Proof.* The equivalence of (ii) and (iii) is clear, since  $X$  and  $J^m X$  are strict. Let us prove the equivalence of (i) and (ii). We know (3.2.3) that (i) is equivalent to the existence of an integer  $r$  such that, for every pair  $(m, n)$  of integers with  $m \geq n + r$ , the transition morphism below is an isomorphism:

$$X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m.$$

Consider the following exact commutative diagram:

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
& & J^{n+1}X_{m+1} & \longrightarrow & J^{n+1}X_m & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F^{m+1}(X_{m+1}) & \longrightarrow & X_{m+1} & \longrightarrow & X_m \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \\
& & X_{m+1}/J^{n+1}X_{m+1} & \longrightarrow & X_m/J^{n+1}X_m & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \\
& & 0 & & 0 & & 
\end{array}$$

It shows the existence of a canonical isomorphism between

$$\mathrm{Ker}(X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m)$$

and

$$F^{m+1}(X_{m+1})/(F^{m+1}(X_{m+1}) \cap J^{n+1}X_{m+1}).$$

In particular, for the transition morphism after reduction modulo  $J^{n+1}$  to be an isomorphism, it is necessary and sufficient that  $F^{m+1}(X_{m+1})$  be contained in  $J^{n+1}X_{m+1}$ , which is precisely assertion (ii).

**Theorem 4.2.6.** Artin-Rees lemma. Let  $X$  be a projective system indexed by  $\mathbb{N}$ , with values in  $\mathcal{C}$ . Suppose that:

- (i)  $X$  satisfies (MLAR);
- (ii)  $X$  is adapted to  $J$ ;
- (iii)  $\mathrm{grs}(X)$  is a graded  $(\mathrm{gr}_J(A), \mathcal{C})$ -module of finite type, i.e. for every increasing sequence  $(M_n)_{n \in \mathbb{N}}$  of graded sub- $\mathrm{gr}_J(A)$ -modules of  $\mathrm{grs}(X)$  whose union is  $\mathrm{grs}(X)$ , there exists an integer  $p$  such that

$$\mathrm{grs}(X) = M_p.$$

Then  $X$  is AR- $J$ -adic.

*Proof.* Recall that if  $X'$  is the projective system of universal images of  $X$ , we have set

$$\text{grs}(X) = \text{gr}(X').$$

Moreover,  $X'$  is adapted to  $J$ , so it remains to show that  $X'$  is AR- $J$ -adic. Suppose that the following lemma has been proved.

**Lemma 4.2.7.** Let  $X$  be a strict projective system adapted to  $J$ . For every integer  $s$ , the graded subobject  $((M_s)_n)_{n \in \mathbb{N}}$  of  $\text{grs}(X)$  defined by the equalities

$$(M_s)_n = J^{n-s} X_n \cap F^n(X_n) \quad (J^i = A \text{ if } i < 0),$$

is a graded  $\text{gr}_J(A)$ -submodule of  $\text{grs}(X)$ .

To prove the theorem, we may suppose  $X$  strict. Then  $\text{grs}(X)$  is the union of the increasing sequence of the  $M_s$ , hence is equal to one of them. This means that there exists an integer  $s_0$  such that

$$F^n(X_n) \subset J^{n-s_0} X_n$$

for every integer  $n$ . We conclude by (4.2.5).

*Proof of 4.2.7.* Return to the notation of (4.2.3). We have to see that the morphism

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q})$$

sends

$$J^p/J^{p+1} \otimes_A (J^{q-s} X_q \cap F^q(X_q))$$

into

$$J^{p+q-s} X_{p+q} \cap F^{p+q}(X_{p+q}).$$

Now the canonical morphism

$$\rho : X_{p+q}/F^{q+1}(X_{p+q}) \longrightarrow X_q$$

is an isomorphism and sends

$$(J^{q-s} X_{p+q} + F^{q+1}(X_{p+q}))/F^{q+1}(X_{p+q})$$

onto  $J^{q-s} X_q$ . It follows that the inverse image under  $\text{id}(J^p/J^{p+1}) \otimes \varphi$  (4.2.3) of

$$J^p/J^{p+1} \otimes_A (J^{q-s} X_q \cap F^q(X_q))$$

is

$$J^p/J^{p+1} \otimes_A \frac{(J^{q-s} X_{p+q} \cap F^q(X_{p+q})) + F^{q+1}(X_{p+q})}{F^{q+1}(X_{p+q})}.$$

It is now clear that the image under  $\beta$  (4.2.3) of this last object is contained in

$$J^{p+q-s} X_{p+q} \cap J^p F^q(X_{p+q}),$$

hence a fortiori in

$$J^{p+q-s} X_{p+q} \cap F^{p+q}(X_{p+q}).$$

This proves the lemma.

## 5. Noetherian $J$ -adic and AR- $J$ -adic projective systems

Throughout this paragraph, a noetherian commutative ring  $A$ , an ideal  $J$  of  $A$ , and an abelian  $A$ -category  $\mathcal{C}$  are fixed.

### 5.1. Generalities

**Definition 5.1.1.** An object  $X$  of the category  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$  is said to be noetherian  $J$ -adic (resp. artinian) if it is  $J$ -adic and if its components are noetherian (resp. artinian) objects of the category  $\mathcal{C}$ .

We shall denote by  $J\text{-adn}(\mathcal{C})$  (resp.  $J\text{-adf}(\mathcal{C})$ ) the full subcategory of  $\mathcal{P} = \text{Hom}(\mathbb{N}^\circ, \mathcal{C})$  whose objects are noetherian (resp. artinian)  $J$ -adic projective systems. It is clear (3.1.3) that an extension of noetherian (resp. artinian)  $J$ -adic projective systems is of the same type.

**Proposition 5.1.2.** Let  $X$  be a  $J$ -adic object of  $\mathcal{P}$ . In order for  $X$  to be noetherian (resp. artinian)  $J$ -adic, it is necessary and sufficient that it be  $J$ -adic and that  $X_0$  be noetherian (resp. artinian).

*Proof.* Indeed, let  $X$  be a  $J$ -adic projective system such that  $X_0$  is noetherian (resp. artinian). Consider the canonical epimorphism (4.2.4):

$$\text{gr}_J(A) \otimes_A X_0 \longrightarrow \text{grs}(X).$$

Since  $X_0$  is noetherian (resp. artinian) and  $J^n$  is of finite type for every  $n$ , the existence of this epimorphism shows that for every integer  $n$ , the kernel of the canonical morphism

$$X_n \longrightarrow X_{n-1}$$

is noetherian (resp. artinian). It follows at once by induction on  $n$  that the  $X_n$  are noetherian (resp. artinian).

**Definition 5.1.3.** Let  $X$  be a projective system indexed by  $\mathbb{N}$ , with values in  $\mathcal{C}$ . It is said to be noetherian (resp. artinian) AR- $J$ -adic if it is AR- $J$ -adic and if its components are noetherian (resp. artinian) objects.

We shall denote by  $\text{AR-}J\text{-adn}(\mathcal{C})$  (resp.  $\text{AR-}J\text{-adf}(\mathcal{C})$ ) the full subcategory of  $\text{Hom}_{\text{AR}}(\mathbb{N}^\circ, \mathcal{C})$  (2.4) generated by the images of noetherian (resp. artinian) AR- $J$ -adic projective systems.

It is then clear that the canonical functors

$$P_{\text{AR}}^n : J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{C}),$$

$$P_{\text{AR}}^f : J\text{-adf}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adf}(\mathcal{C}),$$

obtained by restriction of  $P_{\text{AR}}$  (2.4), are equivalences of categories.

Let us briefly recall the statement and proof of Hilbert's theorem.

**Theorem 5.1.4.** Hilbert's theorem. Let  $A$  be a commutative ring,  $\mathcal{C}$  an abelian  $A$ -category, and  $M$  an object of finite type (resp. noetherian) of  $\mathcal{C}$ . For every  $A$ -algebra of finite type  $B$ , graded by  $\mathbb{N}$ , the graded  $(B, \mathcal{C})$ -module

$$B \otimes_A M$$

is of finite type (resp. noetherian).

N.B. If  $B = \bigoplus_{n \geq 0} B_n$ , then  $B \otimes_A M$  denotes the graded object  $(B_n \otimes_A M)_{n \in \mathbb{N}}$ .

*Proof.* One reduces immediately to the case where  $B$  is the polynomial ring  $A[T]$ , endowed with its usual grading. The  $(B, \mathcal{C})$ -module  $A[T] \otimes_A M = (M_n)_{n \in \mathbb{N}}$  is then obtained by putting  $M_n = M$  for every  $n$ , with  $T$  sending  $M_n$  to  $M_{n+1}$  by the identity of  $M$ , for every integer  $n$ .

Let  $N$  be a graded sub- $(A[T], \mathcal{C})$ -module of  $A[T] \otimes_A M$ . The components  $(N_n)_{n \in \mathbb{N}}$  of  $N$  define an increasing sequence

$$N_0 \subset N_1 \subset \dots \subset N_n \subset \dots$$

of subobjects of  $M$ . Suppose now that  $M$  is noetherian (resp. of finite type), and let

$$N^1 \subset N^2 \subset \dots \subset N^p \subset \dots$$

be an increasing sequence of graded sub- $(A[T], \mathcal{C})$ -modules of  $A[T] \otimes_A M$ , whose union, in the first hypothesis, when  $M$  is of finite type, is  $A[T] \otimes_A M$ .

Consider the following double-entry table:

$$\begin{array}{ccccccc} N_0^1 & \subset & N_0^2 & \subset \dots \subset & N_0^p & \subset \dots \\ \cap & & \cap & & \cap & \\ N_1^1 & & N_1^2 & & N_1^p & \\ \vdots & & \vdots & & \vdots & \\ \cap & & \cap & & \cap & \\ N_n^1 & \subset & N_n^2 & \subset \dots \subset & N_n^p & \subset \dots \end{array}$$

It is clear by hypothesis on  $M$  that there exists a pair  $(p_0, n_0)$  such that  $N_{n_0}^{p_0}$  is maximal (resp. equal to  $M$ ). If  $p_1$  is an integer  $\geq p_0$  such that the  $N_n^{p_1}$  ( $n \leq n_0$ ) are the largest elements of the sequences  $(N_n^p)_{p \in \mathbb{N}}$ , it is clear that  $N^p = N^{p_1}$  as soon as  $p \geq p_1$ , which allows one to conclude in both cases.

**Corollary 5.1.5.** Suppose moreover that the category  $\mathcal{C}$  has denumerable direct sums. Then for every object  $M$  of finite type (resp. noetherian) of  $\mathcal{C}$ , and every  $A$ -algebra of finite type  $B$ , the  $(B, \mathcal{C})$ -module  $B \otimes_A M$  is of finite type (resp. noetherian).

*Proof.* One reduces to the case where  $B = A[T]$ . Let  $N$  be a sub- $(A[T], \mathcal{C})$ -module of  $A[T] \otimes_A M$ , considered as an object of  $\mathcal{C}$ . We shall denote by  $\text{gr}(N)$  the associated graded of  $N$  for the filtration induced by the natural filtration of the graded object  $A[T] \otimes_A M$ :  $\text{gr}(N)$  is a graded sub- $(A[T], \mathcal{C})$ -module of the graded  $(A[T], \mathcal{C})$ -module  $A[T] \otimes_A M$ .

Now let

$$N^0 \subset N^1 \subset \dots \subset N^p \subset \dots$$

be an increasing sequence of sub- $(A[T], \mathcal{C})$ -modules of  $A[T] \otimes_A M$ , whose union is  $A[T] \otimes_A M$  in the first hypothesis. In the first hypothesis ( $M$  of finite type), there exists an integer  $i$  such that  $\text{gr}(N^i) = A[T] \otimes_A M$ , whence

$$N^i = A[T] \otimes_A M.$$

In the second hypothesis ( $M$  noetherian), the sequence of the  $\text{gr}(N^i)$  is stationary, hence the sequence of the  $N^i$  is stationary as well.

In what follows, Hilbert's theorem will be useful to us especially through the following proposition.

**Proposition 5.1.6.** Let  $A$  be a noetherian commutative ring,  $J$  an ideal of  $A$ , and  $\mathcal{C}$  an abelian  $A$ -category. For every noetherian  $J$ -adic projective system  $X$  (5.1.1), the graded object  $\text{gr}_J(X)$  is a noetherian  $(\text{gr}_J(A), \mathcal{C})$ -module.

*Proof.* Since  $\text{gr}_J(A)$  is an  $A$ -algebra of finite type, Hilbert's theorem (5.1.4) implies that

$$\text{gr}_J(A) \otimes_A X_0$$

is a noetherian graded  $(\text{gr}_J(A), \mathcal{C})$ -module. The assertion then follows from the existence of the epimorphism (4.2.4):

$$\text{gr}_J(A) \otimes_A X_0 \longrightarrow \text{gr}_J(X).$$

## 5.2. Structure of $J\text{-adn}(\mathcal{C})$ and $\text{AR-}J\text{-adn}(\mathcal{C})$

In this section, we shall denote by  $\mathcal{E}_{\mathcal{C}}$ , or by  $\mathcal{E}$  if there is no possible confusion, the full subcategory of  $\text{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$  whose objects are noetherian  $\text{AR-}J$ -adic projective systems.

**Proposition 5.2.1.** The category  $\mathcal{E}_{\mathcal{C}}$  is stable under kernels and cokernels in  $\text{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$ . In other words,  $\mathcal{E}_{\mathcal{C}}$  is an abelian category and the inclusion functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \text{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$$

is exact.

*Proof.* Let  $u : X \rightarrow Y$  be a morphism of  $\mathcal{E}_{\mathcal{C}}$ . We shall successively see that the cokernel, the image, and the kernel of  $u$  are noetherian  $\text{AR-}J$ -adic. Of course, the proof that the image of  $u$  is noetherian  $\text{AR-}J$ -adic is only a technical intermediate step.

a)  $\text{Coker}(u)$  is noetherian  $\text{AR-}J$ -adic. Since  $X$  satisfies (MLAR) and  $Y$  is  $\text{AR-}J$ -adic,  $\text{Coker}(u)$  is  $\text{AR-}J$ -adic (3.2.4 (ii)).

b)  $\text{Im}(u)$  is noetherian  $\text{AR-}J$ -adic. From the exact sequence

$$0 \longrightarrow \text{Im}(u) \longrightarrow Y \longrightarrow \text{Coker}(u) \longrightarrow 0,$$

the assertion will follow from Lemma 5.2.1.1 below.

**Lemma 5.2.1.1.** Let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be an exact sequence in the category  $\text{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$ . If  $M$  and  $N$  are noetherian  $\text{AR-}J$ -adic, then  $L$  is noetherian  $\text{AR-}J$ -adic.

*Proof.* It is plainly enough to see that it is  $\text{AR-}J$ -adic. One reduces, in the same way as in the proof of (3.2.4), to the case where  $M$  and  $N$  are strict, and then even noetherian  $J$ -adic. In this case, it follows from 3.1.3 (ii) that  $L$  is strict. The projective system  $L$  is strict, adapted to  $J$ , and its associated graded, as a graded subobject of  $\text{gr}_J(M)$ , is noetherian:  $\text{gr}_J(M)$ , a quotient of  $\text{gr}_J(A) \otimes_A M_0$ , is noetherian (5.1.6). The assertion then follows from the Artin-Rees lemma (4.2.6).

c)  $\text{Ker}(u)$  is noetherian AR- $J$ -adic. It is enough to apply (5.2.1.1) to the evident exact sequence below:

$$0 \longrightarrow \text{Ker}(u) \longrightarrow X \longrightarrow \text{Im}(u) \longrightarrow 0.$$

**Proposition 5.2.2.** The objects of the abelian category  $\text{AR-}J\text{-adn}(\mathcal{C})$  are noetherian.

*Proof.* One reduces immediately to proving the following. Suppose we have a noetherian  $J$ -adic object  $X$ , a sequence of noetherian  $J$ -adic objects  $(Y^n)_{n \in \mathbb{N}}$ , and morphisms in  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$

$$v_n : Y^n \longrightarrow X$$

such that

- (i) the  $v_n$  become monomorphisms in  $\text{AR-}J\text{-adn}(\mathcal{C})$ ;
- (ii) for every integer  $n$ ,  $v_n$  factors through  $v_{n+1}$  in  $\text{AR-}J\text{-adn}(\mathcal{C})$ ,

then the  $v_n$  remain stationary, up to isomorphism, for sufficiently large  $n$ , in  $\text{AR-}J\text{-adn}(\mathcal{C})$ .

Now, to say that  $v_n$  factors through  $v_{n+1}$  in  $\text{AR-}J\text{-adn}(\mathcal{C})$  means that there exists an integer  $r$  and a morphism

$$f : Y^n[r] \longrightarrow Y^{n+1}$$

such that, with  $a$  denoting the canonical morphism

$$Y^n[r] \longrightarrow Y^n,$$

the following equality holds:

$$v_n \circ a = v_{n+1} \circ f.$$

Since the morphism  $a$  is strict, it follows in particular that the image of  $v_n$  is contained in that of  $v_{n+1}$ . Replacing each  $Y^n$  by  $\text{Im}(v_n)$ , we are reduced to seeing that every increasing sequence of strict projective subsystems of  $X$  is stationary. But if  $(Z^n)_{n \in \mathbb{N}}$  is such a sequence, then, since  $\text{gr}_s(X)$  is noetherian, the increasing sequence of the  $\text{gr}_s(Z^n)$  is stationary, hence so is the sequence of the  $Z^n$ .

Summarizing:

**Theorem 5.2.3.** The categories  $J\text{-adn}(\mathcal{C})$  and  $\text{AR-}J\text{-adn}(\mathcal{C})$  are abelian and noetherian.

As for  $\text{AR-}J\text{-adn}(\mathcal{C})$ , this is what we have just proved. As for  $J\text{-adn}(\mathcal{C})$ , it is equivalent to it.

**Proposition 5.2.4.** Stability under extension. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ , with  $J^{n+1}Y_n = 0$  for every  $n$ . Suppose that:

- (i)  $Z$  is AR- $J$ -adic;
- (ii)  $X$  is artinian AR- $J$ -adic.

Then  $Y$  is AR- $J$ -adic.

*Proof.* As in the proof of (3.2.4), one reduces to the case where  $Y$  and  $Z$  are strict and  $Z$  is  $J$ -adic. Then  $X$  is strict by (3.1.3 (ii)). There then exists an integer  $r$  such that the projective system  $(X_{n+r}/J^{n+1}X_{n+r})_{n \in \mathbb{N}}$  is artinian  $J$ -adic. Since  $X$  is a quotient of it and is AR-isomorphic to it, we are reduced to proving the following lemma.

**Lemma 5.2.4.1.** Let

$$0 \longrightarrow T \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ , with  $J^{n+1}Y_n = 0$  for every  $n$ . If  $X$  is artinian  $J$ -adic,  $Z$  is  $J$ -adic, and  $T$  is AR-zero, then  $Y$  is AR- $J$ -adic.

*Proof.* To prove the lemma, we first show that, for every integer  $n$ , the evident projective system

$$(Y_{n+k}/J^{n+1}Y_{n+k})_{k \in \mathbb{N}}$$

is essentially constant. Let  $S_n^k$  denote the kernel of the evident morphism

$$X_{n+k}/J^{n+1}X_{n+k} \longrightarrow Y_{n+k}/J^{n+1}Y_{n+k}.$$

Consider the following evident exact commutative diagram:

$$\begin{array}{ccccccccccc} 0 & \rightarrow & S_n^k & \rightarrow & X_{n+k}/J^{n+1}X_{n+k} & \rightarrow & Y_{n+k}/J^{n+1}Y_{n+k} & \rightarrow & Z_{n+k}/J^{n+1}Z_{n+k} & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & T_n & \rightarrow & X_n & \rightarrow & Y_n & \rightarrow & Z_n & \rightarrow & 0. \end{array}$$

With  $n$  fixed, the objects  $S_n^k$  are identified with subobjects of  $T_n$  and form a decreasing sequence. Since  $T_n$  is artinian, there exists an integer  $k_0$  such that  $S_n^k = S_n^{k_0}$  for every integer  $k \geq k_0$ ; it follows at once that the canonical morphisms

$$Y_{n+k}/J^{n+1}Y_{n+k} \longrightarrow Y_{n+k_0}/J^{n+1}Y_{n+k_0}$$

are isomorphisms.

This being said, denote by  $Y'$  the projective system defined by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}$$

on objects, and in the evident way on morphisms. The projective system  $Y'$  is  $J$ -adic. Moreover, by passage to the limit one obtains an exact commutative diagram of the following type:

$$\begin{array}{ccccccccc} 0 & \rightarrow & S & \rightarrow & X & \rightarrow & Y' & \rightarrow & Z & \rightarrow & 0 \\ & & \downarrow & \downarrow & \downarrow \text{id} & & \downarrow \text{id} & & & & \\ 0 & \rightarrow & T & \rightarrow & X & \rightarrow & Y & \rightarrow & Z & \rightarrow & 0. \end{array}$$

Since  $T$  is AR-zero, the snake lemma applied to this diagram shows that the evident morphism  $Y' \rightarrow Y$  is an AR-isomorphism, whence  $Y$  is AR- $J$ -adic.

**Corollary 5.2.5.** Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ , with  $J^{n+1}Y_n = 0$  for every  $n$ . Suppose that:

- (i)  $X$  is artinian AR- $J$ -adic (resp. artinian and noetherian);
- (ii)  $Z$  is artinian AR- $J$ -adic (resp. noetherian).

Then  $Y$  is artinian AR- $J$ -adic (resp. noetherian).

### 5.3. Noetherian AR- $J$ -adic projective systems and $\delta$ -functors

Let  $\mathcal{C}$  and  $\mathcal{D}$  be two abelian  $A$ -categories, and let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be an additive functor. The functor  $F$  extends in the clear way to a functor, again denoted  $F$ ,

$$F : \text{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \text{Hom}(\mathbb{N}^\circ, \mathcal{D}),$$

which is additive and “commutes with translations”.

**Proposition 5.3.1.** Let  $\mathcal{C}$  and  $\mathcal{D}$  be two abelian  $A$ -categories and

$$T = (T^i)_{i \in \mathbb{N}}$$

an exact  $A$ -linear  $\delta$ -functor from  $\mathcal{C}$  to  $\mathcal{D}$ . Suppose that:

- (i) For every noetherian object  $M$  of  $\mathcal{C}$ , annihilated by a power of  $J$ , and every integer  $n$ ,  $T^n(M)$  is noetherian.
- (ii) For every graded  $A$ -algebra  $B$  of finite type, annihilated by  $J$ , and every noetherian graded  $(B, \mathcal{C})$ -module  $N$ , the graded  $(B, \mathcal{D})$ -modules  $T^n(N)$ ,  $n \in \mathbb{N}$ , are noetherian.

Then, for every noetherian AR- $J$ -adic object  $X$  of  $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ , the  $T^n(X)$  are noetherian AR- $J$ -adic.

*Proof.* Since the functors  $T^n$  commute with translations, it is enough to prove the assertion when  $X$  is noetherian  $J$ -adic. In that case, since  $\text{grs}(X)$  is a noetherian graded  $\text{gr}_J(A)$ -module (5.1.6), it follows from hypothesis (ii) and from a variant of a corollary of Shih’s theorem (cf. Appendix A.3) that for every integer  $n$  the following assertions hold:

$$(a) \quad T^n(X) \text{ satisfies (MLAR),}$$

$$(b) \quad \text{grs } T^n(X) \text{ is a noetherian graded } (\text{gr}_J(A), \mathcal{D})\text{-module.}$$

The Artin-Rees lemma (4.2.6) allows us to conclude.

**Corollary 5.3.2.** The  $\delta$ -functor

$$T = (T^i)_{i \in \mathbb{N}} : \text{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \text{Hom}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact  $A$ -linear  $\delta$ -functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathcal{E}_{\mathcal{D}},$$

commuting with translations, and hence, by passage to the quotient, a new exact  $A$ -linear  $\delta$ -functor, again denoted  $(T^i)_{i \in \mathbb{N}}$ ,

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

## Appendix: Shih's theorem

The purpose of the present appendix is to give a proof of Shih's theorem in the form used in the proof of (5.3.1). This form being more general than the one appearing in EGA 0<sub>III</sub>, a direct proof, adapted from that of loc. cit., has proved necessary.

**Terminology.** Let  $r_0$  be an integer  $\geq 1$ . The datum, for every integer  $r \geq r_0$ , of a bigraded object

$$E_r = (E_r^{pq})_{p \in \mathbb{Z}, q \in \mathbb{Z}},$$

endowed with a differential  $d_r$  of bidegree  $(r, 1 - r)$ , and, for every  $r$ , of an isomorphism

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called below a spectral filter. The notion of a morphism of spectral filters is defined in the evident way.

### A.1. Recollections on the spectral sequences associated with $\delta$ -functors

Let  $\mathcal{C}$  be an abelian category and  $X$  an object of  $\mathcal{C}$ , endowed with a decreasing filtration  $(F^p X)_{p \in \mathbb{Z}}$ . For every exact  $\delta$ -functor  $T$  from  $\mathcal{C}$  to an abelian category  $\mathcal{D}$ , the method of “exact couples” makes it possible to describe a spectral sequence

$$E(X) : \quad E_1^{pq} = T^{p+q}(F^p X / F^{p+1} X) \Rightarrow E^n = T^n(X),$$

for which the filtration of  $E^n$  is defined as follows:

$$F^p(E^n) = \text{Im}(T^n(F^p X) \longrightarrow T^n(X)) = \text{Ker}(T^n(X) \longrightarrow T^n(X/F^p X)).$$

Moreover, if the filtration of  $X$  is finite, the spectral sequence so obtained is weakly convergent (EGA 0<sub>III</sub> 11.1.3).

**Functoriality.** If  $f : X \rightarrow Y$  is a morphism of filtered objects of  $\mathcal{C}$ , one constructs functorially a morphism of spectral sequences

$$E(f) : E(X) \longrightarrow E(Y),$$

coinciding with  $T^{p+q}(\text{gr}^p f)$  (resp.  $T^n(f)$ ) on  $E_1^{pq}(X)$  (resp.  $E^n(X)$ ). In particular, if  $u : X \rightarrow X$  is an endomorphism of  $X$  such that

$$u(F^p X) \subset F^{p+h}(X)$$

for an integer  $h$  independent of  $p$ , then  $E(u)$  is an endomorphism of bidegree  $(h, -h)$  of the spectral filter associated with  $E(X)$ , commuting with the differentials.

### A.2. The case of projective systems

Let now  $X = (X_n)_{n \in \mathbb{Z}}$  be a projective system of objects of an abelian category  $\mathcal{C}$ , such that

$$X_i = 0 \quad (i < 0).$$

If  $T$  denotes an exact  $\delta$ -functor from  $\mathcal{C}$  to an abelian category  $\mathcal{D}$ ,  $T$  extends naturally to an exact  $\delta$ -functor, again denoted  $T$ ,

$$\mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{D}).$$

Thus, endowing  $X$  with its natural filtration as a projective system (EGA 0<sub>III</sub> 13.4.1), one obtains a spectral sequence

$$E(X) = E(X_n)_{n \in \mathbb{Z}},$$

where the  $E(X_n)$  denote the spectral sequences associated with  $T$  and the  $X_n$  endowed with the induced filtration.

**Properties of  $E(X)$ .** a) For every integer  $n$ , the filtration of the abutment  $E^n(X)$  in the spectral sequence  $E(X)$  is identical with its natural filtration as a projective system when  $X$  is strict.

Indeed, denoting by the letter  $F$  the first filtration and by the letter  $\Phi$  the second, one has for every integer  $\alpha$ :

$$\begin{aligned} {}^p F E^n(X)_\alpha &= \mathrm{Ker}(T^n(X_\alpha) \longrightarrow T^n(X_\alpha / F^p X_\alpha)) \\ &= \begin{cases} 0, & \alpha \leq p-1, \\ \mathrm{Ker}(T^n(X_\alpha) \longrightarrow T^n(X_{p-1})), & \alpha \geq p-1. \end{cases} \end{aligned}$$

In both cases one recognizes the expression of  $\Phi^p E^n(X)$ .

This shows in particular that, for  $\beta \geq \alpha \geq p$ , the transition morphism

$$E_\infty^{pq}(X_\beta) \longrightarrow E_\infty^{pq}(X_\alpha)$$

which is identified with

$$\mathrm{gr}_T^p T^{p+q}(X_\beta) \longrightarrow \mathrm{gr}_T^p T^{p+q}(X_\alpha),$$

is a monomorphism, by EGA 0<sub>III</sub> 13.4.1.3.

b) Fixing  $\alpha \in \mathbb{Z}$ , it is clear that, since  $F^i(X_\alpha) = 0$  for  $i \geq \alpha + 1$ , one has

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \quad \text{for } r > p,$$

$$Z_r^{pq}(X_\alpha) = Z_{r+1}^{pq}(X_\alpha) \quad \text{for } r > \alpha - p.$$

c) When  $X$  satisfies the Mittag-Leffler condition, the various projective systems  $\mathrm{gr}^p(X)$  are essentially constant (EGA 0<sub>III</sub> 13.4.3). It follows at once that, for  $r < +\infty$ , the projective systems  $E_r^{pq}(X)$  are essentially constant. We shall denote by

$$\mathcal{E}_r^{pq}(X)$$

their projective limits. Since the differential operators pass to the limit, one thus obtains a spectral filter  $\mathcal{E}_r^{pq}(X)$ ,  $p, q \in \mathbb{Z}$ ,  $r \geq 1$ . The objects

$$\mathcal{Z}_r^{pq}(X), \quad \mathcal{B}_r^{pq}(X) \quad (r < +\infty)$$

associated with this spectral filter are precisely the limits of the essentially constant projective systems  $Z_r^{pq}(X)$  and  $B_r^{pq}(X)$ .

It also follows from part b) that

$$B_\infty^{pq}(X) = B_{p+1}^{pq}(X),$$

which allows one to put

$$\mathcal{B}_\infty^{pq}(X) = \mathcal{B}_{p+1}^{pq}(X).$$

Still supposing that  $X$  satisfies (ML), since, for every  $\alpha$ ,

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \quad (r > p),$$

one has a sequence of canonical monomorphisms

$$\cdots \subset \mathcal{E}_r^{pq}(X) \subset \cdots \subset \mathcal{E}_{p+2}^{pq}(X) \subset \mathcal{E}_{p+1}^{pq}(X). \quad (\text{S})$$

Suppose from now on that  $X$  is strict. The following expressions for  $Z_r^{pq}(X_n)$  and  $B_r^{pq}(X_n)$ :

$$Z_r^{pq}(X_n) = \text{Im} \left[ T^{p+q-1}(F^{p-r+1}X_n/F^pX_n) \xrightarrow{\partial} T^{p+q}(F^pX_n/F^{p+1}X_n) \right],$$

$$B_r^{pq}(X_n) = \text{Im} \left[ T^{p+q}(F^pX_n/F^{p+r}X_n) \longrightarrow T^{p+q}(F^pX_n/F^{p+1}X_n) \right],$$

show that

$$\begin{cases} Z_r^{pq}(X_m) \xrightarrow{\sim} Z_r^{pq}(X_n) & \text{for } m \geq n \geq p, \\ B_r^{pq}(X_m) \xrightarrow{\sim} B_r^{pq}(X_n) & \text{for } m \geq n \geq p+r-1, \end{cases} \quad (d_1)$$

where the canonical morphisms are isomorphisms. In particular, the canonical morphism

$$\mathcal{E}_r^{pq}(X) \longrightarrow E_r^{pq}(X_{p+r-1})$$

is an isomorphism. Moreover, it follows from b) that

$$\begin{cases} Z_\infty^{pq}(X_{p+r-1}) = Z_r^{pq}(X_{p+r-1}) & (r \geq 1), \\ B_\infty^{pq}(X_{p+r-1}) = B_r^{pq}(X_{p+r-1}) & (r > p). \end{cases} \quad (d_2)$$

This makes it possible to define, for every  $r$ , a canonical epimorphism

$$\theta_r^{pq} : \mathcal{E}_r^{pq}(X) \longrightarrow E_\infty^{pq}(X_{p+r-1})$$

which is an isomorphism when  $r > p$ . In addition, if  $a$  is an integer such that

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \quad (r \geq a),$$

for example  $a = p+1$  suffices, so that

$$\cdots \hookrightarrow \mathcal{E}_r^{pq}(X) \hookrightarrow \cdots \hookrightarrow \mathcal{E}_{a+1}^{pq}(X) \hookrightarrow \mathcal{E}_a^{pq}(X), \quad (\text{S bis})$$

then, for  $s \geq r \geq a$ , the diagrams

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow & & \downarrow \text{can} \\ \mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array} \quad (d_3)$$

are commutative.

### A.3. A variant of Shih's theorem

Suppose given a ring  $S$  filtered by ideals  $F^i(S)$ ,  $i \geq 0$ , acting on  $X$  compatibly with its filtration as a projective system.

It follows from the generalities of A.1 that, for every integer  $i$ ,  $\text{gr}^i(S)$  acts on the bigraded objects  $E_r(X)$  by morphisms of bidegree  $(i, -i)$ . In particular, for every integer  $n$ ,

$$E_r^{(n)}(X) = (E_r^{p,q}(X))_{p+q=n}$$

is canonically endowed with the structure of a graded  $(\text{gr}^\bullet(S), \mathcal{D})$ -module. The same holds by passage to the limit for

$$\mathcal{E}_r^{(n)}(X) = (\mathcal{E}_r^{p,q}(X))_{p+q=n}.$$

The ring  $S$ , acting on  $X$  compatibly with its filtration as a projective system, likewise acts on  $T^n(X)$  for every integer  $n$ . Consequently, the strict associated graded  $\text{grs}(X)$  is canonically endowed with the structure of a graded  $(\text{gr}^\bullet(S), \mathcal{C})$ -module, and similarly, when  $T^n(X)$  satisfies (ML), the strict associated graded  $\text{grs } T^n(X)$  is canonically endowed with the structure of a graded  $(\text{gr}^\bullet(S), \mathcal{D})$ -module.

**Theorem A.3.** Suppose that, for an integer  $n$ , the graded  $(\text{gr}^\bullet(S), \mathcal{D})$ -modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\text{grs}^\bullet X) \quad \text{and} \quad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\text{grs}^\bullet X)$$

are noetherian. Then:

- (i)  $T^n(X)$  satisfies (MLAR);
- (ii)  $\text{grs } T^n(X)$  is a noetherian  $(\text{gr}^\bullet(S), \mathcal{D})$ -module.

*Proof.* i) For every pair  $(m, r)$  of integers,

$$\mathcal{B}_r^{(m)}(X) = (\mathcal{B}_r^{p,m-p}(X))_{p \in \mathbb{N}}$$

is a graded sub- $(\text{gr}^\bullet(S), \mathcal{D})$ -module of  $\mathcal{E}_1^{(m)}(X)$ . In particular, for every integer  $m$ , the  $\mathcal{B}_r^{(m)}(X)$  form an increasing sequence, indexed by  $r$ , of graded sub- $(\text{gr}^\bullet(S), \mathcal{D})$ -modules of  $\mathcal{E}_1^{(m)}(X)$ . Using the fact that  $\mathcal{E}_1^{(n)}(X)$  and  $\mathcal{E}_1^{(n-1)}(X)$  are noetherian graded  $(\text{gr}^\bullet(S), \mathcal{D})$ -modules, one deduces that the sequences

$$[\mathcal{B}_r^{(n)}(X)]_{r \in \mathbb{N}} \quad \text{and} \quad [\mathcal{B}_r^{(n-1)}(X)]_{r \in \mathbb{N}}$$

are stationary.

Thus there exists an integer  $a \geq 1$  such that, for every  $(p, q)$ , with  $p + q = n$  or  $n - 1$ ,

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \quad (r \geq a). \tag{A.3.1}$$

Consequently, for  $r \geq a + 1$ , the arrows  $d_r$  with source and target  $\mathcal{E}_r^{pq}(X)$ ,  $p + q = n$ , are zero. In other words, the monomorphisms

$$\mathcal{E}_s^{pq}(X) \hookrightarrow \mathcal{E}_r^{pq}(X) \quad (s \geq r \geq a + 1)$$

deduced from (A.3.1) with  $p + q = n$  are in fact isomorphisms. For  $s \geq r \geq a + 1$ , the commutativity of diagram  $(d_3)$ , with  $p + q = n$ ,

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow \sim & & \downarrow \\ \mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array}$$

shows that the transition morphism

$$E_\infty^{pq}(X_{p+s-1}) \longrightarrow E_\infty^{pq}(X_{p+r-1}),$$

which is in any case a monomorphism (cf. a)), is an epimorphism, hence an isomorphism. In other words, the projective system

$$r \longmapsto \left[ \bigoplus_p \text{gr}^p T^n(X_{p+r}) \right]_{r \geq 1}$$

is constant from  $r = a$  on, and therefore  $T^n(X)$  satisfies (MLAR), by (4.1.3).

ii) Denoting by

$$\mathcal{E}_\infty^{(n)}(X), \quad \mathcal{Z}_\infty^{(n)}(X), \quad \mathcal{B}_\infty^{(n)}(X)$$

the “limits” of the various projective systems indexed by  $r$  considered above, we see that

$$\mathcal{E}_\infty^{(n)}(X) = \mathcal{Z}_\infty^{(n)}(X) / \mathcal{B}_\infty^{(n)}(X)$$

is a noetherian  $(\text{gr}^\bullet(S), \mathcal{D})$ -module, since  $\mathcal{Z}_\infty^{(n)}(X)$  is contained in  $\mathcal{E}_1^{(n)}(X)$ . Moreover, it follows from A.2 a) that

$$\mathcal{E}_\infty^{(n)}(X) = \text{grs } T^n(X).$$